Engineering Mathematics - I · Unit_05 · MATLAB

DRAFT — UE26MA141A · Unit_05 — MATLAB · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_05 · MATLAB

1 UE26MA141A · Unit_05: MATLAB

This unit master organizes source material for Engineering Mathematics - I, Unit_05, officially titled MATLAB. It includes 6 video transcripts and 13 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26MA141A, Unit_05 (MATLAB). Bucket folder present: yes. Videos: 6; slide decks: 13.

EV0000001

2 Introduction to MATLAB Software

This lesson is indexed from the source video titled Introduction to MATLAB Software. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

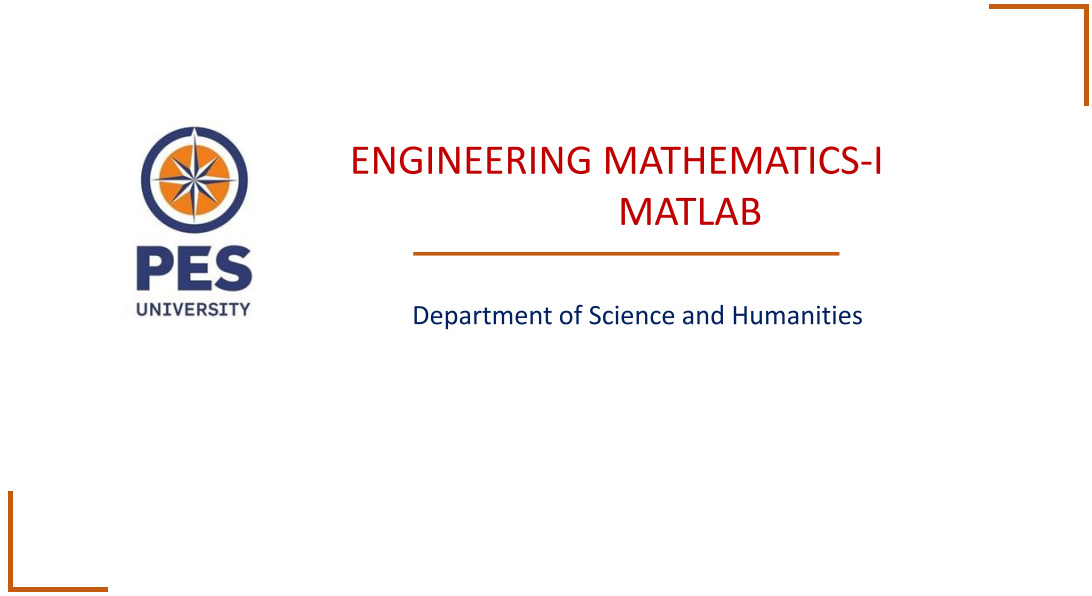

Figure G000001. Slides · source slide page 1.

EV0000111

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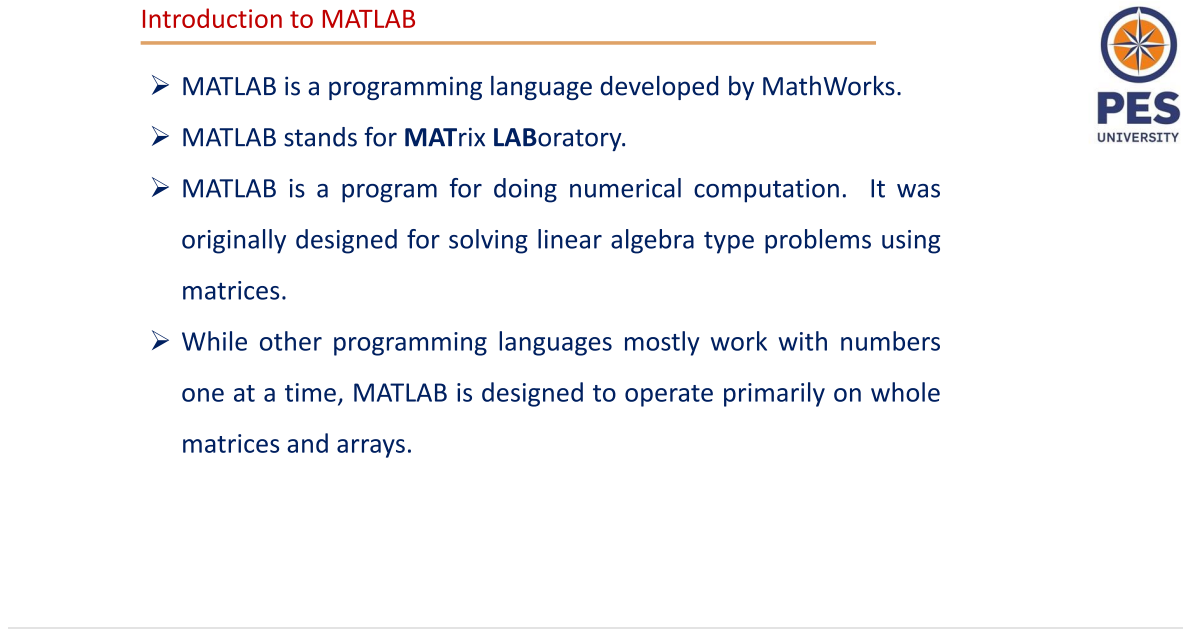

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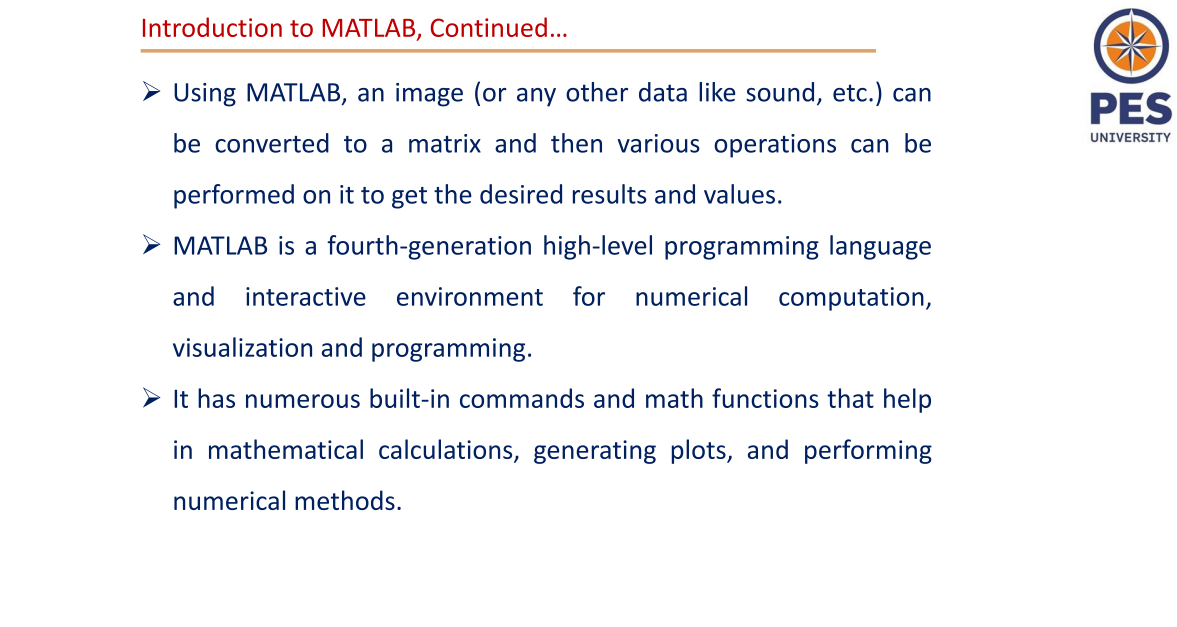

Figure G000003. Slides · source slide page 3.

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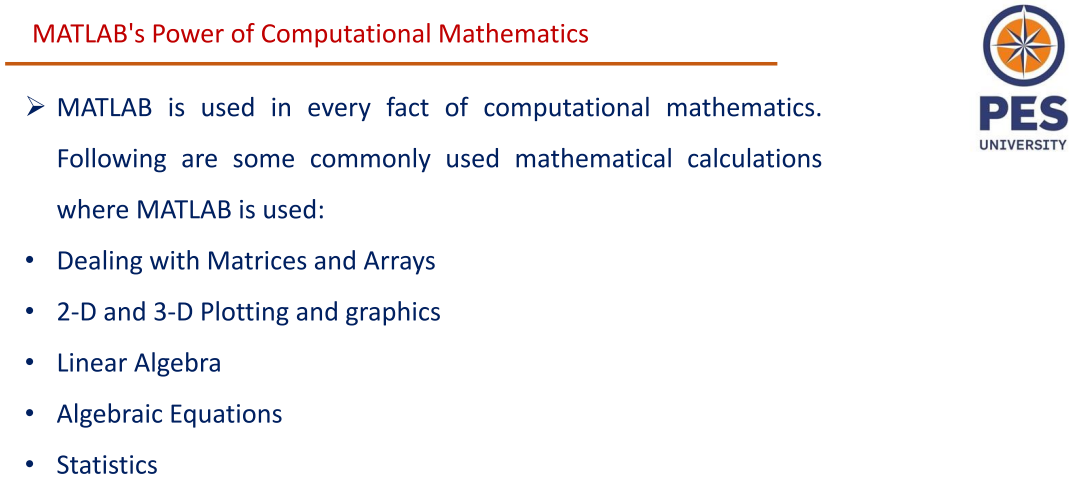

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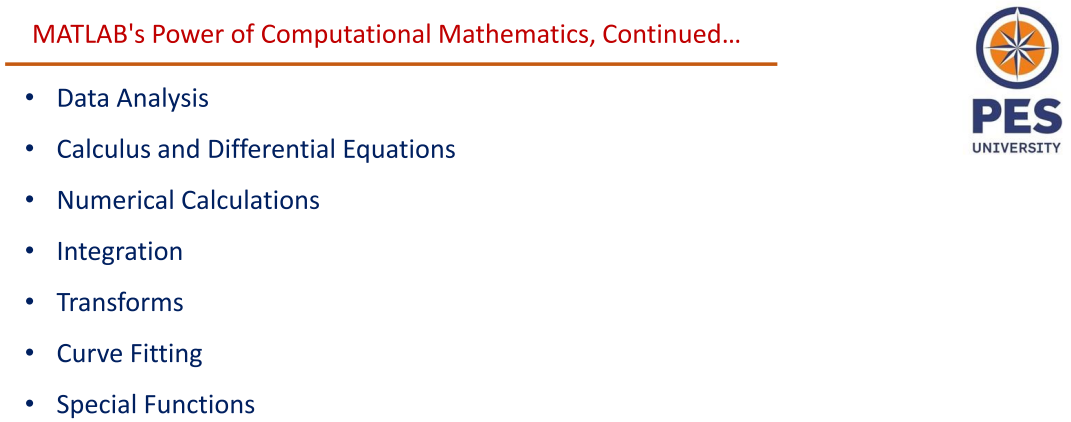

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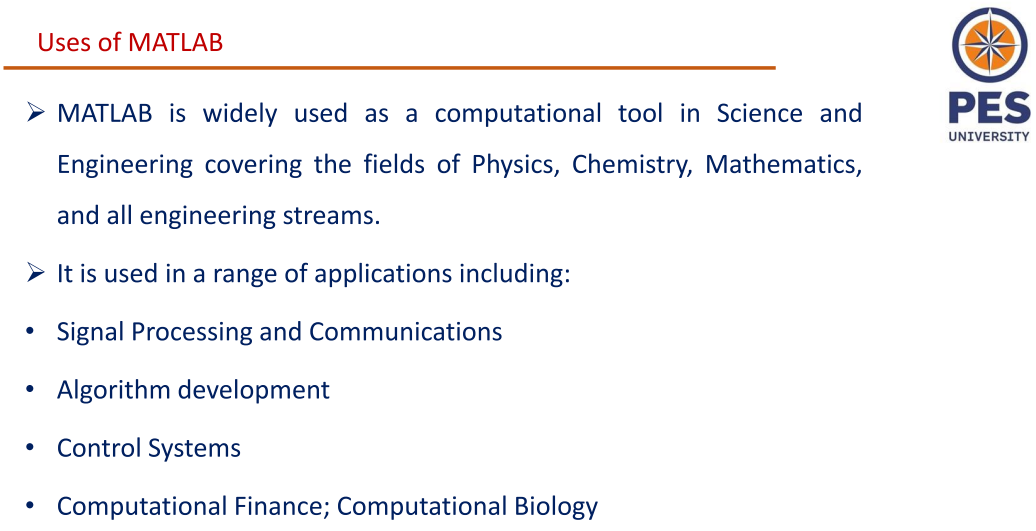

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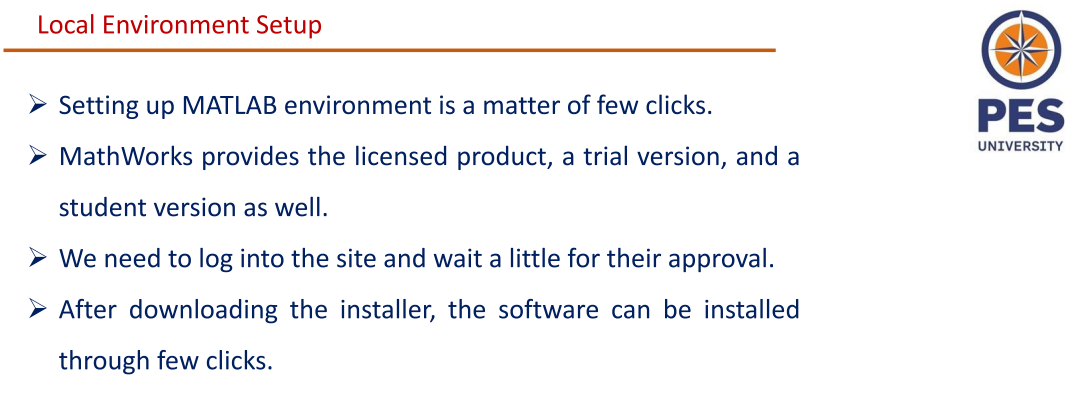

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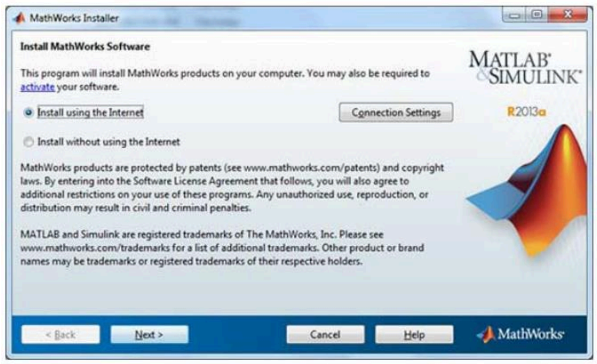
Local Environment Setup, Continued...



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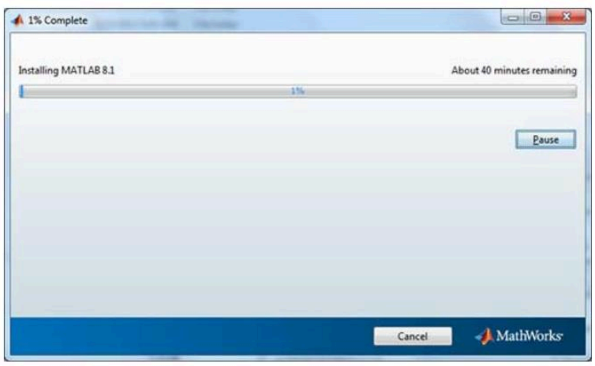

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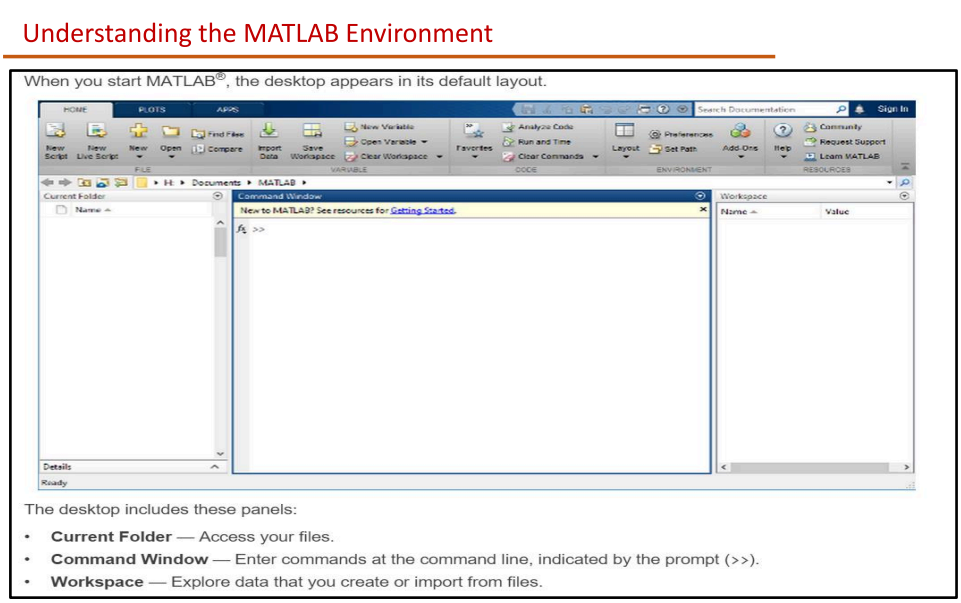

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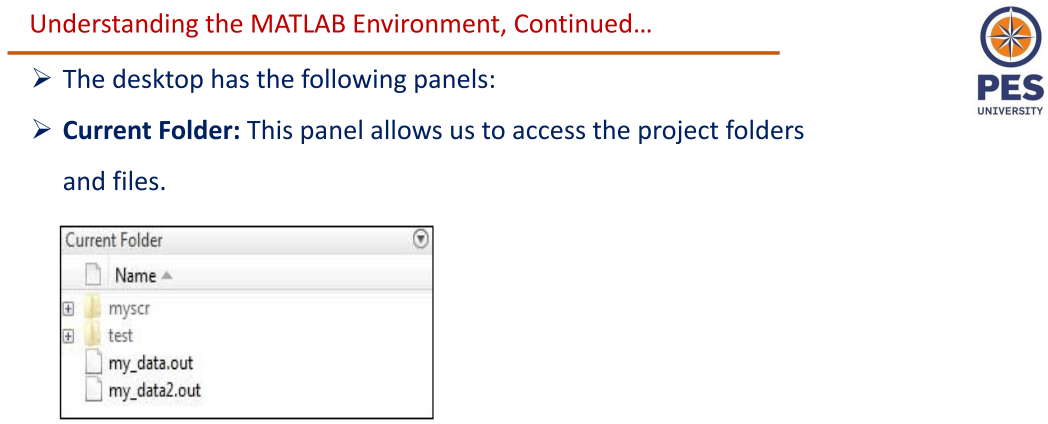

Figure G000011. Slides · source slide page 11.

EV0000141

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Understanding the MATLAB Environment, Continued...

- **Command Window:** This is the main area where commands can be entered at the command line. It is indicated by the command prompt (>>).

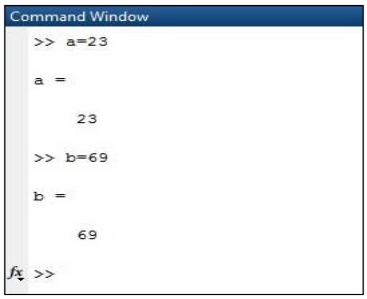

Figure G000012. Slides · source slide page 12.

EV0000144

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Understanding the MATLAB Environment, Continued...

- **Workspace:** The workspace shows all the variables created and/or imported from files.

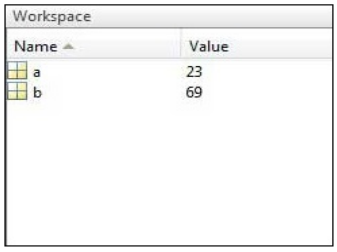

Figure G000013. Slides · source slide page 13.

EV0000147

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Understanding the MATLAB Environment, Continued...

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- **Command History:** This panel shows or return commands that are entered at the command line.

```
%-- 7/14/2013 5:58 PM --%
%-- 7/15/2013 9:01 AM --%
  simulink
%-- 7/15/2013 6:09 PM --%
  simulink
%-- 7/25/2013 7:57 AM --%
%-- 7/25/2013 7:58 AM --%
  chdir test
  prog4
%-- 7/29/2013 8:55 AM --%
  a=23
  b=69
```

Figure G000014. Slides · source slide page 14.

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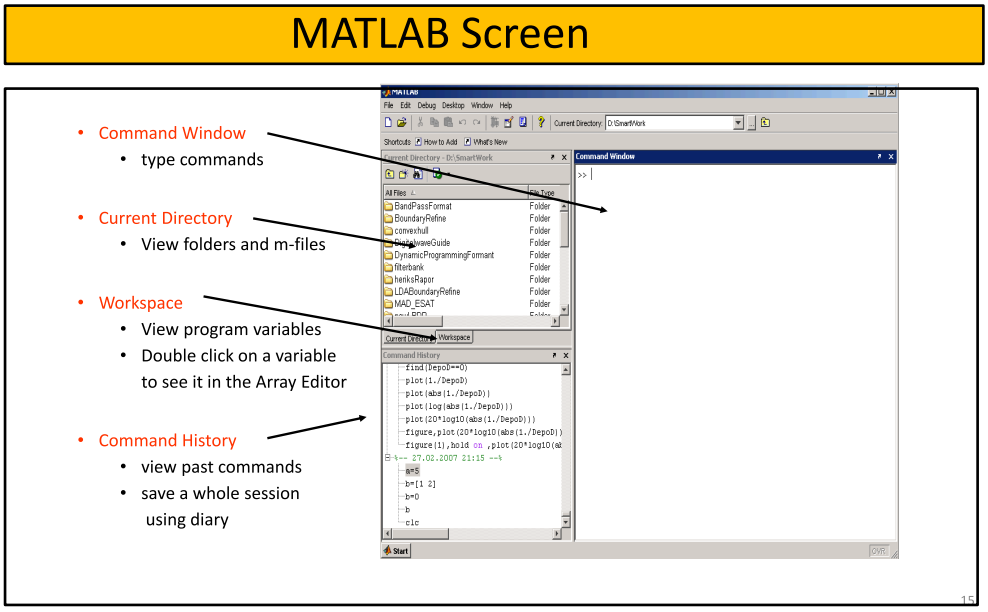

Figure G000015. Slides · source slide page 15.

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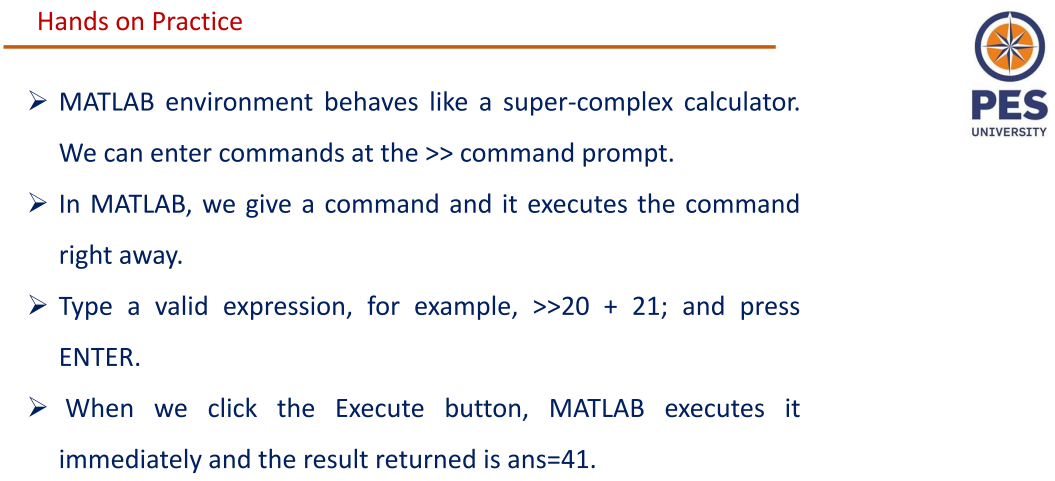

Figure G000016. Slides · source slide page 16.

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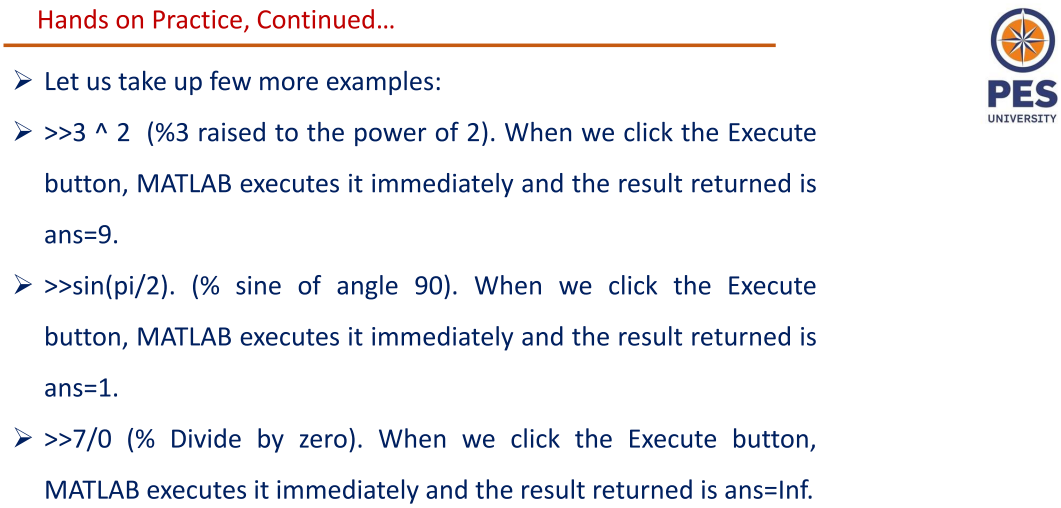

Figure G000017. Slides · source slide page 17.

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Hands on Practice, Continued…

- >>732 * 20.3. When we click the Execute button, MATLAB executes it immediately and the result returned is ans=1.4860e+04.
- MATLAB provides some special expressions for some mathematical symbols, like pi for π, Inf for ∞, i (and j) for $\sqrt{-1}$.
- In MATLAB, **NaN** stands for 'not a number'.
  For example, >> 0/0; -inf/inf

Figure G000018. Slides · source slide page 18.

EV0000162

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Use of Semicolon (;) in MATLAB

- Semicolon (;) indicates end of statement. However, if we want to suppress and hide the MATLAB output for an expression, add a semicolon after the expression.
- For example, >>x = 5; y = x + 5. When we click the Execute button, MATLAB executes it immediately and the result returned is y=8.

Figure G000019. Slides · source slide page 19.

EV0000165

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Adding Comments in MATLAB

- To add comments to MATLAB code, we use the percent (%) symbol. Comment lines can appear anywhere in a program file, and we can add comments to the end of a line of code.

  For example, y = sum(x) % Use the sum function

Figure G000020. Slides · source slide page 20.

EV0000168

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Commonly used Operators and Special Characters

Operation	Symbol	Example
Addition	+	5+3=8
Subtraction	-	5-3=2
Multiplication	*	5*3=15
Right Division	/	5/3=1.6667
Left Division	\	5/3=3\5=1.667
Exponentiation	^	5^3=125

Figure G000021. Slides · source slide page 21.

EV0000171

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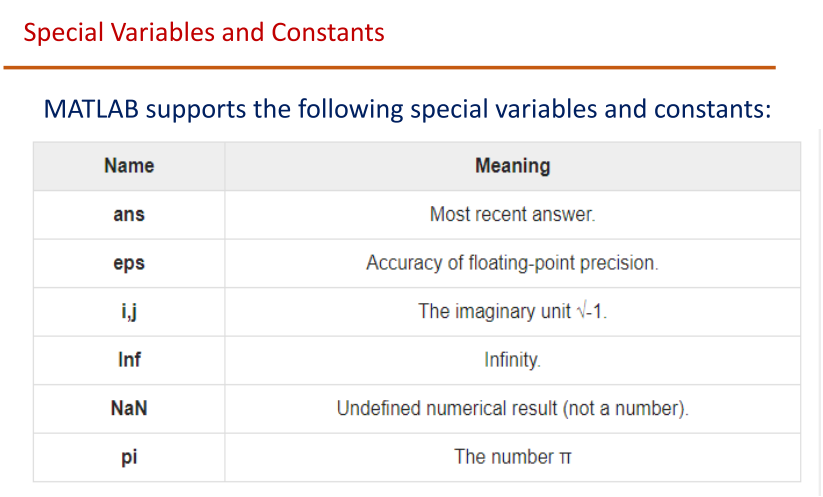

Figure G000022. Slides · source slide page 22.

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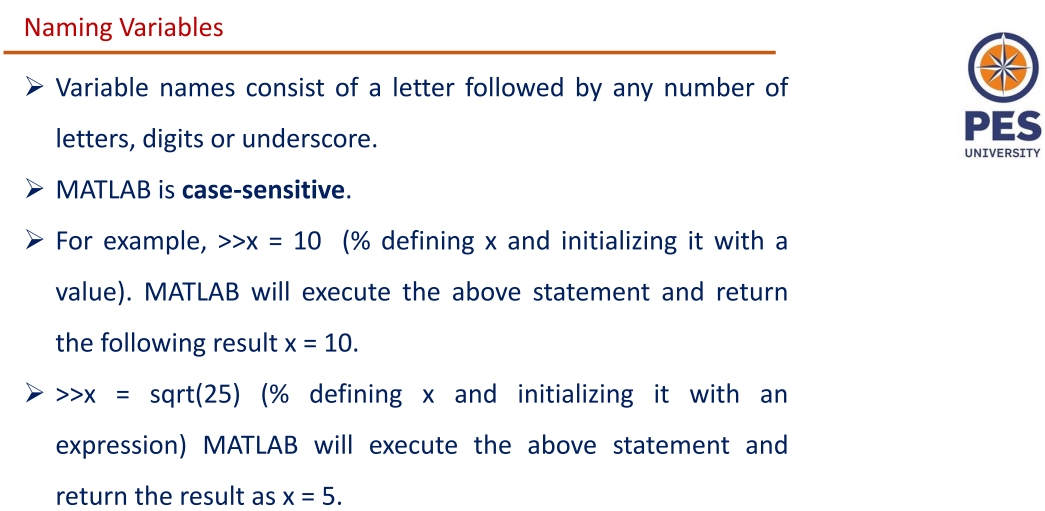

Figure G000023. Slides · source slide page 23.

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Naming Variables, Continued…

- Note that once a variable is entered into the system, we can refer to it later. Variables must have values before they are used. When an expression returns a result that is not assigned to any variable, the system assigns it to a variable named answer, which can be used later.
- For example, >>x = 7 * 8; y = x * 7.89. MATLAB will execute the above statement and return the following result y = 441.8400.

Figure G000024. Slides · source slide page 24.

EV0000180

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Multiple assignments

- We can have multiple assignments on the same line.
- For example, >> a = 2; b = 7; c = a * b
- MATLAB will execute the above statement and return the result c = 14.

Figure G000025. Slides · source slide page 25.

EV0000183

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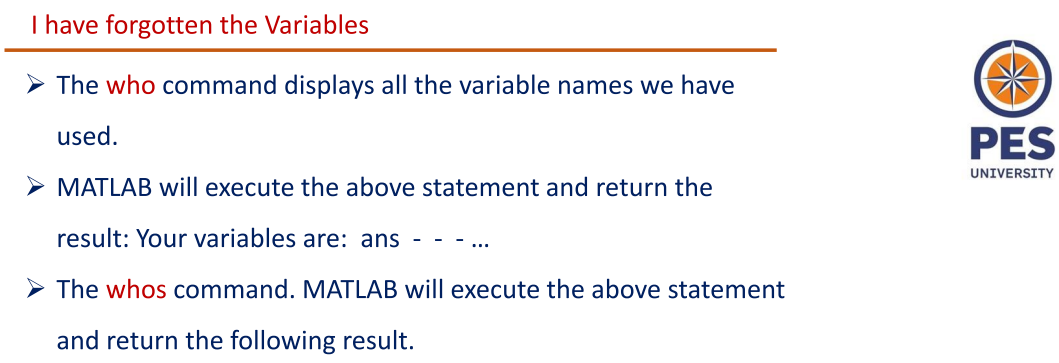

Figure G000026. Slides · source slide page 26.

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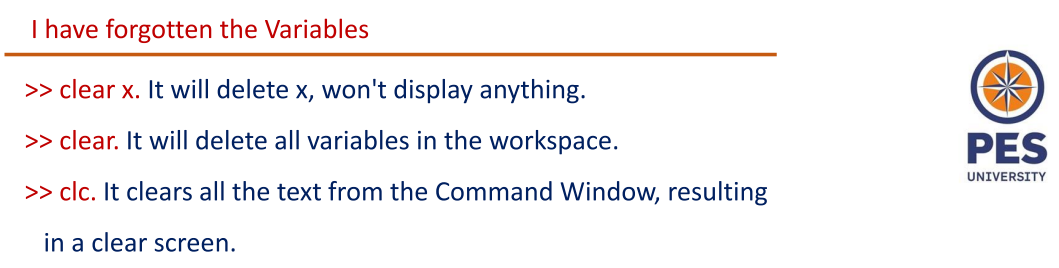

Figure G000027. Slides · source slide page 27.

EV0000189

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Long Assignments

- Long assignments can be extended to another line as follows:

```
>> initial_velocity = 0;
>> acceleration = 9.8;
>> time = 20;
>> Final_velocity = initial_velocity + acceleration * time
```

- MATLAB will execute the above statement and return the following result: final velocity = 196.

Figure G000028. Slides · source slide page 28.

EV0000192

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The format Command

- By default, MATLAB displays numbers with four decimal place values. This is known as **short format**.
- However, if we want more precision, then we need to use the **format** command.
- The **format long** command displays 15 digits after the decimal point.
- For example: >> format long

```
>> x = 7 + 10/3 + 5 ^ 1.2
```

- MATLAB will execute the above statement and return the following result: x = 17.231981640639408.

Figure G000029. Slides · source slide page 29.

EV0000195

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The format Command, Continued...

Another example,

```
>> format short
>> x = 7 + 10/3 + 5 ^ 1.2
```

- MATLAB will execute the above statement and return the following result: x = 17.232
- The **format bank** command rounds numbers to two decimal places.

Figure G000030. Slides · source slide page 30.

EV0000198

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The format Command, Continued...

For example,

```
>> format bank
>> daily_wage = 177.45; weekly_wage = daily_wage * 6
```

- MATLAB will execute the above statement and return the following result: weekly_wage = 1064.70
- MATLAB displays large numbers using exponential notation.
- The **format short e** command allows displaying in exponential form with four decimal places plus the exponent.

Figure G000031. Slides · source slide page 31.

EV0000201

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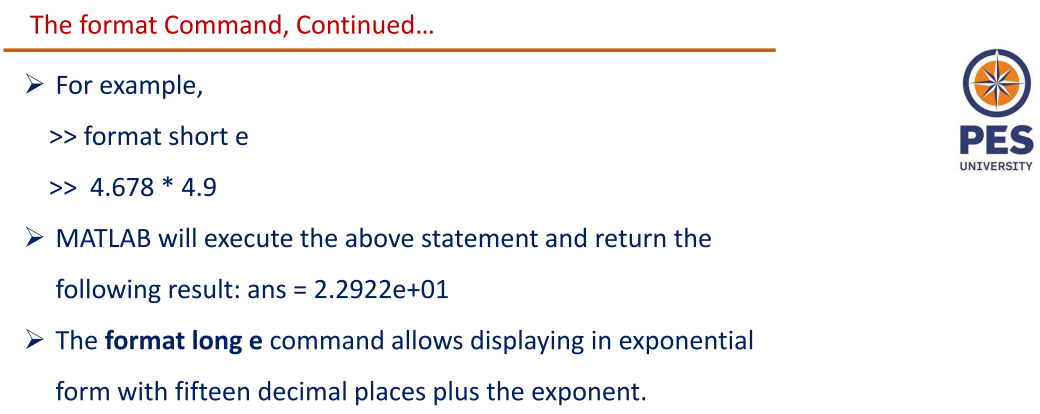

Figure G000032. Slides · source slide page 32.

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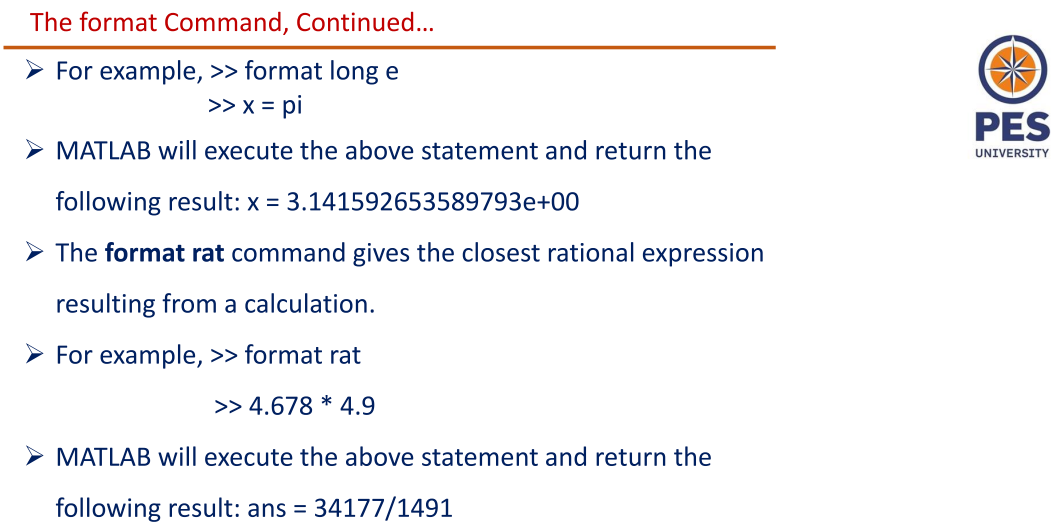

Figure G000033. Slides · source slide page 33.

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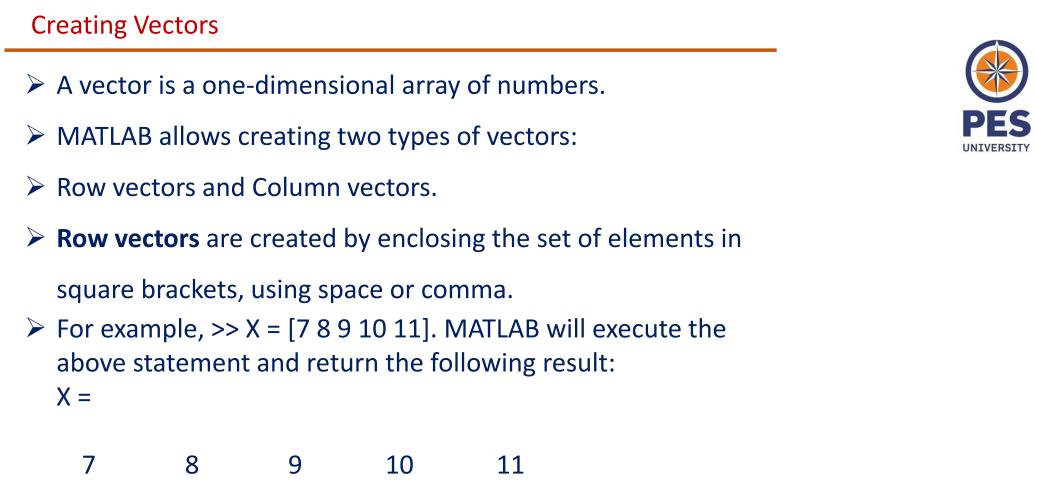

Figure G000034. Slides · source slide page 34.

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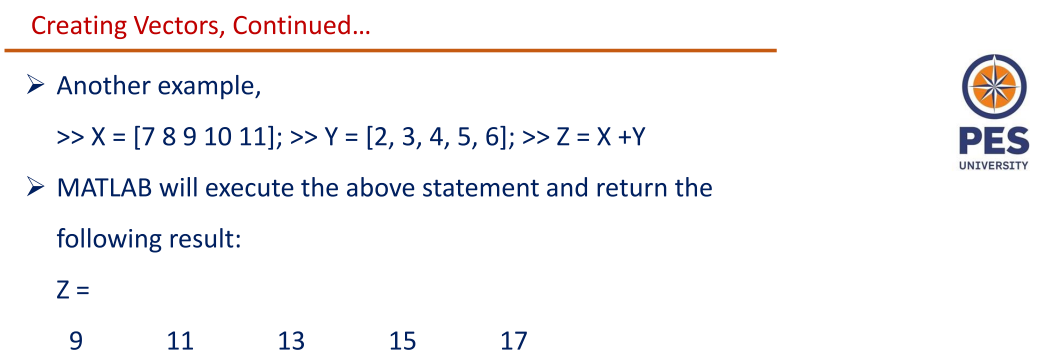

Figure G000035. Slides · source slide page 35.

EV0000213

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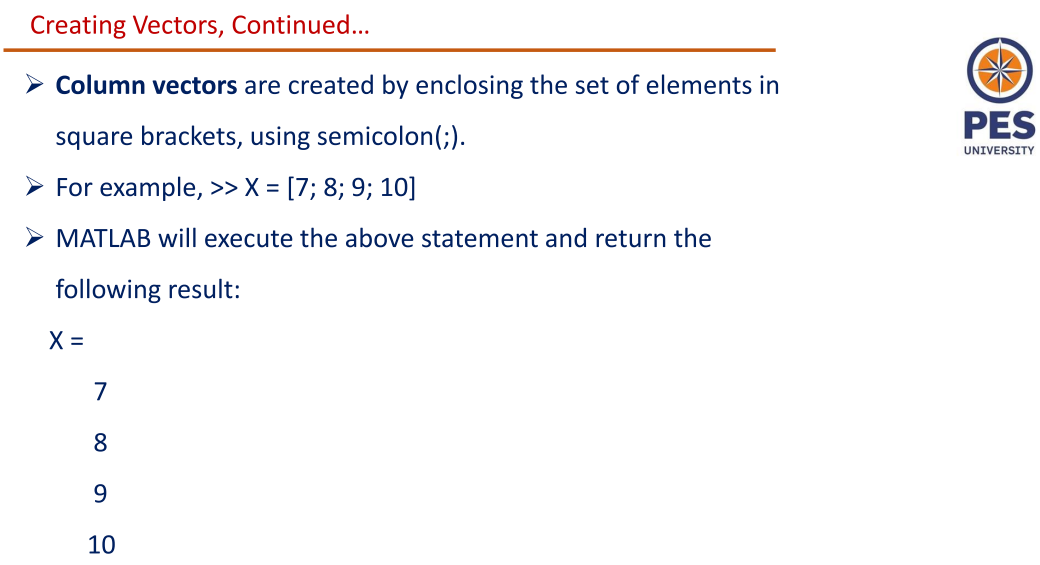

Figure G000036. Slides · source slide page 36.

EV0000216

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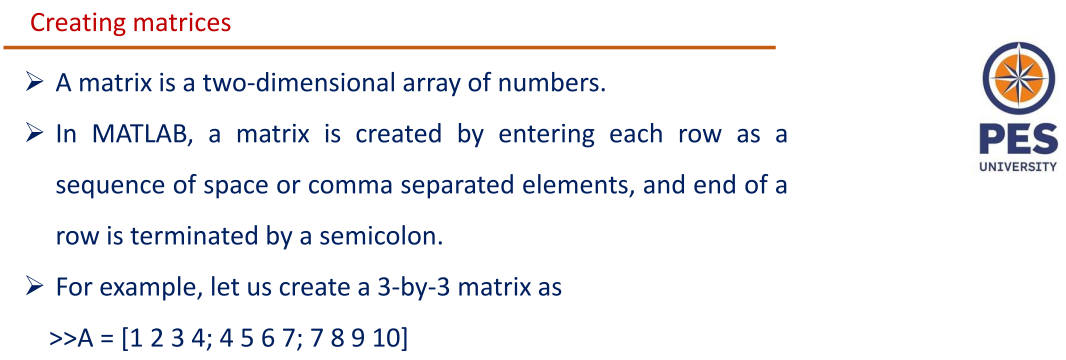

Figure G000037. Slides · source slide page 37.

EV0000219

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Creating matrices, Continued…

- MATLAB will execute the above statement and return the following result:

A =

1	2	3	4
4	5	6	7
7	8	9	10

Figure G000038. Slides · source slide page 38.

EV0000222

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Elementary Math Built – In Functions

Function	Description	Examples
sqrt(x)	Square root	>>sqrt(81) ans=9
nthroot(x,n)	Real nth root of a real number x. (If x is negative n must be an odd integer).	>>nthroot(8,3) ans=2
exp(x)	Exponential(e^x)	>>exp(5) ans=1.484131591025766e+02
abs(x)	Absolute value	>>abs(-24) ans=24
log(x)	Natural logarithm. Base e logarithm (ln)	>>log(1000) ans=6.907755278982137e+00
log10(x)	Base 10 logarithm	>>log10(1000) ans=3
factorial(x)	The factorial function x! (x must be a psitive integer)	>>factorial(5) ans=120

Figure G000039. Slides · source slide page 39.

EV0000225

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Trigonometric Math Functions		
Function	**Description**	**Examples**
sin(x) sind(x)	Sine of angle x (x in radians) Sine of angle x (x in degrees)	>>sin(pi/6) ans=0.5000 >>sind(30) ans=0.5000
cos(x) cosd(x)	Cosine of angle x (x in radians) Cosine of angle x (x in degrees)	>>cos(0.5) ans=0.8776 >>cosd(30) ans=0.8660
tan(x) tand(x)	Tangent of angle x (x in radians) Tangent of angle x (x in degrees)	>>tan(pi/6) ans=0.5774 >>tand(45) ans=1
cotx) cotd(x)	Cotangent of angle x (x in radians) Cotangent of angle x (x in degrees)	>>cot(0.5) ans=1.8305 >>cotd(90) ans=1
asin(x) asind(x)	Inverse of sine of angle x (x in radians) Inverse of sine angle x (x in degrees)	>>asin(0.5) ans=0.5236 >>asin(0.8660) ans=59.9971
acos(x) acosd(x)	Inverse of cosine of angle x (x in radians) Inverse of cosine angle x (x in degrees)	>>acos(0.3) ans=1.2661 >>acosd(0.5) ans=60

Figure G000040. Slides · source slide page 40.

EV0000228

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Rounding Functions		
Function	**Description**	**Examples**
round(x)	Round to the nearest integer	>>round(25/3) ans=8 >>round(4.49) ans=4 >>round(4.5) ans=5 >>round(4.9999) ans=5
fix(x)	Round towards zero. In other words, chops off the fraction part.	>>fix(17/5) ans=3 >>fix(4.49) ans=4 >>fix(4.9999) ans=4 >>fix(-4.6674) ans=-4
ceil(x)	Round towards positive infinity.	>>ceil(27/2) ans=14 >>ceil(2.1) ans=3 >>ceil(2.9) ans=3 >>ceil(-4.99) ans=-4
floor(x)	Round towards minus infinity	>>floor(-9/4) ans=-3 >>floor(2.1) ans=2 >>floor(2.99) ans=2
rem(x,y)	Returns the remainder after x is divided by y	>>rem(13,5) ans=3

Figure G000041. Slides · source slide page 41.

EV0000231

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Examples for Arithmetic operators

1) Calculate the following using MATLAB.

$$\frac{1}{2+3^2}+\frac{4}{5}*\frac{6}{7}$$

```
>>a=1/(2+3^2)
a=
    0.0909
>>b=4/5
b=
     0.8000
>>c=6/7
c=
    0.8571
>>a+b*c
ans=
      0.7766
```

Figure G000042. Slides · source slide page 42.

EV0000233

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2. Find the value of $y = e^{-a}\sin(x) + 10\sqrt{y}$ *if* $a = 5, x = 2, y = 8$.

```
>>a=5;  x=2;   y=8;
y=exp(-a)*sin(x)+10*sqrt(y)
y=
    2.829039804533026e+013.
```

3. Compute $\sin\left(\frac{\pi}{4}\right), e^{10}$.

```
>>sin(pi/4)
ans=0.7071
>>exp(10)
ans=2.2026e+004
```

Figure G000043. Slides · source slide page 43.

EV0000234

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Reference:

1. Getting started with MATLAB, Rudra Pratap, Oxford University Press, 7th Edition, 2016.
2. https://www.tutorialspoint.com/matlab/matlab_data_import.htm

Figure G000044. Slides · source slide page 44.

EV0000235

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

THANK YOU

Figure G000045. Slides · source slide page 45.

EV0000236

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000110 · EV0000111 · EV0000112
2. Source slide page 2 (extracted text): Introduction to MATLAB ➢MATLAB is a programming language developed by MathWorks. ➢MATLAB stands for MATrix LABoratory. ➢MATLAB is a program for doing numerical computation. It was originally designed for solving linear algebra type problems using matrices. ➢While other programming languages mostly work with numbers one at a time, MATLAB is designed to operate primarily on whole matrices and arrays. EV0000113 · EV0000114 · EV0000115
3. Source slide page 3 (extracted text): Introduction to MATLAB, Continued... ➢Using MATLAB, an image (or any other data like sound, etc.) can be converted to a matrix and then various operations can be performed on it to get the desired results and values. ➢MATLAB is a fourth-generation high-level programming language and interactive environment for numerical computation, visualization and programming. ➢It has numerous built-in commands and math functions that help in mathematical calculations, generating plots, and performing numerical methods. EV0000116 · EV0000117 · EV0000118
4. Source slide page 4 (extracted text): MATLAB's Power of Computational Mathematics ➢MATLAB is used in every fact of computational mathematics. Following are some commonly used mathematical calculations where MATLAB is used: • Dealing with Matrices and Arrays • 2-D and 3-D Plotting and graphics • Linear Algebra • Algebraic Equations • Statistics EV0000119 · EV0000120 · EV0000121
5. Source slide page 5 (extracted text): MATLAB's Power of Computational Mathematics, Continued... • Data Analysis • Calculus and Differential Equations • Numerical Calculations • Integration • Transforms • Curve Fitting • Special Functions EV0000122 · EV0000123 · EV0000124
6. Source slide page 6 (extracted text): Uses of MATLAB ➢MATLAB is widely used as a computational tool in Science and Engineering covering the fields of Physics, Chemistry, Mathematics, and all engineering streams. ➢It is used in a range of applications including: • Signal Processing and Communications • Algorithm development • Control Systems • Computational Finance; Computational Biology EV0000125 · EV0000126 · EV0000127

7. Source slide page 7 (extracted text): Local Environment Setup ➢Setting up MATLAB environment is a matter of few clicks. ➢MathWorks provides the licensed product, a trial version, and a student version as well. ➢We need to log into the site and wait a little for their approval. ➢After downloading the installer, the software can be installed through few clicks. ## Page / slide 8 (OCR) Local Environment Setup, Continued... PES MathWorksInstaller ☐ UNIVERSITY InstallMathWorksSoftware MATLAB' This program willinstall MathWorks products onyour computer.Youmayalso berequired to SIMULINK' activateyoursoftware. Installusing theIntenet Connection Settings R2013a Install withoutusing theInternet MathWorksproducts areprotectedby patents(seewww.mathworks.com/patents) andcopyright laws.By entering into theSoftware LicenseAgreement thatfollows,youwillalso agreeto additionalrestrictionsonyouruse of theseprograms.Anyunauthorized use,reproduction,or distribution may result in civl and criminal penalties. MATLABandSimulink areregistered trademarks of TheMathWorks,Inc.Please see www.mathworks.com/trademarksforalistofadditional trademarks.Otherproductorbrand namesmay be trademarks orregistered trademarksof theirrespective holders. Back Not> Cancel Help MathWorks ## Page / slide 9 (OCR) Local Environment Setup, Continued... PES 1%Complete UNIVERSITY InstallingMATLAB8.1 About40minutesremaining Pause Cancel MathWorks ## Page / slide 10 (OCR) Understanding the MATLAB Environment WhenyoustartMATLAB?,thedesktopappearsinitsdefaultlayout. PES HONE PLOTS APOS Sign In UNIVERSITY ewVaatte AmheCocs Cortunly 1ew Runane Open RequestSuppot New New Comoore rport Save raverdes Layout SetPatn Add.0ns Hep Sorot Lve Serpn Dero Wodapice CieRr Cicar Leom WATLAB Doruments MATLAB Currcnt Folder ComenandWndow Workeosce Nai Newto MATLAB? SeerescurcesforGetingSated nmc valuc Detsils Ready Thedesktopincludesthesepanels: CurrentFolder-Accessyourfiles. CommandWindow—Entercommandsatthecommand line,indicatedbytheprompt(>>). Workspace-Exploredatathatyoucreateorimportfromfiles. EV0000128 · EV0000129 · EV0000130
8. Source slide page 8: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000131 · EV0000132 · EV0000133
9. Source slide page 9: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000134 · EV0000135 · EV0000136
10. Source slide page 10: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000137 · EV0000138 · EV0000139
11. Source slide page 11 (extracted text): Understanding the MATLAB Environment, Continued... ➢The desktop has the following panels: ➢Current Folder: This panel allows us to access the project folders and files. EV0000140 · EV0000141 · EV0000142
12. Source slide page 12 (extracted text): Understanding the MATLAB Environment, Continued... ➢Command Window: This is the main area where commands can be entered at the command line. It is indicated by the command prompt (>>). EV0000143 · EV0000144 · EV0000145
13. Source slide page 13 (extracted text): Understanding the MATLAB Environment, Continued... ➢Workspace: The workspace shows all the variables created and/or imported from files. EV0000146 · EV0000147 · EV0000148
14. Source slide page 14 (extracted text): Understanding the MATLAB Environment, Continued... ➢Command History: This panel shows or return commands that are entered at the command line. EV0000149 · EV0000150 · EV0000151

15. Source slide page 15 (extracted text): MATLAB Screen • Command Window • type commands • Current Directory • View folders and m-files • Workspace • View program variables • Double click on a variable to see it in the Array Editor • Command History • view past commands • save a whole session using diary 15 EV0000152 · EV0000153 · EV0000154

16. Source slide page 16 (extracted text): Hands on Practice ➢MATLAB environment behaves like a super-complex calculator. We can enter commands at the >> command prompt. ➢In MATLAB, we give a command and it executes the command right away. ➢Type a valid expression, for example, >>20 + 21; and press ENTER. ➢ When we click the Execute button, MATLAB executes it immediately and the result returned is ans=41. EV0000155 · EV0000156 · EV0000157

17. Source slide page 17 (extracted text): Hands on Practice, Continued... ➢Let us take up few more examples: ➢>>3 ^ 2 (%3 raised to the power of 2). When we click the Execute button, MATLAB executes it immediately and the result returned is ans=9. ➢>>sin(pi/2). (% sine of angle 90). When we click the Execute button, MATLAB executes it immediately and the result returned is ans=1. ➢>>7/0 (% Divide by zero). When we click the Execute button, MATLAB executes it immediately and the result returned is ans=Inf. EV0000158 · EV0000159 · EV0000160

18. Source slide page 18 (extracted text): Hands on Practice, Continued... ➢>>732 * 20.3. When we click the Execute button, MATLAB executes it immediately and the result returned is ans=1.4860e+04. ➢MATLAB provides some special expressions for some mathematical symbols, like pi for π, Inf for ∞, i (and j) for -1 . ➢In MATLAB, NaN stands for 'not a number'. For example, >> 0/0; -inf/inf EV0000161 · EV0000162 · EV0000163

19. Source slide page 19 (extracted text): Use of Semicolon (;) in MATLAB ➢Semicolon (;) indicates end of statement. However, if we want to suppress and hide the MATLAB output for an expression, add a semicolon after the expression. ➢For example, >>x = 5; y = x + 5. When we click the Execute button, MATLAB executes it immediately and the result returned is y=8. EV0000164 · EV0000165 · EV0000166

20. Source slide page 20 (extracted text): Adding Comments in MATLAB ➢To add comments to MATLAB code, we use the percent (%) symbol. Comment lines can appear anywhere in a program file, and we can add comments to the end of a line of code. For example, y = sum(x) % Use the sum function EV0000167 · EV0000168 · EV0000169

21. Source slide page 21 (extracted text): Commonly used Operators and Special Characters Operation Symbol Example Addition + 5+3=8 Subtraction - 5-3=2 Multiplication * 5*3=15 Right Division / 5/3=1.6667 Left Division \ 5/3=3\5=1.667 Exponentiation ^ 5^3=125 EV0000170 · EV0000171 · EV0000172

22. Source slide page 22 (extracted text): Special Variables and Constants MATLAB supports the following special variables and constants: EV0000173 · EV0000174 · EV0000175

23. Source slide page 23 (extracted text): Naming Variables ➢Variable names consist of a letter followed by any number of letters, digits or underscore. ➢MATLAB is case-sensitive. ➢For example, >>x = 10 (% defining x and initializing it with a value). MATLAB will execute the above statement and return the following result x = 10. ➢>>x = sqrt(25) (% defining x and initializing it with an expression) MATLAB will execute the above statement and return the result as x = 5. EV0000176 · EV0000177 · EV0000178

24. Source slide page 24 (extracted text): Naming Variables, Continued... ➢Note that once a variable is entered into the system, we can refer to it later. Variables must have values before they are used. When an expression returns a result that is not assigned to any variable, the system assigns it to a variable named answer, which can be used later. ➢For example, >>x = 7 * 8; y = x * 7.89. MATLAB will execute the above statement and return the following result y = 441.8400. EV0000179 · EV0000180 · EV0000181

25. Source slide page 25 (extracted text): Multiple assignments ➢We can have multiple assignments on the same line. ➢For example, >> a = 2; b = 7; c = a * b ➢MATLAB will execute the above statement and return the result c = 14. EV0000182 · EV0000183 · EV0000184

26. Source slide page 26 (extracted text): I have forgotten the Variables ➢The who command displays all the variable names we have used. ➢MATLAB will execute the above statement and return the result: Your variables are: ans - - - ... ➢The whos command. MATLAB will execute the above statement and return the following result. EV0000185 · EV0000186 · EV0000187

27. Source slide page 27 (extracted text): I have forgotten the Variables >> clear x. It will delete x, won't display anything. >> clear. It will delete all variables in the workspace. >> clc. It clears all the text from the Command Window, resulting in a clear screen. EV0000188 · EV0000189 · EV0000190

28. Source slide page 28 (extracted text): Long Assignments ➢Long assignments can be extended to another line as follows: >> initial_velocity = 0; >> acceleration = 9.8; >> time = 20; >> Final_velocity = initial_velocity + acceleration * time ➢MATLAB will execute the above statement and return the following result: final velocity = 196. EV0000191 · EV0000192 · EV0000193

29. Source slide page 29 (extracted text): The format Command ➢By default, MATLAB displays numbers with four decimal place values. This is known as short format. ➢However, if we want more precision, then we need to use the format command. ➢The format long command displays 15 digits after the decimal point. ➢For example: >> format long >> x = 7 + 10/3 + 5 ^ 1.2 ➢MATLAB will execute the above statement and return the following result: x = 17.231981640639408. EV0000194 · EV0000195 · EV0000196

30. Source slide page 30 (extracted text): The format Command, Continued... Another example, >> format short >> x = 7 + 10/3 + 5 ^ 1.2 ➢MATLAB will execute the above statement and return the following result: x = 17.232 ➢The format bank command rounds numbers to two decimal places. EV0000197 · EV0000198 · EV0000199

31. Source slide page 31 (extracted text): The format Command, Continued... For example, >> format bank >> daily_wage = 177.45; weekly_wage = daily_wage * 6 ➢MATLAB will execute the above statement and return the following result: weekly_wage = 1064.70 ➢MATLAB displays large numbers using exponential notation. ➢The format short e command allows displaying in exponential form with four decimal places plus the exponent. EV0000200 · EV0000201 · EV0000202

32. Source slide page 32 (extracted text): The format Command, Continued... ➢For example, >> format short e >> 4.678 * 4.9 ➢MATLAB will execute the above statement and return the following result: ans = 2.2922e+01 ➢The format long e command allows displaying in exponential form with fifteen decimal places plus the exponent. EV0000203 · EV0000204 · EV0000205

33. Source slide page 33 (extracted text): The format Command, Continued... ➢For example, >> format long e >> x = pi ➢MATLAB will execute the above statement and return the following result: x = 3.141592653589793e+00 ➢The format rat command gives the closest rational expression resulting from a calculation. ➢For example, >> format rat >> 4.678 * 4.9 ➢MATLAB will execute the above statement and return the following result: ans = 34177/1491 EV0000206 · EV0000207 · EV0000208

34. Source slide page 34 (extracted text): Creating Vectors ➢A vector is a one-dimensional array of numbers. ➢MATLAB allows creating two types of vectors: ➢Row vectors and Column vectors. ➢Row vectors are created by enclosing the set of elements in square brackets, using space or comma. ➢For example, >> X = [7 8 9 10 11]. MATLAB will execute the above statement and return the following result: X = 7 8 9 10 11 EV0000209 · EV0000210 · EV0000211

35. Source slide page 35 (extracted text): Creating Vectors, Continued... ➢Another example, >> X = [7 8 9 10 11]; >> Y = [2, 3, 4, 5, 6]; >> Z = X +Y ➢MATLAB will execute the above statement and return the following result: Z = 9 11 13 15 17 EV0000212 · EV0000213 · EV0000214

36. Source slide page 36 (extracted text): Creating Vectors, Continued... ➢Column vectors are created by enclosing the set of elements in square brackets, using semicolon(;). ➢For example, >> X = [7; 8; 9; 10] ➢MATLAB will execute the above statement and return the following result: X = 7 8 9 10 EV0000215 · EV0000216 · EV0000217

37. Source slide page 37 (extracted text): Creating matrices ➢A matrix is a two-dimensional array of numbers. ➢In MATLAB, a matrix is created by entering each row as a sequence of space or comma separated elements, and end of a row is terminated by a semicolon. ➢For example, let us create a 3-by-3 matrix as >>A = [1 2 3 4; 4 5 6 7; 7 8 9 10] EV0000218 · EV0000219 · EV0000220

38. Source slide page 38 (extracted text): Creating matrices, Continued... ➢MATLAB will execute the above statement and return the following result: A = 1 2 3 4 4 5 6 7 7 8 9 10 EV0000221 · EV0000222 · EV0000223

39. Source slide page 39 (extracted text): Elementary Math Built – In Functions Function Description Examples sqrt(x) Square root >>sqrt(81) ans=9 nthroot(x,n) Real nth root of a real number x. (If x is negative n >>nthroot(8,3) must be an odd integer). ans=2 exp(x) Exponential(e^x) >>exp(5) ans=1.484131591025766e+02 abs(x) Absolute value >>abs(-24) ans=24 log(x) Natural logarithm. Base e logarithm (ln) >>log(1000) ans=6.907755278982137e+00 log10(x) Base 10 logarithm >>log10(1000) ans=3 factorial(x) The factorial function x! >>factorial(5) (x must be a psitive integer) ans=120 EV0000224 · EV0000225 · EV0000226

40. Source slide page 40 (extracted text): Trigonometric Math Functions Function Description Examples sin(x) Sine of angle x (x in radians) >>sin(pi/6) ans=0.5000 sind(x) Sine of angle x (x in degrees) >>sind(30) ans=0.5000 cos(x) Cosine of angle x (x in radians) >>cos(0.5) ans=0.8776 cosd(x) Cosine of angle x (x in degrees) >>cosd(30) ans=0.8660 tan(x) Tangent of angle x (x in radians) >>tan(pi/6) ans=0.5774 tand(x) Tangent of angle x (x in degrees) >>tand(45) ans=1 cotx) Cotangent of angle x (x in radians) >>cot(0.5) ans=1.8305 cotd(x) Cotangent of angle x (x in degrees) >>cotd(90) ans=1 asin(x) Inverse of sine of angle x (x in radians) >>asin(0.5) ans=0.5236 asind(x) Inverse of sine angle x (x in degrees) >>asin(0.8660) ans=59.9971 acos(x) Inverse of cosine of angle x (x in radians) >>acos(0.3) ans=1.2661 acosd(x) Inverse of cosine angle x (x in degrees) >>acosd(0.5) ans=60 EV0000227 · EV0000228 · EV0000229

41. Source slide page 41 (extracted text): Rounding Functions Function Description Examples round(x) Round to the nearest integer >>round(25/3) ans=8 >>round(4.49) ans=4 >>round(4.5) ans=5 >>round(4.9999) ans=5 fix(x) Round towards zero. In other words, chops off the >>fix(17/5) ans=3 fraction part. >>fix(4.49) ans=4 >>fix(4.9999) ans=4 >>fix(-4.6674) ans=-4 ceil(x) Round towards positive infinity. >>ceil(27/2) ans=14 >>ceil(2.1) ans=3 >>ceil(2.9) ans=3 >>ceil(-4.99) ans=-4 floor(x) Round towards minus infinity >>floor(-9/4) ans=-3 >>floor(2.1) ans=2 >>floor(2.99) ans=2 rem(x,y) Returns the remainder after x is divided by y >>rem(13,5) ans=3 EV0000230 · EV0000231 · EV0000232

42. Source slide page 42 (extracted text): Examples for Arithmetic operators 1) Calculate the following using MATLAB. 1 4 + ∗6 2 + 32 5 7 >>a=1/(2+3^2) a= 0.0909 >>b=4/5 b= 0.8000 >>c=6/7 c= 0.8571 >>a+b*c ans= 0.7766 EV0000233

43. Source slide page 43 (extracted text): 2. Find the value of $y = e - a\sin x + 10\ yif\ a = 5,\ x = 2,\ y = 8$. >>a=5; x=2; y=8; y=exp(-a)*sin(x)+10*sqrt(y) y= 2.829039804533026e+013. π 3. Compute sin , e10. 4 >>sin(pi/4) ans=0.7071 >>exp(10) ans=2.2026e+004 EV0000234

44. Source slide page 44 (extracted text): Reference: 1. Getting started with MATLAB, Rudra Pratap, Oxford University Press, 7th Edition, 2016. 2. https://www.tutorialspoint.com/matlab/matlab_data_import. htm ## Page / slide 45 (OCR) THANK YOU PES UNIVERSITY EV0000235

45. Source slide page 45: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000236

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.290] Hello, everyone. Welcome to MATLAB session. [00:00:05.040] I'm Sibashankari from the Department of Science and Humanities. [00:00:10.380] For the semester, in the lab session, we will cover basics of MATLAB, [00:00:16.770] programming in MATLAB, plotting 2D and 3D, and then linear algebra, [00:00:22.951] partial differentiation, and solving differential equation. [00:00:27.901] And the reference books are [00:00:31.050] "Getting Started with MATLAB" by Rudra Pratap, and the [00:00:36.735] "MATLAB for Engineers" by Holly Moore, and then "MATLAB Programming for Engineers" [00:00:42.690] by Stephen J. Chapman. So throughout this course, we will be referring to the [00:00:48.449] highlighted textbook, that is, "Getting Started with MATLAB" by Rudra [00:00:53.011] Pratap. The whole syllabus will be covered in 12 sessions, and this is [00:00:58.590] the first session in MATLAB. [00:01:02.310] So in this introductory session, we will be covering what is MATLAB, [00:01:08.190] how to work with MATLAB user interface, and then the basic command, and the assign [00:01:14.010] variable, and operation with variable. [00:01:18.690] Hope everyone would have downloaded the MATLAB software. [00:01:22.530] In case if you have not downloaded, then you can use the installation [00:01:27.990] guidelines link that is attached along with this video. [00:01:32.610] As you know that MATLAB is a paid software, and so if you have institution mail [00:01:38.130] ID, then you can get the access to licensed version. [00:01:42.720] Otherwise, you need to download the trial version. [00:01:46.590] So how to download the trial version and the licensed version, both the links are [00:01:51.420] attached along with this video. So please refer to that and download the MATLAB [00:01:56.820] software. So let's get started. [00:02:00.180] So what is MATLAB? [00:02:02.370] MATLAB is a software package for numerical

computation, visualization, and [00:02:07.950] programming. It provides an interactive environment with hundreds of built-in [00:02:12.660] function for technical computation, graphics, and animation. [00:02:18.270] The name MATLAB stands for Matrix Laboratory because the basic building [00:02:24.270] block of MATLAB is the matrix. [00:02:27.900] Some basic features and commands of MATLAB. [00:02:31.470] To begin with, let us look at the general structure of MATLAB environment. [00:02:37.050] It consists of MATLAB desktop, a figure window, and an [00:02:42.810] editor window. [00:02:44.880] The figure window opens automatically when graphic commands are executed and [00:02:50.790] contains graph created by these command. [00:02:54.300] So we will discuss about this in our plotting session. [00:02:58.590] The editor window is used for writing and editing programs. [00:03:04.050] This is the place where you can write, edit, create, or save your own program in [00:03:09.180] files called M files. So we will discuss about this in programming session. [00:03:15.690] Now coming to the MATLAB desktop. So when we start MATLAB, this special window [00:03:21.330] opens up, and by default it has three sub-windows, namely [00:03:27.330] the command window, the current folder, and the workspace. [00:03:32.460] So we will discuss about this in detail now. [00:03:36.150] So I will switch to MATLAB environment, click on MATLAB [00:03:41.640] icon. We can see the MATLAB desktop, which contains three [00:03:47.671] sub-windows, command window, current folder, and the workspace. [00:03:53.130] This is the default layout. If you want to change the layout, click on the [00:03:58.890] layout option here and change as per your preference. [00:04:04.260] Suppose if I select the three column, then the current folder, command window, and [00:04:10.110] the workspace appear in column-wise. [00:04:14.250] Suppose if I select it as command window only, then you could see only the command [00:04:19.531] window. Other two windows are not displayed. [00:04:24.330] So now I will select the default one, [00:04:28.920] and this is the command window. This is the main window where we write the commands [00:04:34.440] as well as see the outputs. In other words, here is the interaction with the [00:04:39.390] software. So to type a command, the cursor must be placed next to the command [00:04:45.150] prompt. So this is the command prompt. So I place the cursor here. [00:04:51.451] To start with, we can use this command window as a calculator where you can type in [00:04:57.420] commands line by line. So let me type 53 minus

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.270] 42. [00:05:05.730] Once a command is typed, to execute the command, Enter key is pressed. [00:05:12.480] Whenever a calculation is performed, MATLAB will assign the result to the built-in [00:05:17.760] variable ans, unless specified. Now if you observe in the workspace, [00:05:24.540] ans variable is created, and the value 11 is stored in it, [00:05:31.620] and it can be used for further calculation. [00:05:35.550] So suppose if I type answer plus [00:05:41.640] 32 [00:05:44.550] and press the Enter key. [00:05:48.300] Now if you observe in the workspace, the new value is [00:05:54.031] replaced for the ans variable. That is the answer variable. [00:06:02.281] And ans is the default output variable. So we can also create variable. [00:06:09.000] The easy way to do is, is to use an assignment operator. [00:06:16.020] So type a variable name, [00:06:19.230] say a. [00:06:21.810] Use the assignment operator, which is equal sign, and then any value or [00:06:27.689] mathematical expression we can give. So let me type 74. [00:06:35.280] So now if you see that, [00:06:37.800] variable a is created, and the value 74 is stored in the variable a. [00:06:45.614] And you should remember that, or make a note of it, the variable must be always [00:06:51.570] on the left, and the expression or the value

should be on the right. [00:06:56.401] Otherwise, an error will occur if they are reversed. [00:07:00.151] And one more thing is that if I type b plus four [00:07:05.790] equal to 66, [00:07:10.470] see, this is an incorrect assignment statement. [00:07:13.800] So always remember that variable must be on left side, and the expression [00:07:19.800] or the values have to be in the right side. [00:07:24.270] Now, if you see here, a variable in the workspace. [00:07:34.201] Now, if you observe that, a variable is taking the value 74. [00:07:39.780] Now I want to replace it by an another number. [00:07:42.990] So I'll put [00:07:45.030] a equal to 46. [00:07:51.811] Now if you observe that, in the workspace, small a is not [00:07:57.570] replaced and capital A variable is created, and it takes the value 46. [00:08:03.990] So from this, we have to understand that MATLAB is case sensitive. [00:08:08.954] It takes small a and capital A as two different variable. [00:08:14.070] So there are some rules for naming variable. [00:08:17.401] That is, the variable should not start with a number. [00:08:21.811] It should always start with the alphabet. [00:08:24.630] So you cannot give 4s, let 4s equal to 20. [00:08:30.990] I cannot assign it this way. [00:08:34.290] Instead, if I assign [00:08:37.651] a4 equal to 20, it will accept. So remember that [00:08:43.920] variable should not start with number. It should start with alphabet. [00:08:49.500] And after alphabet, a number can be given, any symbol can be given, even underscore [00:08:55.110] can be given, but a space should not be there. [00:08:58.110] And we should avoid using the names of built-in function for variable. [00:09:03.660] Suppose if I use [00:09:07.681] sin [00:09:09.894] As my variable, sin equal to 66, [00:09:14.994] then sin variable is created and the value 66 is stored in that [00:09:21.084] sin variable. Now, if I want to find out the value of sin 13, [00:09:30.054] see what happens, sin appears to be both a function and a variable. [00:09:35.904] So [00:09:37.854] once a function name is used to define a variable, the function cannot be used for [00:09:42.744] its purpose, so it is better not to use the built-in function. [00:09:47.604] So if you have already used it, then how to remove it is just type clear [00:09:54.354] sin. [00:09:56.874] So if you see here in the workspace, that sin variable is removed.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.124] So remember that clear command removes all the variables from the memory. [00:10:08.154] So as far as possible, do not use built-in functions for naming the variable. [00:10:15.024] And how do we know that it is a built-in function? [00:10:18.744] So to check whether the variable is a built-in function or not, what we can do is [00:10:23.424] that type [00:10:25.164] typesin. [00:10:32.604] So if you could see here, the color changes, that indicates that it is a built-in [00:10:37.314] function, or you can press Enter, so it will say that sin is a built-in function. [00:10:44.874] So the list of built-in functions are given in the PPT, and also regarding the [00:10:49.944] rules for naming the variable, the do's and the do not's are also given in the PPT. [00:10:55.014] So please go through that so that you will have a clear idea about it. [00:10:59.994] Now I want to clear this command window, so what I will do is that I will type [00:11:05.424] clc. [00:11:10.164] So the command window is cleared now. [00:11:14.184] Suppose when you are assigning a value to your variable and if you want to check [00:11:18.774] whether you have already used this variable name, or if you want to have the list [00:11:24.084] of variables used, just type who. [00:11:29.604] This will give you what are the variables that are there in the workspace. [00:11:34.734] Or if you want to know about the size of the data type, then you can just type [00:11:40.164] whos. [00:11:43.705] Then it will give you the variable along with the size and the data type. [00:11:50.545] And coming to the next one, several commands can be typed in the same line. [00:11:55.764] So this is done by typing a comma between

the commands. [00:12:01.075] So how do we do is that suppose if I want to give, say, a equal to four [00:12:07.134] and b equal to five, now I want to perform some operations on this. [00:12:13.734] So what I can do is that, but I want to perform a series of operation. [00:12:17.664] That is, I want to find out the sum of it, [00:12:21.295] then I want to find out the difference, I want to find out a/b, [00:12:26.874] then I want to find out a star b. a star b indicates the [00:12:32.064] multiplication. So when the enter key is pressed, you could observe that [00:12:38.064] the commands are executed in the order from left to right. [00:12:42.864] So first this is a+b, and this is a-b, then this is a/b, [00:12:49.284] and this is a*b. [00:12:53.320] Now, if I want to give some other value for a and b, so let me change it. [00:12:58.494] So a equal to, [00:13:01.404] I'll give it as 33, [00:13:04.854] then, and I don't want to display every time when I give a equal to 33, and when I [00:13:10.015] press the enter key, that value is displayed. [00:13:14.094] If I don't want to see that it is being displayed on the [00:13:19.734] screen, what we can do is that for assigning b value, so b equal [00:13:25.734] to, say, if I write 44, I can put one semicolon. [00:13:30.594] What is this semicolon will do is that it will suppress the output. [00:13:35.034] So if I press enter here, one more time, that b equal to 44 will not be displayed. [00:13:40.314] But in the workspace, if you see, b takes the value 44. It is stored there. [00:13:45.504] Now I want to perform the same operation, but it is not possible to go back to the [00:13:51.325] previous line that is displayed in the command window and then re-execute the [00:13:56.664] command. So either what we can do is that you can copy-paste those [00:14:03.564] operation, or you can keep the cursor next to the [00:14:09.354] command prompt and click the up arrow key in your keyboard. [00:14:15.414] Then you can see that command. You can recall that command. [00:14:21.204] Suppose if you want to modify or make a correction, you can do it. [00:14:26.064] So now I will change the order. So first is addition, subtraction, [00:14:32.844] then I will do multiplication, [00:14:37.644] then I will do the division. So if you observe here, each command [00:14:43.464] is separated by a comma. Now, when I execute it by pressing the enter key, [00:14:49.614] you can see the output. This is a+b, a-b, [00:14:55.015] a*b, and a/b.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.174] Now we will clear the screen by typing clc. [00:15:05.545] We will do some arithmetic operation in MATLAB. [00:15:10.825] Percentage symbol, [00:15:14.814] arithmetic [00:15:17.604] operations. [00:15:22.404] So anything after the percentage symbol on the same line is neglected by MATLAB [00:15:28.015] compiler. Usually, there's no need for commands in the command window, but [00:15:33.924] this is very useful when we are doing programs. [00:15:37.134] So to add description or explain the program, we have to use this percentage [00:15:42.804] symbol. Now we will move on to the problem. [00:15:46.644] We will take first problem, and we will execute this. [00:15:50.484] Calculate the following using MATLAB command: [00:15:54.924] one by two plus three square, plus four by five [00:16:00.684] into six by seven. So actually, we have to do term by term, first is one [00:16:06.774] by two plus three square, [00:16:09.894] and then four by five into six by seven. [00:16:15.835] So you should remember that while doing arithmetic operation, some operators have [00:16:21.474] precedence over the other. The first preference is given to the parenthesis, next [00:16:27.325] is the exponential, then comes the multiplication and division, and finally [00:16:32.874] addition and subtraction. Suppose if you see multiplication and division [00:16:38.844] or addition and subtraction on the same line or on the simultaneously if it is [00:16:44.724] present, then it is solved from left to right. [00:16:49.674] Now I will copy this

problem or type this problem in MATLAB. [00:16:57.954] So it is one by two plus three square. [00:17:00.984] So I have given the parenthesis here for four by five also. [00:17:07.374] And for multiplication, I should use [00:17:11.154] asterisk symbol [00:17:14.004] six by seven. [00:17:16.225] Now, when you execute this, you get the answer to be [00:17:21.384] 0.7766. [00:17:27.085] Usually, MATLAB by default displays number with four decimal [00:17:32.814] points. Suppose if you want to be more accurate, then what you can do is [00:17:38.754] that you can type format long. [00:17:43.434] If you do so, now if you calculate any problem, say two star sin of [00:17:50.304] 1.4. [00:17:53.634] So look at the answer, 15 decimal points are there. [00:17:58.704] So format long will give you the result in 15 decimal places, [00:18:05.334] and format short will give you in four decimal places. [00:18:13.225] So all these display formats are given in the PPT, so you can go through that. [00:18:18.804] Now we go to one more problem. [00:18:23.184] Take on the next problem. Find the value of Y equal to E power minus A [00:18:29.424] sine X plus 10 root Y, if A equal to 5, X is equal to [00:18:35.784] 2, and Y is equal to 8. [00:18:39.114] So first step, we have to declare these variables. [00:18:43.314] So you have to assign A as 5, X as 2, and Y as 8. [00:18:49.284] So we will do that now. [00:18:52.374] A is 5. [00:18:56.064] So I don't want to display this on the screen, so I will suppress the output by [00:19:01.194] putting a semicolon. A equal to 5. X equal to 2. [00:19:08.364] And then Y is equal to 8. [00:19:17.124] Now look at the problem. I need to find out Y is equal to E power minus A sine X [00:19:23.004] plus 10 root Y. [00:19:26.784] So if you see here, Y is equal to E power minus A. [00:19:31.734] So how to write down this exponential is exp. [00:19:36.774] So you will write it as Y equal to [00:19:40.914] exp, exponential of minus A [00:19:48.774] into sine of X, so into sine [00:19:53.754] of [00:19:58.524] X

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:00.744] plus 10 into square root of Y. So whenever you have the 10 [00:20:06.204] into square root of Y, this 10 is a constant coefficient along with the variable. [00:20:12.024] So whenever you have like this, you have to put it as 10 star square root of, [00:20:17.725] square root it as an input function is sqrt of Y. [00:20:24.714] So now I have typed it as Y is equal to E power minus A sine X plus [00:20:31.015] 10 square root of Y. So now if I execute the command, you can see that it [00:20:37.015] is 28.2904. [00:20:42.234] Now we will do one more problem to have a better clarity. [00:20:47.784] Compute sine pi by 4 [00:20:51.834] and 4 into E power 10 sine 30 degree cos inverse pi by [00:20:57.924] 4 and log 10 of exponential E power [00:21:03.384] 2 plus cos pi. So we will do one by one. First, we will do sine pi by 4. [00:21:09.234] So you remember this, always sine is an inbuilt function, the sine, log 10, [00:21:15.654] and for cos inverse, the inbuilt function is acos. [00:21:19.284] So whenever you are typing the inbuilt function, remember that you have to use only [00:21:25.164] the lowercase alphabets. Don't use the uppercase. [00:21:30.864] So it is sine of, and when you are using the inbuilt function, it should be within [00:21:36.265] the open brackets. So sine of pi by 4, and if you observe that, pi is [00:21:42.265] also an already defined built-in variable, that is pI. [00:21:46.944] So sine by pI by 4. [00:21:52.614] So we get the value sine pi by 4. The next is four into exponential [00:21:58.524] E power 10. So whenever I told you, whenever you have a constant coefficient, so [00:22:04.674] 4, you have to put star into. For exponential, you have to type it as the inbuilt [00:22:10.404] function as exp of 10. So this [00:22:16.314] is four into E power 10. And if you observe this, the answer is [00:22:22.464] 8.8106 into N power 4. So that what it this [00:22:28.134] means. [00:22:30.038] Now the next is, we need to find out the value of sine 30 degree. [00:22:36.774] So this is one important one you have to make a note of it. [00:22:40.164]

You cannot find out for 30 degree. You can only find out for the radian. [00:22:46.225] So what you need to do is that you have to convert this degree into radian. [00:22:51.624] So how do we convert this degree into radian is sine of... [00:22:57.714] You know that to convert the degree into radian, you have to multiply by pi by 180. [00:23:02.904] So 30 by 180 [00:23:08.034] star pi. [00:23:17.334] So you know that sine 30 value is half, 0.5. [00:23:22.404] So remember that when you use the built-in function sine, S-I-N, you have [00:23:28.434] to convert the degree into radian mode, and you have to do it. [00:23:33.534] Suppose if you want to do only in the degree mode, then you have to use another [00:23:38.574] built-in function, which is S-I-N-D. [00:23:42.714] So if you do so, sine d of 30, then you will get the answer [00:23:48.804] straight away. You need not convert it to radian mode. [00:23:52.765] So the list of all the math built-in functions are given in the PPT, so that you [00:23:58.704] can go through that, [00:24:01.074] you will have a better idea about it. [00:24:03.864] Now coming to the next problem, we need to find out cos inverse of pi by 4. [00:24:10.644] So how to find out cos inverse of pi by 4 is, the built-in function for cos inverse [00:24:16.164] is acos. [00:24:19.374] This also it is there in the PPT. And pi by 4, as I have told you before, pI [00:24:25.314] is the built-in variable by 4. So this give you the [00:24:31.194] value for cos inverse of pi by 4. And the last problem, [00:24:37.254] log 10 of E power 2 plus cos pi. [00:24:42.714] So how to do this is log, L-O-G. So if you type LOG, [00:24:48.594] it represents log to the base E. So if you want log to the base 10, it [00:24:54.624] is LOG 10. [00:24:58.209] And always you know that built-in function has to be within the open brackets

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:04.254] into of E, exponential is exp, [00:25:10.015] 2 plus, [00:25:12.564] cos is also a built-in function, I need to open the bracket, pI. [00:25:18.144] So now I will close the bracket for cos, and this is for exponential, and this is [00:25:24.024] for the log 10. [00:25:28.074] So the answer you get here is 0.4343. [00:25:33.594] So now we have used the command window as a calculator and performed some [00:25:38.484] operations. [00:25:40.524] Hope everyone would have got some idea how to work with MATLAB. [00:25:44.754] Some more problems are given in the assignment section. [00:25:48.264] Practice those problems and make yourself familiar with MATLAB interface. [00:25:53.064] So we will meet you in the next session. Thank you all.

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Full transcript: 3223 normalized words, 303 timestamped segments, 1556.181 seconds duration. Low-confidence segments flagged: 0.

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3 Data types; relational operators; logical operators

This lesson is indexed from the source video titled Data types; relational operators; logical operators. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000026 · EV0000027

Graphs and diagrams

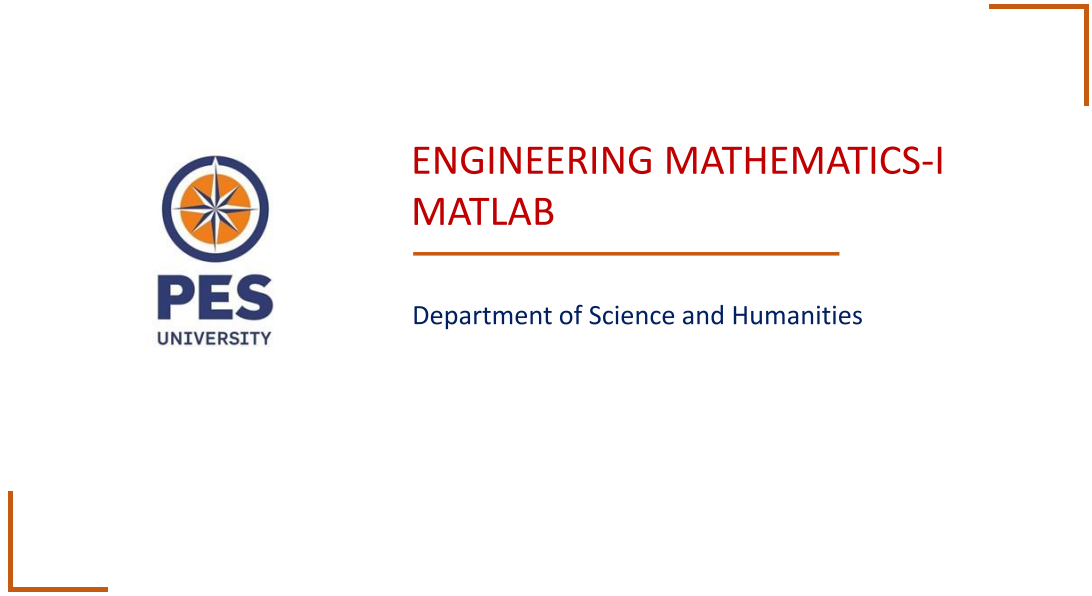

Figure G000046. Slides · source slide page 1.

EV0000238

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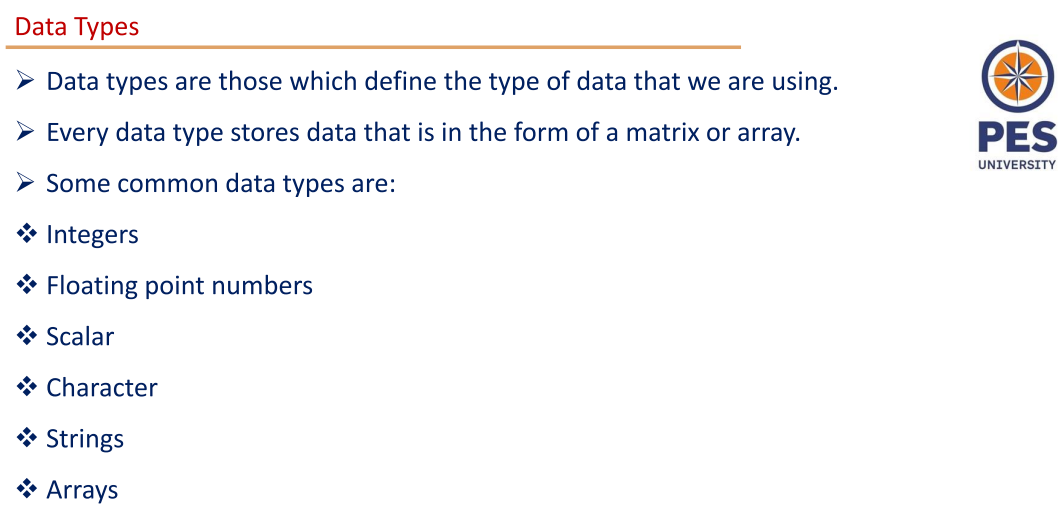

Figure G000047. Slides · source slide page 2.

EV0000241

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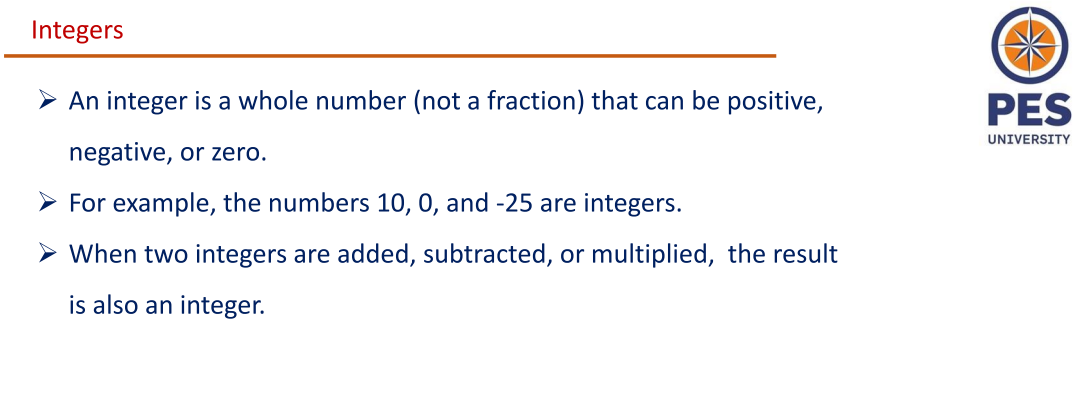

Figure G000048. Slides · source slide page 3.

EV0000244

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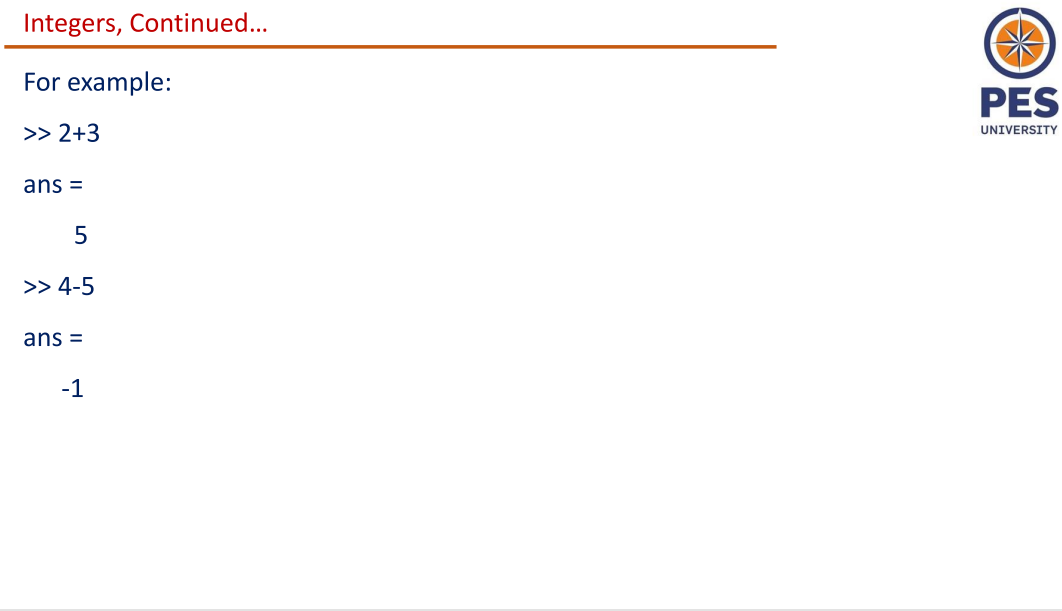

Figure G000049. Slides · source slide page 4.

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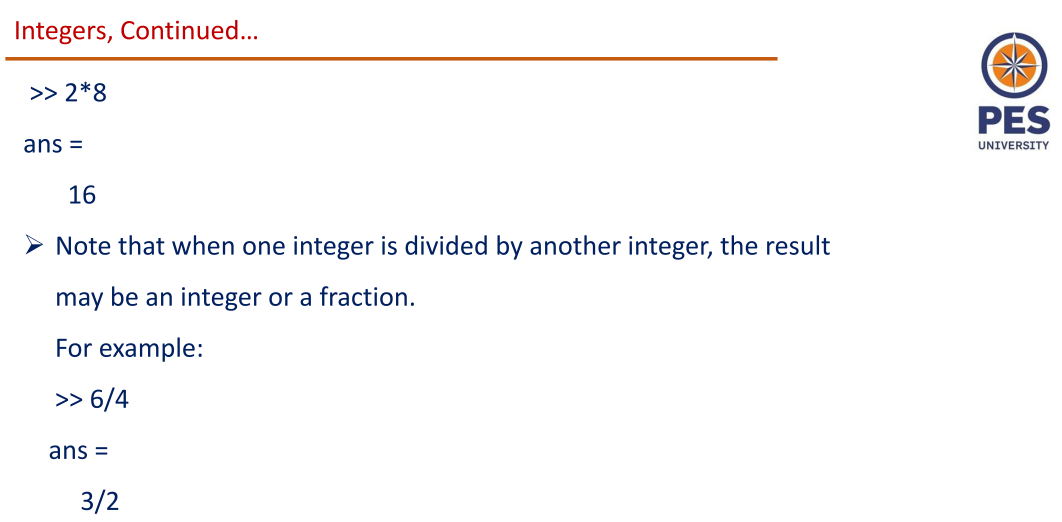

Figure G000050. Slides · source slide page 5.

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Figure G000051. Slides · source slide page 6.

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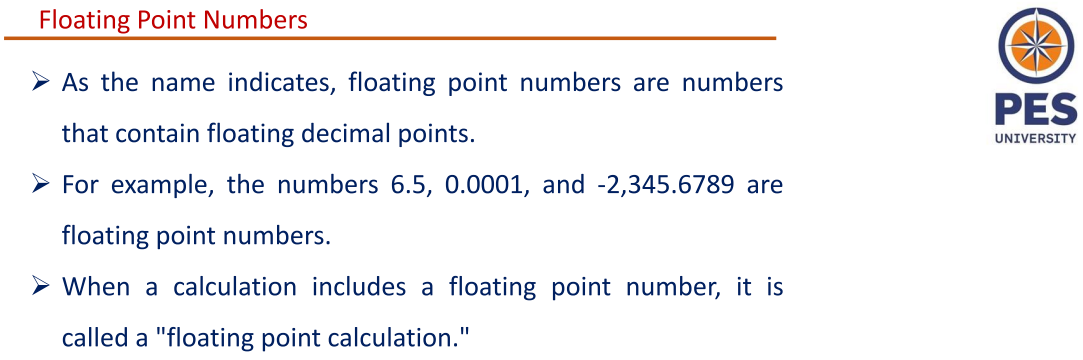

Figure G000052. Slides · source slide page 7.

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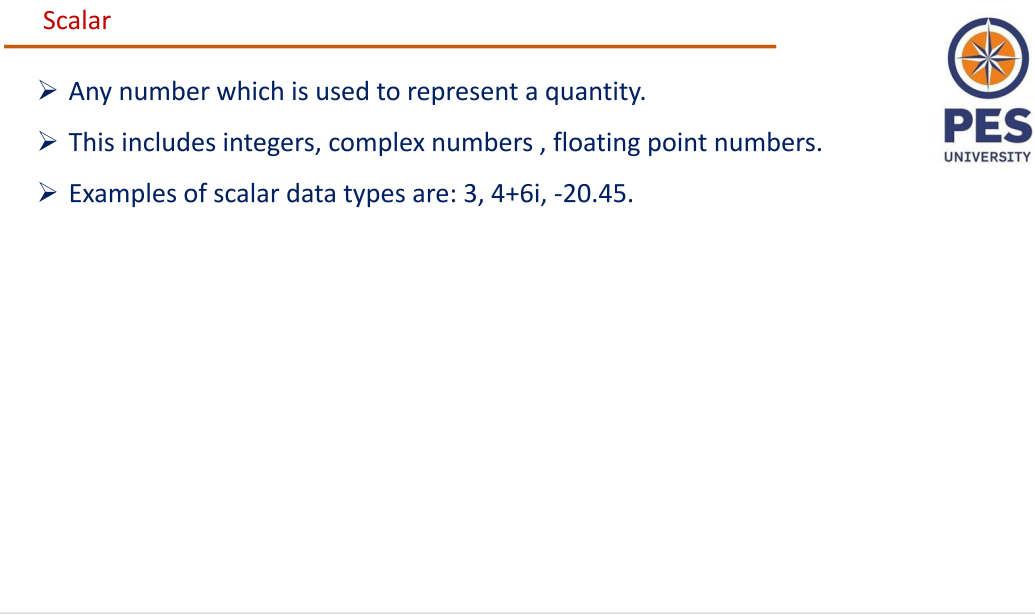

Figure G000053. Slides · source slide page 8.

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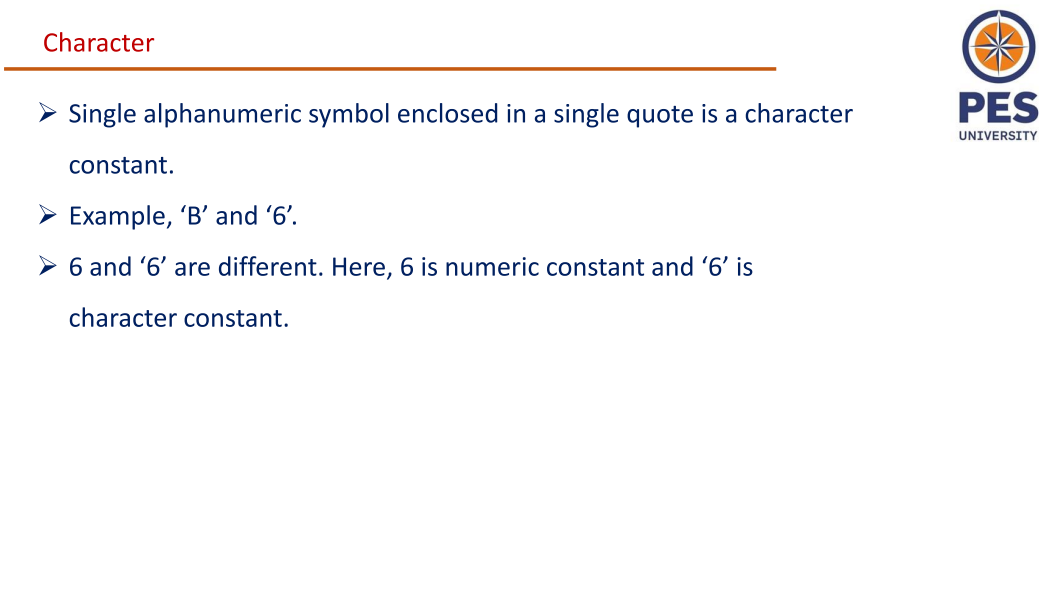

Figure G000054. Slides · source slide page 9.

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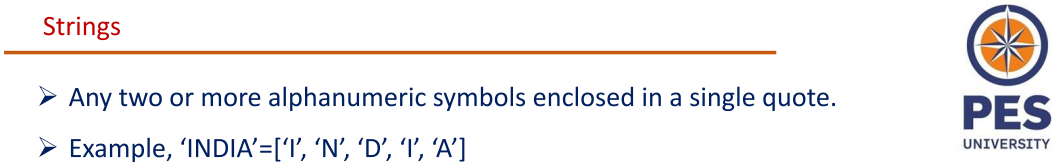

Figure G000055. Slides · source slide page 10.

EV0000265

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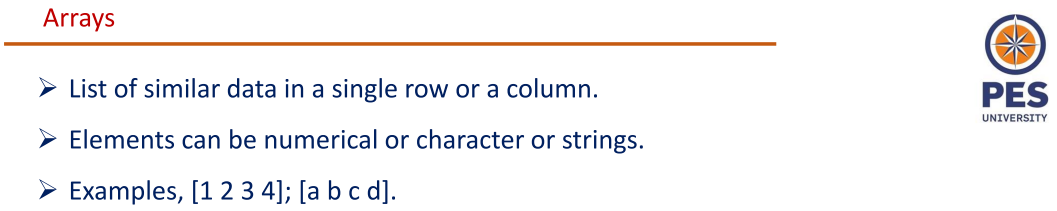

Figure G000056. Slides · source slide page 11.

EV0000268

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Figure G000057. Slides · source slide page 12.

EV0000271

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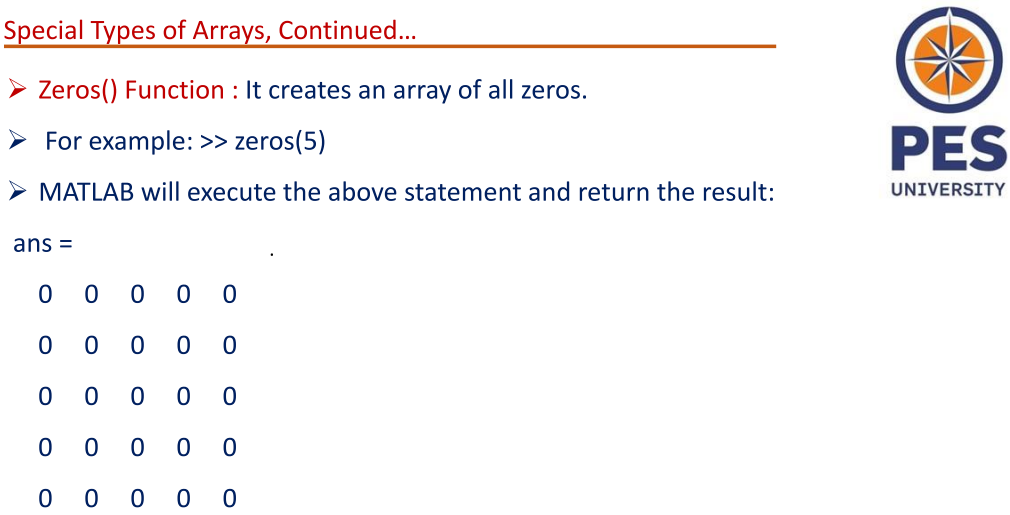

Figure G000058. Slides · source slide page 13.

EV0000274

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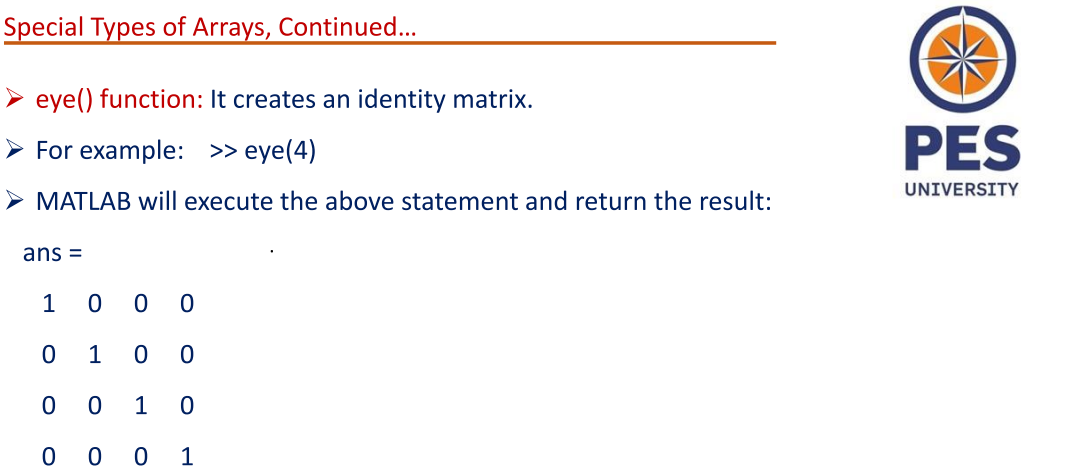


Figure G000059. Slides · source slide page 14.

EV0000277

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

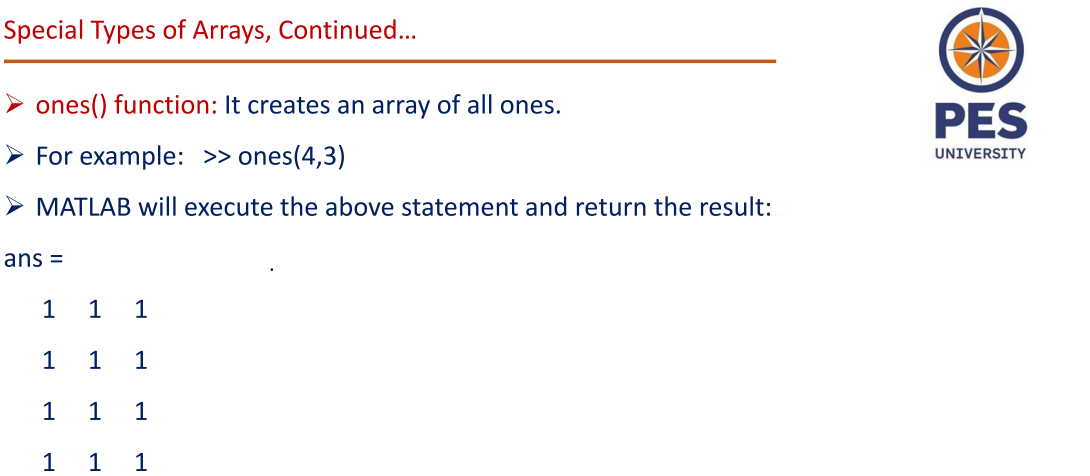


Figure G000060. Slides · source slide page 15.

EV0000280

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Special Types of Arrays, Continued…

- rand() function: It generates random numbers between 0 and 1 that are distributed uniformly.
- For example: >> rand(3, 5)
- MATLAB will execute the above statement and return the result

ans =

0.8147	0.9134	0.2785	0.9649	0.9572
0.9058	0.6324	0.5469	0.1576	0.4854
0.1270	0.0975	0.9575	0.9706	0.8003

Figure G000061. Slides · source slide page 16.

EV0000283

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Relational Operators:

- Relational operator compare the elements in two arrays and return logical true or false values to indicate where the relation holds.

==	Determine the equality
>=	Determine greater than or equal to
>	Determine greater than
<=	Determine less than or equal to
<	Determine less than
~=	Determine inequality

Figure G000062. Slides · source slide page 17.

EV0000286

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

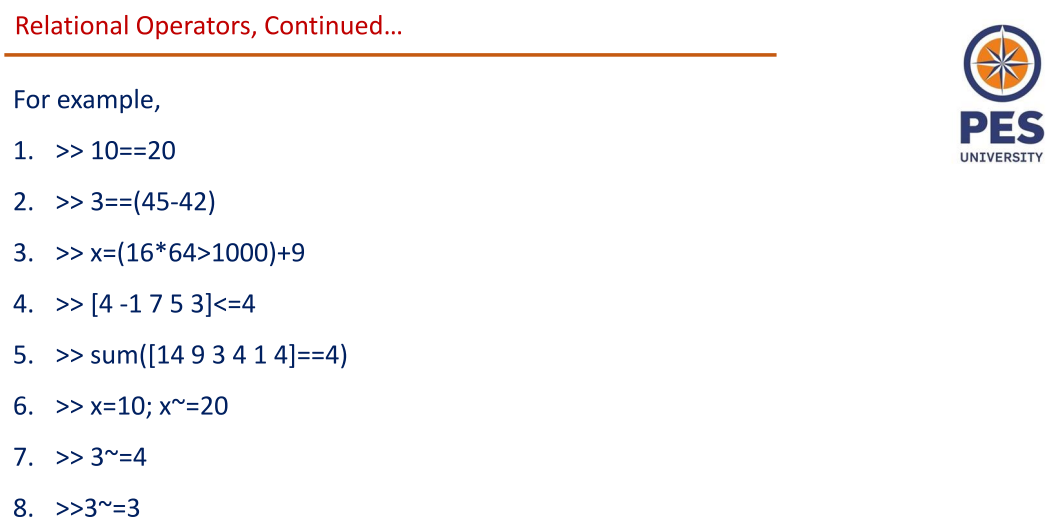


Figure G000063. Slides · source slide page 18.

EV0000289

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

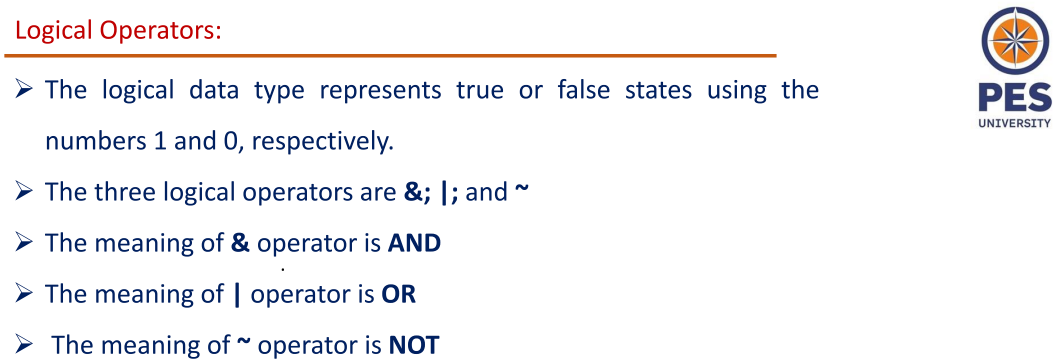


Figure G000064. Slides · source slide page 19.

EV0000292

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

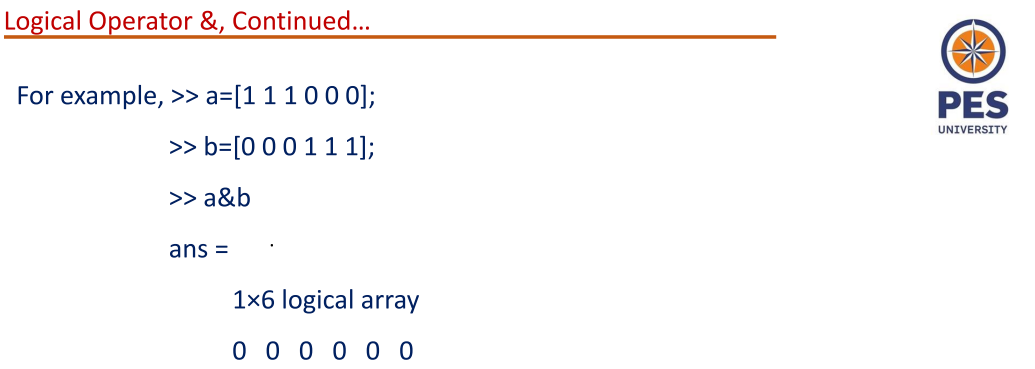


Figure G000065. Slides · source slide page 20.

EV0000295

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

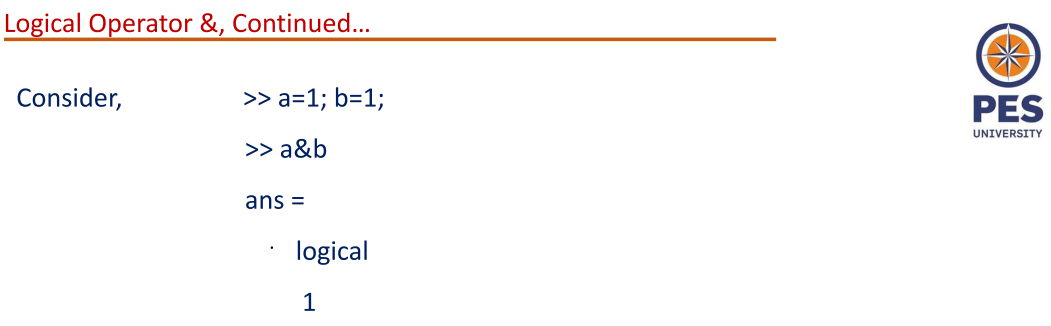


Figure G000066. Slides · source slide page 21.

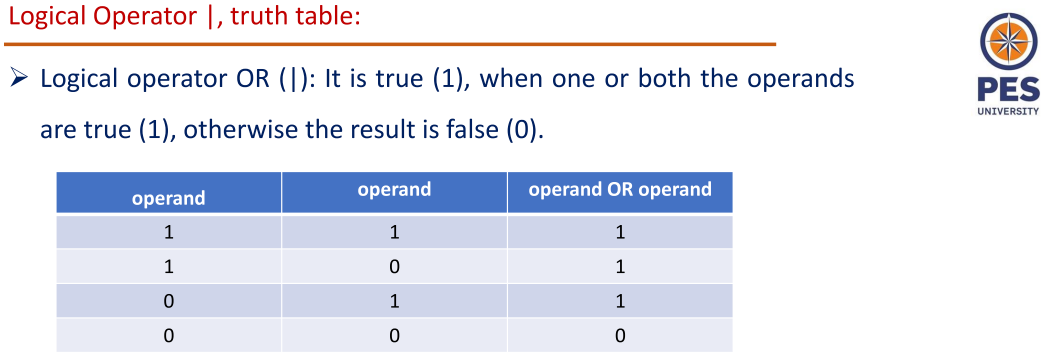
EV0000298

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000067. Slides · source slide page 22.

EV0000301

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



operand	operand	operand OR operand
1	1	1
1	0	1
0	1	1
0	0	0

Figure G000068. Slides · source slide page 23.

EV0000304

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

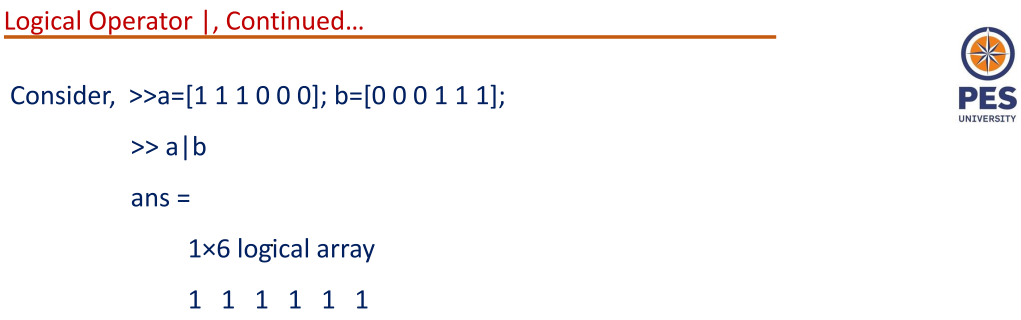


Figure G000069. Slides · source slide page 24.

EV0000307

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

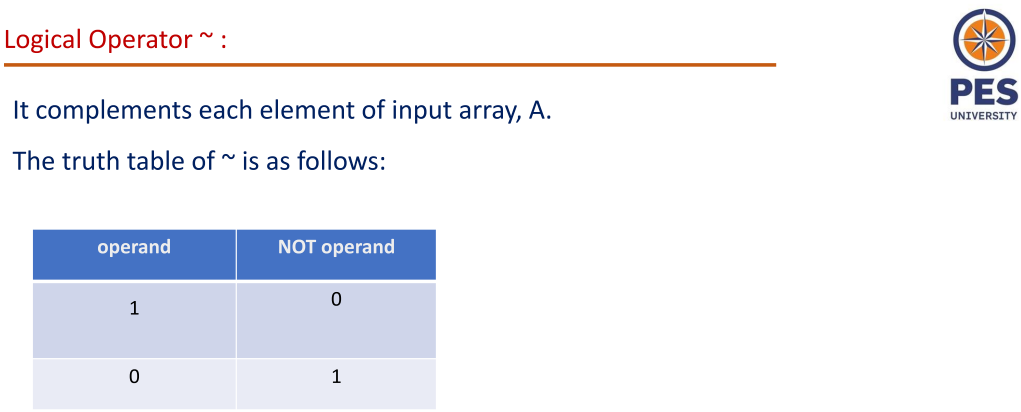


Figure G000070. Slides · source slide page 25.

EV0000310

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

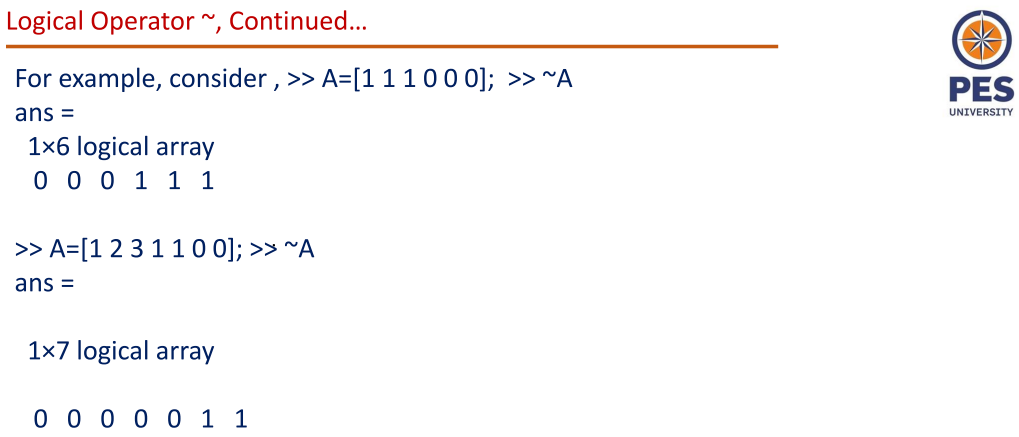


Figure G000071. Slides · source slide page 26.

EV0000312

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000072. Slides · source slide page 27.

EV0000313

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000237 · EV0000238 · EV0000239
2. Source slide page 2 (extracted text): Data Types ➢Data types are those which define the type of data that we are using. ➢Every data type stores data that is in the form of a matrix or array. ➢Some common data types are: ❖Integers ❖Floating point numbers ❖Scalar ❖Character ❖Strings ❖Arrays EV0000240 · EV0000241 · EV0000242
3. Source slide page 3 (extracted text): Integers ➢An integer is a whole number (not a fraction) that can be positive, negative, or zero. ➢For example, the numbers 10, 0, and -25 are integers. ➢When two integers are added, subtracted, or multiplied, the result is also an integer. EV0000243 · EV0000244 · EV0000245
4. Source slide page 4 (extracted text): Integers, Continued... For example: >> 2+3 ans = 5 >> 4-5 ans = -1 EV0000246 · EV0000247 · EV0000248
5. Source slide page 5 (extracted text): Integers, Continued... >> 2*8 ans = 16 ➢Note that when one integer is divided by another integer, the result may be an integer or a fraction. For example: >> 6/4 ans = 3/2 ## Page / slide 6 (OCR) Integers, Continued... >> 6/3 PES UNIVERSITY ans = 2 EV0000249 · EV0000250 · EV0000251
6. Source slide page 6: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000252 · EV0000253 · EV0000254
7. Source slide page 7 (extracted text): Floating Point Numbers ➢As the name indicates, floating point numbers are numbers that contain floating decimal points. ➢For example, the numbers 6.5, 0.0001, and -2,345.6789 are floating point numbers. ➢When a calculation includes a floating point number, it is called a "floating point calculation." EV0000255 · EV0000256 · EV0000257
8. Source slide page 8 (extracted text): Scalar ➢Any number which is used to represent a quantity. ➢This includes integers, complex numbers , floating point numbers. ➢Examples of scalar data types are: 3, 4+6i, -20.45. EV0000258 · EV0000259 · EV0000260
9. Source slide page 9 (extracted text): Character ➢Single alphanumeric symbol enclosed in a single quote is a character constant. ➢Example, ‘B’ and ‘6’. ➢6 and ‘6’ are different. Here, 6 is numeric constant and ‘6’ is character constant. EV0000261 · EV0000262 · EV0000263
10. Source slide page 10 (extracted text): Strings ➢Any two or more alphanumeric symbols enclosed in a single quote. ➢Example, ‘INDIA’=[‘I’, ‘N’, ‘D’, ‘I’, ‘A’] EV0000264 · EV0000265 · EV0000266
11. Source slide page 11 (extracted text): Arrays ➢List of similar data in a single row or a column. ➢Elements can be numerical or character or strings. ➢Examples, [1 2 3 4]; [a b c d]. EV0000267 · EV0000268 · EV0000269
12. Source slide page 12 (extracted text): Special Types of Arrays ➢The four types of arrays are: ❖ zeros() function ❖ eye() function ❖ ones() function . ❖ rand() function EV0000270 · EV0000271 · EV0000272
13. Source slide page 13 (extracted text): Special Types of Arrays, Continued... ➢Zeros() Function : It creates an array of all zeros. ➢ For example: >> zeros(5) ➢MATLAB will execute the above statement and return the result: ans = . 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 EV0000273 · EV0000274 · EV0000275

14. Source slide page 14 (extracted text): Special Types of Arrays, Continued... ➢eye() function: It creates an identity matrix. ➢For example: >> eye(4) ➢MATLAB will execute the above statement and return the result: ans = . 1 0 0 0 0 1 0 0 0 0 1 0 0 0 0 1 EV0000276 · EV0000277 · EV0000278

15. Source slide page 15 (extracted text): Special Types of Arrays, Continued... ➢ones() function: It creates an array of all ones. ➢For example: >> ones(4,3) ➢MATLAB will execute the above statement and return the result: ans = . 1 1 1 1 1 1 1 1 1 1 1 1 EV0000279 · EV0000280 · EV0000281

16. Source slide page 16 (extracted text): Special Types of Arrays, Continued... ➢rand() function: It generates random numbers between 0 and 1 that are distributed uniformly. ➢For example: >> rand(3, 5) ➢MATLAB will execute the above statement and return the result . ans = 0.8147 0.9134 0.2785 0.9649 0.9572 0.9058 0.6324 0.5469 0.1576 0.4854 0.1270 0.0975 0.9575 0.9706 0.8003 EV0000282 · EV0000283 · EV0000284

17. Source slide page 17 (extracted text): Relational Operators: ➢Relational operator compare the elements in two arrays and return logical true or false values to indicate where the relation holds. == Determine the equality >= Determine greater than or equal to . > Determine greater than <= Determine less than or equal to < Determine less than ~= Determine inequality EV0000285 · EV0000286 · EV0000287

18. Source slide page 18 (extracted text): Relational Operators, Continued... For example, 1. >> 10==20 2. >> 3==(45-42) 3. >> x=(16*64>1000)+9 4. >> [4 -1 7 5 3]<=4 5. >> sum([14 9 3 4 1 4]==4) 6. >> x=10; x~=20 7. >> 3~=4 8. >>3~=3 EV0000288 · EV0000289 · EV0000290

19. Source slide page 19 (extracted text): Logical Operators: ➢The logical data type represents true or false states using the numbers 1 and 0, respectively. ➢The three logical operators are &; |; and ~ ➢The meaning of & operator is AND . ➢The meaning of | operator is OR ➢ The meaning of ~ operator is NOT EV0000291 · EV0000292 · EV0000293

20. Source slide page 20 (extracted text): Logical Operator &, Continued... For example, >> a=[1 1 1 0 0 0]; >> b=[0 0 0 1 1 1]; >> a&b . ans = 1×6 logical array 0 0 0 0 0 0 EV0000294 · EV0000295 · EV0000296

21. Source slide page 21 (extracted text): Logical Operator &, Continued... Consider, >> a=1; b=1; >> a&b ans = . logical 1 EV0000297 · EV0000298 · EV0000299

22. Source slide page 22 (extracted text): Logical Operator &, Continued... Consider, >> a=0; b=1; >> a&b ans = . logical 0 EV0000300 · EV0000301 · EV0000302

23. Source slide page 23 (extracted text): Logical Operator |, truth table: ➢Logical operator OR (|): It is true (1), when one or both the operands are true (1), otherwise the result is false (0). operand operand OR operand operand 1 1 1 . 1 0 1 0 1 1 0 0 0 EV0000303 · EV0000304 · EV0000305

24. Source slide page 24 (extracted text): Logical Operator |, Continued... Consider, >>a=[1 1 1 0 0 0]; b=[0 0 0 1 1 1]; >> a|b ans = . 1×6 logical array 1 1 1 1 1 1 EV0000306 · EV0000307 · EV0000308

25. Source slide page 25 (extracted text): Logical Operator ~ : It complements each element of input array, A. The truth table of ~ is as follows: operand NOT operand 0 1 0 1 EV0000309 · EV0000310 · EV0000311

26. Source slide page 26 (extracted text): Logical Operator ~, Continued... For example, consider , >> A=[1 1 1 0 0 0]; >> ~A ans = 1×6 logical array 0 0 0 1 1 1 . >> A=[1 2 3 1 1 0 0]; >> ~A ans = 1×7 logical array 0 0 0 0 0 1 1 ## Page / slide 27 (OCR) THANK YOU PES UNIVERSITY EV0000312

27. Source slide page 27: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000313

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hi, welcome to PES University Online Teaching Learning Platform. I am Professor Nagesh Ham, Faculty Member, Department of Science and Humanities. This is the second session of MadLab. [00:00:17.956] In this session we shall discuss the concept of data types, relational operators and logical operators. Let's start with data types. Data types are those which define the type of data that we are using. [00:00:39.140] Every data type stores data that is in a form of a matrix or array. Some common data types are integers, floating point numbers, scalar, character, strings and arrays. Let's look at this one by one. [00:01:00.036] Integers. An integer is a whole number. There is not a fraction that can be positive, negative or zero. For example, the numbers 10, 0, and minus 25 are integers. [00:01:20.164] Note that when two integers are added, subtracted or multiplied, then the corresponding result is also an integer. For example, 2 plus 3, 4 minus 5. [00:01:39.236] 2 multiplied by 8, etc When one integer is divided by another integer, the corresponding result may be an integer or a fraction. For example, six by four plus three by two [00:01:58.212] Six by three, that is two. Floating point numbers. As a name indicates, floating point numbers are numbers that contain floating decimal points. For example, the numbers six point five [00:02:17.700] Zero point zero zero one etcetera are floating point numbers. [00:02:24.964] When a calculation includes a floating point number, it is called a floating point calculation. [00:02:34.564] Scalar. Any number which is used to represent a quantity is called a scalar. This includes integers, complex numbers, floating point numbers. For example, three [00:02:53.636] four plus six i minus twenty point four five r scalars. [00:03:02.020] character single alpha numeric symbol enclosed in a single code is a character constant. For example B and six note that these two six are different [00:03:20.548] The first one is a numerical constant. The one which is enclosed in a single code is a character constant. [00:03:31.908] Strings. Any two or more alpha numeric symbols enclosed in a single code is called a string. Example you can see India. [00:03:47.524] Arrays List of similar data in a single row or a column is called an array. Elements can be numerical or character or strings. For example 1, 2, 3, 4, ABC and so on. [00:04:09.572] After understanding data types and the different types of data types, let's look at some of the special types of arrays. The four types of arrays are zeros function, i function, once function and ran function. [00:04:31.908] Zero's function. As the name indicates, it creates a matrix of all zeros. I function. It creates an identity matrix. One's function. [00:04:52.356] It creates a matrix of all ones. Rand function. It generates random numbers between 0 and 1 that are distributed uniformly. Let's look at the illustration.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.868] Zero's function. [00:05:27.492] Zero's function, it creates a matrix of all zeros. For example [00:05:35.300] zeros let me give some eight is it output it creates a matrix of order eight whose elements are zeros next function [00:05:54.052] Function. [00:06:00.388] It creates an matrix. Example I sub seven.

[00:06:12.676] It creates an identity matrix of order seven. Next one's function. [00:06:26.468] It creates a matrix of all ones. For example, you can give 10. It creates a matrix whose elements are 1. It's a matrix of order 10 plus 10. Next rank function. [00:06:52.964] Rand function is used to generate random numbers between 0 and 1 that are distributed uniformly. For example, RAN you can say 3 cover 5. Here the first number 3 is number of rows and the second number 5 is number of columns. [00:07:12.548] This is the corresponding output. Let us use the same command and you can see that the numbers are different. The numbers are different, but they are between 0 and 1. Again use the same command. Again you can see that the numbers are different. So this is how we can generate random numbers. [00:07:32.676] between zero and one that are distributed uniformly using rand function. [00:07:40.196] After understanding the special types of arrays, let's move on to relational operators. [00:07:52.804] There are six relational operators equal to determine the equality greater than or equal to determine greater than or equal to greater than determine greater than greater than [00:08:08.516] and less than or equal to less than and determined inequally. These are the six relational operators we have in MATLAB. Relational operator compare the elements in two arrays and return logical true or false values to indicate where the relation holds. It means [00:08:28.548] If the result is true, then it returns the value 1. If it is false, then it gives the value 0. Let's look at the illustration part. Let us take 10 [00:08:45.956] equal to 20. Remember in MATLAB when we compare two things we use double equal sign. Right? So 10 is not equal to 20, there will answer is zero. Right? So let us take [00:09:04.996] three equal to forty five minus forty two it is true therefore the answer is one let us take same example three equal to [00:09:22.436] 42 minus 45. 3 is not equal to minus 3. Therefore the answer is 0. Let us take x equals to [00:09:35.396] Sixteen star sixty four greater than thousand less nine. [00:09:53.796] Look at the answer. How the answer is in here.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.228] 16 multiplied by 64. It is more than thousand. So this thousand is more than this. So therefore it gives a value one. That one is added with nine. Therefore the answer is ten. Right? Let us take another example. [00:10:18.372] four space minus one space seven five three less than or equal to four. Look at the answer. It is one one zero zero one because four [00:10:35.972] Less than or equal to four, it is true. Therefore the answer is one. Minus one less than or equal to four. True. The value is one. Seven is not equal to is not less than or equal to four. Therefore 0, 0 and 1. Right? Let us take another example. Sum [00:10:57.188] you can take fourteen then nine three four one four like this equal to four now look at the answer the answer is two here how is it possible [00:11:14.212] Now 14 is compared with 4. So 14 is not equal to 4. It gives the value 0. 9 not equal to 4, another 0. 3 is equal to 4, another 0. There are 3 zeros. When I add again I get 0. Now 4 equal to 4, you will get 1. [00:11:33.796] One not equal to four get zero, four equal to four another one. So you got two ones. It is added because we have used some, therefore the answer is two. Is it clear? Let us take another example. X equals to 10. CB colon. [00:11:53.092] X till a equal to twenty. It means see the answer. X is ten and X is not equal to twenty. It is true the answer is one. Now three [00:12:13.764] is not equal to four. It is true, therefore it returns the value one. Three not equal to three. It is false. Therefore the answer is zero. So this is how relational operators work in MATLAB. [00:12:33.892] Let's look at logical operators. The logical data type represents true or false states using the numbers 1 and 0 respectively. Right? It means we have uh three logical operators [00:12:52.356] The first one is and second one is r and the last one is negation. Okay now let's look at one by one. The logical

operator and it means when both operates are true then it gives the value one, otherwise it is zero. [00:13:13.348] It is true, that is one when both the operates are true, otherwise the value is 0. The result is false. Let's look at the example. Let us take A equals to 1. Semicolon. [00:13:30.564] B equal to one semicolon a and b and b both are one it is true therefore the answer is one let's take another example a equal to zero semicolon [00:13:51.332] B equal to one semi-golen a and b it compares it is not true therefore the answer is zero that's it another example a equals to one zero [00:14:10.692] Let us take B. [00:14:16.420] Zero one one [00:14:22.116] [ASR low confidence; check audio] E and B. [00:14:25.412] No the answer is zero. How it happens? One it compares one and zero. And one and zero, one it compares element by element, then it gives you corresponding answer. It means the logical and it is true when both are true. Otherwise the result is false. [00:14:44.356] Let's look at logical operator R [00:14:52.580] The logical operator R it is true when one or both operands are true. You can see when both when one or both operands are true, it is one, otherwise it is zero. Therefore got three ones here and one zero. So let's look at the same example to understand this.

EV0000034 · EV0000035 · EV0000036 · EV0000037

Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.028] Let us take A equals to one, B is also one, then A B is one, A is zero, and B is one then A one B [00:15:32.196] That's it still one equal to one zero then it take me to be zero one one [00:15:50.244] then A or B the answer is six ones right so it is true when one or both are true otherwise it is false lastly we have this logical operator [00:16:09.444] negation till then if one then is zero zero and one let's look at the example for this let us take uh one by zero and minus two right take the not [00:16:29.604] operation on this it is one by zero minus two then this is the corresponding research thank you

EV0000038 · EV0000039 · EV0000040 · EV0000041

Full transcript: 1610 normalized words, 61 timestamped segments, 1004.715 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:14:22.116

EV0000026 · EV0000027 · EV0000028 · EV0000029

4 M-Files; Loop Control Statements; Conditional Statements

This lesson is indexed from the source video titled M-Files; Loop Control Statements; Conditional Statements. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000042 · EV0000043

Graphs and diagrams

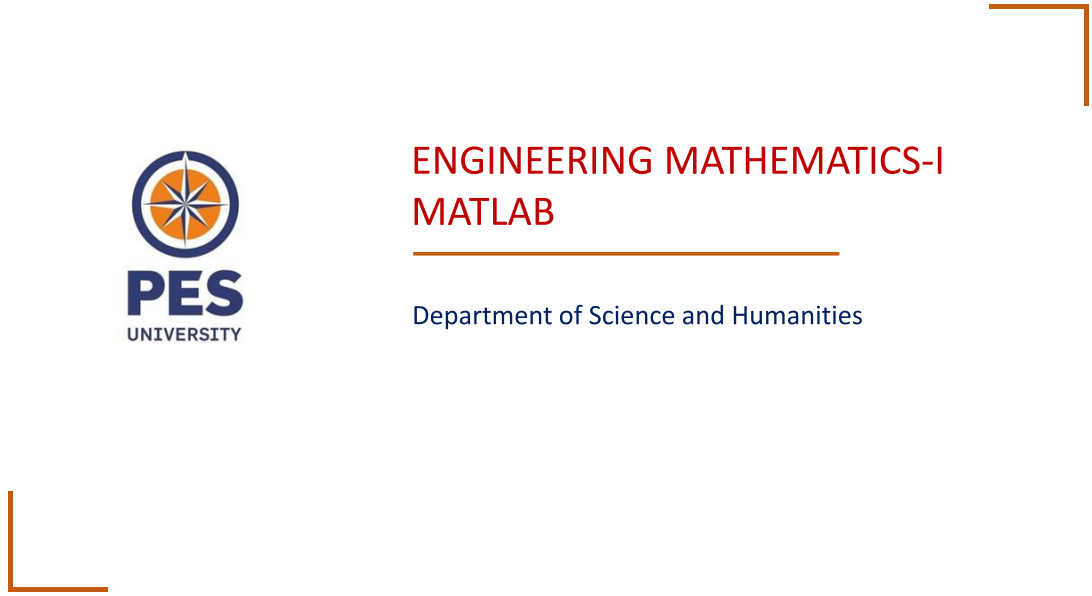

Figure G000073. Slides · source slide page 1.

EV0000315

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

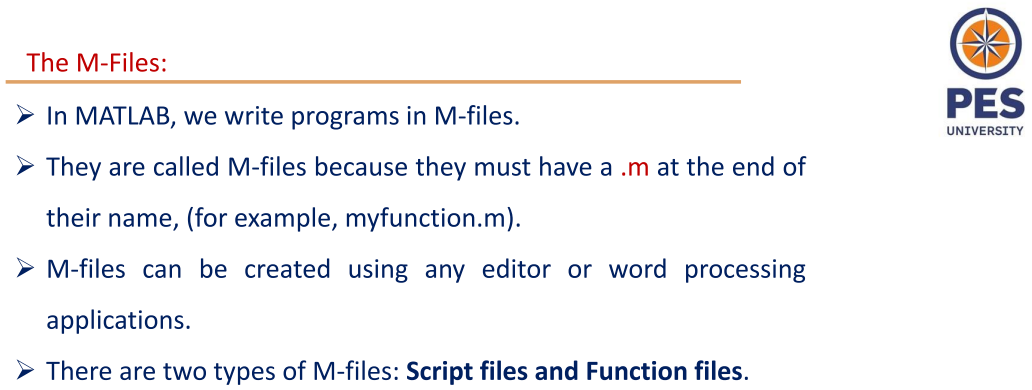

Figure G000074. Slides · source slide page 2.

EV0000318

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

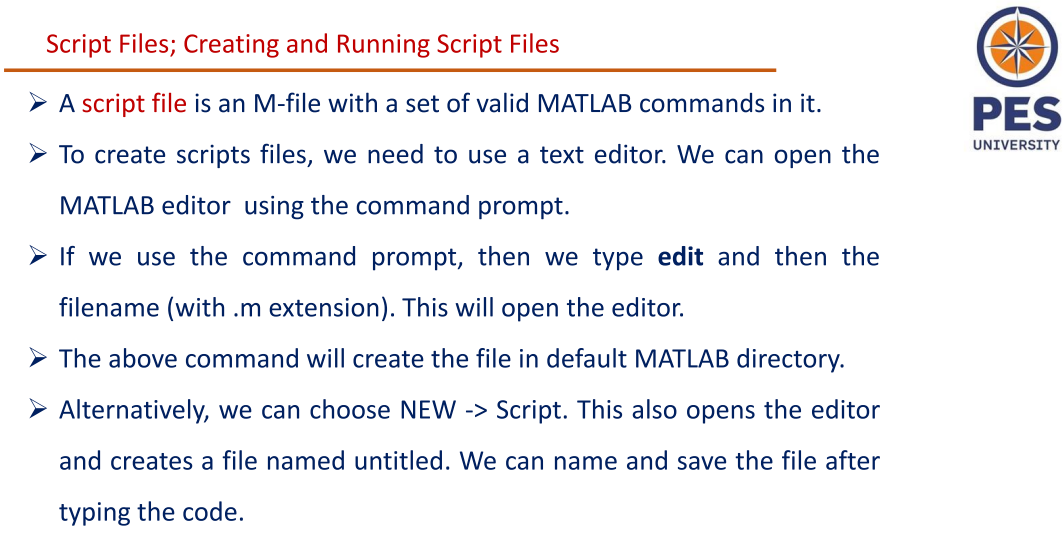

Figure G000075. Slides · source slide page 3.

EV0000321

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

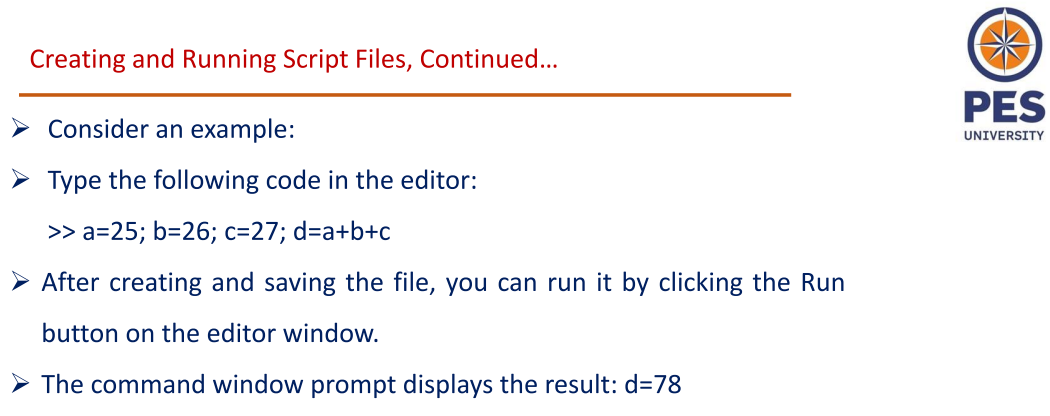

Figure G000076. Slides · source slide page 4.

EV0000324

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

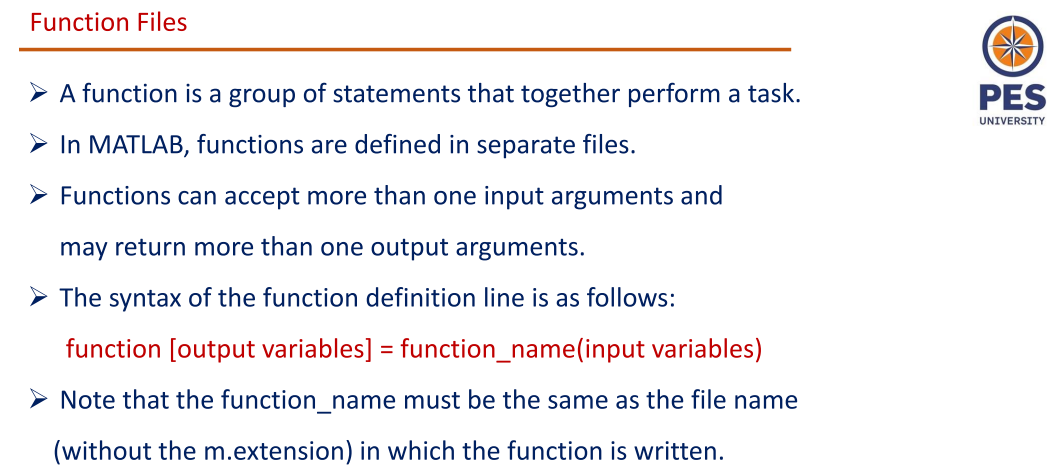

Figure G000077. Slides · source slide page 5.

EV0000327

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

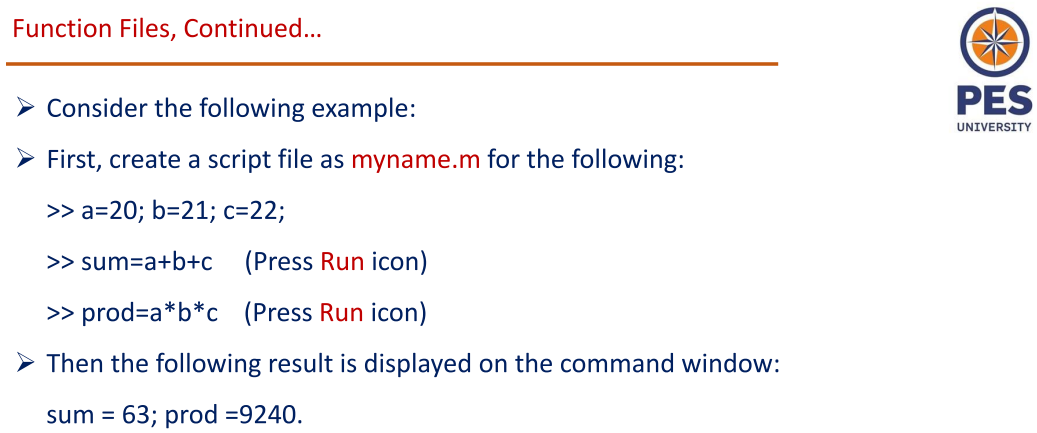

Figure G000078. Slides · source slide page 6.

EV0000330

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

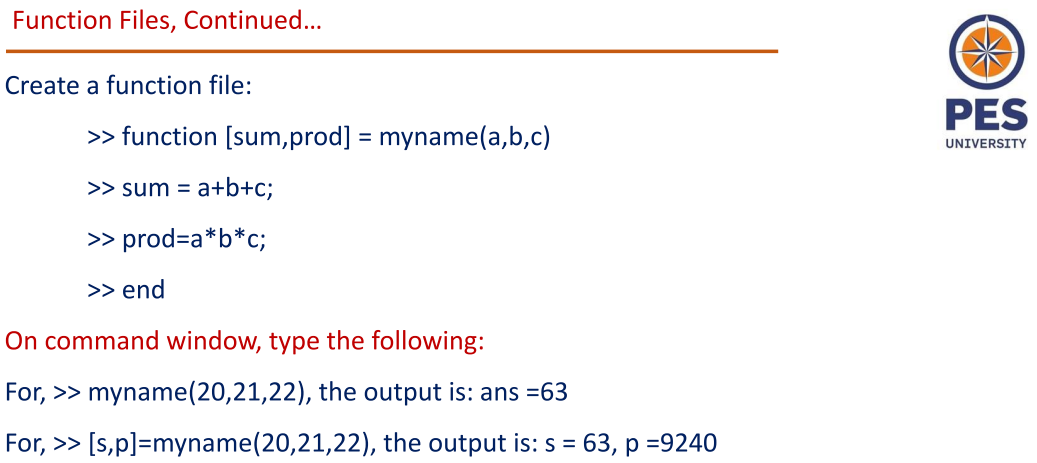

Figure G000079. Slides · source slide page 7.

EV0000333

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Loop Control Statements

- With loop control statements, we can repeatedly execute a block of code. There are two types of loops: for loops and while loops.
- for loop: A for loop is used to repeat a statement or a group of statements for a fixed number of times.
- The syntax is:

```
for index = values
<program statements>
.....
end
```

Figure G000080. Slides · source slide page 8.

EV0000336

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

for loop, Continued…

- For example, consider the following code:

```
for m=1:3
num=1/(m+1)
end
```

- When we run the file, it displays the following result:

```
num = 0.5000
num = 0.3333
num = 0.2500
```

Figure G000081. Slides · source slide page 9.

EV0000339

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

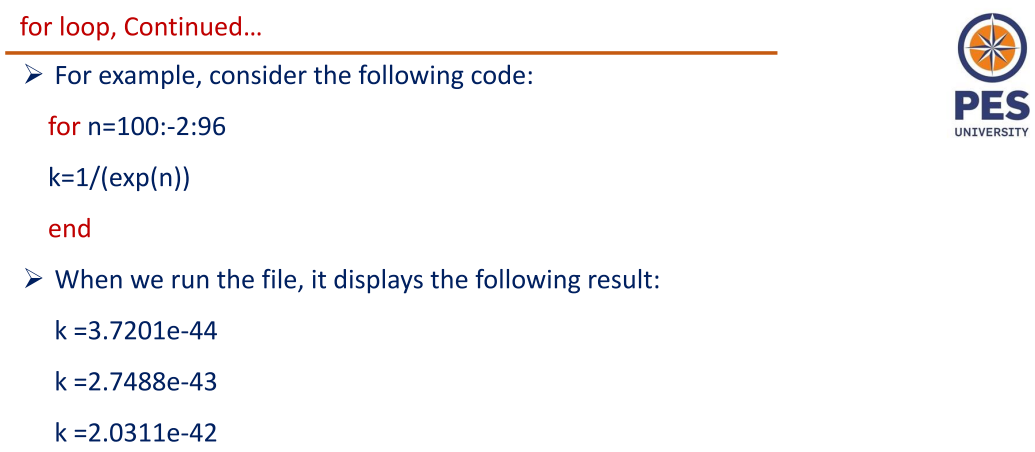

Figure G000082. Slides · source slide page 10.

EV0000342

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

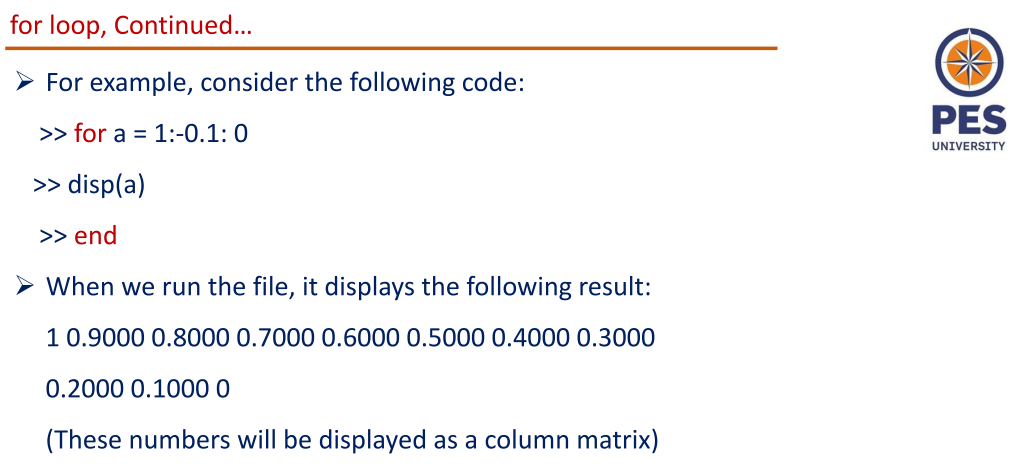

Figure G000083. Slides · source slide page 11.

EV0000345

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

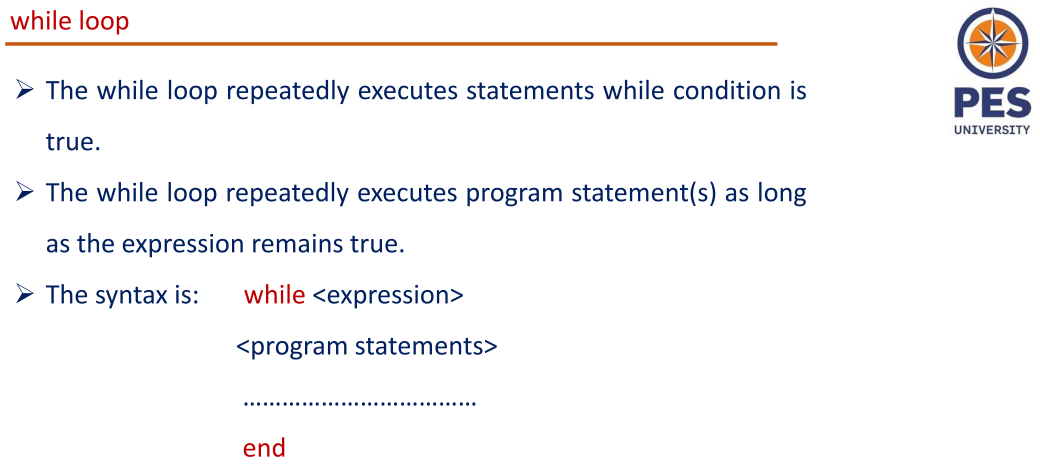

Figure G000084. Slides · source slide page 12.

EV0000348

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

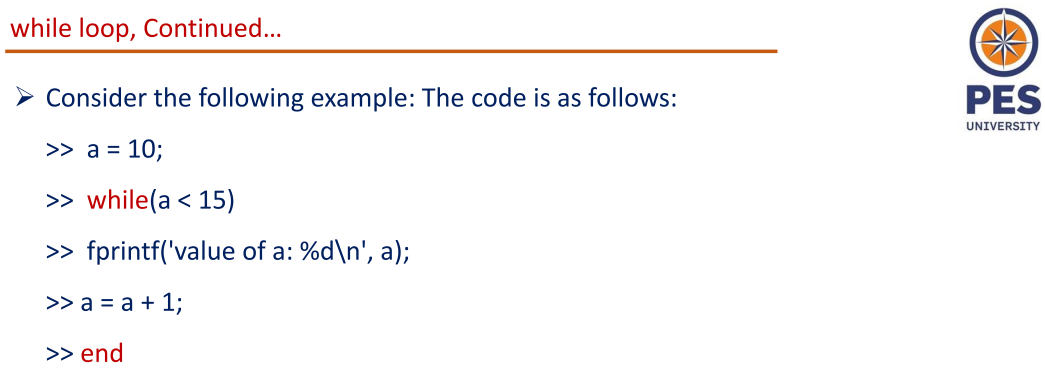

Figure G000085. Slides · source slide page 13.

EV0000351

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

while loop, Continued…

- When we run the file, it displays the following result:

  value of a: 10

  value of a: 11

  value of a: 12

  value of a: 13

  value of a: 14

Figure G000086. Slides · source slide page 14.

EV0000354

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Conditional Statements

- There will be some situation where a program or a particular block has to be executed only when a specific condition is true. The conditional statements will be useful in such situations.
- The following are the conditional statements that we can use in MATLAB. if-end; if-else-end; if-elseif-elseif-else-end; switch case.

Figure G000087. Slides · source slide page 15.

EV0000357

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

if-end statement

- if-end statement: It decides whether a particular block of code has to be executed or not, based on the given Boolean condition. When the given condition is true, only then it executes the statements inside the block otherwise not.

Figure G000088. Slides · source slide page 16.

EV0000360

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

if-end statement, Continued…

- For example, consider the following:

```
>> number = 25;
>> if number > 20
>> fprintf('The number is greater than 20');
>> end
```

- When we run the file, it displays the following result:

```
The number is greater than 20.
```

Figure G000089. Slides · source slide page 17.

EV0000363

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

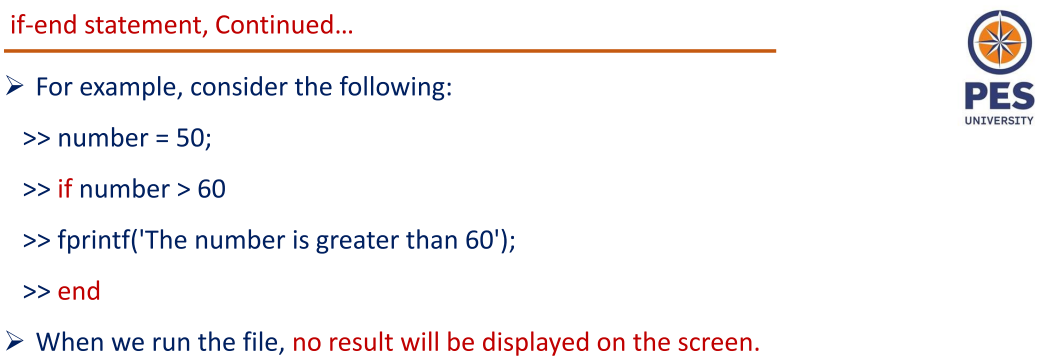

Figure G000090. Slides · source slide page 18.

EV0000366

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

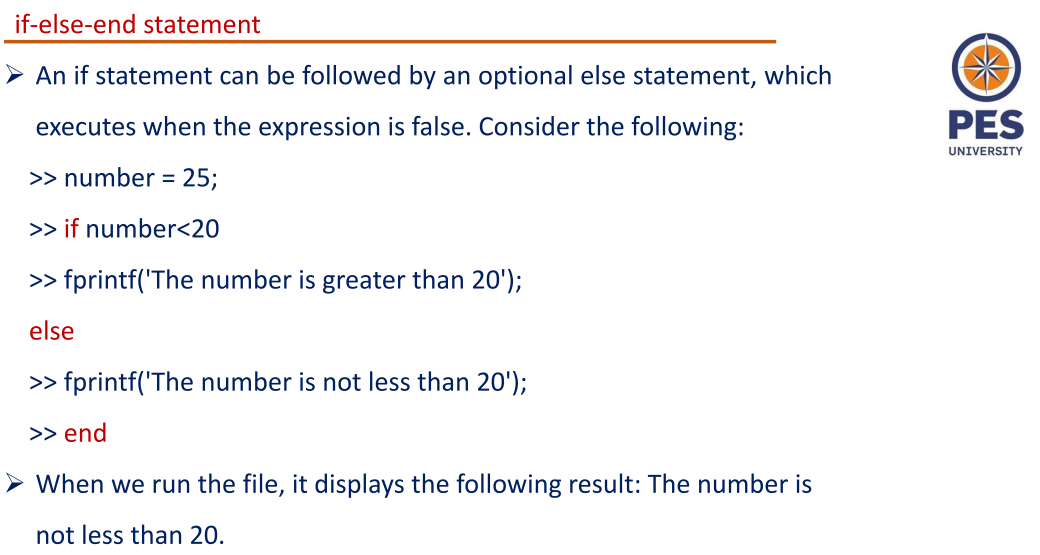

Figure G000091. Slides · source slide page 19.

EV0000369

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

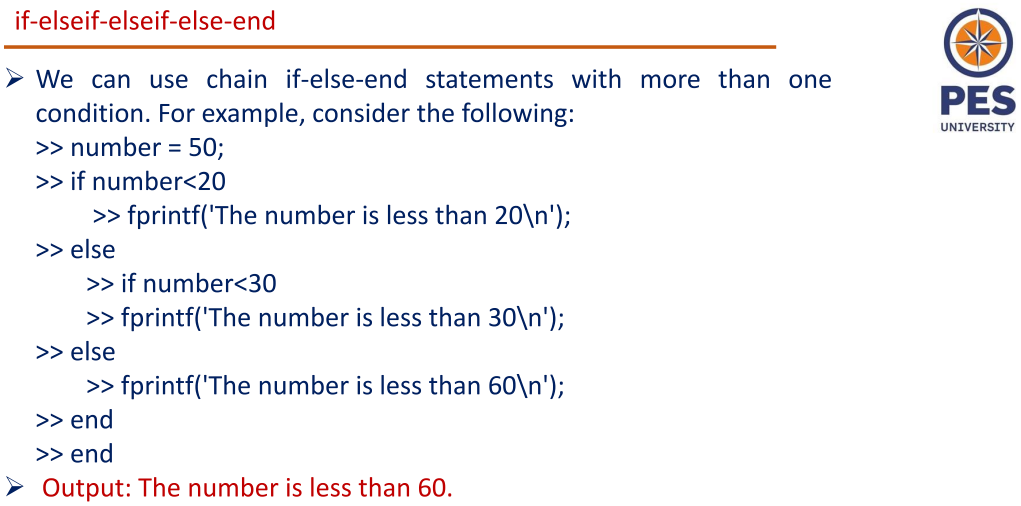

Figure G000092. Slides · source slide page 20.

EV0000372

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

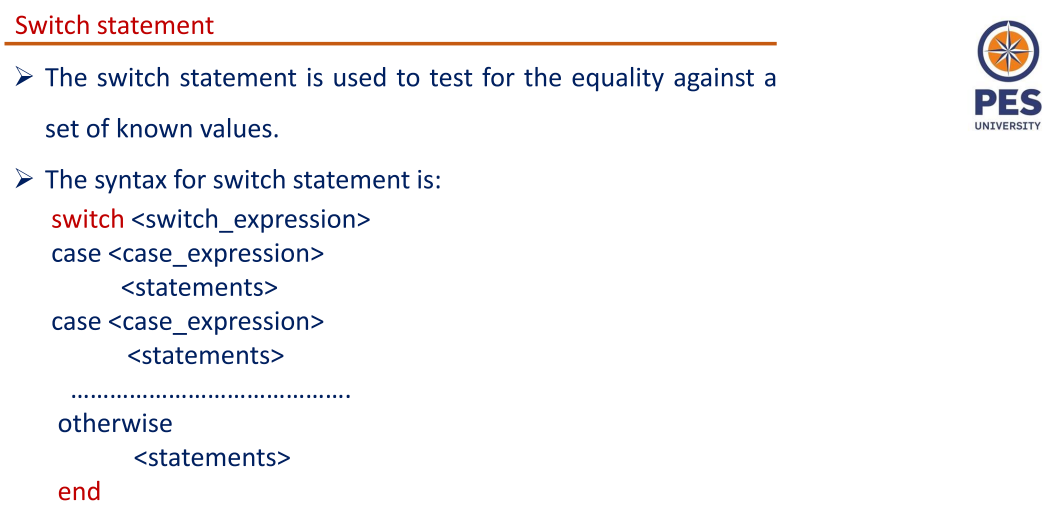

Figure G000093. Slides · source slide page 21.

EV0000375

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

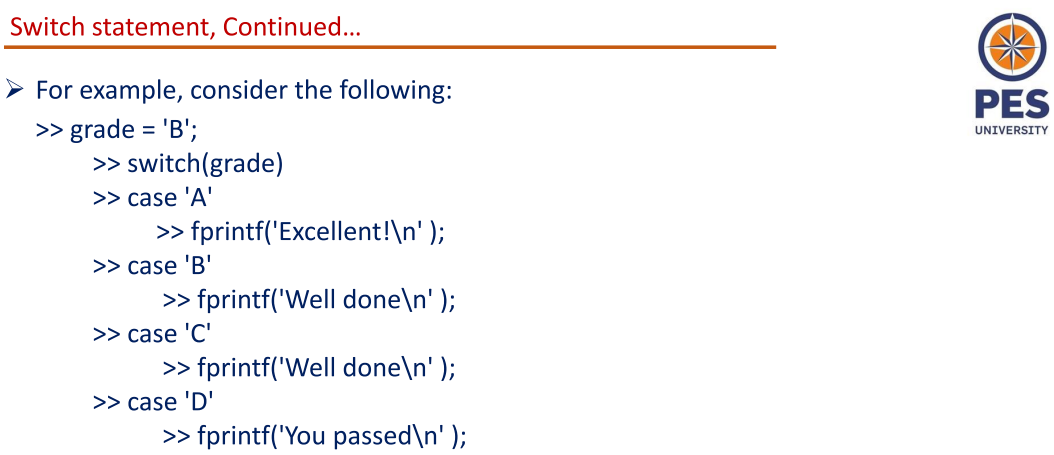

Figure G000094. Slides · source slide page 22.

EV0000378

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

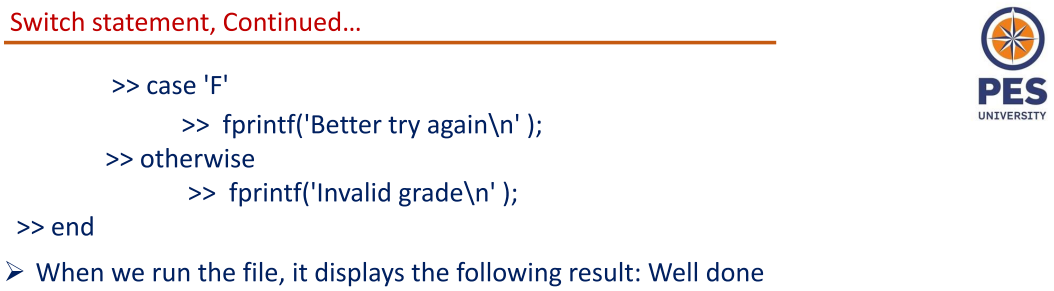

Figure G000095. Slides · source slide page 23.

EV0000381

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000096. Slides · source slide page 24.

EV0000383

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000314 · EV0000315 · EV0000316
2. Source slide page 2 (extracted text): The M-Files: ➢In MATLAB, we write programs in M-files. ➢They are called M-files because they must have a .m at the end of their name, (for example, myfunction.m). ➢M-files can be created using any editor or word processing applications. ➢There are two types of M-files: Script files and Function files. EV0000317 · EV0000318 · EV0000319
3. Source slide page 3 (extracted text): Script Files; Creating and Running Script Files ➢A script file is an M-file with a set of valid MATLAB commands in it. ➢To create scripts files, we need to use a text editor. We can open the MATLAB editor using the command prompt. ➢If we use the command prompt, then we type edit and then the filename (with .m extension). This will open the editor. ➢The above command will create the file in default MATLAB directory. ➢Alternatively, we can choose NEW -> Script. This also opens the editor and creates a file named untitled. We can name and save the file after typing the code. EV0000320 · EV0000321 · EV0000322
4. Source slide page 4 (extracted text): Creating and Running Script Files, Continued... ➢ Consider an example: ➢ Type the following code in the editor: >> a=25; b=26; c=27; d=a+b+c ➢After creating and saving the file, you can run it by clicking the Run button on the editor window. ➢The command window prompt displays the result: d=78 EV0000323 · EV0000324 · EV0000325

5. Source slide page 5 (extracted text): Function Files ➢A function is a group of statements that together perform a task. ➢In MATLAB, functions are defined in separate files. ➢Functions can accept more than one input arguments and may return more than one output arguments. ➢The syntax of the function definition line is as follows: function [output variables] = function_name(input variables) ➢Note that the function_name must be the same as the file name (without the m.extension) in which the function is written. EV0000326 · EV0000327 · EV0000328
6. Source slide page 6 (extracted text): Function Files, Continued... ➢Consider the following example: ➢First, create a script file as myname.m for the following: >> a=20; b=21; c=22; >> sum=a+b+c (Press Run icon) >> prod=a*b*c (Press Run icon) ➢Then the following result is displayed on the command window: sum = 63; prod =9240. EV0000329 · EV0000330 · EV0000331
7. Source slide page 7 (extracted text): Function Files, Continued... Create a function file: >> function [sum,prod] = myname(a,b,c) >> sum = a+b+c; >> prod=a*b*c; >> end On command window, type the following: For, >> myname(20,21,22), the output is: ans =63 For, >> [s,p]=myname(20,21,22), the output is: s = 63, p =9240 EV0000332 · EV0000333 · EV0000334
8. Source slide page 8 (extracted text): Loop Control Statements ➢With loop control statements, we can repeatedly execute a block of code. There are two types of loops: for loops and while loops. ➢for loop: A for loop is used to repeat a statement or a group of statements for a fixed number of times. ➢The syntax is: for index = values <program statements> ..... end EV0000335 · EV0000336 · EV0000337
9. Source slide page 9 (extracted text): for loop, Continued... ➢For example, consider the following code: for m=1:3 num=1/(m+1) end ➢When we run the file, it displays the following result: num = 0.5000 num = 0.3333 num = 0.2500 EV0000338 · EV0000339 · EV0000340
10. Source slide page 10 (extracted text): for loop, Continued... ➢For example, consider the following code: for n=100:-2:96 k=1/(exp(n)) end ➢When we run the file, it displays the following result: k =3.7201e-44 k =2.7488e-43 k =2.0311e-42 EV0000341 · EV0000342 · EV0000343
11. Source slide page 11 (extracted text): for loop, Continued... ➢For example, consider the following code: >> for a = 1:-0.1: 0 >> disp(a) >> end ➢When we run the file, it displays the following result: 1 0.9000 0.8000 0.7000 0.6000 0.5000 0.4000 0.3000 0.2000 0.1000 0 (These numbers will be displayed as a column matrix) EV0000344 · EV0000345 · EV0000346
12. Source slide page 12 (extracted text): while loop ➢The while loop repeatedly executes statements while condition is true. ➢The while loop repeatedly executes program statement(s) as long as the expression remains true. ➢The syntax is: while <expression> <program statements> ..... end EV0000347 · EV0000348 · EV0000349
13. Source slide page 13 (extracted text): while loop, Continued... ➢Consider the following example: The code is as follows: >> a = 10; >> while(a < 15) >> fprintf('value of a: %d\n', a); >> a = a + 1; >> end EV0000350 · EV0000351 · EV0000352
14. Source slide page 14 (extracted text): while loop, Continued... ➢When we run the file, it displays the following result: value of a: 10 value of a: 11 value of a: 12 value of a: 13 value of a: 14 EV0000353 · EV0000354 · EV0000355

15. Source slide page 15 (extracted text): Conditional Statements ➢There will be some situation where a program or a particular block has to be executed only when a specific condition is true. The conditional statements will be useful in such situations. ➢The following are the conditional statements that we can use in MATLAB. if-end; if-else-end; if-elseif-elseif-else-end; switch case. EV0000356 · EV0000357 · EV0000358

16. Source slide page 16 (extracted text): if-end statement ➢if-end statement: It decides whether a particular block of code has to be executed or not, based on the given Boolean condition. When the given condition is true, only then it executes the statements inside the block otherwise not. EV0000359 · EV0000360 · EV0000361

17. Source slide page 17 (extracted text): if-end statement, Continued... ➢For example, consider the following: >> number = 25; >> if number > 20 >> fprintf('The number is greater than 20'); >> end ➢When we run the file, it displays the following result: The number is greater than 20. EV0000362 · EV0000363 · EV0000364

18. Source slide page 18 (extracted text): if-end statement, Continued... ➢For example, consider the following: >> number = 50; >> if number > 60 >> fprintf('The number is greater than 60'); >> end ➢When we run the file, no result will be displayed on the screen. EV0000365 · EV0000366 · EV0000367

19. Source slide page 19 (extracted text): if-else-end statement ➢An if statement can be followed by an optional else statement, which executes when the expression is false. Consider the following: >> number = 25; >> if number<20 >> fprintf('The number is greater than 20'); else >> fprintf('The number is not less than 20'); >> end ➢When we run the file, it displays the following result: The number is not less than 20. EV0000368 · EV0000369 · EV0000370

20. Source slide page 20 (extracted text): if-elseif-elseif-else-end ➢We can use chain if-else-end statements with more than one condition. For example, consider the following: >> number = 50; >> if number<20 >> fprintf('The number is less than 20\n'); >> else >> if number<30 >> fprintf('The number is less than 30\n'); >> else >> fprintf('The number is less than 60\n'); >> end >> end ➢ Output: The number is less than 60. EV0000371 · EV0000372 · EV0000373

21. Source slide page 21 (extracted text): Switch statement ➢The switch statement is used to test for the equality against a set of known values. ➢The syntax for switch statement is: switch <switch_expression> case <case_expression> <statements> case <case_expression> <statements> ..... otherwise <statements> end EV0000374 · EV0000375 · EV0000376

22. Source slide page 22 (extracted text): Switch statement, Continued... ➢For example, consider the following: >> grade = 'B'; >> switch(grade) >> case 'A' >> fprintf('Excellent!\n'); >> case 'B' >> fprintf('Well done\n'); >> case 'C' >> fprintf('Well done\n'); >> case 'D' >> fprintf('You passed\n'); EV0000377 · EV0000378 · EV0000379

23. Source slide page 23 (extracted text): Switch statement, Continued... >> case 'F' >> fprintf('Better try again\n'); >> otherwise >> fprintf('Invalid grade\n'); >> end ➢When we run the file, it displays the following result: Well done ## Page / slide 24 (OCR) THANK YOU PES UNIVERSITY EV0000380 · EV0000381 · EV0000382

24. Source slide page 24: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000383

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.772] Hi, welcome to PS University Online Teaching Learning Platform. I am Professor Nagesh Ham, faculty member, Department of Science and Humanities. This is the third session of MATLAB. In this session we learn M files. [00:00:19.876] That is script files and function files, loop control statements, that is for loops and while loops. Finally condition statements. So let's start with M files. [00:00:44.708] In MATLAB we write programs in M files. They are called M files because they must have dot M at the end of their name. For example it should be like India.m [00:01:07.108] then searching dot m and so on right so they are called m files because they must have dot m at the end of their file name there are two types of m files [00:01:24.708] The first one script files and then function files. Later we shall see what's the advantage of scries as well as function files. Okay. Let's start with [00:01:45.732] Script files. [00:01:52.548] A script file is an M file with a set of valid MATLAB commands in it. Let us see how to create a script file. Remember there are two ways to create a script file. The first one is this. [00:02:16.548] On the command window we just write like this edit and then the file name which we want. For example let me take ecampus.m [00:02:30.980] press enter now you can see a script file has been created by the name ecval.m now you can type the command whatever you want let me say a equal to 100 then b equal to 200 [00:02:49.028] then c equal to 300. Right? So I want to find out the sum of this. So sum equal to A plus B plus C. Now there is a small difference here in order to compile this we just run this. You can see the output is going to be 600. [00:03:09.604] We also find out the product of these three numbers PR O D A star B star C. Again, use this button run and you can see the answer is going to be this much. Okay, now you just cancel this and just clear the screen. [00:03:29.828] Now what the advantage of this for example if you want to use the same file in future you can use script file like this for example either you can use ECK directly from here or you can use edit then file name [00:03:47.684] eccampus.m so when you do like this you can see the comments what we have used now what the second way of creating a script file now let us go to new then write script [00:04:07.780] Again you give the command whatever you want. Let me find out a equal to 10 colon b equal to [00:04:21.828] Sign of y by two sine of pi by two then c equal to twenty right let me find out b equal to a plus b plus [00:04:39.620] C. Now if you want to compile this. So before you run this, we have to save this file. Again go to this save. Again you can give. Let me give EC campus one. EC campus one dot m right now. [00:04:59.684] the file has been created by the name ecampus1.m. Now if you want to compile this I can use this run. You can see the answer is going to be 31. Now you cancel this and you clear the screen

EV0000042 · EV0000043 · EV0000044 · EV0000045

Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.700] Assume you want to use this after one month of time. You want to compile this. So what you do? Either you can choose the file from here from the current folder or you just type edit then the file name ecampus1.m again you can see the comments on the screen. [00:05:32.932] So this is the advantage of script files and it is very easy to create. Now what are the use of having function files? Now clear this, clear the screen. So let us understand function files. [00:05:52.548] function points A function is a group of statements that together perform a task. A function is a group of statements that together perform a task. [00:06:05.060] So let's understand how to create a function file. So remember before we create function file, we have to create a script file, right? So let us take an example. Just clear the screen. So let me take another different name. It let me take such in. [00:06:24.676] such in dot m

let's create a script file for this searching now you give the command e equal to [00:06:39.268] one B equal to two C equal to hundred then find out the sum equal to A plus B plus C [00:06:56.900] then run this against answer is 103 then find out your product equal to a star b star c [00:07:12.740] We run this, you can see answer is going to be 200. Now, how we compile the same thing using function file, right? Okay, let us create a function file. So for that, what you need to do, you just uh clear the screen and remember for this particular example. [00:07:32.612] You have three inputs, A, B, C are the inputs, and the sum and product are the output. Clear. A B C are the input variables and sum and product are the output variables here. Clear. Now let us create the function for the same example. So clear the screen. [00:07:53.796] Can I cancel this. Okay. [00:08:00.196] We just use uh new, let us choose function. Is it clear? Now this is the syntax of the function file. As we know that from the previous example we have uh output variable as summon product su m then product. [00:08:20.292] is clear then you can see the input variables are a comma b comma c a comma b comma c like this now remember the file name should be same as the one which we have used for creating the script file clear so this should be same as such in [00:08:40.676] without M we should not use dot M here. Is it clear? Then we can use sum equal to A plus B plus C [00:08:55.748] Then Product PROD equal to A star B star C, B star C. Okay, now, [00:09:05.636] Remember before we run this we have to say this. Remember again the file m should be same as the one what we have used for creating script file. So let us save this. Yes. Now [00:09:24.324] We have uh saved a function file by searching.m. Now in order to compile this we should use command window. Remember in order to compile this we should use command window. So on the command window we just give like this function name searching. [00:09:44.068] search in of you can use whatever variables you want you can give uh okay I gave one comma one comma hundred right okay this is equal you can write it as s for sum p for product

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.556] Equal to like this. Now shift enter. You can see S is 102, the sum is 102, and the product is 100. Now what the advantage of having this function five? See if you want to [00:10:20.772] Add some of the three numbers. What we should do? We have to manually update a script file every time. So when you use function file, it is not necessary. Now for example, for the same command, we can give some number like this. Just copy paste here. You can change the number here. 200 [00:10:41.348] Then hundred, then thousand. Use we can see the output. It means instead of manually updating the script each time, we can make program more flexible by converting it to a function. Is it clear? So functions are more flexible than [00:11:00.356] extendable. So we can use however we want. So this is the advantage of having uh function files as well as script files. Okay [00:11:16.260] Clear the screen. Next we shall move on to low control statements. [00:11:30.404] Using loop control statements, we can repeatedly execute a block of code. There are two types of loops. One is for loops and while loops. Let's talk about for loops. For loops [00:11:50.692] For loop is used to repeat a statement or a group of statements for a specific number of times. For loop is used to repeat a statement or a group of statements for a specific number of times. For example, we consider this. [00:12:10.116] Let's see and see let us take four let us take a equals to one comma two comma three comma four right the display [00:12:29.540] Remember here what we see DISP of A, this is not a display command. It displays the values of A without a variable name. Is it clear? You can see the this is A is a variable whose values are 1, 2, 3, 4. Now what DISP of A does is it [00:12:49.252] Displays the values of A without printing a variable name that is A. Now you can see the output is

1234. Let us take another example for A equal to 1 colon. [00:13:09.412] minus 0 point one colon zero then display then n now here a equal to minus 1 colon minus 0 point one zero see here minus stands for decreasing [00:13:27.300] decrement order decreasing order it means a start from zero goes up to a starts from one goes up to zero with a decrement value of 0.21 you can see this is the output is that clear let us take another example for [00:13:47.620] m equal to 1 colon 3. Here m start from 1 goes up to 3 so you can write number equal to 1 divided by m plus 1 then n [00:14:06.884] So these are the values. So when m is one the number is one by m plus one. When m is two number is uh one zero point double three double three and so on. Is it clear? So for loop is used to repeat a statement or a group of statements for a specific number of times. [00:14:25.412] Okay let's understand what is while loop while loop repeatedly execute program statements as long as the expression remains true [00:14:39.876] Is it clear? While loop repeatedly execute program statements as long as the expression remains true. Let us take an example. Let us define the variable A equals to 10. Right? Then now the while loop starts. [00:14:59.396] And the condition here is A is less than fifteen. It means

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.196] the while will run till the value of A is less than 15. So currently it is 10. Next we print the value F P R I N D print f printf [00:15:23.140] value of e colon percentage D slash N [00:15:34.788] It means it prints a current value of A and then A equal to A plus 1 is executed. Then N [00:15:49.540] It means this is the output for the following command. It means we first define a equals to 10. We define the variable a equal to 10. Now the while loop starts and the condition for a is a is less than 15. [00:16:04.484] It means that the while loop will run till the value of A is less than 15. Currently the value of A is 10. Now this line you just print the current value of A and then A equal A plus 1 is executed. The line A equal to A plus 1 [00:16:23.332] add 1 to A and then while conduction is checked. Now the value of A is 11 which is less than 15 so loop runs again. So finally the loop runs until A gets 14 and after that it stops. That's why the output is like this 10 to 14 [00:16:43.652] After understanding loop control statements, so let's uh move on to conditional statements. [00:16:53.828] So this is what we have discussed, you know m files, how to create function script file, function files, loop control statements, for loops and while loops. And these are the some of the examples what we have discussed. [00:17:10.692] Let us model the loop a condition statements. There will be some situation where a program or a particular block has to be executed only when a specific condition is true. The condition statements will be useful in such situations. [00:17:28.580] The following are the condition statements that we can use in MATLAB. If and if else and if else if else if else and switch case. So these are the some of the condition statements what we use in MATLAB. If and statement. [00:17:49.060] It decides whether a particular block of code has to be executed or not based on the given Boolean condition that is 0 or 1. When the given condition is true, only then it executes a statements inside the block. Otherwise, no. Now let us take an example. Here [00:18:08.900] Number equal to 25. So if number is more than 20, it prints the value, the number is greater than 20. Otherwise no. Here you can see that number is 25. [00:18:20.644] So this 25 is more than 20, so it is true, therefore it prints the output like this. So when we run the file, it displays the result, the number is greater than 20. Because the condition is true here. Number is 25. So this 25 is more than 20. Look at this second example. [00:18:39.940] Here the condition is false, that's why it displays nothing on the screen. For example, the number is 50, so number is greater than 60, it is not true, therefore when you run this file, it displays nothing on the screen. So

next we have uh if else and statement. [00:18:57.700] An if and statement can be followed by an optional L statement which executes when the expression is false. For example, you consider the number is 25 here. If the number is less than 20, so this is false, therefore it prints the second line. [00:19:15.844] The number is not less than 20. So when I run the file, it displays the result. The number is not less than 20. So number is 25 here. So if this condition is not true, therefore it executes the next line. Next if else if else if else and so this is a chain. [00:19:34.724] If L Z and statements which with more than one condition, for example, number is 50, number less than it is not true, this is not true, therefore the last line is printed, the number is less than 60. So next we have a switch statement. [00:19:52.324] The switch statement is used to test for equality against a set of known values. This is the syntax of switch statement and this is an example. For example, consider the following grade B, switch grade A, and these are the particular cases. So the output is going to be

EV0000054 · EV0000055 · EV0000056 · EV0000057

Transcript block 5 (source interval 001200-001499 seconds): [00:20:12.292] This one when we run the file it displays the following, is it? Well done. Thank you.

EV0000058 · EV0000059 · EV0000060 · EV0000061

Full transcript: 2472 normalized words, 70 timestamped segments, 1220.139 seconds duration. Low-confidence segments flagged: 0.

EV0000042 · EV0000043 · EV0000044 · EV0000045

5 Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling

This lesson is indexed from the source video titled Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000062 · EV0000063

Graphs and diagrams

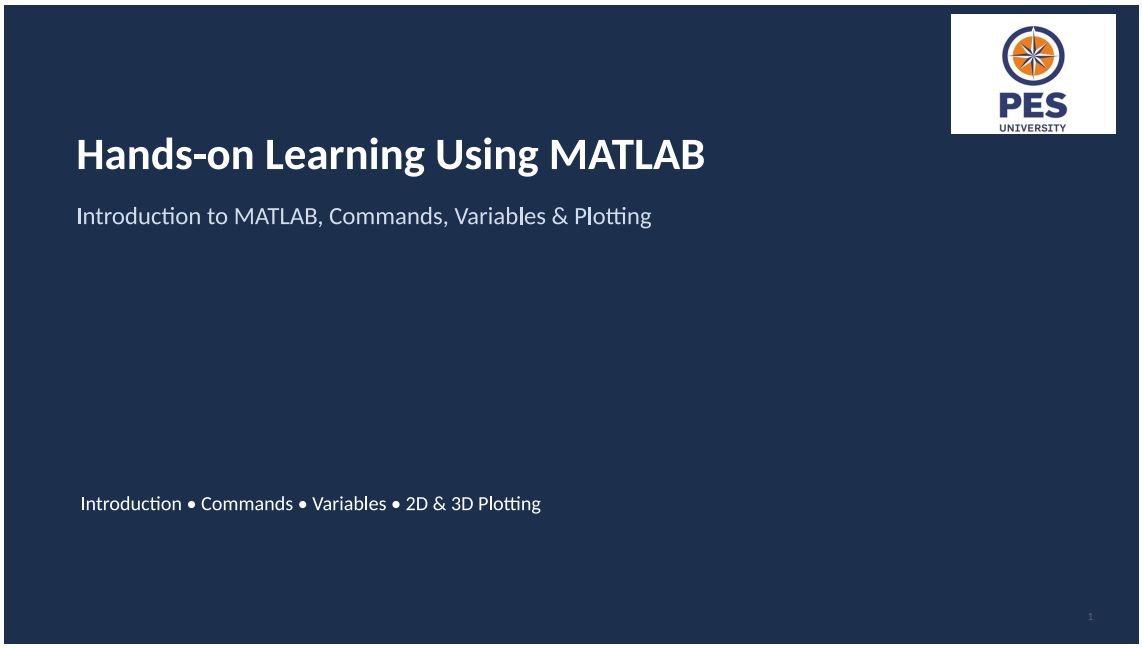

Figure G000127. MATLAB INTRODUCTION · source slide page 1.

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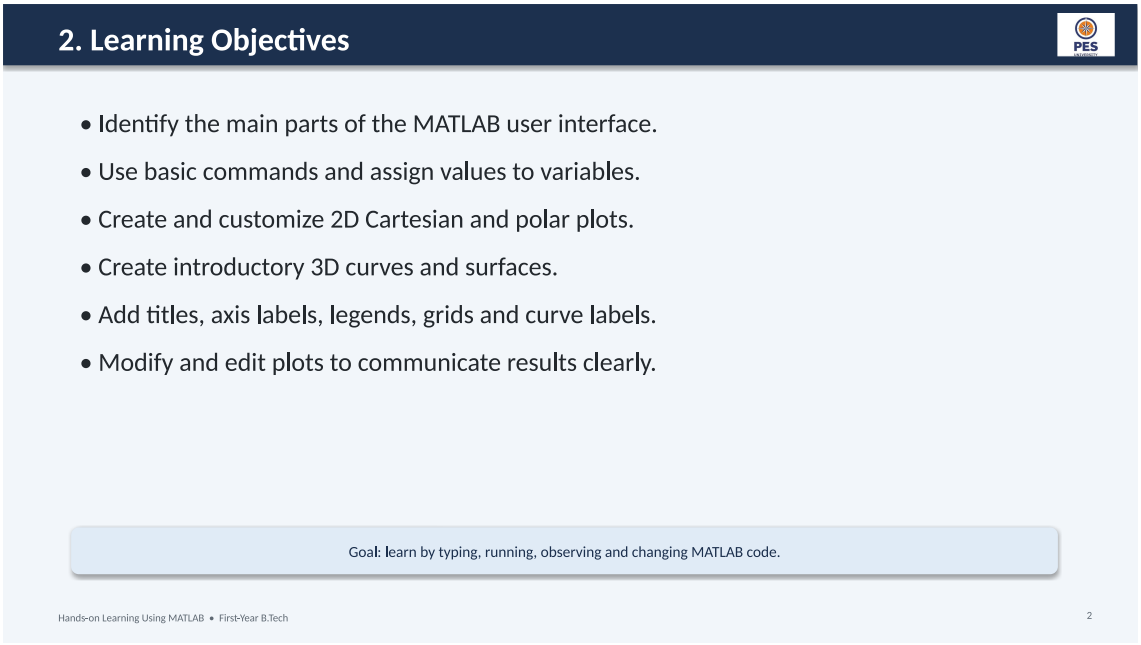

Figure G000128. MATLAB INTRODUCTION · source slide page 2.

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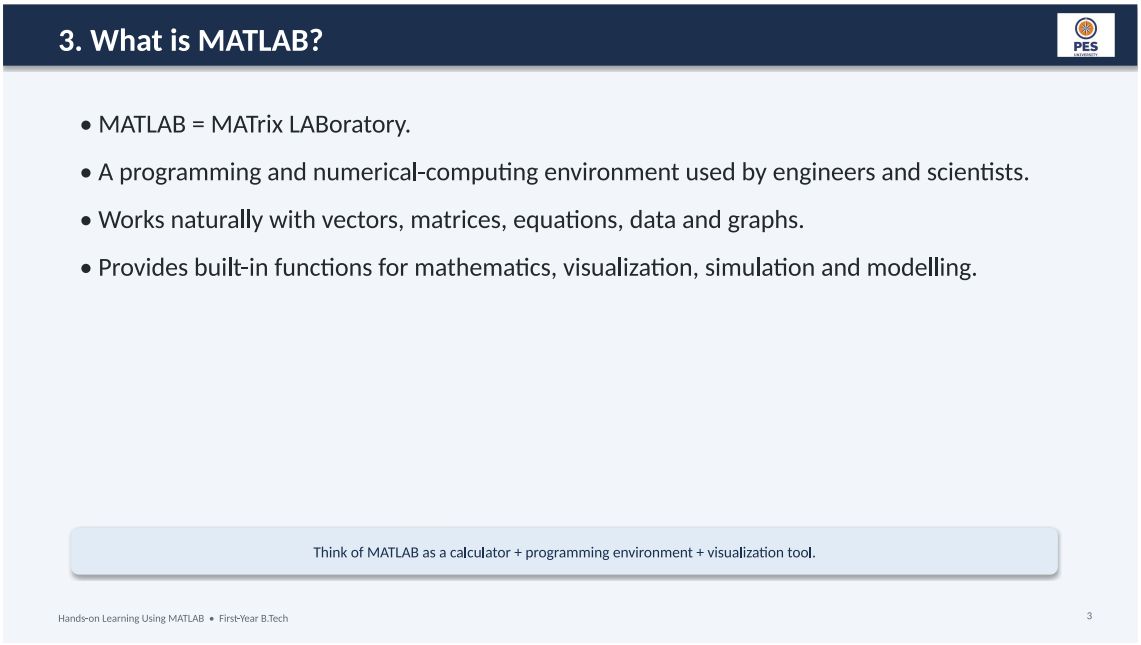

Figure G000129. MATLAB INTRODUCTION · source slide page 3.

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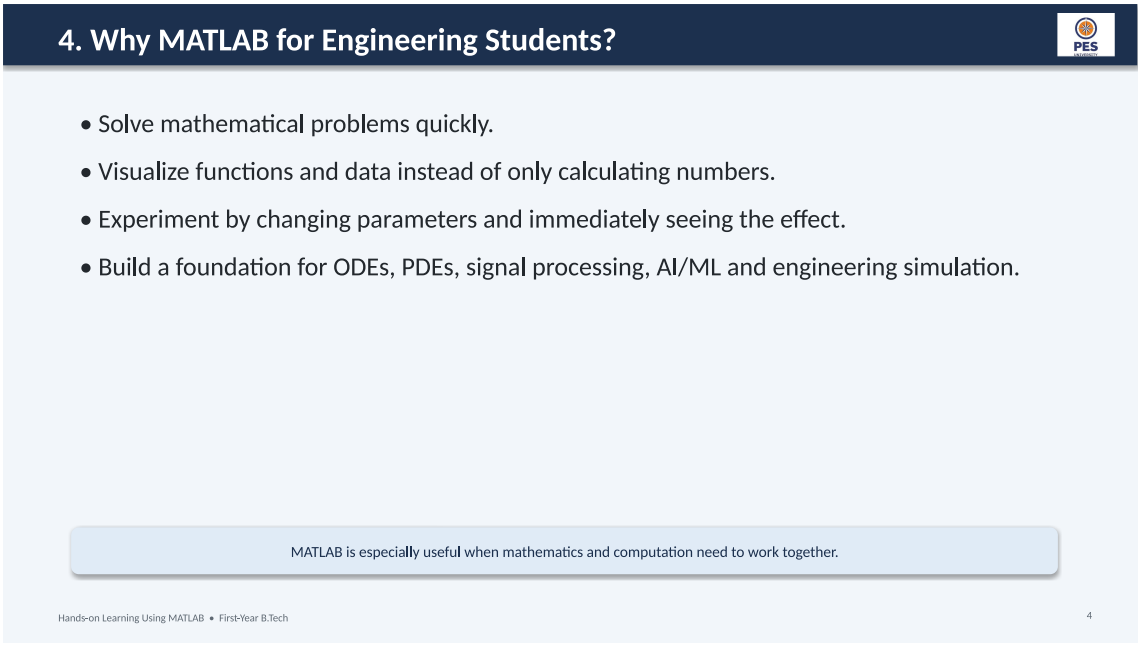

Figure G000130. MATLAB INTRODUCTION · source slide page 4.

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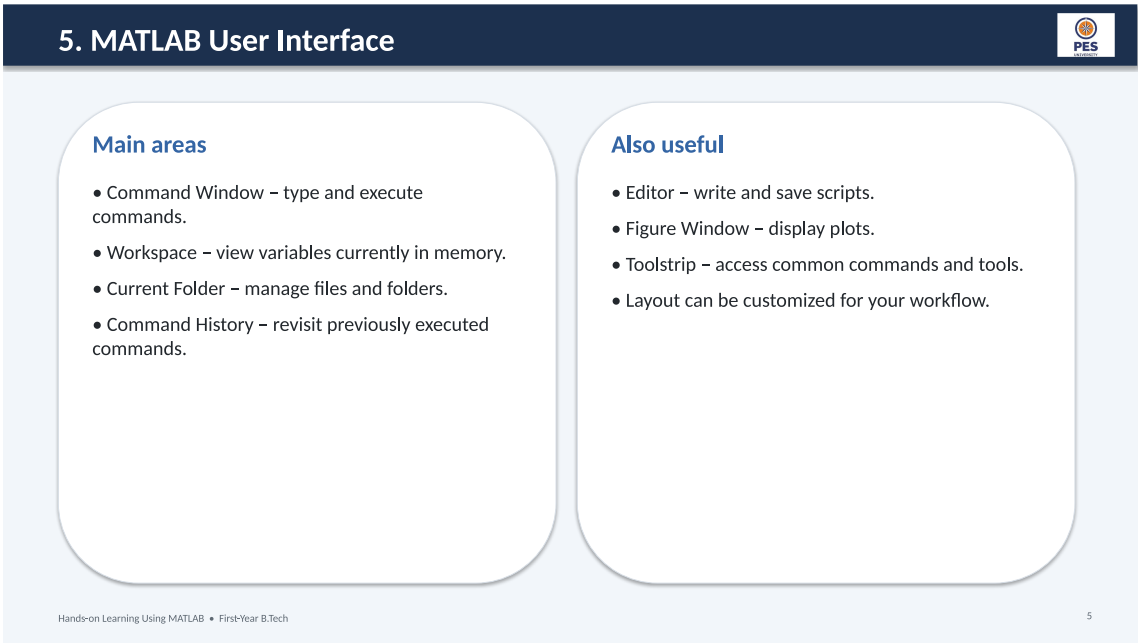

Figure G000131. MATLAB INTRODUCTION · source slide page 5.

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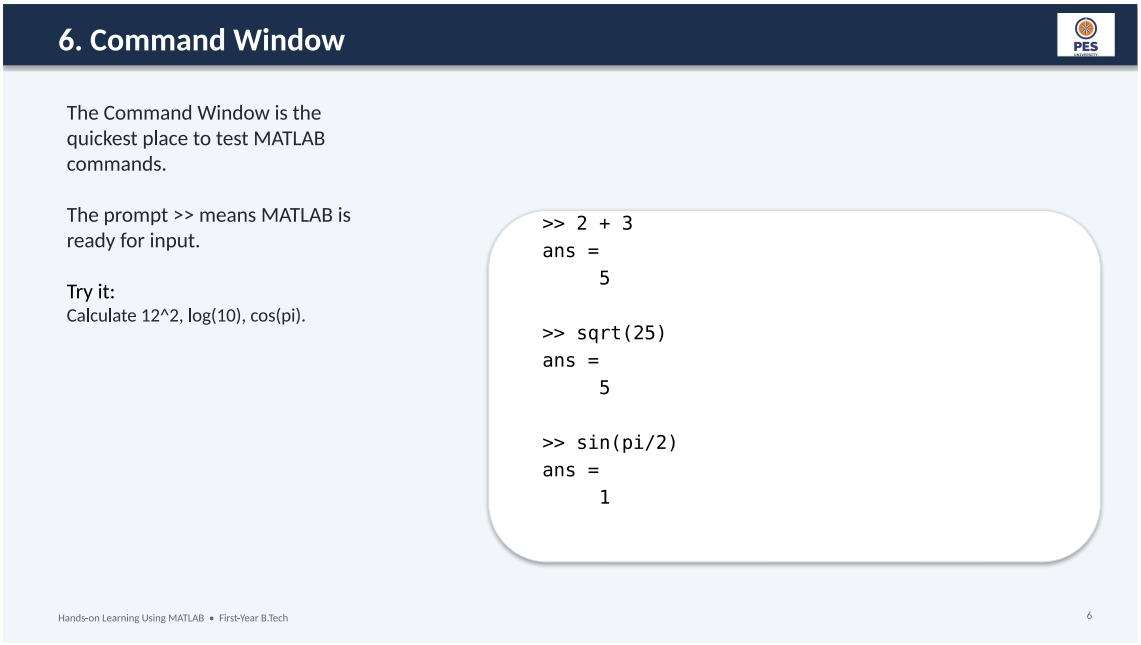

Figure G000132. MATLAB INTRODUCTION · source slide page 6.

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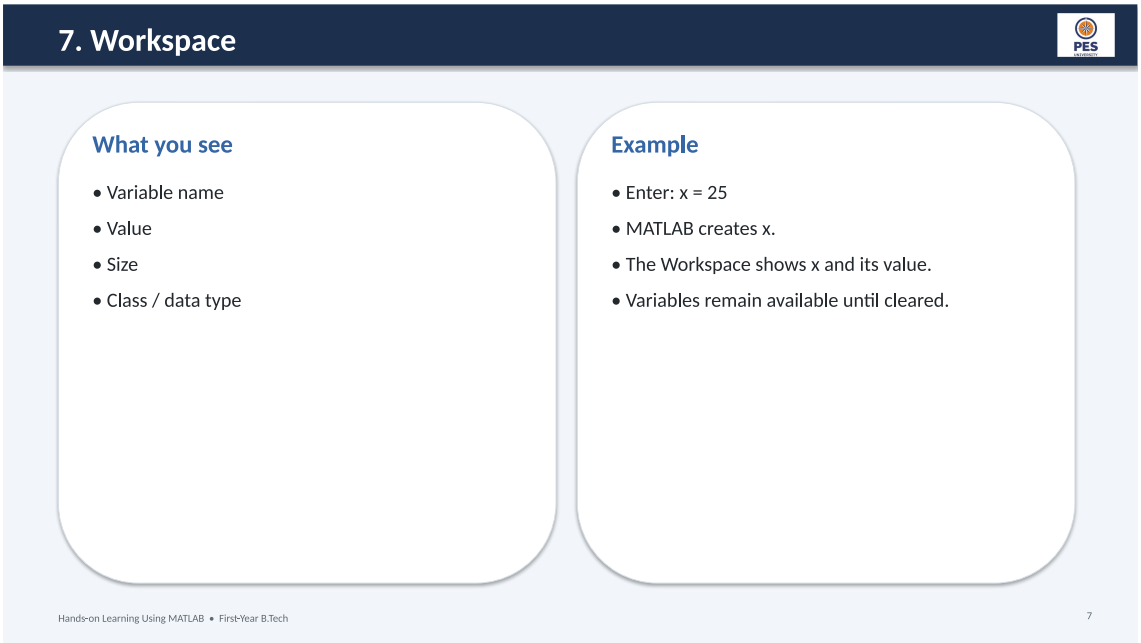

Figure G000133. MATLAB INTRODUCTION · source slide page 7.

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Figure G000134. MATLAB INTRODUCTION · source slide page 8.

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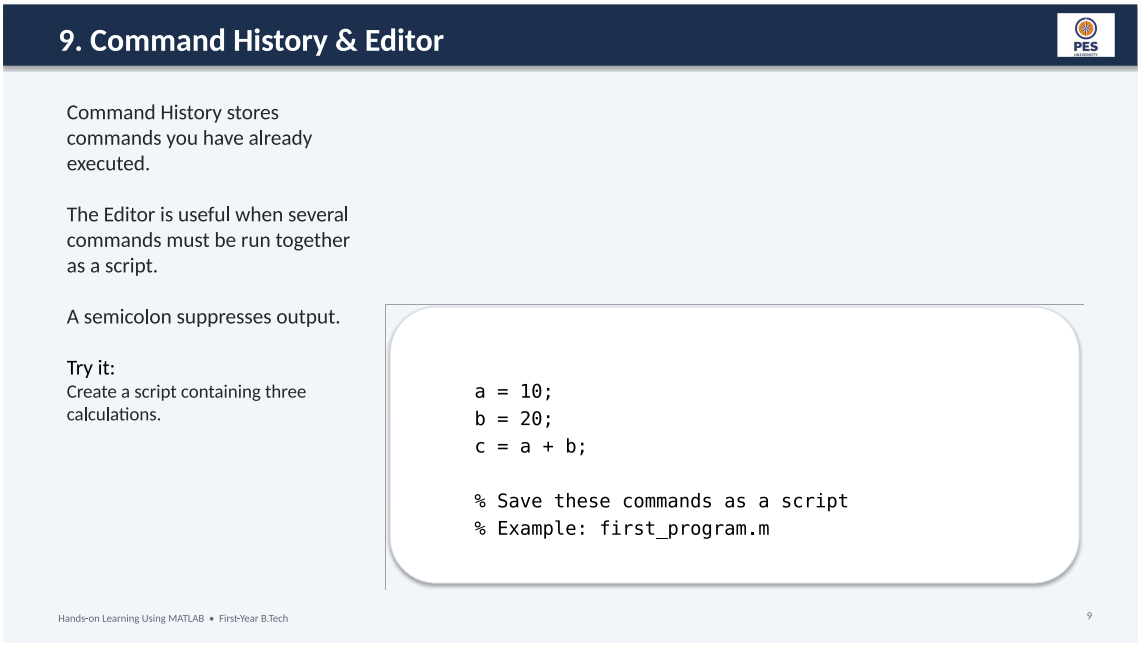

Figure G000135. MATLAB INTRODUCTION · source slide page 9.

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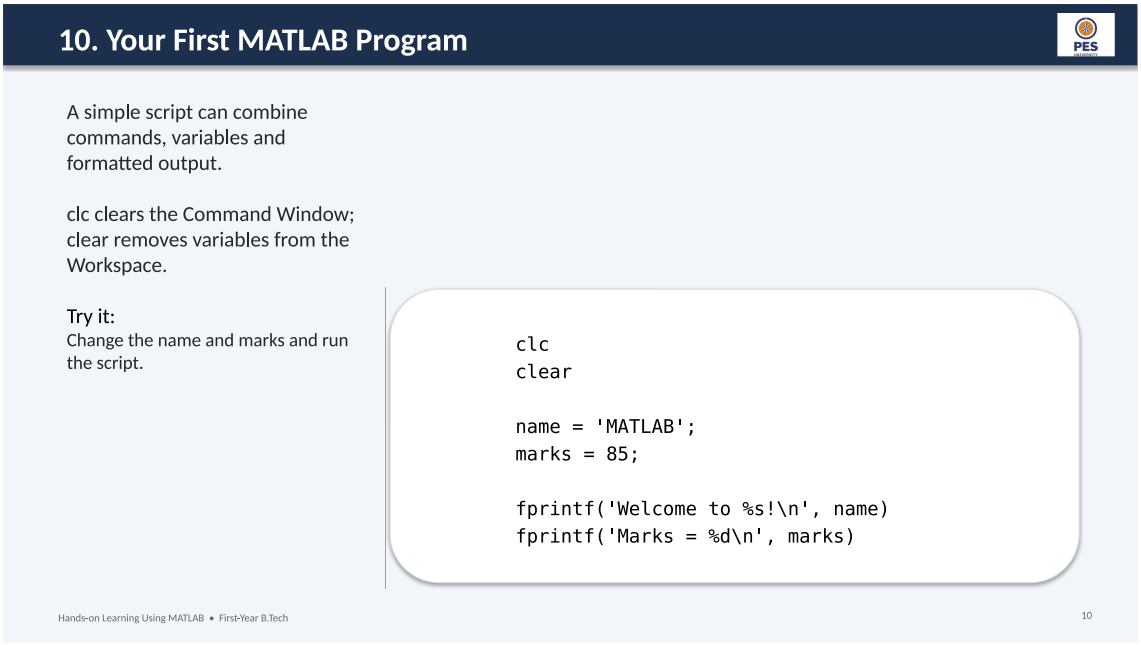

Figure G000136. MATLAB INTRODUCTION · source slide page 10.

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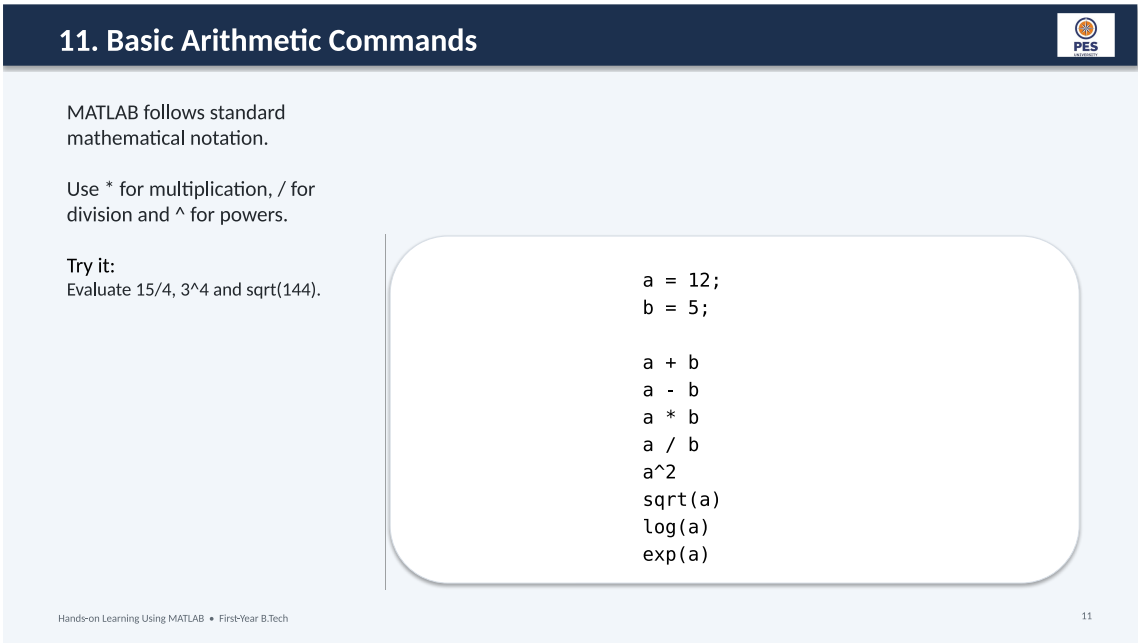

Figure G000137. MATLAB INTRODUCTION · source slide page 11.

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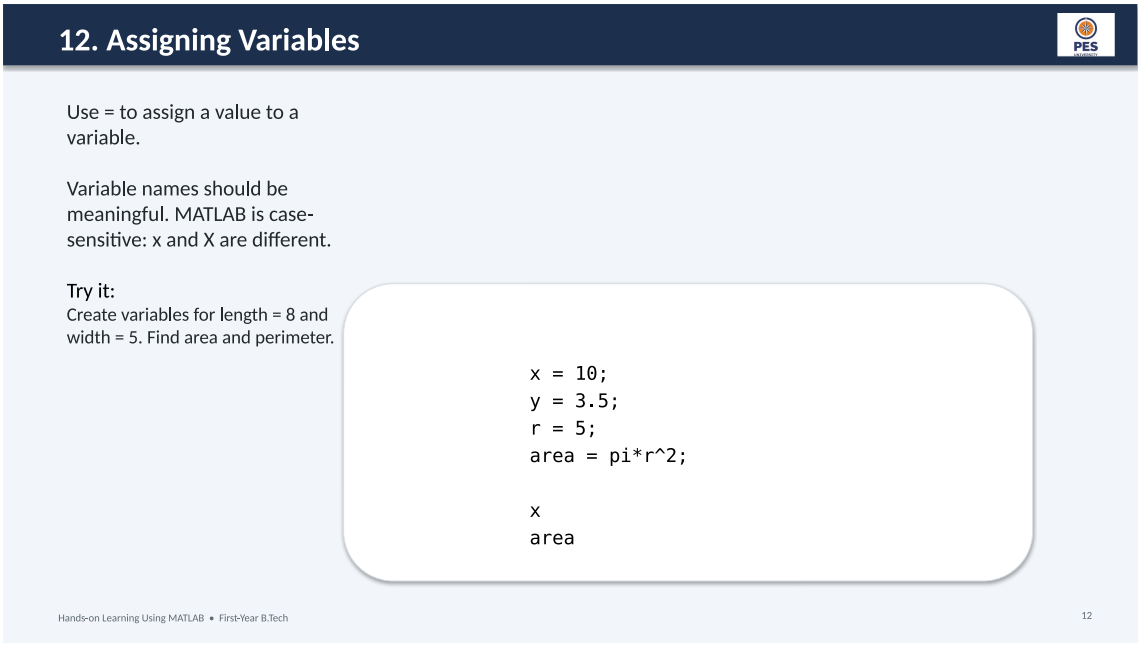

Figure G000138. MATLAB INTRODUCTION · source slide page 12.

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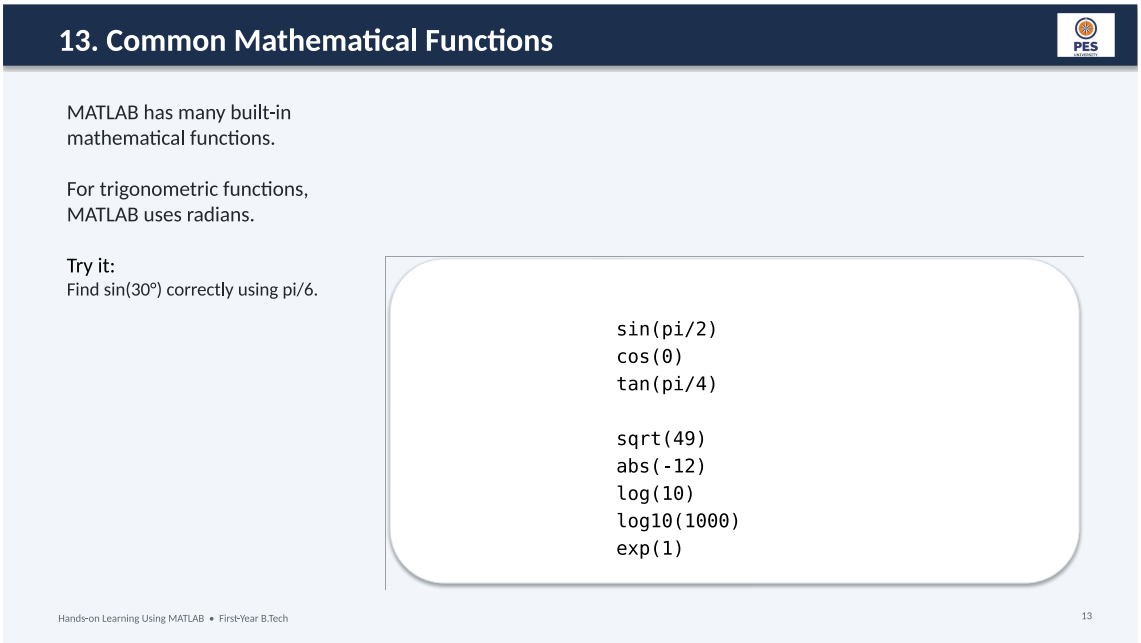

Figure G000139. MATLAB INTRODUCTION · source slide page 13.

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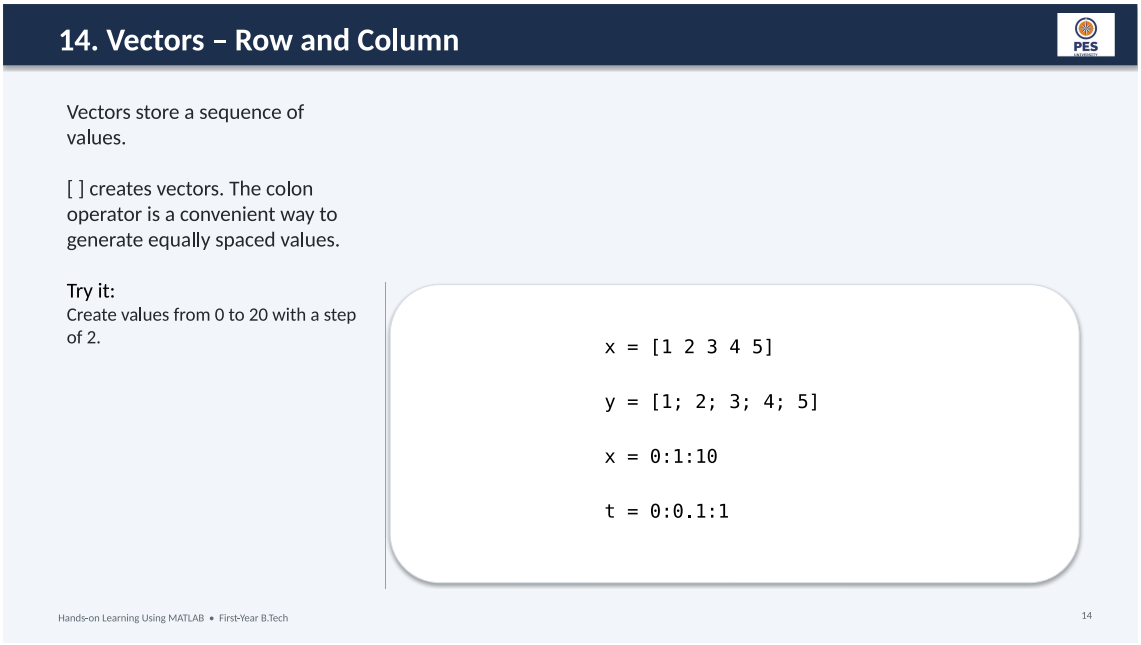

Figure G000140. MATLAB INTRODUCTION · source slide page 14.

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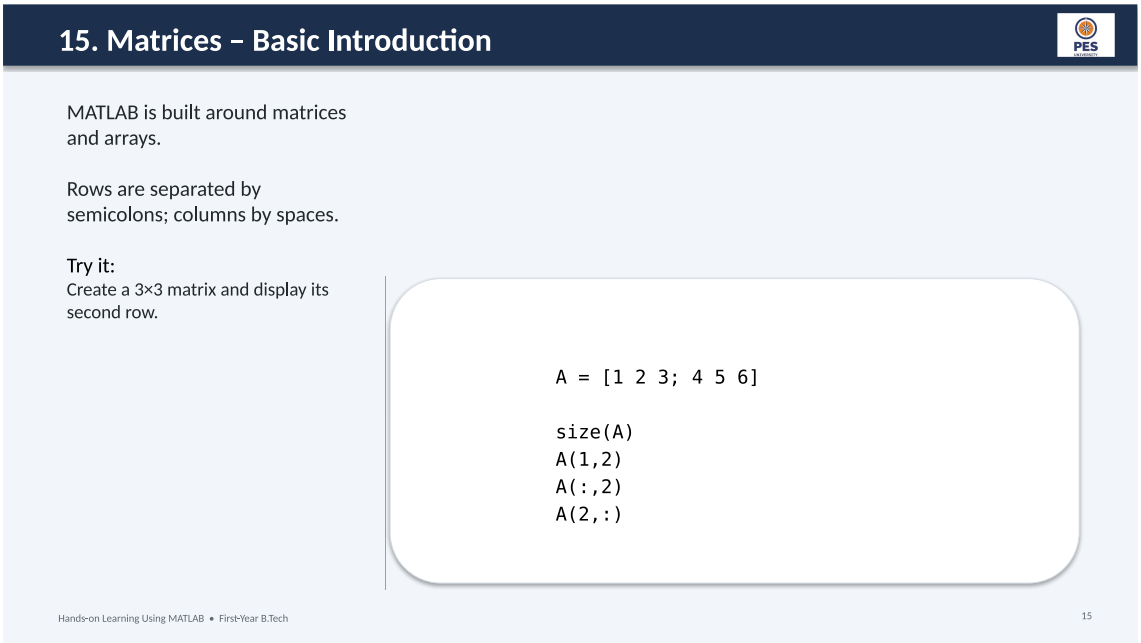

Figure G000141. MATLAB INTRODUCTION · source slide page 15.

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Figure G000142. MATLAB INTRODUCTION · source slide page 16.

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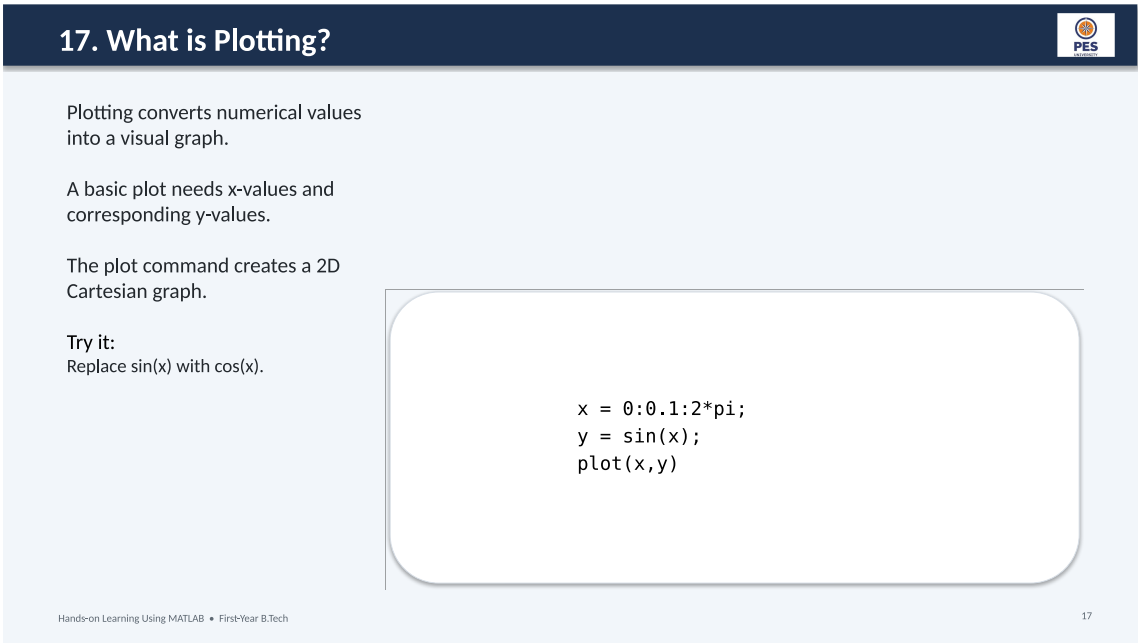

Figure G000143. MATLAB INTRODUCTION · source slide page 17.

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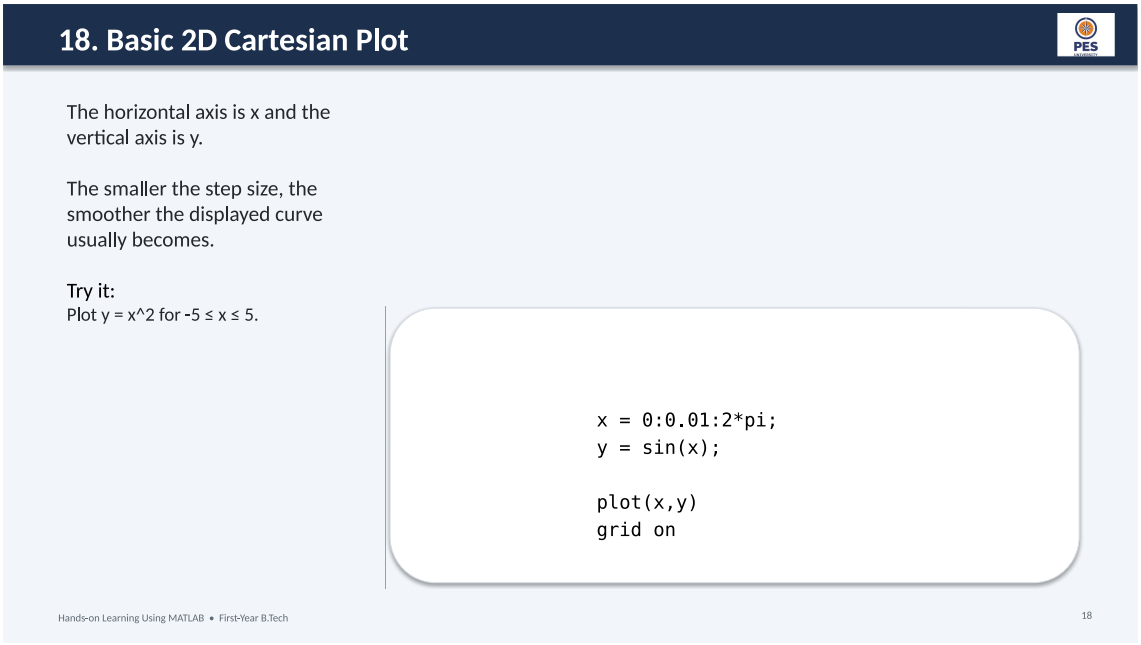

Figure G000144. MATLAB INTRODUCTION · source slide page 18.

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Figure G000145. MATLAB INTRODUCTION · source slide page 19.

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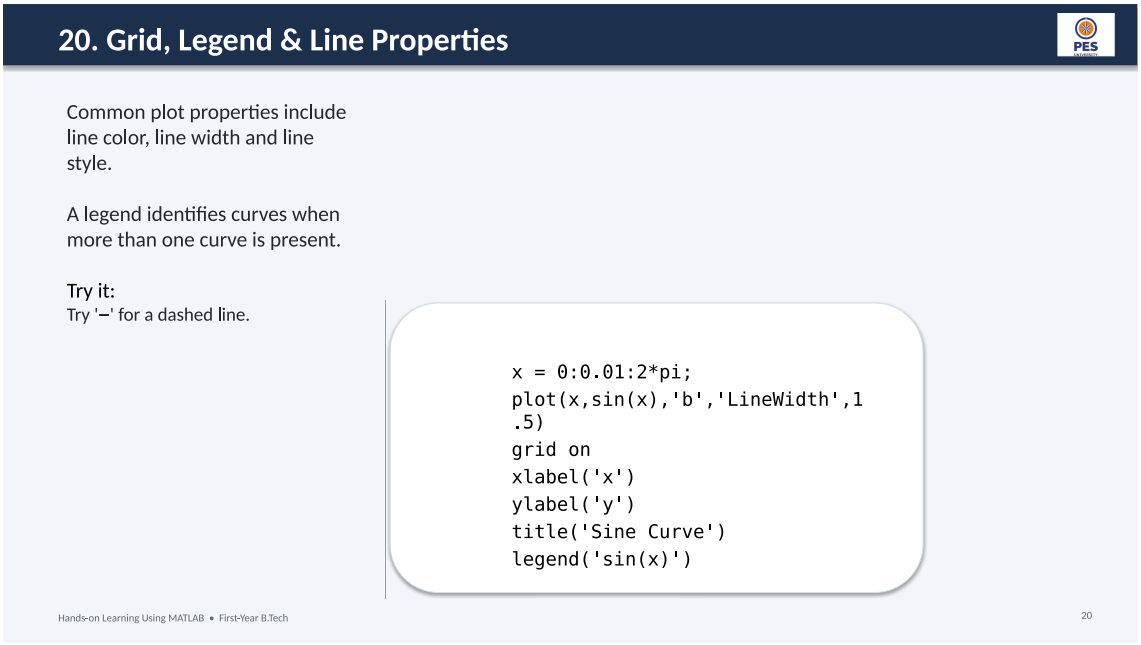

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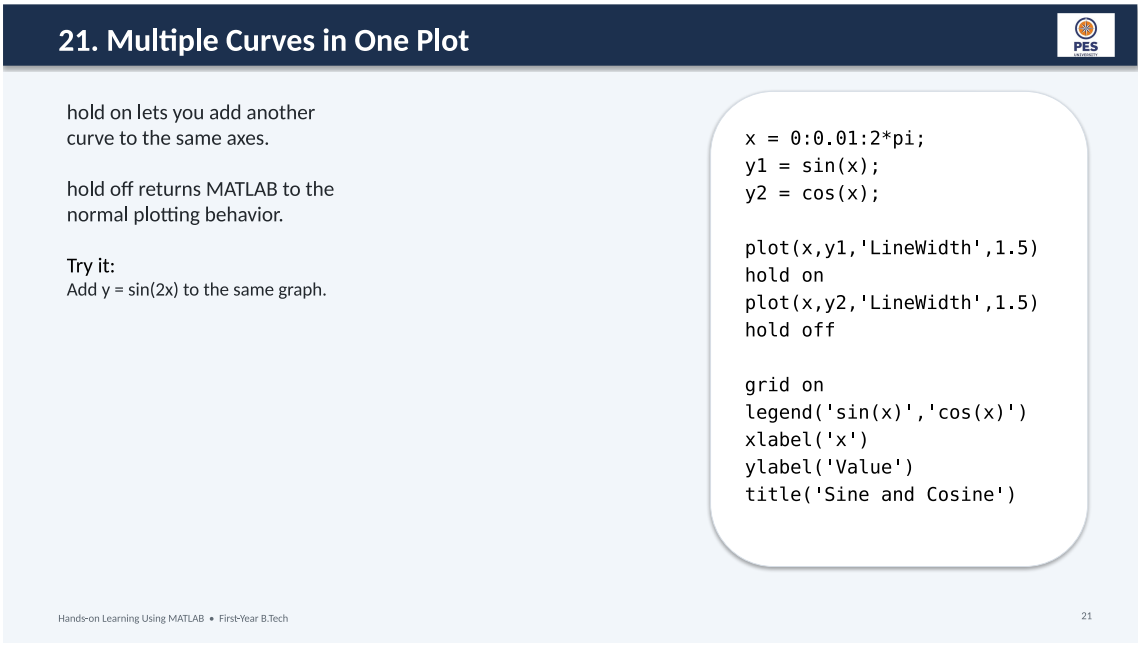

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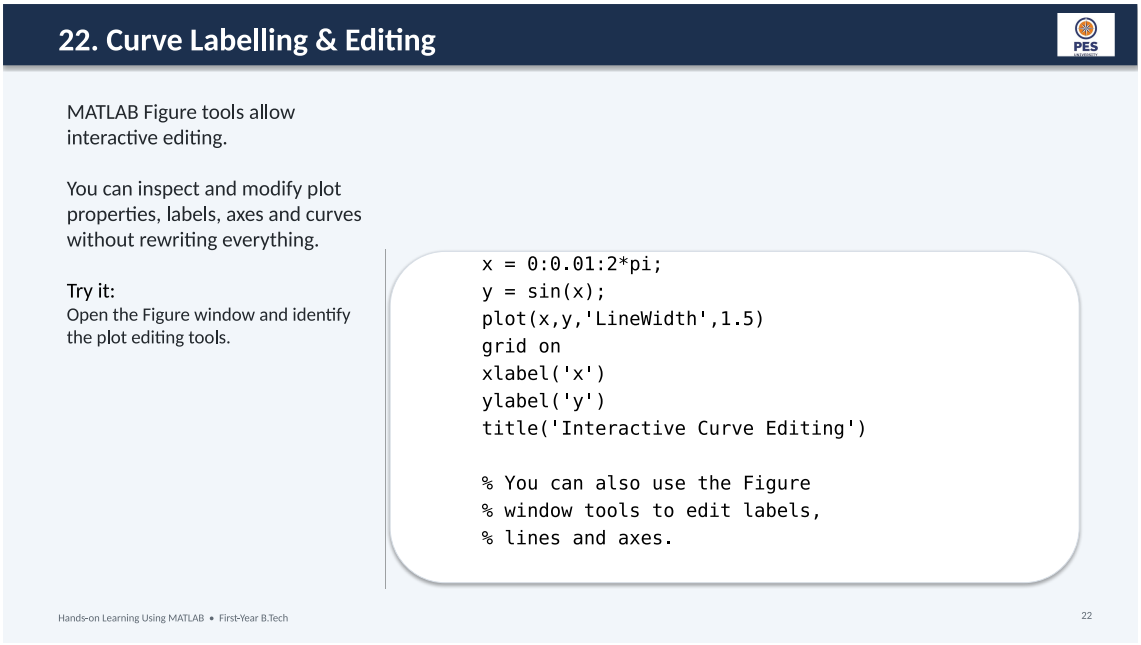

Figure G000148. MATLAB INTRODUCTION · source slide page 22.

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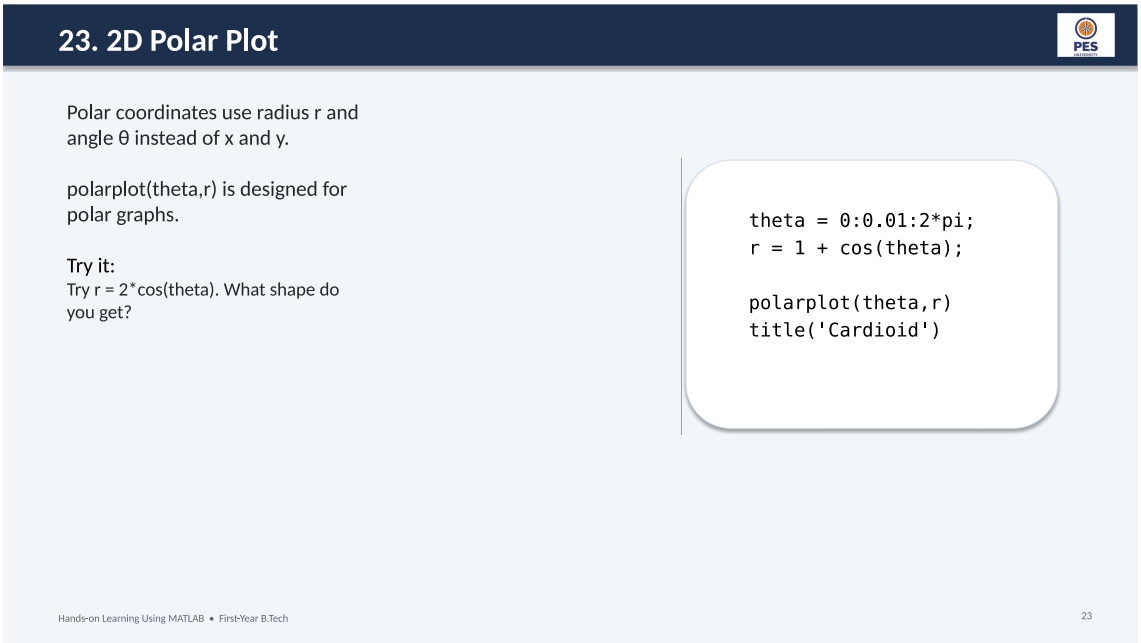

Figure G000149. MATLAB INTRODUCTION · source slide page 23.

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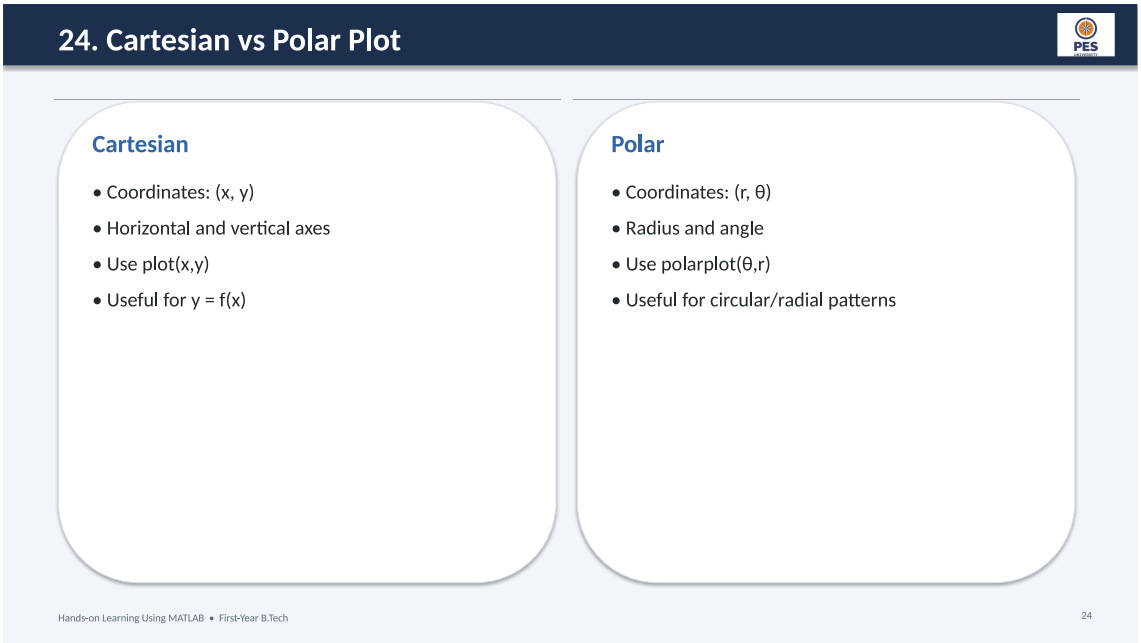

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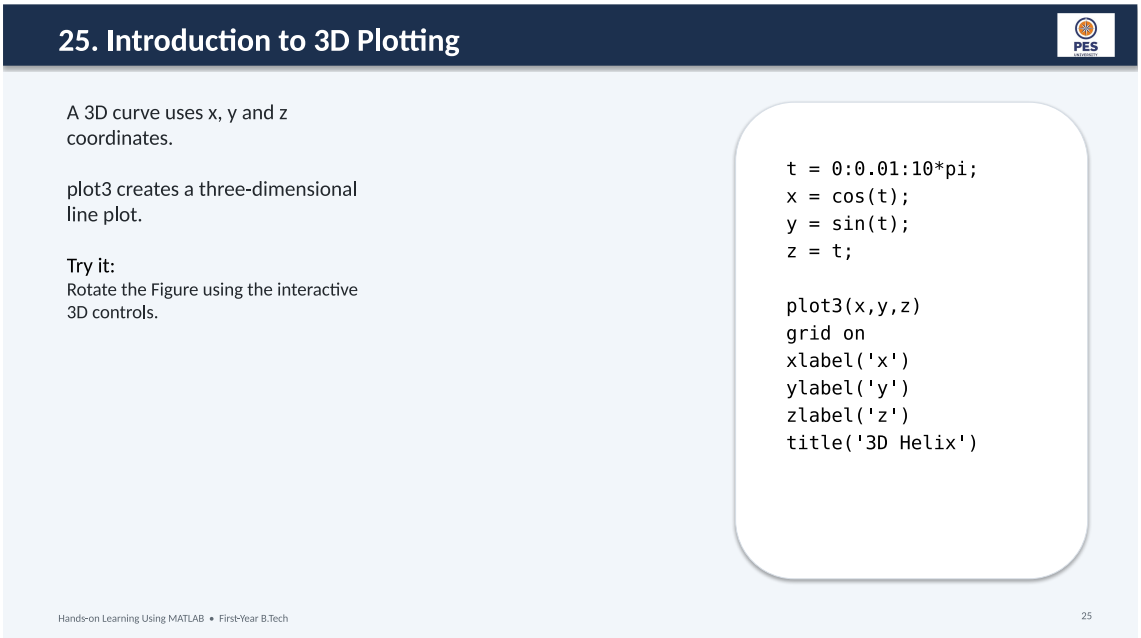

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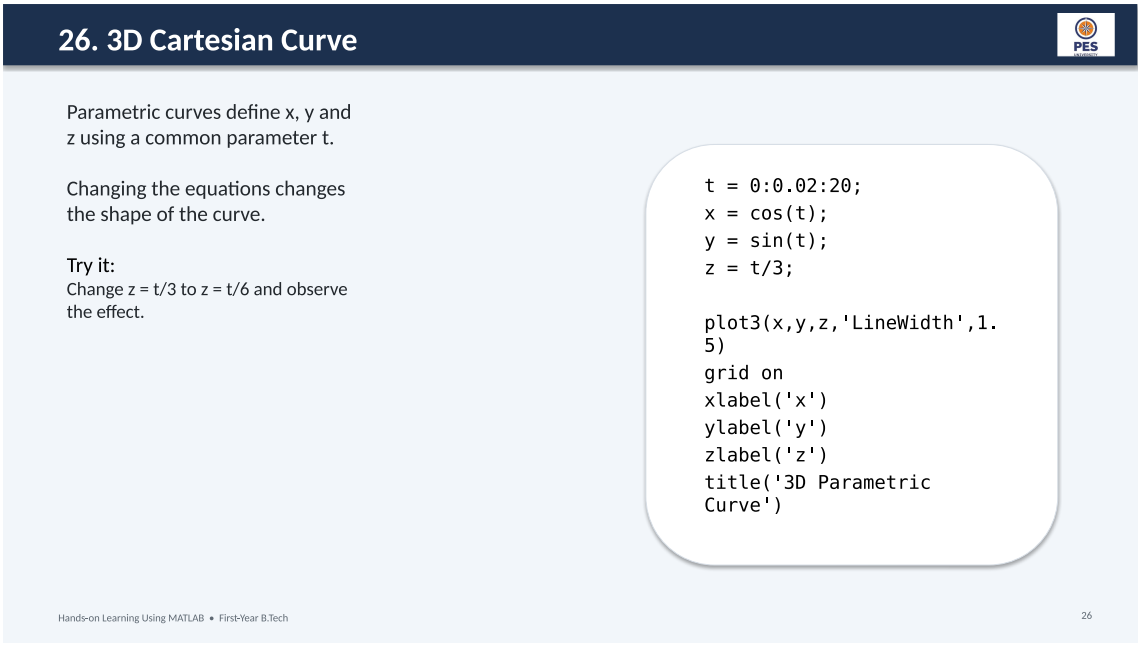

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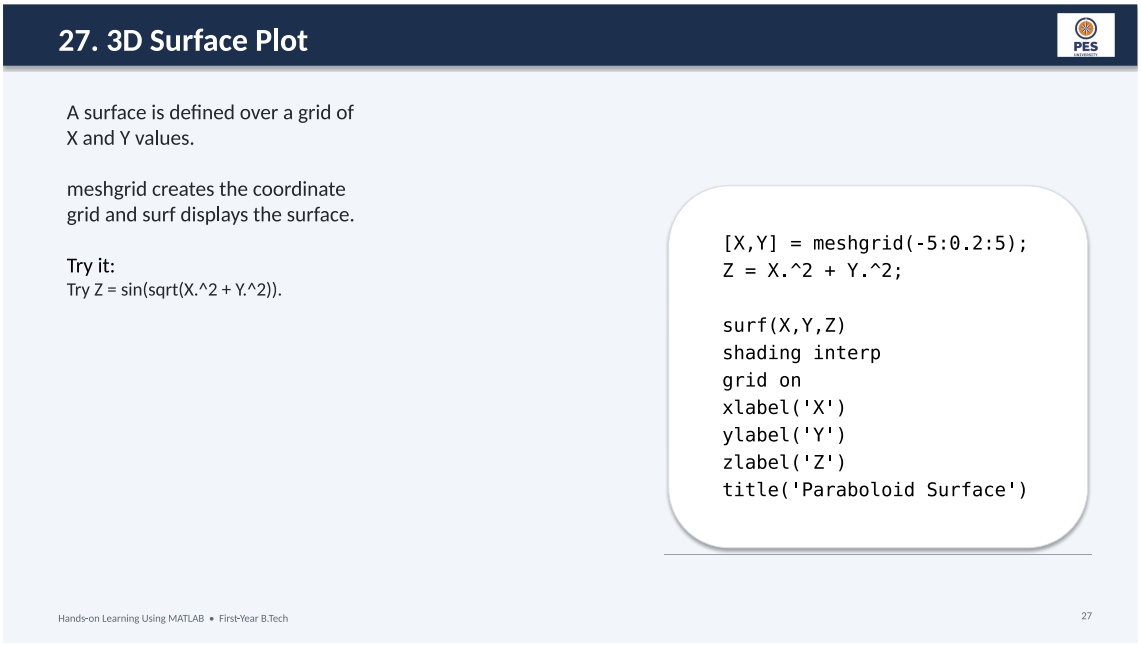

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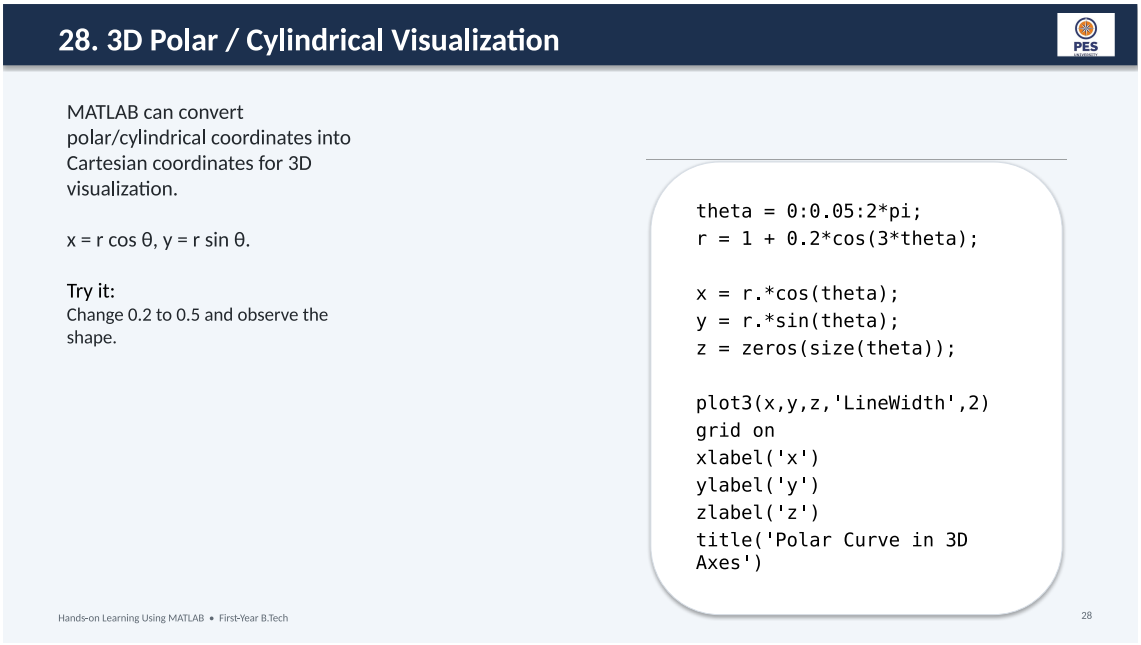

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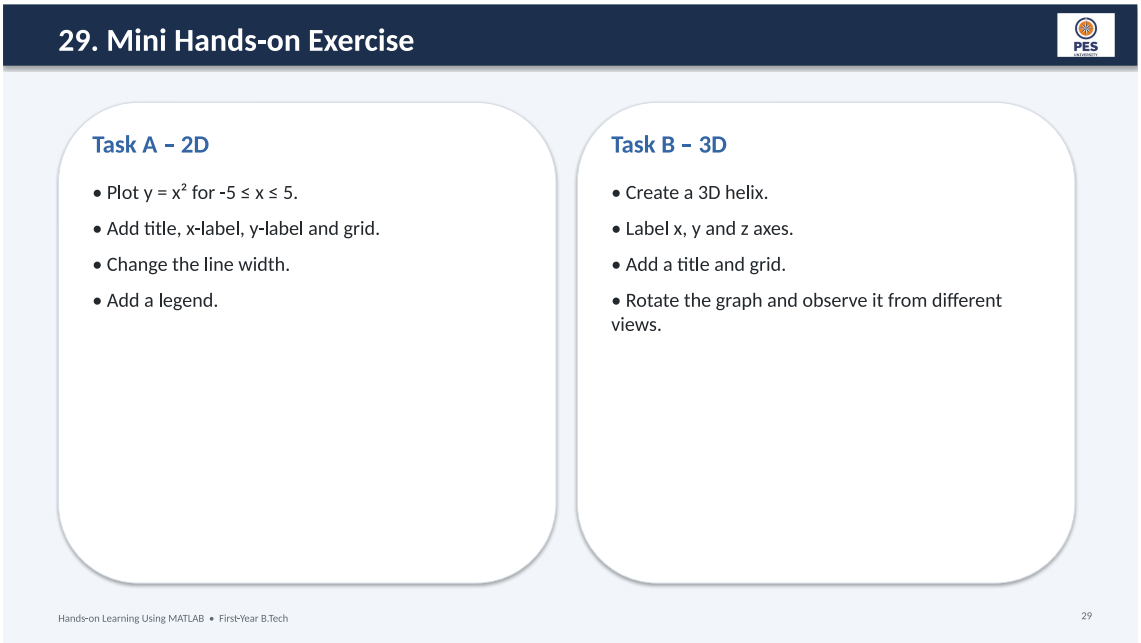

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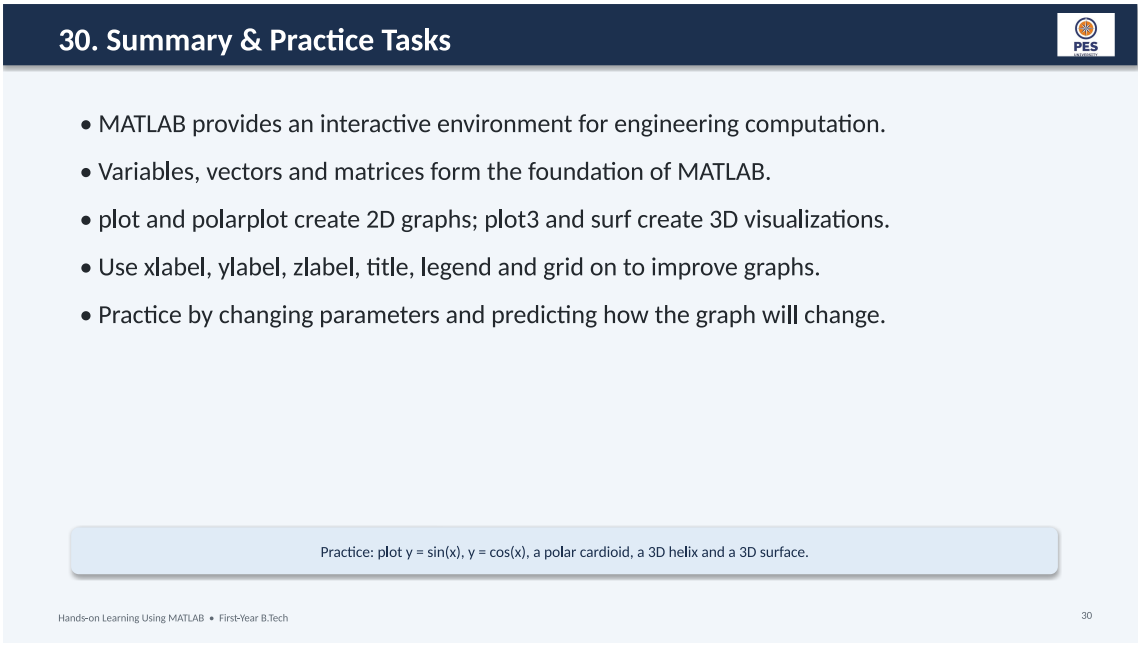

Figure G000156. MATLAB INTRODUCTION · source slide page 30.

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Important points and why

1. Source slide page 1 (extracted text): Hands-on Learning Using MATLAB Introduction to MATLAB, Commands, Variables & Plotting Introduction • Commands • Variables • 2D & 3D Plotting 1 EV0000472 · EV0000473 · EV0000474
2. Source slide page 2 (extracted text): 2. Learning Objectives • Identify the main parts of the MATLAB user interface. • Use basic commands and assign values to variables. • Create and customize 2D Cartesian and polar plots. • Create introductory 3D curves and surfaces. • Add titles, axis labels, legends, grids and curve labels. • Modify and edit plots to communicate results clearly. Goal: learn by typing, running, observing and changing MATLAB code. 2 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000475 · EV0000476 · EV0000477
3. Source slide page 3 (extracted text): 3. What is MATLAB? • MATLAB = MATrix LABoratory. • A programming and numerical-computing environment used by engineers and scientists. • Works naturally with vectors, matrices, equations, data and graphs. • Provides built-in functions for mathematics, visualization, simulation and modelling. Think of MATLAB as a calculator + programming environment + visualization tool. 3 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000478 · EV0000479 · EV0000480

4. Source slide page 4 (extracted text): 4. Why MATLAB for Engineering Students? • Solve mathematical problems quickly. • Visualize functions and data instead of only calculating numbers. • Experiment by changing parameters and immediately seeing the effect. • Build a foundation for ODEs, PDEs, signal processing, AI/ML and engineering simulation. MATLAB is especially useful when mathematics and computation need to work together. 4 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000481 · EV0000482 · EV0000483
5. Source slide page 5 (extracted text): 5. MATLAB User Interface Main areas Also useful • Command Window – type and execute commands. • Workspace – view variables currently in memory. • Current Folder – manage files and folders. • Command History – revisit previously executed commands. • Editor – write and save scripts. • Figure Window – display plots. • Toolstrip – access common commands and tools. • Layout can be customized for your workflow. 5 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000484 · EV0000485 · EV0000486
6. Source slide page 6 (extracted text): 6. Command Window The Command Window is the quickest place to test MATLAB commands. The prompt >> means MATLAB is ready for input. Try it: Calculate 12^2, log(10), cos(pi). >> 2 + 3 ans = 5 >> sqrt(25) ans = 5 >> sin(pi/2) ans = 1 6 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000487 · EV0000488 · EV0000489
7. Source slide page 7 (extracted text): 7. Workspace What you see Example • Variable name • Value • Size • Class / data type • Enter: x = 25 • MATLAB creates x. • The Workspace shows x and its value. • Variables remain available until cleared. 7 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000490 · EV0000491 · EV0000492
8. Source slide page 8 (extracted text): 8. Current Folder The Current Folder tells MATLAB where files are being stored or searched for. Use pwd to display the current folder and ls to list its contents. Try it: Use pwd. Identify your current working folder. pwd ls cd('C:\MATLAB\MyFiles') 8 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000493 · EV0000494 · EV0000495
9. Source slide page 9 (extracted text): 9. Command History & Editor Command History stores commands you have already executed. The Editor is useful when several commands must be run together as a script. A semicolon suppresses output. Try it: Create a script containing three calculations. a = 10; b = 20; c = a + b; % Save these commands as a script % Example: first_program.m 9 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000496 · EV0000497 · EV0000498
10. Source slide page 10 (extracted text): 10. Your First MATLAB Program A simple script can combine commands, variables and formatted output. clc clears the Command Window; clear removes variables from the Workspace. Try it: Change the name and marks and run the script. clc clear name = 'MATLAB'; marks = 85; fprintf('Welcome to %s!\n', name) fprintf('Marks = %d\n', marks) 10 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000499 · EV0000500 · EV0000501
11. Source slide page 11 (extracted text): 11. Basic Arithmetic Commands MATLAB follows standard mathematical notation. Use * for multiplication, / for division and ^ for powers. Try it: Evaluate 15/4, 3^4 and sqrt(144). a = 12; b = 5; a + b a - b a * b a / b a^2 sqrt(a) log(a) exp(a) 11 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000502 · EV0000503 · EV0000504

12. Source slide page 12 (extracted text): 12. Assigning Variables Use = to assign a value to a variable. Variable names should be meaningful. MATLAB is case-sensitive: x and X are different. Try it: Create variables for length = 8 and width = 5. Find area and perimeter. x = 10; y = 3.5; r = 5; area = pi*r^2; x area 12 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000505 · EV0000506 · EV0000507

13. Source slide page 13 (extracted text): 13. Common Mathematical Functions MATLAB has many built-in mathematical functions. For trigonometric functions, MATLAB uses radians. Try it: Find sin(30°) correctly using pi/6. sin(pi/2) cos(0) tan(pi/4) sqrt(49) abs(-12) log(10) log10(1000) exp(1) 13 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000508 · EV0000509 · EV0000510

14. Source slide page 14 (extracted text): 14. Vectors – Row and Column Vectors store a sequence of values. [ ] creates vectors. The colon operator is a convenient way to generate equally spaced values. Try it: Create values from 0 to 20 with a step of 2. x = [1 2 3 4 5] y = [1; 2; 3; 4; 5] x = 0:1:10 t = 0:0.1:1 14 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000511 · EV0000512 · EV0000513

15. Source slide page 15 (extracted text): 15. Matrices – Basic Introduction MATLAB is built around matrices and arrays. Rows are separated by semicolons; columns by spaces. Try it: Create a 3×3 matrix and display its second row. A = [1 2 3; 4 5 6] size(A) A(1,2) A(:,2) A(2,:) 15 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000514 · EV0000515 · EV0000516

16. Source slide page 16 (extracted text): 16. Useful MATLAB Commands These commands help you organize and debug your MATLAB session. Try it: Run who and whos after creating three variables. clc % clear Command Window clear % clear variables who % list variables whos % detailed variable list close all % close figures help plot % get help % Comments begin with % 16 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000517 · EV0000518 · EV0000519

17. Source slide page 17 (extracted text): 17. What is Plotting? Plotting converts numerical values into a visual graph. A basic plot needs x-values and corresponding y-values. The plot command creates a 2D Cartesian graph. Try it: Replace sin(x) with cos(x). x = 0:0.1:2*pi; y = sin(x); plot(x,y) 17 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000520 · EV0000521 · EV0000522

18. Source slide page 18 (extracted text): 18. Basic 2D Cartesian Plot The horizontal axis is x and the vertical axis is y. The smaller the step size, the smoother the displayed curve usually becomes. Try it: Plot y = x^2 for -5 ≤ x ≤ 5. x = 0:0.01:2*pi; y = sin(x); plot(x,y) grid on 18 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000523 · EV0000524 · EV0000525

19. Source slide page 19 (extracted text): 19. Title and Axis Labels Good graphs tell the reader what is being plotted. Always label axes and give the graph a meaningful title. Try it: Label a graph of y = x^2 with units if appropriate. x = 0:0.01:2*pi; y = sin(x); plot(x,y) xlabel('x') ylabel('sin(x)') title('Sine Curve') grid on 19 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000526 · EV0000527 · EV0000528

20. Source slide page 20 (extracted text): 20. Grid, Legend & Line Properties Common plot properties include line color, line width and line style. A legend identifies curves when more than one curve is present. Try it: Try '--' for a dashed line. x = 0:0.01:2*pi; plot(x,sin(x),'b','LineWidth',1.5) grid on xlabel('x') ylabel('y') title('Sine Curve') legend('sin(x)') 20 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000529 · EV0000530 · EV0000531

21. Source slide page 21 (extracted text): 21. Multiple Curves in One Plot hold on lets you add another curve to the same axes. hold off returns MATLAB to the normal plotting behavior. Try it: Add y = sin(2x) to the same graph. x = 0:0.01:2*pi; y1 = sin(x); y2 = cos(x); plot(x,y1,'LineWidth',1.5) hold on plot(x,y2,'LineWidth',1.5) hold off grid on legend('sin(x)','cos(x)') xlabel('x') ylabel('Value') title('Sine and Cosine') 21 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000532 · EV0000533 · EV0000534

22. Source slide page 22 (extracted text): 22. Curve Labelling & Editing MATLAB Figure tools allow interactive editing. You can inspect and modify plot properties, labels, axes and curves without rewriting everything. Try it: Open the Figure window and identify the plot editing tools. x = 0:0.01:2*pi; y = sin(x); plot(x,y,'LineWidth',1.5) grid on xlabel('x') ylabel('y') title('Interactive Curve Editing') % You can also use the Figure % window tools to edit labels, % lines and axes. 22 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000535 · EV0000536 · EV0000537

23. Source slide page 23 (extracted text): 23. 2D Polar Plot Polar coordinates use radius r and angle ϑ instead of x and y. polarplot(theta,r) is designed for polar graphs. Try it: Try r = 2*cos(theta). What shape do you get? theta = 0:0.01:2*pi; r = 1 + cos(theta); polarplot(theta,r) title('Cardioid') 23 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000538 · EV0000539 · EV0000540

24. Source slide page 24 (extracted text): 24. Cartesian vs Polar Plot Cartesian Polar • Coordinates: (x, y) • Horizontal and vertical axes • Use plot(x,y) • Useful for y = f(x) • Coordinates: (r, ϑ) • Radius and angle • Use polarplot(ϑ,r) • Useful for circular/radial patterns 24 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000541 · EV0000542 · EV0000543

25. Source slide page 25 (extracted text): 25. Introduction to 3D Plotting A 3D curve uses x, y and z coordinates. plot3 creates a three-dimensional line plot. Try it: Rotate the Figure using the interactive 3D controls. t = 0:0.01:10*pi; x = cos(t); y = sin(t); z = t; plot3(x,y,z) grid on xlabel('x') ylabel('y') zlabel('z') title('3D Helix') 25 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000544 · EV0000545 · EV0000546

26. Source slide page 26 (extracted text): 26. 3D Cartesian Curve Parametric curves define x, y and z using a common parameter t. Changing the equations changes the shape of the curve. Try it: Change z = t/3 to z = t/6 and observe the effect. t = 0:0.02:20; x = cos(t); y = sin(t); z = t/3; plot3(x,y,z,'LineWidth',1.5) grid on xlabel('x') ylabel('y') zlabel('z') title('3D Parametric Curve') 26 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000547 · EV0000548 · EV0000549

27. Source slide page 27 (extracted text): 27. 3D Surface Plot A surface is defined over a grid of X and Y values. meshgrid creates the coordinate grid and surf displays the surface. Try it: Try Z = sin(sqrt(X.^2 + Y.^2)). [X,Y] = meshgrid(-5:0.2:5); Z = X.^2 + Y.^2; surf(X,Y,Z) shading interp grid on xlabel('X') ylabel('Y') zlabel('Z') title('Paraboloid Surface') 27 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000550 · EV0000551 · EV0000552

28. Source slide page 28 (extracted text): 28. 3D Polar / Cylindrical Visualization MATLAB can convert polar/cylindrical coordinates into Cartesian coordinates for 3D visualization. x = r cos ϑ, y = r sin ϑ. Try it: Change 0.2 to 0.5 and observe the shape. theta = 0:0.05:2*pi; r = 1 + 0.2*cos(3*theta); x = r.*cos(theta); y = r.*sin(theta); z = zeros(size(theta)); plot3(x,y,z,'LineWidth',2) grid on xlabel('x') ylabel('y') zlabel('z') title('Polar Curve in 3D Axes') 28 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000553 · EV0000554 · EV0000555

29. Source slide page 29 (extracted text): 29. Mini Hands-on Exercise Task A – 2D Task B – 3D • Plot y = x² for -5 ≤ x ≤ 5. • Add title, x-label, y-label and grid. • Change the line width. • Add a legend. • Create a 3D helix. • Label x, y and z axes. • Add a title and grid. • Rotate the graph and observe it from different views. 29 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000556 · EV0000557 · EV0000558

30. Source slide page 30 (extracted text): 30. Summary & Practice Tasks • MATLAB provides an interactive environment for engineering computation. • Variables, vectors and matrices form the foundation of MATLAB. • plot and polarplot create 2D graphs; plot3 and surf create 3D visualizations. • Use xlabel, ylabel, zlabel, title, legend and grid on to improve graphs. • Practice by changing parameters and predicting how the graph will change. Practice: plot y = sin(x), y = cos(x), a polar cardioid, a 3D helix and a 3D surface. 30 Hands-on Learning Using MATLAB • First-Year B.Tech EV0000559 · EV0000560 · EV0000561

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Hi, welcome to PES University Online Teaching Learning Platform. I am Professor Nagesh Ham, faculty member Department of Science and Humanities. [00:00:14.436] This is the fourth session of MATLAB. In this session we learn plotting using MATLAB. There are three cases. Case number one, Cartesian curves that is y equal to f of x. [00:00:27.524] Case number two polar curves that is R equal to F of theta. Case number three parametric curves x equal to f of t and y equal to g of t. Let's start with the Cartesian case that is uh y equal to f of x. [00:00:46.436] To plot the graph of a function, we need to take the following steps. We first define the variable x, then specify the range of values for the variable x. [00:01:00.516] We define the function y equal to f of x. Finally we use plot command in order to plot the Cartesian curve that is y equal to f of x. The plot command is used to create two dimensional plots. [00:01:17.028] In order to understand this concept, let us look at the example y equal to 3.5 power 0.5 x into cos of 6x where x lies between minus 2 and plus 4. Before we plot this graph, we have to use the first one. [00:01:35.236] Let's understand the concept of dot operator in Matlab. [00:01:48.900] In order to understand this concept, let us take a simple example. Let us take a matrix to be 1, 2, 3, then 4, 5, 6. Then as you can see, we have to have a 1. [00:02:06.692] this is a three crossary matrix. Similarly look at the matrix B equals to four five six eight nine zero minus one two four right again this is a three crossing matrix [00:02:26.116] Now, let's see what is A star B. [00:02:35.140] So this is the answer. So A star b is going to be 17, 29 and 18. And this is a normal or usual matrix multiplication what we know A star b. But if you look at A dot star B [00:02:52.420] So this is the answer. So what do we observe? As you can see the first element of this A is multiplied with this 4. Then similarly, multiplication of 2 and 5, so the corresponding number is 10. [00:03:09.540] This is how the dot operator works. So what do you are what do you understand from this? The dot operator is used with multiplication, division and exponentiation to create element wise operation. That's what we see here. [00:03:26.788] If you look at A star b this is a usual matrix multiplication what we know, right? So if you look at A dot star B this is element wise matrix multiplication. For example, here as I said 1 and 4 multiplication, similarly a 6 and 0, therefore the answer is 0 here. [00:03:45.668] So this is how dot operator works in MATLAB. I repeat, the dot operator is used with multiplication, division, exponentiation to create element wise operation. So let's clear the screen. [00:04:07.268] So after understanding the concept of dot operator let us move on to how to plot this graph that is y equal to 3.5 power 0.5 x

into cos of 6x where x lies between minus 2 and plus 4. [00:04:22.980] And we know this is a procedure. We first define the variable x and then we specify the range of values for the variable x. Then we define the function y equal to f of x for which we need to plot the graph. Then finally we use plot of x comma y command. [00:04:41.924] So let look at the illustration part. Let x equals two so minus 100. Right? This is colon. [00:04:56.004] So let's start with minus 2 then 0.01 colon 4. It means x starts with minus 2, it goes up to 4 with a step size 0.01. Then define the function y equal to

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:16.420] three point five dot this is a dot operator remember cap minus zero point five x again this has to be star x then dot operator star cos [00:05:35.844] of six times x right then finally use the command plot of x comma y. So this is a graph of the function what we get right okay now. [00:05:55.460] What happens if you give plot of y comma x [00:06:01.956] This is shape of the graph. So this is plot of [00:06:08.324] [ASR low confidence; check audio] Way come my X. [00:06:13.284] This is the shape of the curve. Is it clear? So we have to use plot of x comma y. Now what happens if you don't leave the dot operator here? For example, let me cancel this, let me delete this, right? Now [00:06:31.972] This is the error what we get. Is it clear? So whenever we deal with multiplication, division, and exponentiation, we use dot operator. So this is how we can plot this graph. [00:06:47.012] y equal to 3.5 powers minus 0.5 x into cos of 6x. So this is the coding part and this is a result what we got. So here plot of x comma y here y is plotted as a function of x. So y is plotted as a function of x. [00:07:06.980] So let us take another example y equal to f of x square. So this is a parabola. So in this example we will draw two graphs with the same function, but in the second time we will reduce the value of increment. I mean to say that in the second case we just [00:07:24.356] reduce this increment then we we look at the shape of the graph right so let's look at illustration so clear the screen so let us take x equal to minus 100 [00:07:40.164] colon twenty colon hundred then y equal to see you have x square right see you cannot simply give x cap to you will get an error you have to use dot operator remember [00:07:58.660] Right? Square then plot of X comma y. Right? So this is the graph. Now what happens if we reduce the decrement? If we reduce [00:08:15.172] the increment twenty uh let us replace it by ten or five. Let's look at the answer. So let us uh you can copy paste [00:08:28.612] so just make it as 10 right then uh just you can copy paste this that is y equal x square [00:08:39.876] Then plot of x comma y. So this is the shape of the graph. So what do you observe here? [00:08:49.092] When we decrease the increment, the graph becomes smoother. When we decrease the increment, the graph becomes smoother. That's it. So this is the graph what we got for minus 10, 20 and 100. [00:09:04.260] and this is a graph we got it for minus hundred phi and hundred is it here now what happens if we [00:09:15.076] delete this so let us copy paste it here so let us make it as fine right then we intentionally delete this uh dot product see what happens you will get an error [00:09:32.452] Is it clear? So this is how we use dot operator in MATLAB. Okay, this is how we can plot partition curves using MATLAB. So let us move on to 3D. We know that [00:09:50.340] A function of two independent random variables defines a surface in three dimensional space. Correct. So let us understand how to plot the surface for a given function of two independent variables. So here also we have a three simple steps. Step number one

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.652] We first create a set of points over the domain of the function using mesh grid command. Next we assign the function

itself. Finally we use a command called surf. [00:10:21.316] to plot the surface. So in order to understand this concept let us take a simple example let us plot the surface for z equal to x into e power minus of x square plus y square. So this is the coding part [00:10:41.476] Said let's look at the illustration. [00:10:46.308] clear the screen we can write it is x y you can also give capital x y there mesh grid mesh grid of minus two right colon [00:11:05.092] minus uh point two that is the step size then colon you can give two. You start from minus two it goes up to four right then you c call the command z equal to the function is x into x dot [00:11:26.212] product exp for exponential minus x square minus x square means dot cap square minus y dot cap square. So this is a function [00:11:45.284] Finally we use a command surf of x comma y comma z right. So this is how the surface looks like for the given function. Is it clear? [00:12:00.548] A function of two independent random variables defines a surface in three-dimensional space. Clearly? So that's why whenever we have a function of two independent variables, we can we can always get a surface like this. You can rotate however you want. Is it clear? See this how you can rotate. [00:12:20.836] Okay, let us take a simple example uh another example you can take uh sine x cos y. So what we can do just uh simply you can copy paste [00:12:35.876] So Z D equal to uh sine right, so it's sine of X plus of Y again surf of X comma Y comma Z. [00:12:56.804] So this is a graph of sin x plus cos y looks like. Is it clear? So this is a graph similarly you can easily plot the graph of this and a graph of this looks like this. So this is how we can plot the surface using MATLAB. [00:13:15.716] So we are done with the 2D and 3D right. So let us move on to how to add title labels and the graph using MATLAB. Using MATLAB we can add title labels along the x axis as well as y axis of the graph. [00:13:33.860] So for this we use two commands here X label and Y label. Commands generate labels along X axis as well as Y axis. Finally the title command allows you to put a title on the graph. In order to understand this, let's look at the simple example. [00:13:55.492] So clear the screen. [00:14:02.724] Let us take x equals to 0 colon 0.01 colon 10 right then you can take your we are just trying to plot sine graph sine of x [00:14:21.412] Now we are going to we need to label the graph. So plot x comma y then you label x axis so x label the code is open bracket single code x right then you can label y axis y label [00:14:41.636] Y label you can say s single code sign of X right then you can uh give title so title it is [00:14:59.972] Sine of X graph. Sign extra. Is it clear? So graph. Single code. Clear. And then you can see the corresponding answer is going to be the corresponding graph is this. As you can see X axis labeled by X.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:19.076] Y axis is labeled by sine of x and this is the name of the graph. Sine of x graph. Is it clear? So this is how we can add title and labels for the graph using Madlab. And this is the corresponding answer. Okay [00:15:39.332] So now we move on to polar curves. Now what is the equation of a polar curve? [00:15:46.660] the equation of a polar curve is given by r equal to f of theta clear so in order to plot a polar curve we use this command polar plot of theta comma rho here theta is the angle in radians [00:16:04.868] and rho is a radius value for each point. Let us take a simple example r equal to sine 2 theta into cos 2 theta right. So let's plot this using MATLAB. [00:16:19.908] So clear the screen, right? So you just define theta equal to here theta is always vary from 0 to 2 pi, right? 0 starting from 0, increment value 0.01 colon 2 pi 2 star pi right [00:16:40.836] Then define the function by rho. Rho equal to we are looking for sine 2 theta sine of 2 star theta right multiplication dot dot operator should be there [00:17:00.356] then cost of two star theta is it clear then use polar plot command polar plot it is simply theta comma row is clear so this theta gamma row [00:17:21.700] So this is

the shape of the graph what we get. Is it clear? [00:17:30.340] Let's look at another example, let us take uh you can take r equal to one minus cos theta, you can take some example. So you can say rho equal to one minus cos theta, cos of theta here right [00:17:50.084] Then uh polar plot. [00:17:54.756] And this is a graph. Clear. [00:17:59.940] Let us take another example. So I can copy paste it here. Let us take r rho equal to cos2 theta. Cas2 star theta here. [00:18:16.164] Right? And then copy paste it here. And this is a graph of cos2 theta looks like. Is it clear? So this is how we can plot uh polar curves using MATLAB. [00:18:34.884] So this is a coding for that and this is the corresponding output. Similarly R equal to 1 minus cos theta and the graph looks like this and finally r equal to cos two theta and the graph looks like this. [00:18:51.620] Okay after understanding Cartesian curves, polar curves, let's move on to parametric curves. So before we understand the plotting of parametric curves using MATLAB, we need to understand what is sim function in MATLAB. SIMS function. [00:19:12.196] Okay. Sims function creates a variable dynamically. For example, if we say S y M S Sims X. [00:19:24.260] So sims x creates the symbolic variable x and automatically assigns it to MATLAB variable with the same name. Is it clear? So sims function creates a variable dynamically. So while plotting parametric curves we use this function called sims function. Clear? [00:19:44.196] Okay. [00:19:47.364] In order to plot parametric curve okay before that what is the equation of a parametric curve? For example, it could be x equal to f of p [00:19:57.796] and uh y equal to g of t where t is a parameter or it could be x equal f of theta y equal to g of theta where theta is a parameter right so this is the equation of a parametric curve. Now in order to plot parametric curve we use two commands

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:14.244] So the f plot 3 x t y t z is used for plotting parametric curve over the default interval minus phi to plus phi. Is it clear? So it is other than minus y to plus phi we use another command which we shall see in the next example. [00:20:31.140] So before that you just under remember that in order to plot parametric curve we use this command f plot 3 x t y t as well as z t. Take an example let us plot a parametric curve x equal to sin t y equal to cos t z equal to t over the default parameter minus y2 plus y. [00:20:51.812] In order to plot this we use this simple command. So let's look at illustration. So since T this has to be there, right? Then we use x t equal to sine t means sin of t. [00:21:11.012] colon then y t equal to cos of t [00:21:18.852] then Z t equal to T right so finally we use F plot three X T Y T Z T clear So this is the shape of the graph what we get [00:21:39.268] Is it clear? Okay. Suppose intentionally we make a mistake. [00:21:47.300] You just ignore these three. [00:21:52.100] Now what happens? You get an error. Is it clear? Therefore, we need to use f plot 3 x t y t and z t. Clear? So what if we have uh the default interval other than minus y2 plus y? Let us take this example. [00:22:10.500] uh parametric curve x equal to e power minus t by 10 sin phi t y equal to e power minus t by 10 cos phi t z equal to t over the interval minus y and plus minus 10 to plus 10. So for this we use this extra command t max and t minimum clear. [00:22:28.964] So for this illustration the code is going to be this sim t these are x y and z this is a command the output is going to be this one. Is it clear? [00:22:41.380] So this is what we wanted to discuss in this class, Cartesian curves, polar curves as well as parametric curves. Thank you.

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Full transcript: 2776 normalized words, 88 timestamped segments, 1370.752 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:06:08.324

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6 Video

This lesson is indexed from the source video titled Video. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000082 · EV0000083

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:09.636] Hello everyone, welcome back to the course on Engineering Mathematics 1. In this MATLAB session, we explore partial derivatives using MATLAB. [00:00:23.844] So these are some of the examples which we are going through in this session. So here we need to find the partial derivative of the given functions. [00:00:35.140] If f is equal to sine of x plus y cube plus x to the power 10 minus y square plus log of x as a given function, then find fx and f y. Here we need to find the partial derivative of the given function f with respect to x and with respect to y also. [00:00:51.952] So firstly we need to find out the symbolic variables mentioned in the given function. So here the symbolic variables in this function are x and y. So we are initiating our MATLAB code by defining the symbolic variables as S Y M S Y. [00:01:11.172] And then after that we need to define the function. So f is equal to sine of x plus y cube plus x to the power 10 minus y square plus log of x. And then the partial derivative of the function f with respect to x is written as dif. [00:01:29.028] of f comma x and with respect to y we write it as dif f of f comma y [00:01:40.548] So we'll write the code for that function that is Syms XY is the step where we define the symbolic variables. [00:01:55.620] And then the function f is equal to sine of x. [00:02:02.692] plus y cube, we're write it as y cap three plus x power ten that is x cap ten minus y square [00:02:18.724] plus log of x. So this is the given function and code to find the partial derivative of the function f with respect to x is dif of f comma x and dif of f comma y [00:02:38.820] is the code to find the partial derivative of the given function with respect to y. So when we run the code we have the output cos of x plus 1 by x plus 10 and to x to the power 9 and for f y we have 3y square minus 2y. [00:03:01.476] We'll move to the next example. If f is equal to x square plus 2y square minus 22, we need to find fxx and f y. [00:03:11.652] The symbolic variables x and y are defined and f is equal to the function x square plus 2y square minus 22 is defined and then dif f of f x2 is nothing but partial derivative of f with respect to x and again also with respect to x. [00:03:28.644] f is differentiated twice that is fxx so it has given us diff of f comma x comma 2 the same way with respect to y that is dif f of f comma y comma 2 so we are getting the output as 2 and 4 [00:03:48.644] If f is equal to x y q plus tan x plus cos of root log x then find dfx here the [00:03:55.172] Given in the function like in the problem to find the partial derivative of f with respect to x. So the symbolic variables x and y are defined and the given function is also defined and then d ff of f comma x is the code to find the partial derivative of f with respect to x. So this is the output we obtain for this problem. [00:04:17.444] If f is equal to xy cube divided by x plus y, then find fx f y fx f x y and f yx. So the symbolic variables s y ms x y are defined and the function is also defined. We know that the code to find the partial derivative of [00:04:37.156] F with respect to x and y are given as dif of f x and dif f of f f and fx that is partial derivative of f with respect to x done twice is given as diff of f x2. [00:04:54.436] And dif of f comma x comma y is nothing but partial derivative of f with respect to x first and then with

respect to y. And at last dif of f f x is nothing but partial derivative of f with respect to y first and then with respect to x.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:12.836] So this is the output we obtain. So here fxy and f y x both are equal. [00:05:20.964] If u is equal to x square y plus y square z plus z square x, we need to find the first order partial derivative of u. So here the symbolic variables are x, y, and z. So we need to define all the three symbolic variables s y um s xy z and then the given function is defined. Dif of u x. [00:05:38.736] one is nothing but partial derivative of the given function u with respect to x and u y and u z is also done in a similar way and this is the output we are obtaining. [00:05:55.108] If z is equal to sin two x plus three y as a given function, we need to find the first order partial derivative of z. So x y are the symbolic variables we are defining and the z as the function. [00:06:06.564] Given function we have defined sin of 2x plus 3y. DIFF of zx1 then dif f of z y 1 are nothing but code to find the partial derivative of the given function z with respect to x and y respectively and we are getting the output as 2 cos of 2x and 3. [00:06:26.468] If u is equal to root of x square plus y square plus z square plus tan inverse of x y, find the first order partial derivative of u. So the symbolic variables here are xyz. We are defining [00:06:40.388] S y ms x y z and then the given function u is equal to square root of x square plus y square plus z square plus a tan of x into y. It is nothing but tan inverse is written as a tan of x into y. And [00:06:55.120] A partial derivative first order partial derivative of u with respect to x y z r also written as d ff of u x1 d ff of u y one df of u z1 respectively. And this is the output we are obtaining for this problem. [00:07:12.548] When all the second order second partial derivatives of f of x y is equal to log of one by x minus one by y at one comma two. They're given at a particular point one comma two we need to find the [00:07:24.608] Partial second order second partial derivatives. So the symbolic variables here are Sy Ms XY. So the given function is also defined. And here we are initially defining [00:07:35.460] A of x y as dif of f of x2, partial derivative of f with respect to x twice, and partial derivative of f with respect to y twice as b of x comma y. [00:07:46.800] And C of x comma y is nothing but partial derivative of f with respect to x first and then with respect to y. D of x y is nothing but partial derivative of uf with respect to y, comma. X is nothing but derivative of f with respect to y first and then with respect to x. [00:08:02.788] And also we are continuing the same code with a of 1 comma 2, b of 1 comma 2, and c of 1 comma 2 d of 1 comma 2 because at upper point 1 comma 2 we need to find the second partial derivatives of the function. And this is the output we obtained. [00:08:21.668] And at 1 comma 2 we are obtaining 0 minus 3 by 4, 1 and 1. If u is equal to e power x y plus y z plus zx, then find u x y uyz and uz. [00:08:35.616] So symbolic variables x y z are defined and also the given function is defined in a similar way and dif f of u x y is nothing but partial derivative of u with respect to x first and then with respect to y. Dif of u y z is nothing but partial derivative of u with respect to y first and then with respect to z. [00:08:52.560] DIFF of U Zx is nothing but partial derivative of U with respect to Z first and then with respect to X. So this is the output we are obtaining. [00:09:04.644] If f is equal to e power x y z then find f x z f y z so symbolic variables x y z are defined and also the even function is defined in the second step. So d f of x f x y z is nothing but partial derivative of f with respect to x first and then with respect to y and at the [00:09:22.416] last with respect to z. In the similar way, f xs z is nothing but partial derivative of f with respect to x first and again also with respect to x and at last with respect to z. The f f of f y z is nothing but partial derivative of uf with respect to [00:09:38.528] Y first and again with respect to y and at last with respect to z.

So this is the output we are obtaining for this given function e power x y z. [00:09:51.108] Next, we'll move on to the Telia series expansion of the function of single variable. So the problem given here is expand f of x is equal to e power x into sine x above the point x is equal to 2 up to 2 degree terms. So here

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:08.740] We need to define the symbolic variables in the initial stage and also the function has to be defined. T is equal to tailor of f of f, comma. Expansion point to order 3. So the g uh given in the problem 2. [00:10:24.388] Expand the function about the point x is equal to 2. So we are defining with a expansion point 2 and it is up to 2 degree terms. So we are giving the order 3. [00:10:42.372] So the symbolic variable here is S Y M S x and function given is f is equal to exponent of [00:10:54.788] x into sine of x is the function given to expand and the code to find the tailor series expansion is t is equal to tailor of the function f. [00:11:13.988] comma with above the point X is equal to 2, so we are defining it as expansion point [00:11:28.132] Two expansion point two [00:11:35.108] And with order. [00:11:44.100] [ASR low confidence; check audio] Street. [00:11:51.396] So this is the output which we are obtaining after expanding the given function about the point x is equal to 2 up to 2 degree terms. [00:12:04.356] Next function f of x is equal to log of cos x about the point x is equal to pi by 3 up to 5th degree terms. It has also worked in a similar way. So the symbolic variable x is defined and the given function log of cos x is defined. The same way t is equal to tailor of the given function f. Here the expansion point is. [00:12:24.164] Pi by 3 and order it is up to 5th degree terms, so we are taking order 6. So we obtain the output. [00:12:36.676] Expand f of x is equal to log of secant x about the origin up to six degree terms. So [00:12:42.916] S y must x and also after that the function log of secant x is defined. T is equal to tailor of f with the order seven. Here it is about the origin, so no need to mention the expansion point. It is up to six degree terms, so we are writing the order seven. So this is the output. [00:13:02.116] f of x is equal to sine of log of x square plus 2x plus 1 about the origin up to 6 degree terms. So same way x is defined and the function f is defined and also it is also about the origin. So t is equal to del r of f comma order 7. [00:13:19.876] Is the code we are writing. That is about the origin. So no need to mention the expansion point here also. So this is all but for the given problem. [00:13:30.820] expand f of x comma y is equal to e power x into cos y above the point x is equal to 1 and y is equal to pi by 4 up to three degree terms. So you can check here this is the function of two variables. [00:14:00.708] So the symbolic variables are S Y M S X and Y R the symbolic variables. [00:14:09.540] And the function is f is equal to exponent of x. [00:14:17.956] into cos y. So this is the given function. So we need to find Taylor series expansion t is equal to [00:14:31.044] Taylor of the function yf [00:14:40.164] It is about the point XY that given it as [00:14:48.804] One comma [00:14:52.868] [ASR low confidence; check audio] I buy sport. [00:14:58.628] Comma. So the order

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:02.948] I have given up to three degrees terms. So we are giving the order as four. [00:15:16.068] So this is the output which we are obtaining. [00:15:20.964] For this given problem. [00:15:27.844] So next problem is f of x y is equal to x cube plus y cube plus xy square about x is equal to 1 and y is equal to 2 up to 4th degree terms. It is also the similar problem to which we just now. [00:15:39.652] So the symbolic variables are x and y. The function is also defined and here the points that are given at as 1 comma 2 and it has up to 4 degree terms. So we are considering the order phi and this is the output we are

obtaining for this problem. [00:15:54.116] Next function e power y into log of 1 plus x above the original to fourth degree terms. So symbolic variables are defined and also the given function is defined. So the code to find the tailor series gives the expansion is t is equal to tailor of your. [00:16:08.228] comma x y so it is about the origin so no need to mention the points over there and uh it is up to four degree terms so the order five and this is the result we are obtaining [00:16:20.452] f of x y is equal to d power x into tan y above the origin up to fifth degree terms. So symbolic variables and functions are defined in the initial steps and the tailor series expansion code is written t is equal to tailor of f, comma xy. That is about the origin no need to mention the points over there in the [00:16:38.788] line in the statement and it is up to fifth degree terms so the order six and this is the output which we are obtaining. [00:16:50.500] Thank you.

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Full transcript: 2495 normalized words, 73 timestamped segments, 1020.4375 seconds duration. Low-confidence segments flagged: 2. Review locators: 00:11:44.100, 00:14:52.868

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7 Video

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SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.956] Welcome back to the course on engineering mathematics 1. In this session we are going to learn about higher order differential equations in MATLAB. [00:00:15.236] Here our first question to solve the higher order differential equations that is y double dash plus phi y which is equals to zero. Things of y of x which is nothing but we are [00:00:26.468] Defining the variables I mean symbolic variables next ODE which is equals to diff of y comma x comma 2 plus phi star fi [00:00:36.688] double is equals to zero. Here O D is nothing but ordinary differential equations and d if f is nothing but differentiation and d if f of y comma x comma two is nothing but second derivy q plus phi into y which is double is equal to zero. Where double is equals to which means [00:00:56.580] defining an equation symbolically that is left hand side which is equals to right hand side. Next to for the solution we are taking y solve of x which is equals to d solve of ODE. Where d solve of [00:01:11.056] Where D solve which is is equal to differential equation solver of ODE function. So moving on to the MATLAB. [00:01:21.444] First we are defining the symbols that is y of x. [00:01:27.396] Next, we are defining the question that ODE which is equals to differentiation of y, comma x, comma 2. [00:01:39.044] Please. [00:01:44.772] 5 star y which is double is equals to 0. Semicolon next y solution of x which is equals to D sol of Ode. [00:02:04.548] So control enter will get the output that is C1 cos root 5x minus C2 side root 5x. Next [00:02:15.076] Moving on to the similar type of the question that is y double dash plus seven y which is equals to zero and we'll get the solution that is y solution of [00:02:25.056] x which is is equals to c1 cos root seven x minus c2

sin root seven x. This is of similar portion instead of 7 I have taken 18y which is equal to 0. Next moving to the higher order higher order differential equation. [00:02:41.540] That is 12y double dash plus 10y dash minus 8y which is equals to 0. Here again we are defining the variables that is of since of y of x and od which is equal to 12 star. The sense 12 into differentiation this is of second derivative plus [00:03:01.444] 10 into first derivative minus 8y which is double is equals to 0 and y solve of x is equals to d solve of od. Next moving on to the matlab I'm taking a section break. This is of defining the symbols Simps of Y of X. [00:03:20.996] Next, defining the question [00:03:28.068] Defining the question as ODE which is equal to 12 star second derivative d i f f of y comma x comma to [00:03:44.292] plus 10 star differentiation of y, comma x comma 1, which is the first derivative minus 8 star y. And double is equal to 0. [00:04:01.636] I am closing with semicolon. Next moving on to the Y solution of X, which is equals to D solve of ODE. So control enter will get an output that is of C2E power. [00:04:22.148] x by 2 plus even e power minus 4x by 3. Next moving on to the next question of similar question that is 8y double dash plus 2y dash minus y is equal to 0 as of same question. Next this is of order 4. So d [00:04:41.444] 4y by dx power 4 plus 8 d y d square y divided by dx square plus 16y which is equals to 0. Hereof also seen we are defining the symbols that is of seams of y of x and od is equals to differentiation of fourth order of

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.516] Fourth the different derivative differentiation of y, comma four plus eight into second. [00:05:07.428] Second derivative plus 16y which is equal to 0 and y solve of x is equals to d solve of ODE. So we will get the output as C1 cos 2x minus C3 sin 2x plus C2X cos 2x minus C4X sin 2x. [00:05:25.412] Here of order 3 of similar question. Next here I have taken d cube minus 3 d square plus 4. [00:05:33.744] into x which is equal to 0. Hereof also same we are defining the symbolic variables that is y of x and o d e which is equals to third order derivative minus 3 into second derivative. [00:05:47.972] Plus 4y which is is equals to double is equals to 0 and y solution which is is equals to desolve of y. Let us see in MATLAB code. [00:06:03.652] Here I am defining the symbols. This is of y of x. Next defining the ODE function ODE which is equal to DIFF of differentiation of Y, X, comma, 3. [00:06:23.364] Minus 3 star d if of differentiation of y, x, 2. [00:06:33.828] Plus 4 star y double is equals to 0. I am closing with semicolon and y solve. Y solve of x which is equals to D solve of Ode. [00:06:54.020] So control enter I'm getting the output [00:07:00.484] I'm getting output as C1 E power 2X plus C3 E power minus X plus C2 X E power 2X. Next, moving on to the next question. Here thereof initial conditions. [00:07:18.116] So that is of y double dash minus 5 y dash plus 6y which is equals to 0 and conditions are y of 0 is equals to minus 1 and y dash of 0 is equals to 0. First here defining as the symbols of variables. [00:07:33.696] Sims of y of x and we are defining the question as ODE which is is equals to second derivative. [00:07:39.920] Minus 5 into first derivative plus 6y which is double is equals to 0 and dy which is equal to first derivative we are defining and d2y which is equal to second derivative we are defining. [00:07:54.532] And C O N D 1, which is of condition 1, which is equal to y of 0, double is equals to minus 1, and condition 2, which is equals to dy of 0, double is equals to 0. And y solution instead [00:08:10.080] We are taking all the three conditions that is of D solve of ODE, condition one, comma condition two. Here Dispov Y solve, which is nothing but we are displaying the. [00:08:21.668] and EZ of EZ plot of Y sort which is nothing but easy to plot the graph. So let me try in the MATLAB. [00:08:34.340] So first we are defining the scenes since of x. Next define the question as ODE which is equal to DIFF of y comma x comma two minus [00:08:54.500] 5 into PIF of y, comma x. This is a this is our first derivative plus 6 star y

which is double is equals to 0. I am again closing with semicolon and [00:09:13.476] First derivative which is equal to dif of y, comma x and d2 y which is equal to second derivative. [00:09:31.268] Dif of y, comma x, comma two and I am defining the conditions. [00:09:47.780] It is of condition one which is equal to dy which is is equals to or y of 0. Y of 0 which is equals to minus 1.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.332] Dou is equals to minus one. [00:10:12.644] So second condition condition two which is equal to D2y. [00:10:22.052] Dy of zero, which double is equals to zero. I'm closing with semicolon. Next Y solve Y solution of X. [00:10:46.628] which is is equal to D solve of ODE, comma, condition one, comma, condition two. I am taking all the three conditions. [00:11:02.212] and closing with semicolon and I am displaying the solution display of y solution so I am getting the output that is 2 e power 3x minus 3 e power 2x and for the plot I am using [00:11:21.828] E Z plot that is of easy to plot the Y solution. So here again Control Enter. [00:11:33.892] So I'm getting the graph like this. [00:11:38.564] Next moving on to the similar questions. So here the conditions are y of 0 is equals to 0, y dash of 0 is equals to 0, and y double dash of 0 is equals to 0. And the question is y triple dash minus 6y double dash plus 1 y dash minus 6y is equals to 0. [00:11:57.520] For this question also, we are going to plot the graph. You are getting this type of graph. Next moving on to the single question. [00:12:08.452] That is of conditions with y of 1 is equals to 0 and y dash of 1 is equals to minus 1. So you are defining the symbols. Next question, next conditions you are defining. Next solution, you are displaying the solution or you are simplify the solution. Next moving [00:12:26.224] on to the a z plot of solution. Here I have taken one step extra that is why solution is equals to simplify of y solve. This is nothing but we are getting the output and we are solving again the same solution and we are displaying the solution. Next [00:12:43.744] We are plotting for the graph for the same question. [00:12:51.460] This type of answer we are going to get that is minus e power 2 minus 2x of x minus 1.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 1539 normalized words, 55 timestamped segments, 791.0625 seconds duration. Low-confidence segments flagged: 0.

EV0000098 · EV0000099 · EV0000100 · EV0000101

8 Source slide deck: Slides

This entry preserves the complete slide deck titled Slides as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000384 · EV0000385

Graphs and diagrams

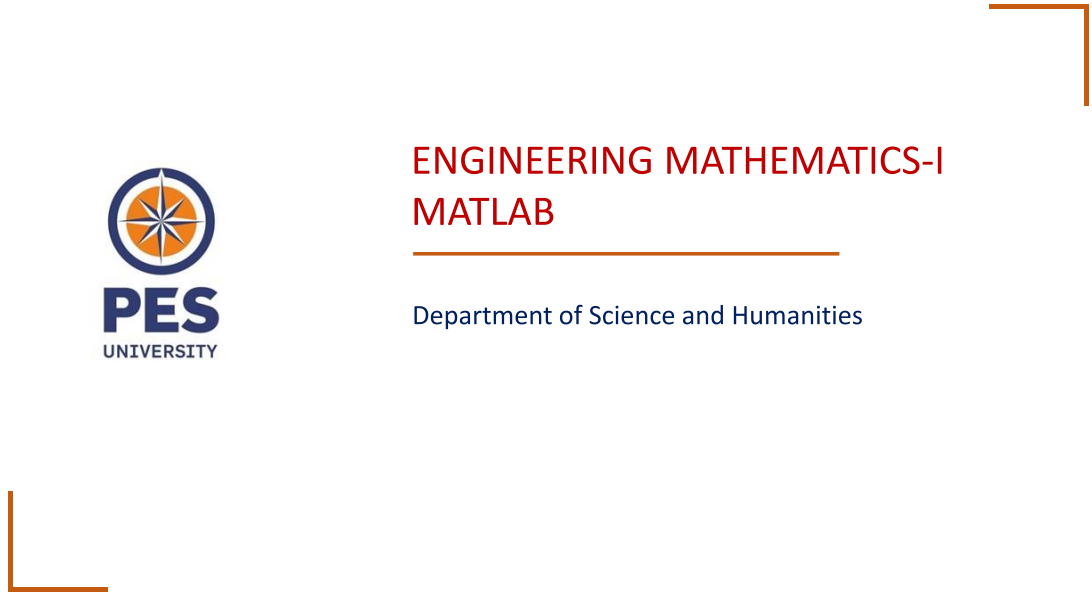

Figure G000097. Slides · source slide page 1.

EV0000385

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Plotting

- To plot the graph of a function, we need to take the following steps:
- Define **x**, by specifying the **range of values** for the variable **x**, for which the function is to be plotted.
- Define the function, **y = f(x).** Call the **plot** command, as **plot(x, y).**
- The **plot** command is used to create two-dimensional plots.
- Following example would demonstrate the concept.
- Let us plot the function $y = 3.5^{-0.5x}\cos(6x)$ for $-2 \leq x \leq 4$.

Figure G000098. Slides · source slide page 2.

EV0000388

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Plotting, Continued…

- The script file is as follows:

```
>> x=[-2:0.01:4];    (Here 0.01 is the spacing value)
>> y=3.5.^(-0.5*x).*cos(6*x);
>> plot(x,y)
```

- Here y is plotted as a function of x.

Figure G000099. Slides · source slide page 3.

EV0000391

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Plotting, Continued…

- When we run the file, MATLAB displays the following plot:

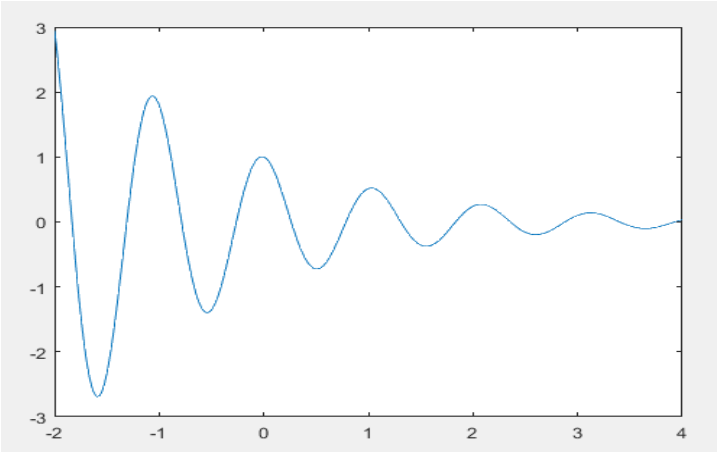

Figure G000100. Slides · source slide page 4.

EV0000394

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MATLAB – Plotting, Continued…

- Let us take another example to plot the function $y = x^2$.
- In this example, we will draw two graphs with the same function, but in second time, we will reduce the value of increment.
- Note that as we decrease the increment, the graph becomes smoother.
- The script file is as follows:

```
>> x = [-100:20:100];
>> y = x.^2;
>> plot(x, y)
```

Figure G000101. Slides · source slide page 5.

EV0000397

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Plotting, Continued…

- When we run the file, MATLAB displays the following plot:

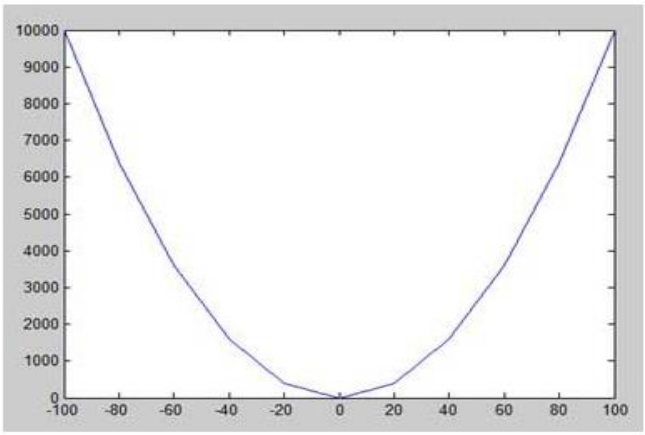

Figure G000102. Slides · source slide page 6.

EV0000400

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MATLAB – Plotting, Continued…

Change the code file a little, reduce the increment to 5.

```
>> x = [-100:5:100];
>> y = x.^2;
```

>> plot(x, y). When you run the file, MATLAB displays the following plot:

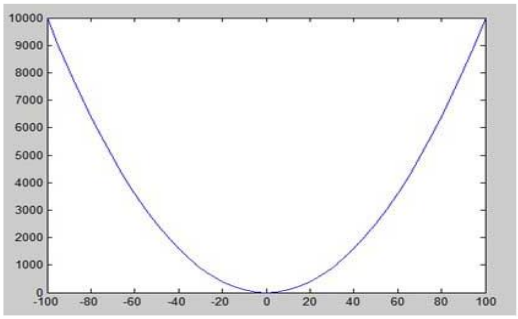

Figure G000103. Slides · source slide page 7.

EV0000403

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MATLAB - Three Dimensional Plots

- Three-dimensional plots display a surface defined by a function in two variables, $g = f(x, y)$.
- To define the function $g = f(x, y)$, we first create a set of (x, y) points over the domain of the function using the **meshgrid** command. Next, we assign the function itself. Finally, we use the **surf** command to create a surface plot.

Figure G000104. Slides · source slide page 8.

EV0000406

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued...

- The following example demonstrates the concept:
- Let us create a 3D surface map for the function $z = xe^{-(x^2+y^2)}$
- The script file is as follows:

```
>> [x,y] = meshgrid(-2:.2:2);
>> z = x.* exp(-x.^2 - y.^2);
>> surf(x,y,z)
```

Figure G000105. Slides · source slide page 9.

EV0000409

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued...

- When we run the file, MATLAB displays the following 3-D map:

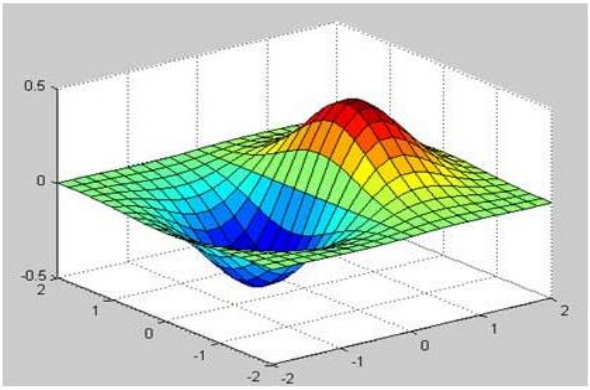

Figure G000106. Slides · source slide page 10.

EV0000412

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued...

- Let us consider another example to create a 3D surface map for the function $z = \sin x + \cos y$
- The script file is as follows:

```
>> [x,y] = meshgrid(-2:.2:2);
>> z=sin(x)+cos(y);
 >> surf(x,y,z)
```

Figure G000107. Slides · source slide page 11.

EV0000415

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued...

- When we run the file, MATLAB displays the following 3-D map:

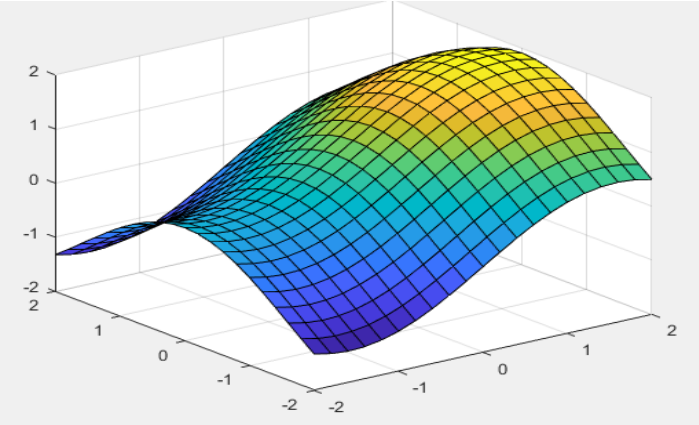

Figure G000108. Slides · source slide page 12.

EV0000418

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued...

- Let us consider another example to create a 3D surface map for the function $z = \frac{xy(x^2-y^2)}{x^2+y^2}$.
- The script file is as follows:

```
>> [x,y] = meshgrid(-2:.2:2);
>> z=x.*y.*(x.^2-y.^2)./(x.^2+y.^2);
>> surf(x,y,z)
```

Figure G000109. Slides · source slide page 13.

EV0000421

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB - Three Dimensional Plots, Continued…

- When we run the file, MATLAB displays the following 3-D map:

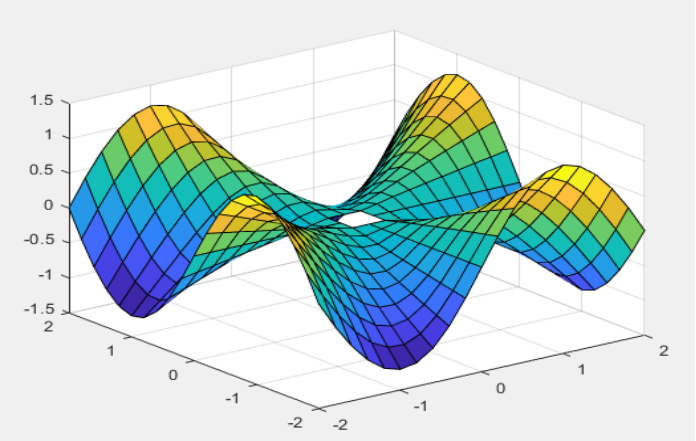

Figure G000110. Slides · source slide page 14.

EV0000424

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Adding Title and Labels on the Graph

- Using MATLAB we can add title, labels along the x-axis and y-axis of the graph.
- The **xlabel** and **ylabel** commands generate labels along x-axis and y-axis.
- The **title** command allows you to put a title on the graph.
- The script file is as follows:

```
>> x = [0:0.01:10];
>> y = sin(x);
>> plot(x, y), xlabel('x'), ylabel('Sin(x)'), title('Sin(x) Graph')
```

Figure G000111. Slides · source slide page 15.

EV0000427

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Adding Title and Labels on the Graph

- When we run the file, MATLAB displays the following plot:

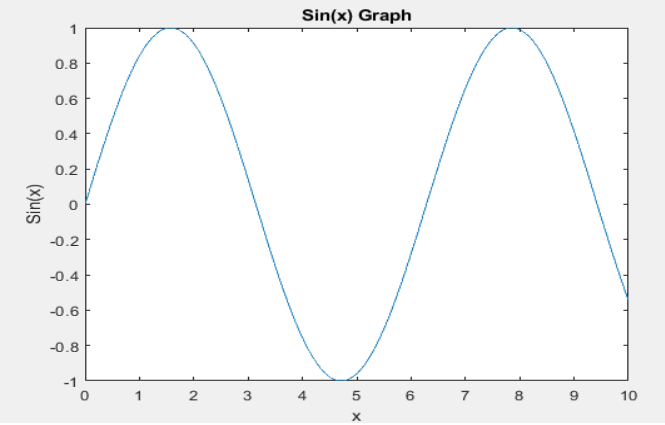

Figure G000112. Slides · source slide page 16.

EV0000430

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots

- To plot a curve in polar coordinates, we use the following command:
- **polarplot(theta,rho)**. Here **theta** is the angle in radians and **rho** is the radius value for each point.
- Following example would demonstrate the concept.
- Let us plot the curve $r = sin2\theta cos2\theta$.

Figure G000113. Slides · source slide page 17.

EV0000433

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

PES UNIVERSITY

- The script file is as follows:

```
>> theta = 0:0.01:2*pi;
>> rho = sin(2*theta).*cos(2*theta);
>> polarplot(theta,rho)
```

Figure G000114. Slides · source slide page 18.

EV0000436

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

PES UNIVERSITY

- When we run the file, MATLAB displays the following plot:

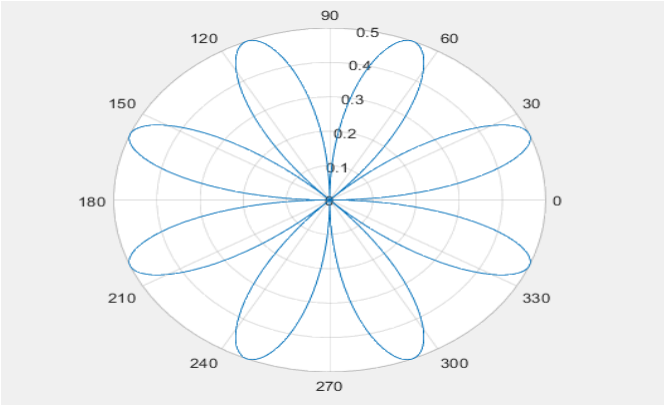

Figure G000115. Slides · source slide page 19.

EV0000439

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

- Consider another example:
- Plot the curve $r = 1 - sin\theta$
- The script file is as follows:

```
>> theta = 0:0.01:2*pi;
>> rho = 1 - sin(theta);
>> polar(theta, rho)
```

Figure G000116. Slides · source slide page 20.

EV0000442

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

- When we run the file, MATLAB displays the following plot:

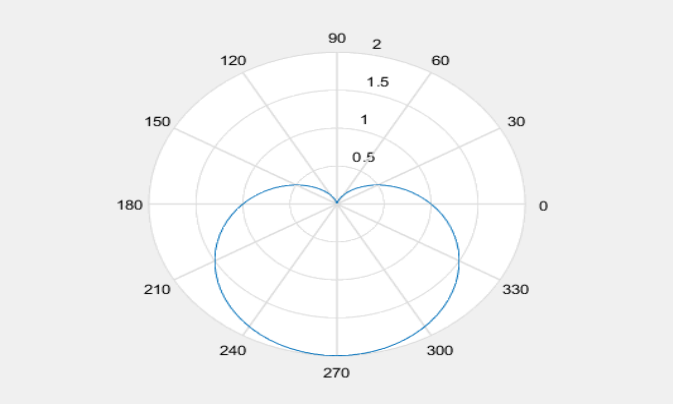

Figure G000117. Slides · source slide page 21.

EV0000445

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

- Consider another example:
- Plot the curve $r = cos2\theta$
- The script file is as follows:

```
>> theta = 0:0.01:2*pi;
>> rho = cos(2*theta);
>> polar(theta, rho)
```

Figure G000118. Slides · source slide page 22.

EV0000448

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

- When we run the file, MATLAB displays the following plot:

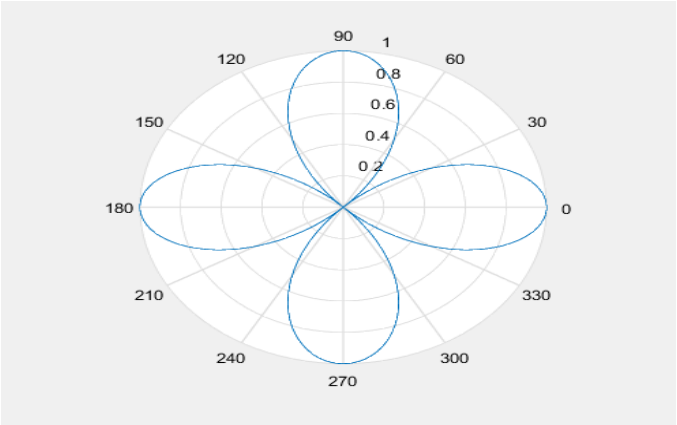

Figure G000119. Slides · source slide page 23.

EV0000451

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

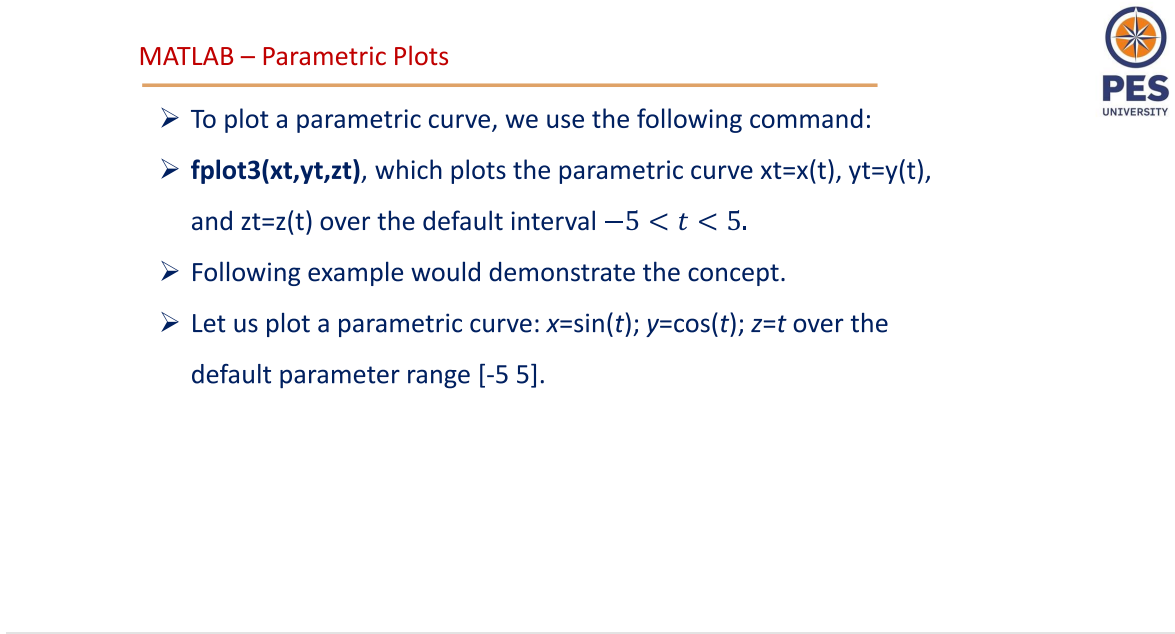

Figure G000120. Slides · source slide page 24.

EV0000454

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

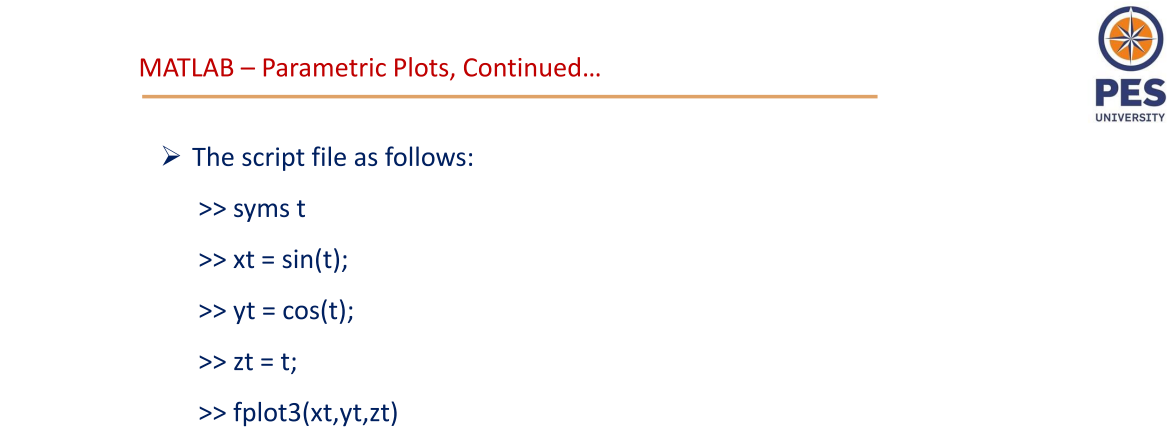

Figure G000121. Slides · source slide page 25.

EV0000457

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Parametric Plots, Continued…

- When we run the file, MATLAB displays the following plot:

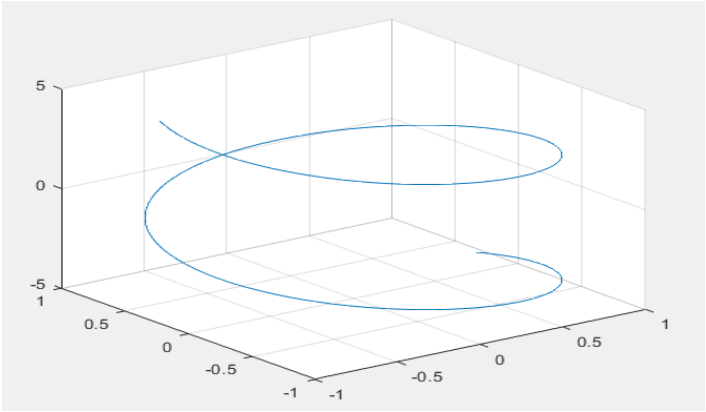

Figure G000122. Slides · source slide page 26.

EV0000460

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Polar Plots, Continued…

- Consider another example:
- Plot the parametric curve $x = e^{-\frac{t}{10}}\sin(5t)\,; y =$

  $e^{-\frac{t}{10}}\cos(5t); z = t$

  over the parameter range [-10 10]

Note that fplot3(xt,yt,zt, [tmin tmax]) plots xt = x(t), yt = y(t), and zt = z(t) over the interval tmin < t < tmax.

Figure G000123. Slides · source slide page 27.

EV0000463

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Parametric Plots, Continued…

- The script file as follows:

```
>> syms t
>> xt = exp(-t/10).*sin(5*t);
>> yt = exp(-t/10).*cos(5*t);
>> zt = t;
>> fplot3(xt,yt,zt,[-10 10])
```

Figure G000124. Slides · source slide page 28.

EV0000466

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB – Parametric Plots, Continued…

- When we run the file, MATLAB displays the following plot:

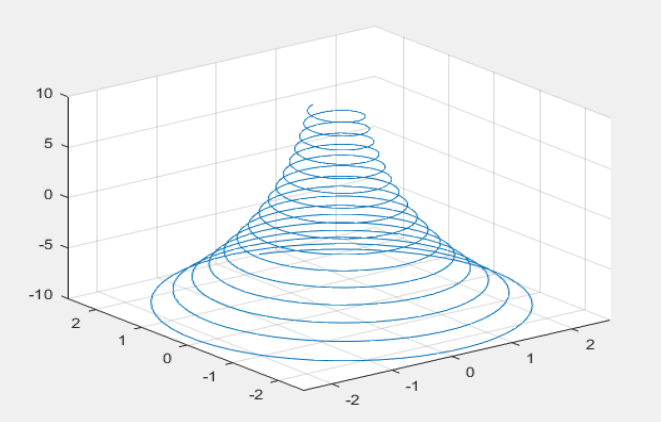

Figure G000125. Slides · source slide page 29.

EV0000469

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000126. Slides · source slide page 30.

EV0000471

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000384 · EV0000385 · EV0000386
2. Source slide page 2 (extracted text): MATLAB – Plotting ➢To plot the graph of a function, we need to take the following steps: ➢Define x, by specifying the range of values for the variable x, for which the function is to be plotted. ➢Define the function, y = f(x). Call the plot command, as plot(x, y). ➢The plot command is used to create two-dimensional plots. ➢Following example would demonstrate the concept. ➢Let us plot the function $y = 3.5 - 0.5x \cos 6x$ for $-2 \le x \le 4$. EV0000387 · EV0000388 · EV0000389
3. Source slide page 3 (extracted text): MATLAB – Plotting, Continued... ➢The script file is as follows: >> x=[-2:0.01:4]; (Here 0.01 is the spacing value) >> y=3.5.^(-0.5*x).*cos(6*x); >> plot(x,y) ➢Here $y$is plotted as a function of x. EV0000390 · EV0000391 · EV0000392
4. Source slide page 4 (extracted text): MATLAB – Plotting, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000393 · EV0000394 · EV0000395
5. Source slide page 5 (extracted text): MATLAB – Plotting, Continued... ➢Let us take another example to plot the function $y = x2$. ➢In this example, we will draw two graphs with the same function, but in second time, we will reduce the value of increment. ➢Note that as we decrease the increment, the graph becomes smoother. ➢The script file is as follows: >> x = [-100:20:100]; >> y = x.^2; >> plot(x, y) EV0000396 · EV0000397 · EV0000398

6. Source slide page 6 (extracted text): MATLAB – Plotting, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000399 · EV0000400 · EV0000401
7. Source slide page 7 (extracted text): MATLAB – Plotting, Continued... Change the code file a little, reduce the increment to 5. >> x = [-100:5:100]; >> y = x.^2; >> plot(x, y). When you run the file, MATLAB displays the following plot: EV0000402 · EV0000403 · EV0000404
8. Source slide page 8 (extracted text): MATLAB - Three Dimensional Plots ➢Three-dimensional plots display a surface defined by a function in two variables, $g= f(x, y)$. ➢To define the function $g= f(x, y)$, we first create a set of x, y points over the domain of the function using the meshgrid command. Next, we assign the function itself. Finally, we use the surf command to create a surface plot. EV0000405 · EV0000406 · EV0000407
9. Source slide page 9 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢The following example demonstrates the concept: ➢Let us create a 3D surface map for the function $z= xe-x2+y2$ ➢The script file is as follows: >> [x,y] = meshgrid(-2:.2:2); >> z = x.* exp(-x.^2 - y.^2); >> surf(x,y,z) EV0000408 · EV0000409 · EV0000410
10. Source slide page 10 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢When we run the file, MATLAB displays the following 3-D map: EV0000411 · EV0000412 · EV0000413
11. Source slide page 11 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢Let us consider another example to create a 3D surface map for the function $z=$ sinx + cosy ➢The script file is as follows: >> [x,y] = meshgrid(-2:.2:2); >> z=sin(x)+cos(y); >> surf(x,y,z) EV0000414 · EV0000415 · EV0000416
12. Source slide page 12 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢When we run the file, MATLAB displays the following 3-D map: EV0000417 · EV0000418 · EV0000419
13. Source slide page 13 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢Let us consider another example to create a 3D surface map for $xyx2-y2$ the function $z=$. $x2+y2$ ➢The script file is as follows: >> [x,y] = meshgrid(-2:.2:2); >> z=x.*y.*(x.^2-y.^2)./(x.^2+y.^2); >> surf(x,y,z) EV0000420 · EV0000421 · EV0000422
14. Source slide page 14 (extracted text): MATLAB - Three Dimensional Plots, Continued... ➢When we run the file, MATLAB displays the following 3-D map: EV0000423 · EV0000424 · EV0000425
15. Source slide page 15 (extracted text): Adding Title and Labels on the Graph ➢Using MATLAB we can add title, labels along the x-axis and y-axis of the graph. ➢The xlabel and ylabel commands generate labels along x-axis and y-axis. ➢The title command allows you to put a title on the graph. ➢The script file is as follows: >> x = [0:0.01:10]; >> y = sin(x); >> plot(x, y), xlabel('x'), ylabel('Sin(x)'), title('Sin(x) Graph') EV0000426 · EV0000427 · EV0000428
16. Source slide page 16 (extracted text): Adding Title and Labels on the Graph ➢When we run the file, MATLAB displays the following plot: EV0000429 · EV0000430 · EV0000431
17. Source slide page 17 (extracted text): MATLAB – Polar Plots ➢To plot a curve in polar coordinates, we use the following command: ➢polarplot(theta,rho). Here theta is the angle in radians and rho is the radius value for each point. ➢Following example would demonstrate the concept. ➢Let us plot the curve $r= sin2\theta cos2\theta$. EV0000432 · EV0000433 · EV0000434

18. Source slide page 18 (extracted text): MATLAB – Polar Plots, Continued... ➢The script file is as follows: >> theta = 0:0.01:2*pi; >> rho = sin(2*theta).*cos(2*theta); >> polarplot(theta,rho) EV0000435 · EV0000436 · EV0000437

19. Source slide page 19 (extracted text): MATLAB – Polar Plots, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000438 · EV0000439 · EV0000440

20. Source slide page 20 (extracted text): MATLAB – Polar Plots, Continued... ➢Consider another example: ➢Plot the curve $r = 1 - sin\theta$ ➢The script file is as follows: >> theta = 0:0.01:2*pi; >> rho = 1 - sin(theta); >> polar(theta, rho) EV0000441 · EV0000442 · EV0000443

21. Source slide page 21 (extracted text): MATLAB – Polar Plots, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000444 · EV0000445 · EV0000446

22. Source slide page 22 (extracted text): MATLAB – Polar Plots, Continued... ➢Consider another example: ➢Plot the curve $r = cos2\theta$ ➢The script file is as follows: >> theta = 0:0.01:2*pi; >> rho = cos(2*theta); >> polar(theta, rho) EV0000447 · EV0000448 · EV0000449

23. Source slide page 23 (extracted text): MATLAB – Polar Plots, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000450 · EV0000451 · EV0000452

24. Source slide page 24 (extracted text): MATLAB – Parametric Plots ➢To plot a parametric curve, we use the following command: ➢fplot3(xt,yt,zt), which plots the parametric curve xt=x(t), yt=y(t), and zt=z(t) over the default interval $-5 < t < 5$. ➢Following example would demonstrate the concept. ➢Let us plot a parametric curve: x=sin(t); y=cos(t); z=t over the default parameter range [-5 5]. EV0000453 · EV0000454 · EV0000455

25. Source slide page 25 (extracted text): MATLAB – Parametric Plots, Continued... ➢The script file as follows: >> syms t >> xt = sin(t); >> yt = cos(t); >> zt = t; >> fplot3(xt,yt,zt) EV0000456 · EV0000457 · EV0000458

26. Source slide page 26 (extracted text): MATLAB – Parametric Plots, Continued... ➢When we run the file, MATLAB displays the following plot: EV0000459 · EV0000460 · EV0000461

27. Source slide page 27 (extracted text): MATLAB – Polar Plots, Continued... ➢Consider another example: 10 ➢Plot the parametric curve $x = e-t$ sin $5t$; $y = 10\ e-t$ cos $5t$; $z = t$ over the parameter range [-10 10] Note that fplot3(xt,yt,zt, [tmin tmax]) plots xt = x(t), yt = y(t), and zt = z(t) over the interval tmin < t < tmax. EV0000462 · EV0000463 · EV0000464

28. Source slide page 28 (extracted text): MATLAB – Parametric Plots, Continued... ➢The script file as follows: >> syms t >> xt = exp(-t/10).*sin(5*t); >> yt = exp(-t/10).*cos(5*t); >> zt = t; >> fplot3(xt,yt,zt,[-10 10]) EV0000465 · EV0000466 · EV0000467

29. Source slide page 29 (extracted text): MATLAB – Parametric Plots, Continued... ➢When we run the file, MATLAB displays the following plot: ## Page / slide 30 (OCR) THANK YOU PES UNIVERSITY EV0000468 · EV0000469 · EV0000470

30. Source slide page 30: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000471

SHORT-NOTE RECAP

Source deck preserved: Slides. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000384

9 Source slide deck: UNIT 1

This entry preserves the complete slide deck titled UNIT 1 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000562 · EV0000563

Graphs and diagrams

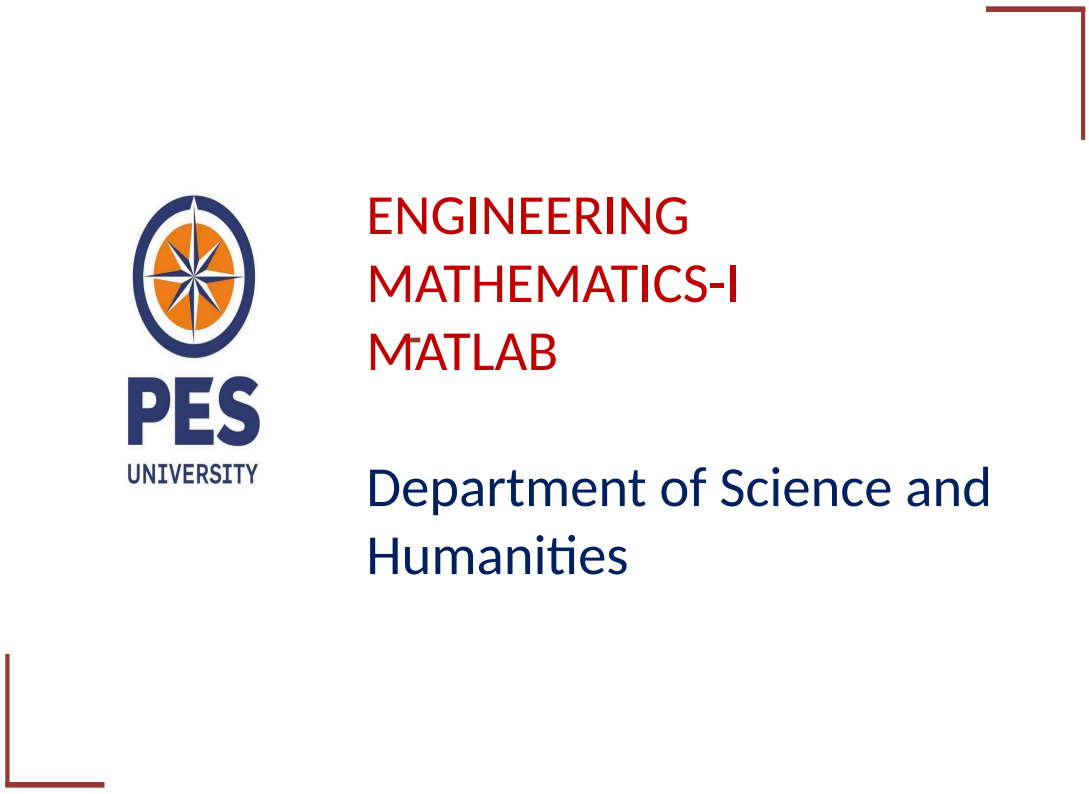

Figure G000157. UNIT 1 · source slide page 1.

EV0000563

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

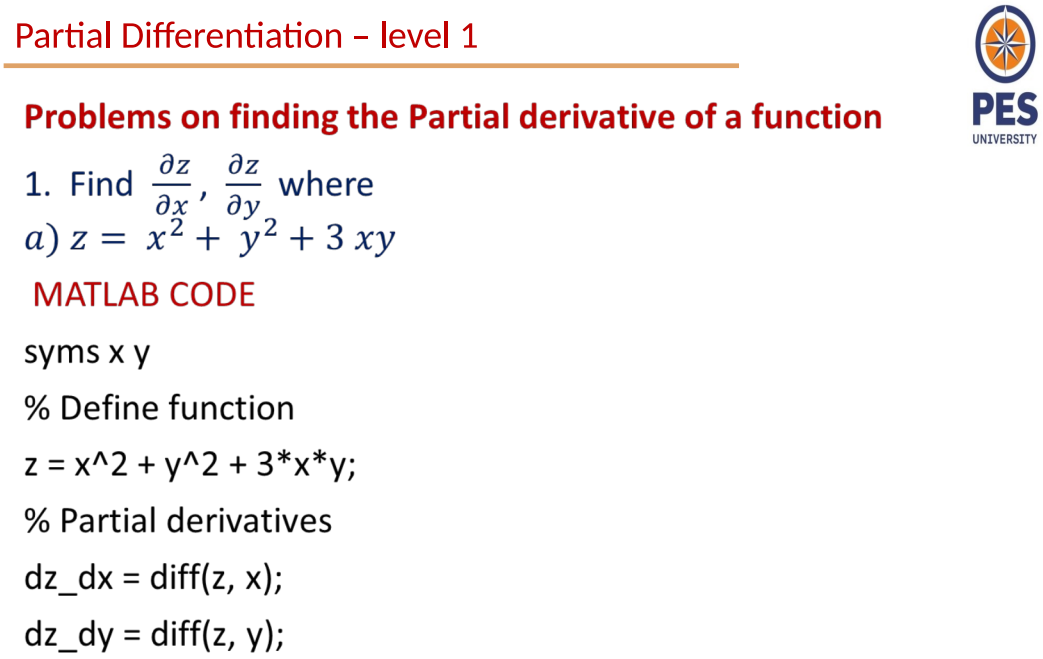

Figure G000158. UNIT 1 · source slide page 2.

EV0000566

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
% Display results
disp('Partial derivative with respect to x:')
disp(dz_dx)
disp('Partial derivative with respect to y:')
disp(dz_dy)
```

OUTPUT

```
Partial derivative with respect to x:
2*x + 3*y
Partial derivative with respect to y:
2*y + 3*x
```

Figure G000159. UNIT 1 · source slide page 3.

EV0000569

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

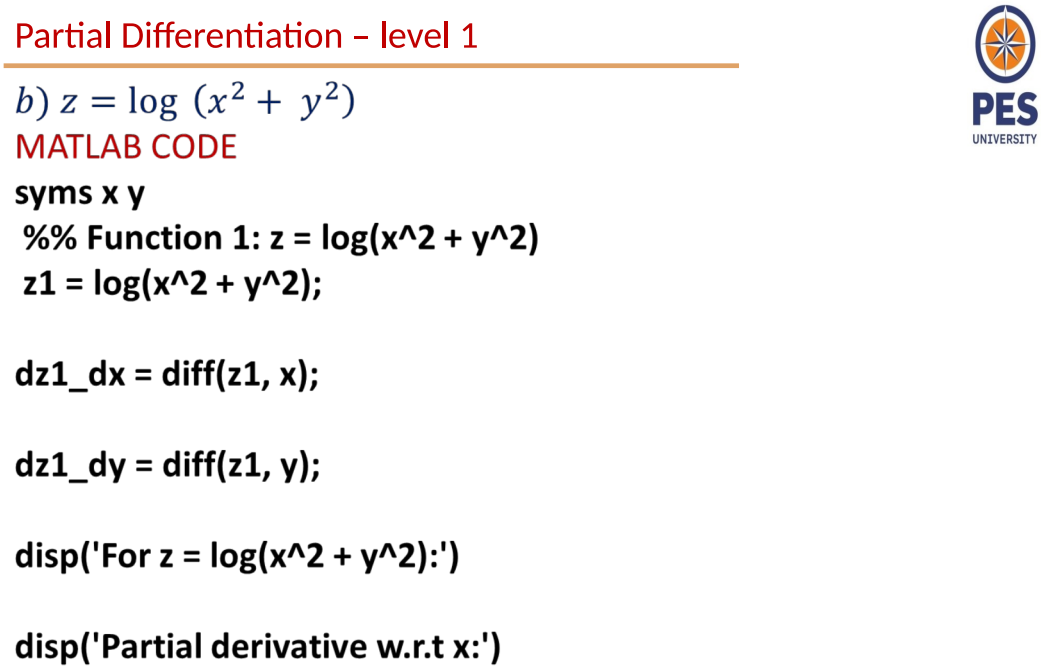

Figure G000160. UNIT 1 · source slide page 4.

EV0000572

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

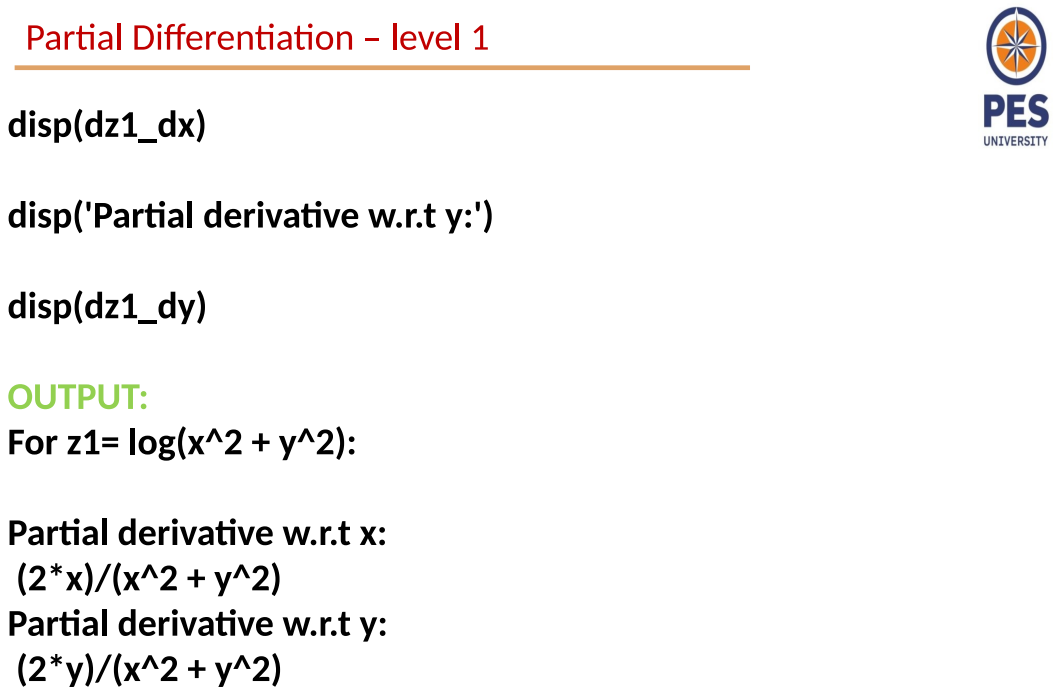

Figure G000161. UNIT 1 · source slide page 5.

EV0000575

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

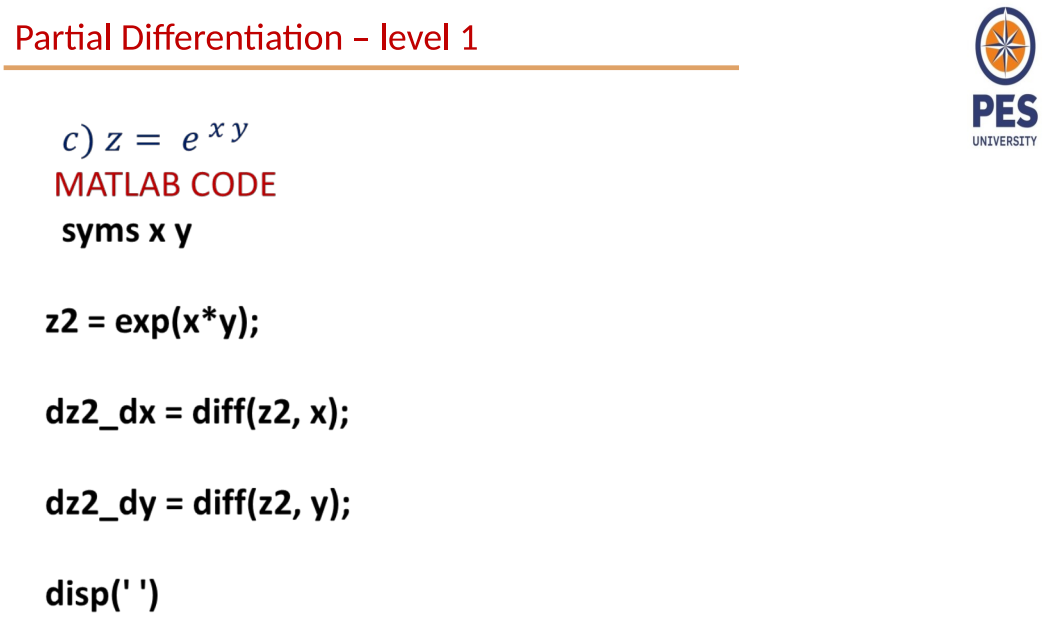

Figure G000162. UNIT 1 · source slide page 6.

EV0000578

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

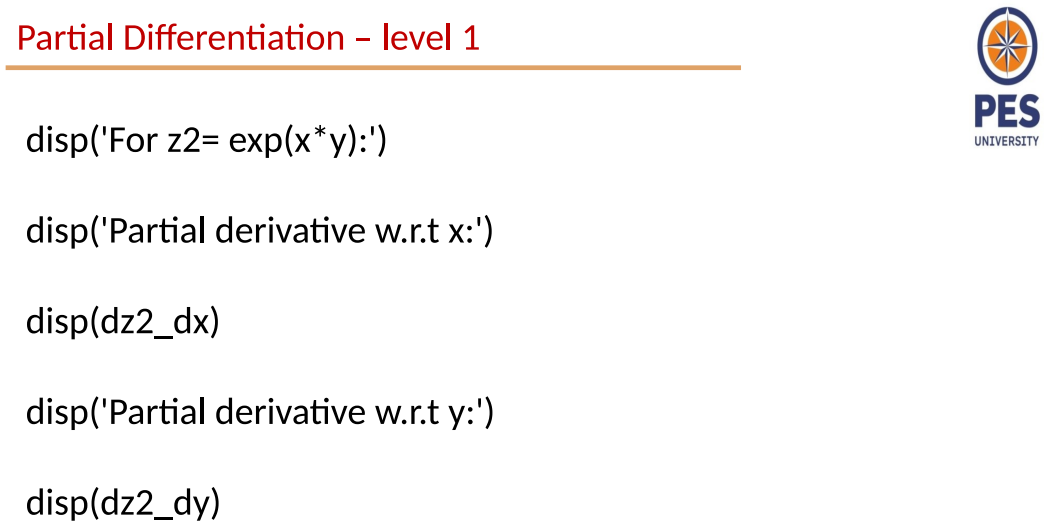

Figure G000163. UNIT 1 · source slide page 7.

EV0000581

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

OUTPUT

For z2 = exp(x*y):

Partial derivative w.r.t x:
y*exp(x*y)

Partial derivative w.r.t y:
x*exp(x*y)

Figure G000164. UNIT 1 · source slide page 8.

EV0000584

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

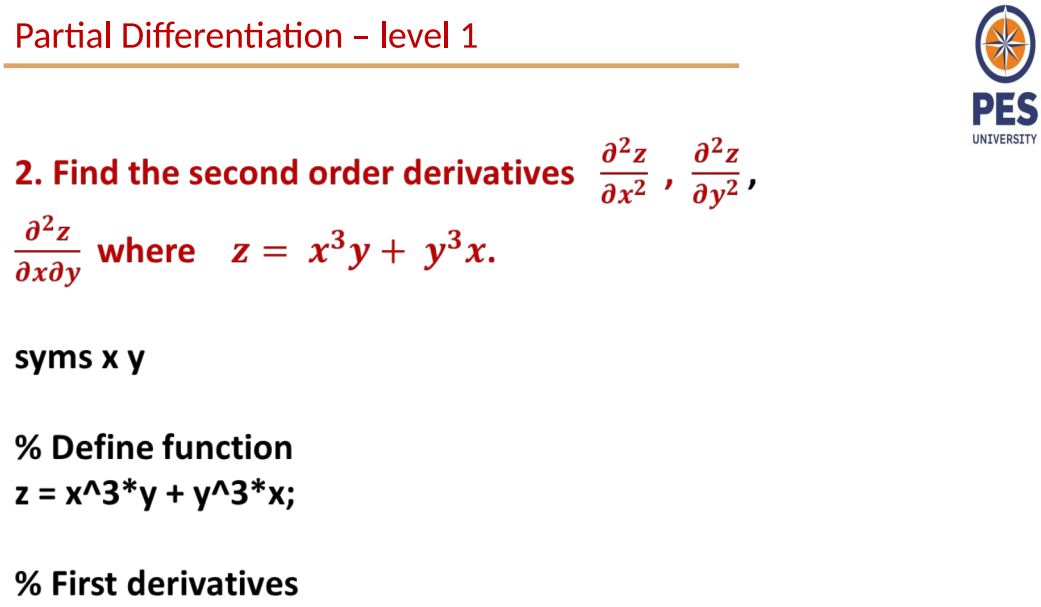

Figure G000165. UNIT 1 · source slide page 9.

EV0000587

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
dz_dx = diff(z, x);

dz_dy = diff(z, y);

% Second derivatives

d2z_dx2 = diff(z, x, 2);

d2z_dy2 = diff(z, y, 2);

d2z_dxdy = diff(diff(z, x), y);

% Display results
```

Figure G000166. UNIT 1 · source slide page 10.

EV0000590

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
disp('Second derivative w.r.t x:');

disp(d2z_dx2);

disp('Second derivative w.r.t y:');

disp(d2z_dy2);


disp('Mixed derivative:');

disp(d2z_dxdy);
```

Figure G000167. UNIT 1 · source slide page 11.

EV0000593

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

Since disp code appear many times, instead you can use fprintf to display results:

fprintf('Second order Partial derivative wrt x =%s\n',char(**d2z_dx2**))

fprintf('Second order Partial derivative wrt y=%s\n',char(**d2z_dy2**))

fprintf('Mixed Partial derivative =%s\n',char(**d2z_dxdy**))

Figure G000168. UNIT 1 · source slide page 12.

EV0000596

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

output

Second derivative w.r.t x:

6*x*y

Second derivative w.r.t y:

6*x*y

Mixed derivative:

3*x^2 + 3*y^2

Figure G000169. UNIT 1 · source slide page 13.

EV0000599

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

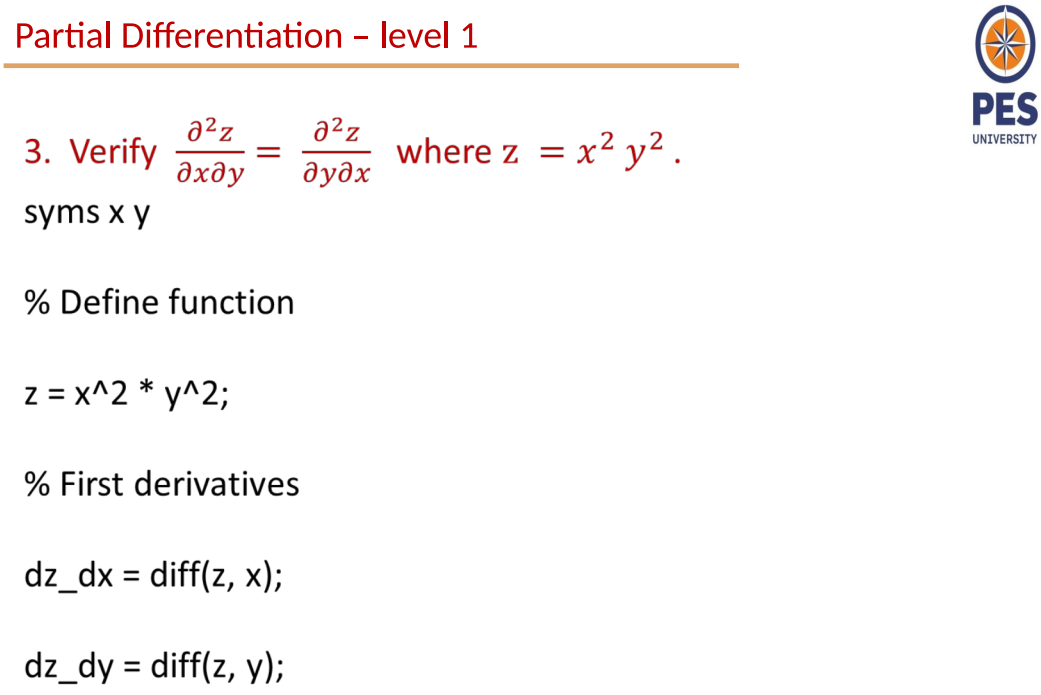

Figure G000170. UNIT 1 · source slide page 14.

EV0000602

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
% Mixed derivatives

d2z_dxdy = diff(dz_dx, y);

d2z_dydx = diff(dz_dy, x);

% Display results

disp('d2z/dxdy = ‘);
disp(d2z_dxdy);
disp('d2z/dydx = ');
disp(d2z_dydx);
```

Figure G000171. UNIT 1 · source slide page 15.

EV0000605

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

output:

```
d2z/dxdy =
4*x*y

d2z/dydx =
4*x*y
```

Figure G000172. UNIT 1 · source slide page 16.

EV0000608

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

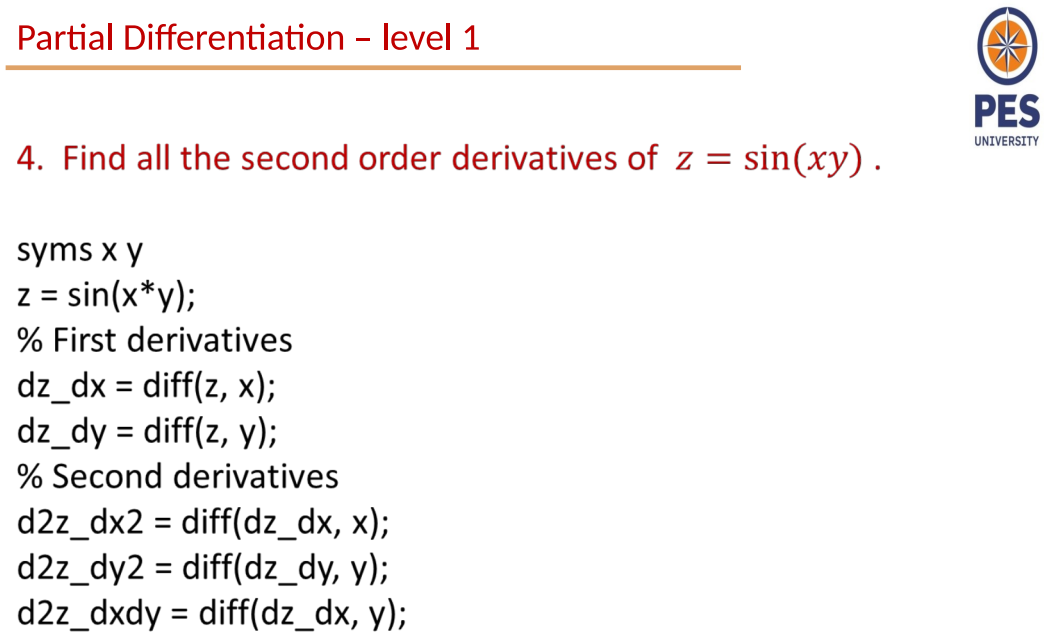

Figure G000173. UNIT 1 · source slide page 17.

EV0000611

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

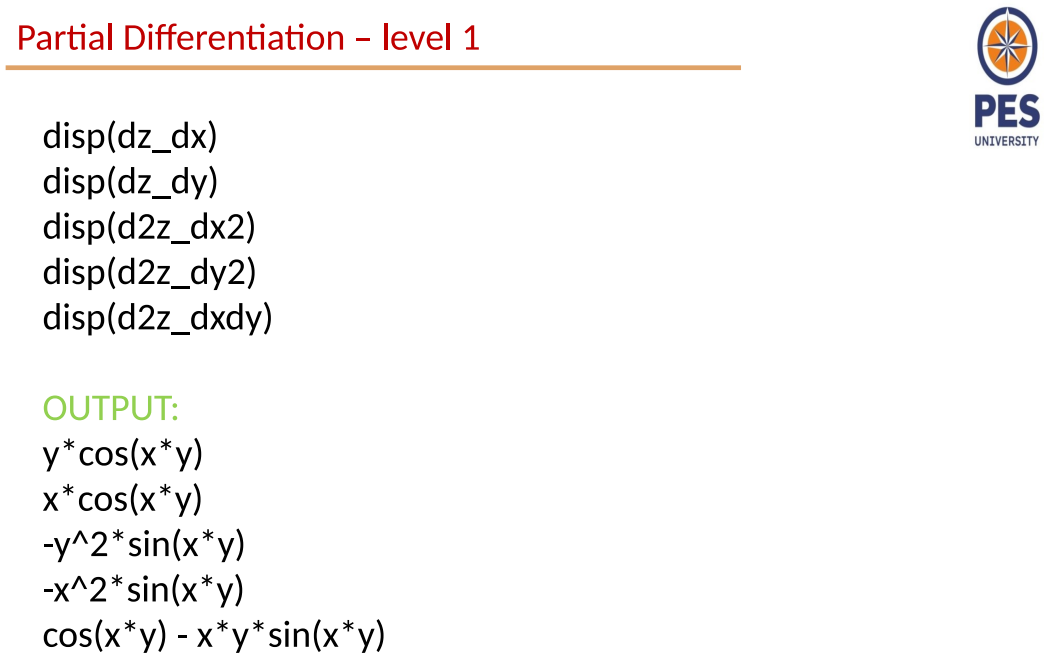

Figure G000174. UNIT 1 · source slide page 18.

EV0000614

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

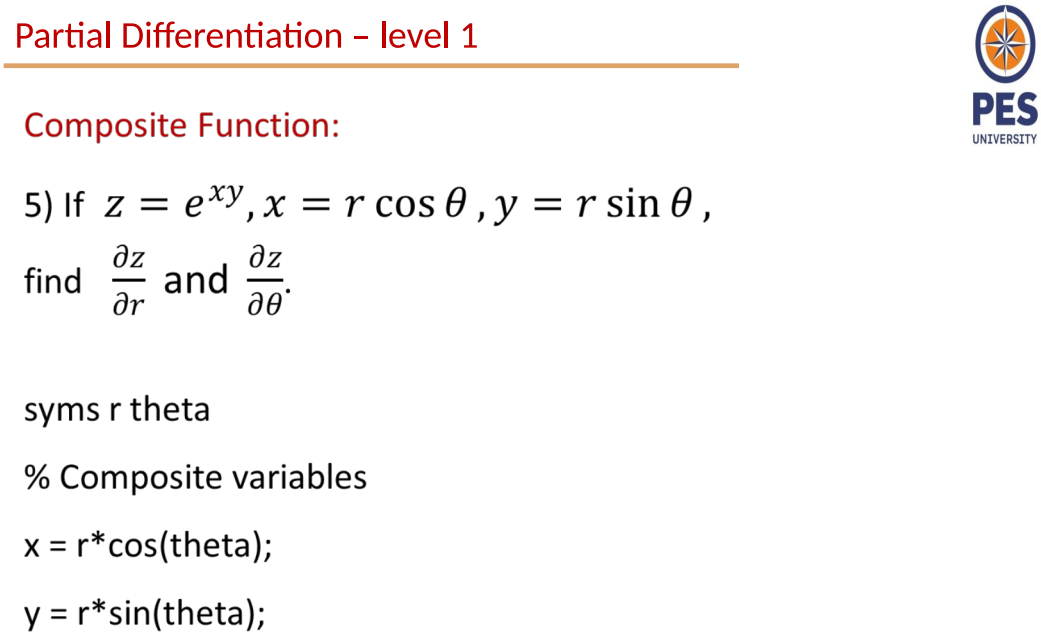

Figure G000175. UNIT 1 · source slide page 19.

EV0000617

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
% Composite function
z = exp(x*y);
% Partial derivatives
dz_dr = simplify(diff(z,r));
dz_dtheta = simplify(diff(z,theta));
disp('dz/dr =')
disp(dz_dr)
disp('dz/dtheta =')
disp(dz_dtheta)
```

Figure G000176. UNIT 1 · source slide page 20.

EV0000620

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

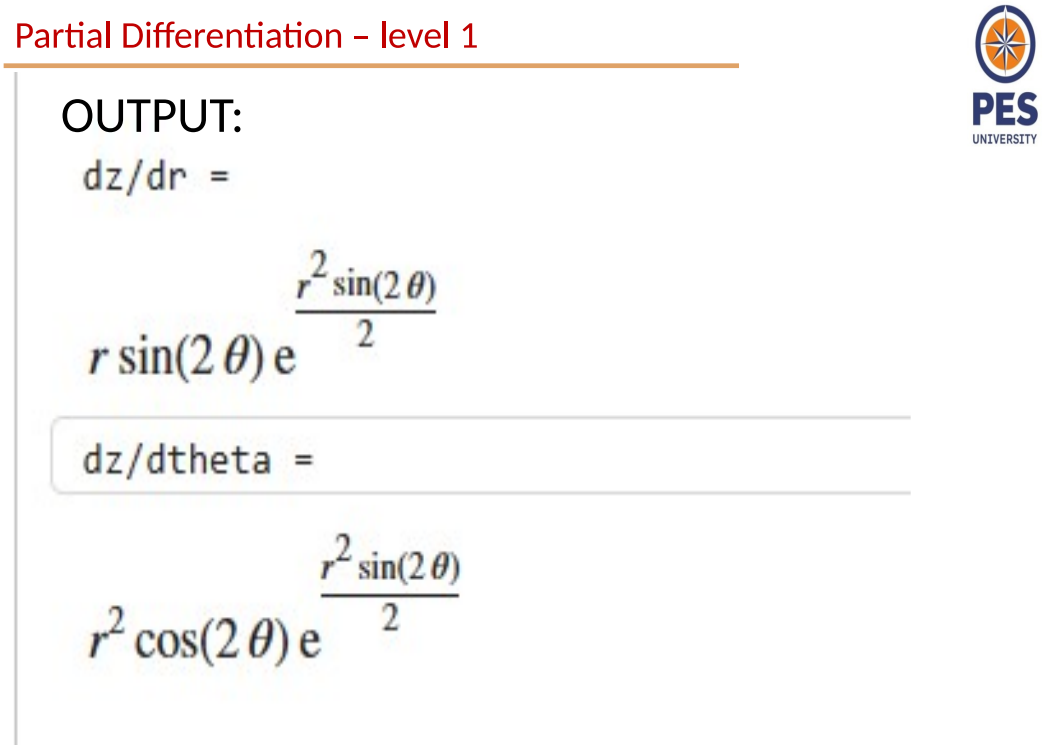

Figure G000177. UNIT 1 · source slide page 21.

EV0000623

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

6) If $z = x^2 + y^2$, where

$$x = r\cos\theta, y = r\sin\theta,$$

verify the Chain Rule

$$\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r}$$

using MATLAB.

```
syms x y r theta

% Original function
z = x^2 + y^2;

% Partial derivatives
dz_dx = diff(z,x);
dz_dy = diff(z,y);
```

Figure G000178. UNIT 1 · source slide page 22.

EV0000626

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
% Composite variables
xr = r*cos(theta);
yr = r*sin(theta); dx_dr = diff(xr,r);
dy_dr = diff(yr,r);

% Chain Rule
Chain = subs(dz_dx,[x y],[xr yr])*dx_dr + ...
      subs(dz_dy,[x y],[xr yr])*dy_dr;

% Direct differentiation
Direct = diff(subs(z,[x y],[xr yr]),r);
```

Figure G000179. UNIT 1 · source slide page 23.

EV0000629

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

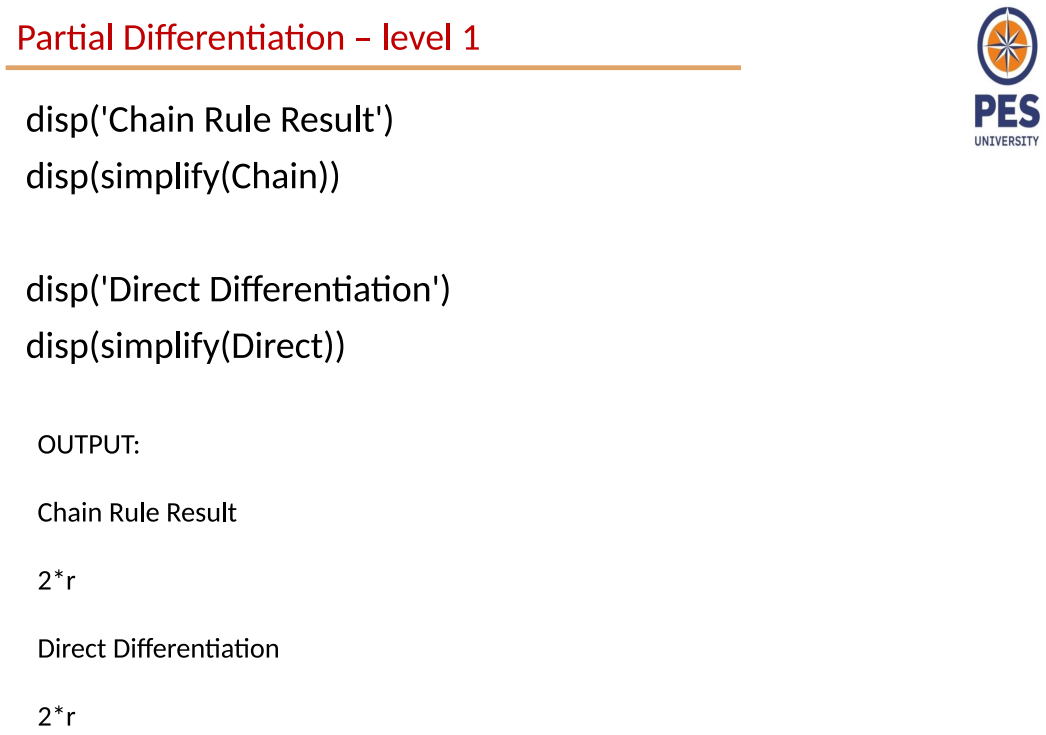

Figure G000180. UNIT 1 · source slide page 24.

EV0000632

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

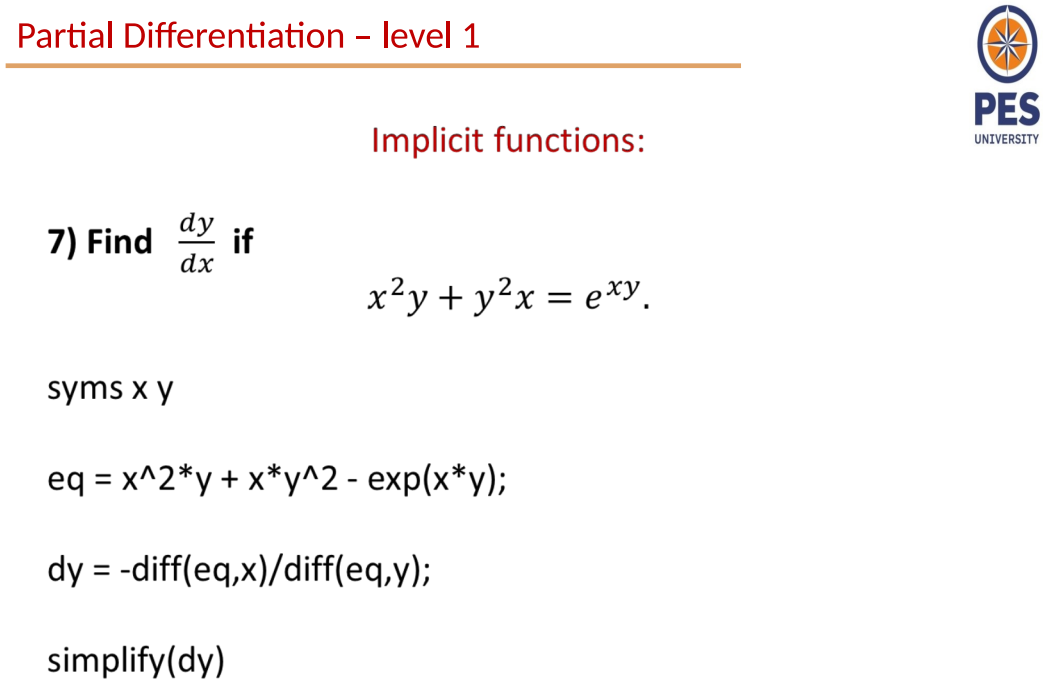

Figure G000181. UNIT 1 · source slide page 25.

EV0000635

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

Output :

```
ans =
```

$$-\frac{y\,(2x+y-e^{xy})}{x\,(x+2y-e^{xy})}$$

Figure G000182. UNIT 1 · source slide page 26.

EV0000638

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

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8). If $z = \frac{x}{y} + \frac{y}{x}$ then find $x \cdot \frac{\partial z}{\partial x} + y \cdot \frac{\partial z}{\partial y}$

```
syms x y

% Define function

z = x/y + y/x;

% Partial derivatives

dz_dx = diff(z, x);
```

Figure G000183. UNIT 1 · source slide page 27.

EV0000641

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
dz_dy = diff(z, y);

% Required expression

result = x*dz_dx + y*dz_dy;

% Simplify result

result = simplify(result);

% Display

disp('Value of x*dz/dx + y*dz/dy = ‘);
disp(result);
```

Figure G000184. UNIT 1 · source slide page 28.

EV0000644

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

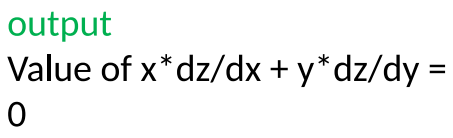

Figure G000185. UNIT 1 · source slide page 29.

EV0000647

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

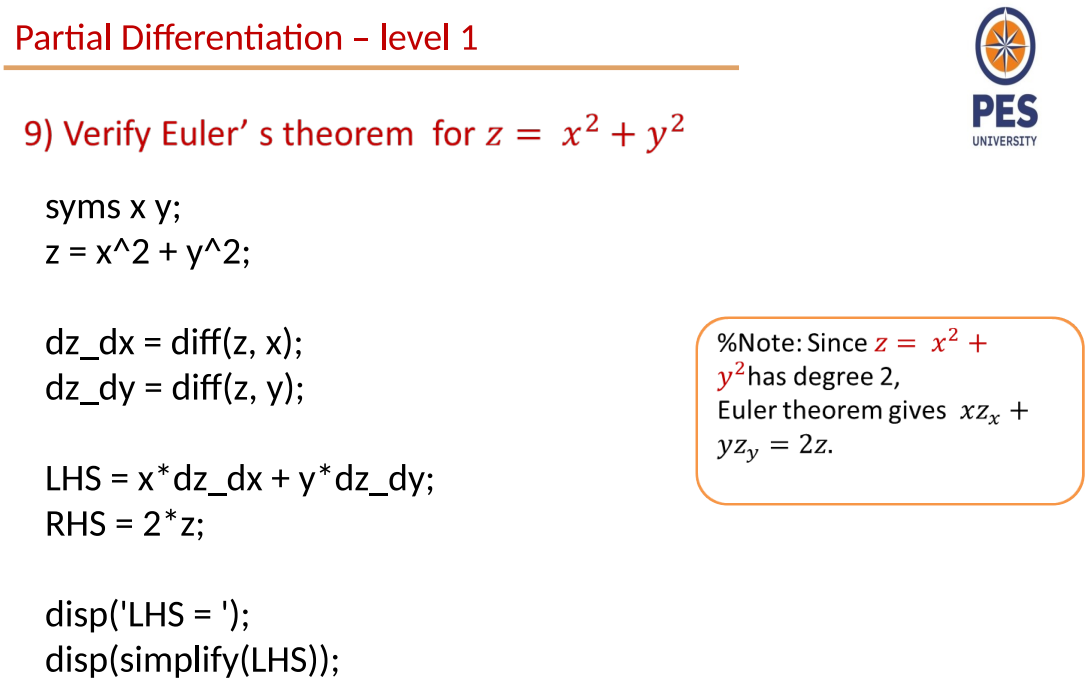

Figure G000186. UNIT 1 · source slide page 30.

EV0000650

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

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```
disp('RHS = ');
disp(simplify(RHS));
OUTPUT
LHS =
2*x^2 + 2*y^2

RHS =
2*x^2 + 2*y^2
```

Figure G000187. UNIT 1 · source slide page 31.

EV0000653

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

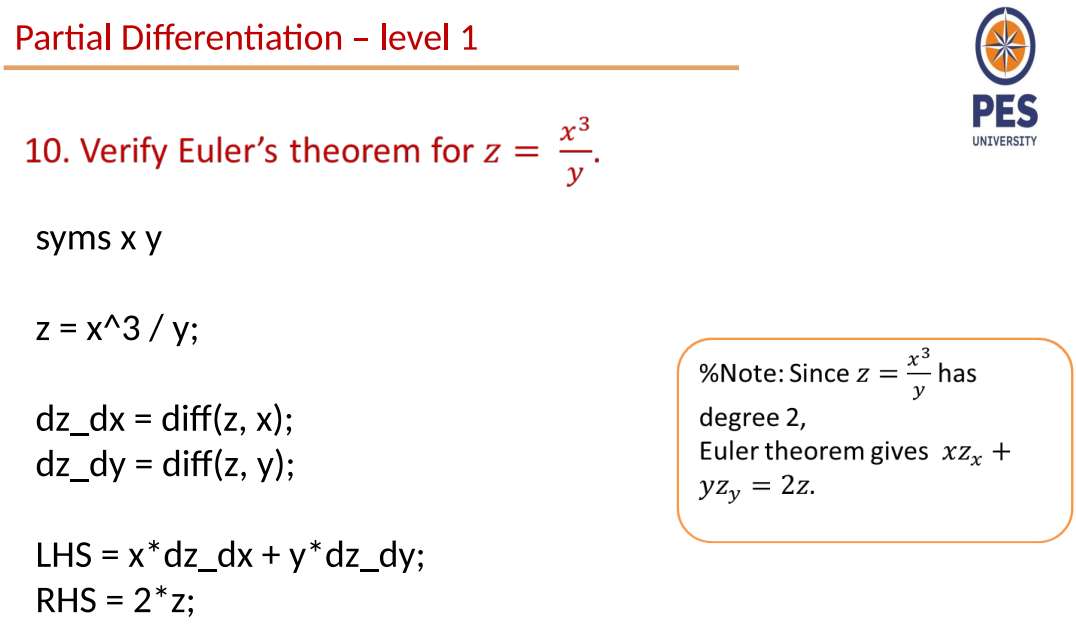

Figure G000188. UNIT 1 · source slide page 32.

EV0000656

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

PES UNIVERSITY

```
disp('LHS = ');
disp(simplify(LHS));

disp('RHS = ');
disp(simplify(RHS));
```

OUTPUT

LHS =2*x^3/y

RHS =2*x^3/y

Figure G000189. UNIT 1 · source slide page 33.

EV0000659

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

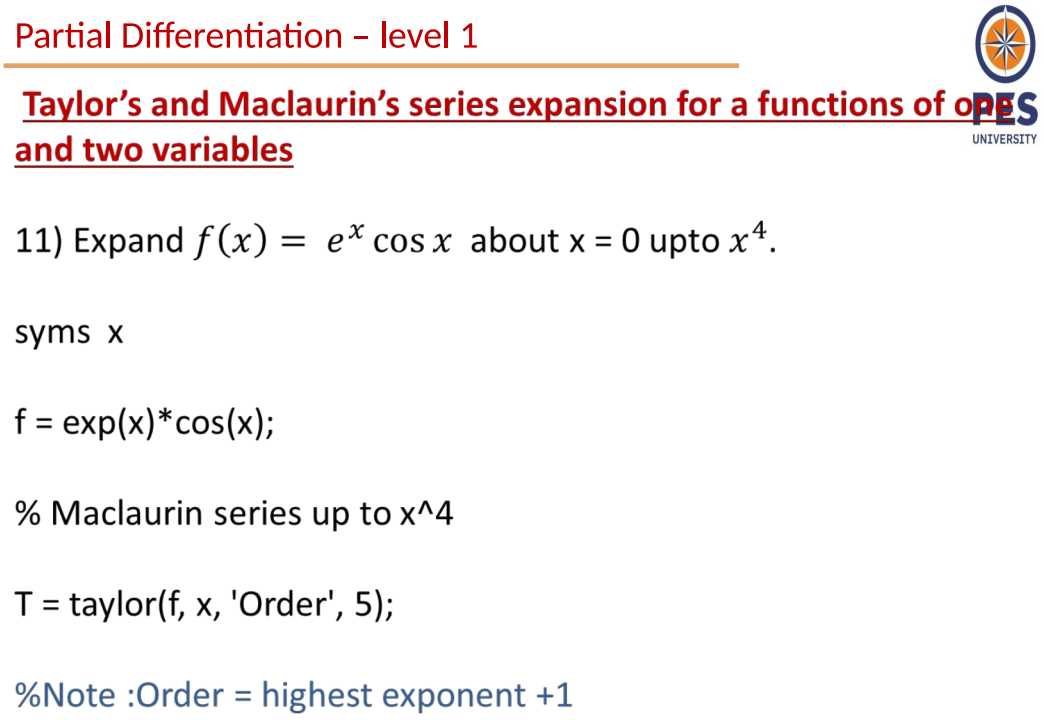

Figure G000190. UNIT 1 · source slide page 34.

EV0000662

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

```
disp('Maclaurin Series:');

disp(T);
```

OUTPUT

```
Maclaurin Series:
1 + x - x^3/3
```

Figure G000191. UNIT 1 · source slide page 35.

EV0000665

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

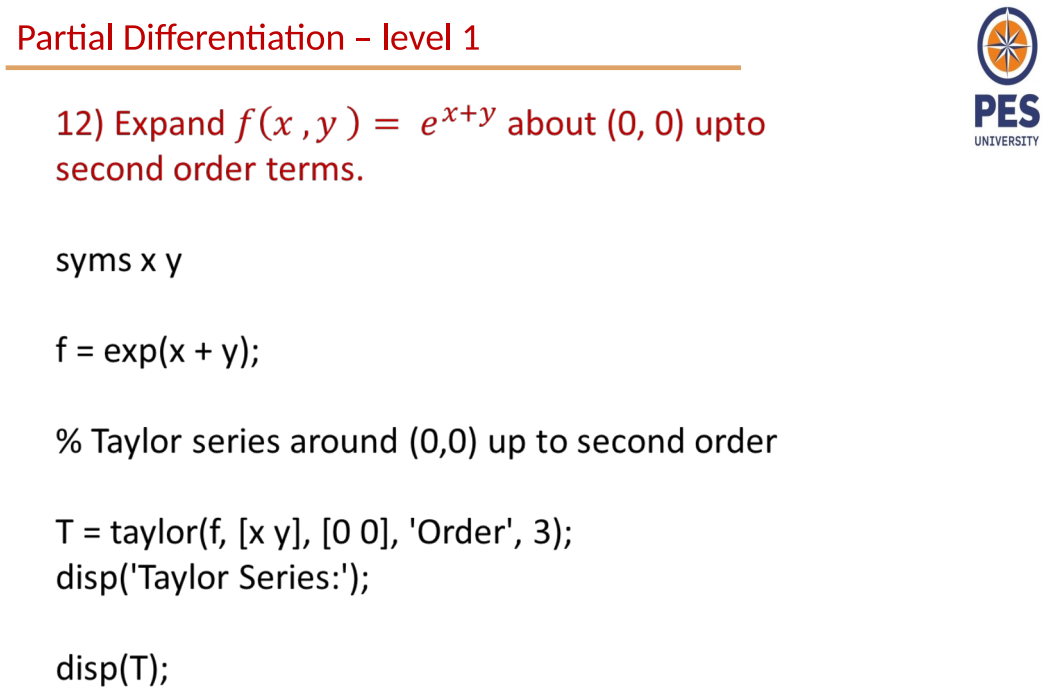

Figure G000192. UNIT 1 · source slide page 36.

EV0000668

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Differentiation – level 1

OUTPUT

Taylor Series:

1 + x + y + x^2/2 + y^2/2 + x*y

Figure G000193. UNIT 1 · source slide page 37.

EV0000671

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000194. UNIT 1 · source slide page 38.

EV0000674

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000562 · EV0000563 · EV0000564
2. Source slide page 2 (extracted text): Partial Differentiation – level 1 EV0000565 · EV0000566 · EV0000567
3. Source slide page 3 (extracted text): Partial Differentiation – level 1 % Display results disp('Partial derivative with respect to x:') disp(dz_dx) disp('Partial derivative with respect to y:') disp(dz_dy) OUTPUT Partial derivative with respect to x: 2*x + 3*y Partial derivative with respect to y: 2*y + 3*x EV0000568 · EV0000569 · EV0000570
4. Source slide page 4 (extracted text): Partial Differentiation – level 1 EV0000571 · EV0000572 · EV0000573
5. Source slide page 5 (extracted text): Partial Differentiation – level 1 disp(dz1_dx) disp('Partial derivative w.r.t y:') disp(dz1_dy) OUTPUT: For z1= log(x^2 + y^2): Partial derivative w.r.t x: (2*x)/(x^2 + y^2) Partial derivative w.r.t y: (2*y)/(x^2 + y^2) EV0000574 · EV0000575 · EV0000576
6. Source slide page 6 (extracted text): Partial Differentiation – level 1 EV0000577 · EV0000578 · EV0000579

7. Source slide page 7 (extracted text): Partial Differentiation – level 1 disp('For z2= exp(x*y):') disp('Partial derivative w.r.t x:') disp(dz2_dx) disp('Partial derivative w.r.t y:') disp(dz2_dy) EV0000580 · EV0000581 · EV0000582
8. Source slide page 8 (extracted text): Partial Differentiation – level 1 OUTPUT For z2 = exp(x*y): Partial derivative w.r.t x: y*exp(x*y) Partial derivative w.r.t y: x*exp(x*y) EV0000583 · EV0000584 · EV0000585
9. Source slide page 9 (extracted text): Partial Differentiation – level 1 EV0000586 · EV0000587 · EV0000588
10. Source slide page 10 (extracted text): Partial Differentiation – level 1 dz_dx = diff(z, x); dz_dy = diff(z, y); % Second derivatives d2z_dx2 = diff(z, x, 2); d2z_dy2 = diff(z, y, 2); d2z_dxdy = diff(diff(z, x), y); % Display results EV0000589 · EV0000590 · EV0000591
11. Source slide page 11 (extracted text): Partial Differentiation – level 1 disp('Second derivative w.r.t x:'); disp(d2z_dx2); disp('Second derivative w.r.t y:'); disp(d2z_dy2); disp('Mixed derivative:'); disp(d2z_dxdy); EV0000592 · EV0000593 · EV0000594
12. Source slide page 12 (extracted text): Partial Differentiation – level 1 Since disp code appear many times, instead you can use fprintf to display results: fprintf('Second order Partial derivative wrt x =%s\n',char(d2z_dx2)) fprintf('Second order Partial derivative wrt y=%s\n',char(d2z_dy2)) fprintf('Mixed Partial derivative =%s\n',char(d2z_dxdy)) EV0000595 · EV0000596 · EV0000597
13. Source slide page 13 (extracted text): Partial Differentiation – level 1 output Second derivative w.r.t x: 6*x*y Second derivative w.r.t y: 6*x*y Mixed derivative: 3*x^2 + 3*y^2 EV0000598 · EV0000599 · EV0000600
14. Source slide page 14 (extracted text): Partial Differentiation – level 1 EV0000601 · EV0000602 · EV0000603
15. Source slide page 15 (extracted text): Partial Differentiation – level 1 % Mixed derivatives d2z_dxdy = diff(dz_dx, y); d2z_dydx = diff(dz_dy, x); % Display results disp('d2z/dxdy = '); disp(d2z_dxdy); disp('d2z/dydx = '); disp(d2z_dydx); EV0000604 · EV0000605 · EV0000606
16. Source slide page 16 (extracted text): Partial Differentiation – level 1 output: d2z/dxdy = 4*x*y d2z/dydx = 4*x*y EV0000607 · EV0000608 · EV0000609
17. Source slide page 17 (extracted text): Partial Differentiation – level 1 EV0000610 · EV0000611 · EV0000612
18. Source slide page 18 (extracted text): Partial Differentiation – level 1 disp(dz_dx) disp(dz_dy) disp(d2z_dx2) disp(d2z_dy2) disp(d2z_dxdy) OUTPUT: y*cos(x*y) x*cos(x*y) -y^2*sin(x*y) -x^2*sin(x*y) cos(x*y) - x*y*sin(x*y) EV0000613 · EV0000614 · EV0000615
19. Source slide page 19 (extracted text): Partial Differentiation – level 1 EV0000616 · EV0000617 · EV0000618
20. Source slide page 20 (extracted text): Partial Differentiation – level 1 % Composite function z = exp(x*y); % Partial derivatives dz_dr = simplify(diff(z,r)); dz_dtheta = simplify(diff(z,theta)); disp('dz/dr =') disp(dz_dr) disp('dz/dtheta =') disp(dz_dtheta) EV0000619 · EV0000620 · EV0000621
21. Source slide page 21 (extracted text): Partial Differentiation – level 1 [Image text, OCR] dz/dr= 2 sin(20) rsin(20)e 2 dz/dtheta: r2 sin(20) r² cos(20)e 2 OUTPUT: EV0000622 · EV0000623 · EV0000624
22. Source slide page 22 (extracted text): Partial Differentiation – level 1 syms x y r theta % Original function z = x^2 + y^2; % Partial derivatives dz_dx = diff(z,x); dz_dy = diff(z,y); EV0000625 · EV0000626 · EV0000627

23. Source slide page 23 (extracted text): Partial Differentiation – level 1 % Composite variables xr = r*cos(theta); yr = r*sin(theta); dx_dr = diff(xr,r); dy_dr = diff(yr,r); % Chain Rule Chain = subs(dz_dx,[x y],[xr yr])*dx_dr + ... subs(dz_dy,[x y],[xr yr])*dy_dr; % Direct differentiation Direct = diff(subs(z,[x y],[xr yr]),r); EV0000628 · EV0000629 · EV0000630
24. Source slide page 24 (extracted text): Partial Differentiation – level 1 disp('Chain Rule Result') disp(simplify(Chain)) disp('Direct Differentiation') disp(simplify(Direct)) OUTPUT: Chain Rule Result 2*r Direct Differentiation 2*r EV0000631 · EV0000632 · EV0000633
25. Source slide page 25 (extracted text): Partial Differentiation – level 1 EV0000634 · EV0000635 · EV0000636
26. Source slide page 26 (extracted text): Partial Differentiation – level 1 Output : [Image text, OCR] ans y(2x+y-e*y) X (x+2y-e*y) EV0000637 · EV0000638 · EV0000639
27. Source slide page 27 (extracted text): Partial Differentiation – level 1 EV0000640 · EV0000641 · EV0000642
28. Source slide page 28 (extracted text): Partial Differentiation – level 1 dz_dy = diff(z, y); % Required expression result = x*dz_dx + y*dz_dy; % Simplify result result = simplify(result); % Display disp('Value of x*dz/dx + y*dz/dy = '); disp(result); EV0000643 · EV0000644 · EV0000645
29. Source slide page 29 (extracted text): Partial Differentiation – level 1 output Value of x*dz/dx + y*dz/dy = 0 EV0000646 · EV0000647 · EV0000648
30. Source slide page 30 (extracted text): Partial Differentiation – level 1 syms x y; z = x^2 + y^2; dz_dx = diff(z, x); dz_dy = diff(z, y); LHS = x*dz_dx + y*dz_dy; RHS = 2*z; disp('LHS = '); disp(simplify(LHS)); EV0000649 · EV0000650 · EV0000651
31. Source slide page 31 (extracted text): Partial Differentiation – level 1 disp('RHS = '); disp(simplify(RHS)); OUTPUT LHS = 2*x^2 + 2*y^2 RHS = 2*x^2 + 2*y^2 EV0000652 · EV0000653 · EV0000654
32. Source slide page 32 (extracted text): Partial Differentiation – level 1 syms x y z = x^3 / y; dz_dx = diff(z, x); dz_dy = diff(z, y); LHS = x*dz_dx + y*dz_dy; RHS = 2*z; EV0000655 · EV0000656 · EV0000657
33. Source slide page 33 (extracted text): Partial Differentiation – level 1 disp('LHS = '); disp(simplify(LHS)); disp('RHS = '); disp(simplify(RHS)); OUTPUT LHS =2*x^3/y RHS =2*x^3/y EV0000658 · EV0000659 · EV0000660
34. Source slide page 34 (extracted text): Partial Differentiation – level 1 EV0000661 · EV0000662 · EV0000663
35. Source slide page 35 (extracted text): Partial Differentiation – level 1 disp('Maclaurin Series:'); disp(T); OUTPUT Maclaurin Series: 1 + x - x^3/3 EV0000664 · EV0000665 · EV0000666
36. Source slide page 36 (extracted text): Partial Differentiation – level 1 EV0000667 · EV0000668 · EV0000669
37. Source slide page 37 (extracted text): Partial Differentiation – level 1 OUTPUT Taylor Series: 1 + x + y + x^2/2 + y^2/2 + x*y EV0000670 · EV0000671 · EV0000672
38. Source slide page 38 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU EV0000673 · EV0000674 · EV0000675

SHORT-NOTE RECAP

Source deck preserved: UNIT 1. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000562

10 Source slide deck: UNIT 1 CONTINUED

This entry preserves the complete slide deck titled UNIT 1 CONTINUED as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000676 · EV0000677

Graphs and diagrams

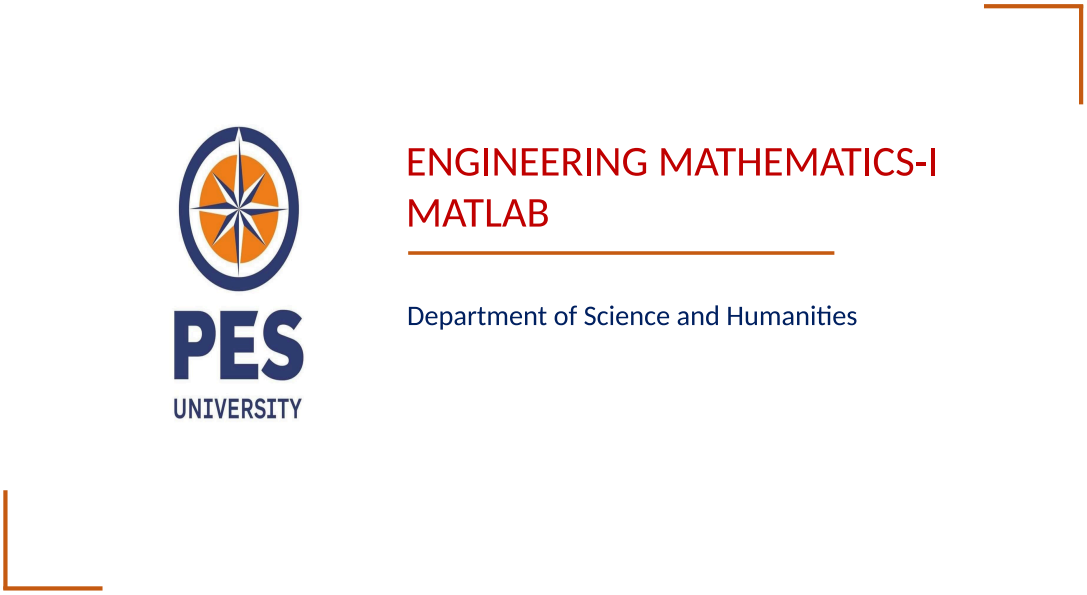

Figure G000195. UNIT 1 CONTINUED · source slide page 1.

EV0000677

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

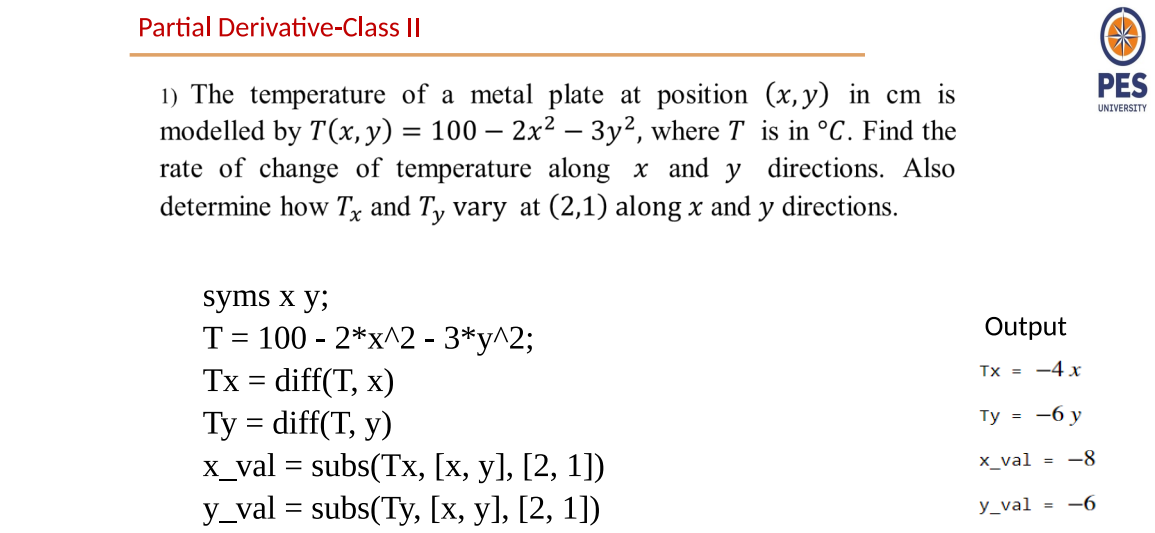

Figure G000196. UNIT 1 CONTINUED · source slide page 2.

EV0000680

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

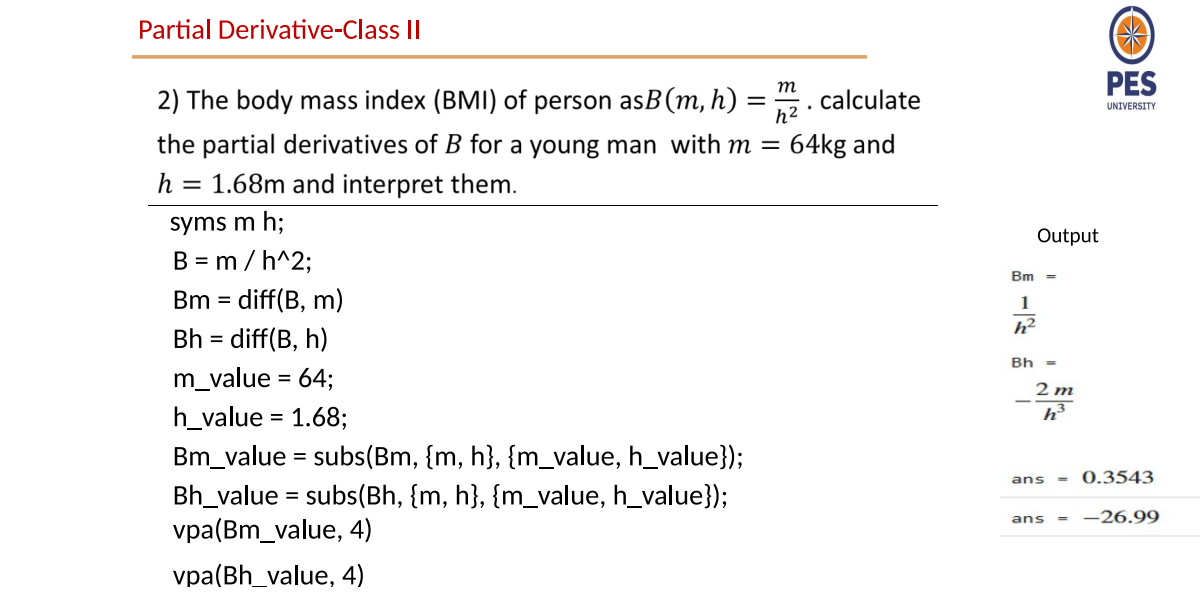

Figure G000197. UNIT 1 CONTINUED · source slide page 3.

EV0000683

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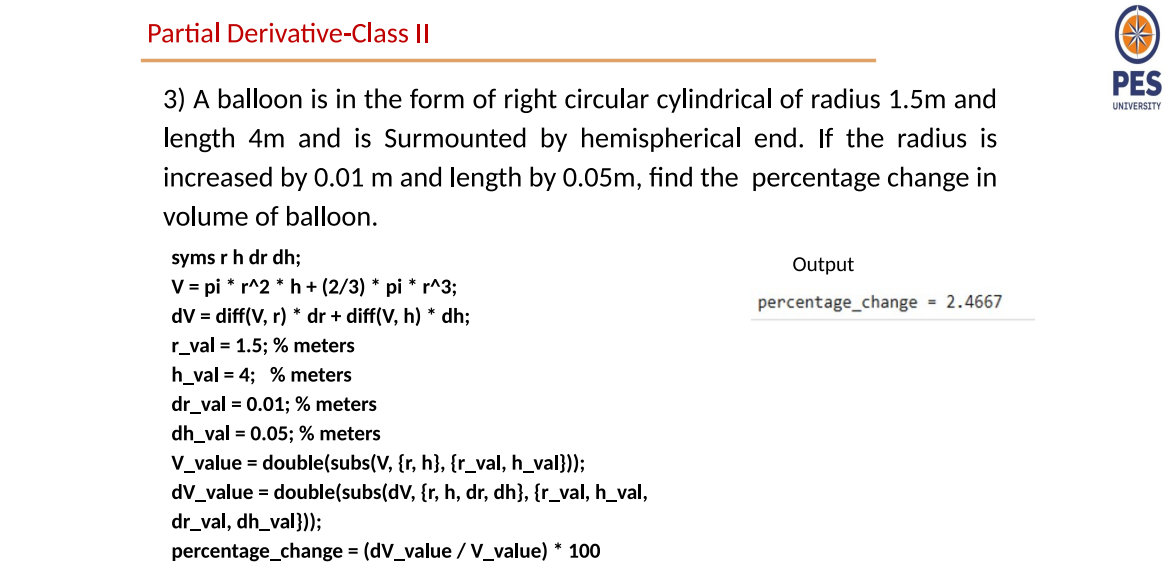

Figure G000198. UNIT 1 CONTINUED · source slide page 4.

EV0000686

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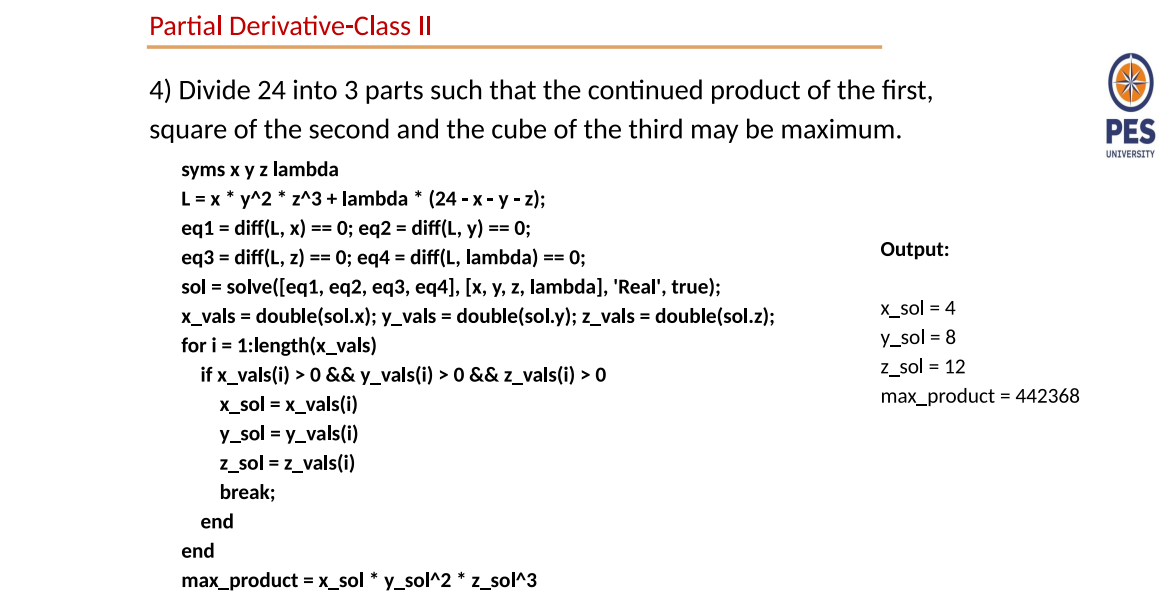

Figure G000199. UNIT 1 CONTINUED · source slide page 5.

EV0000689

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

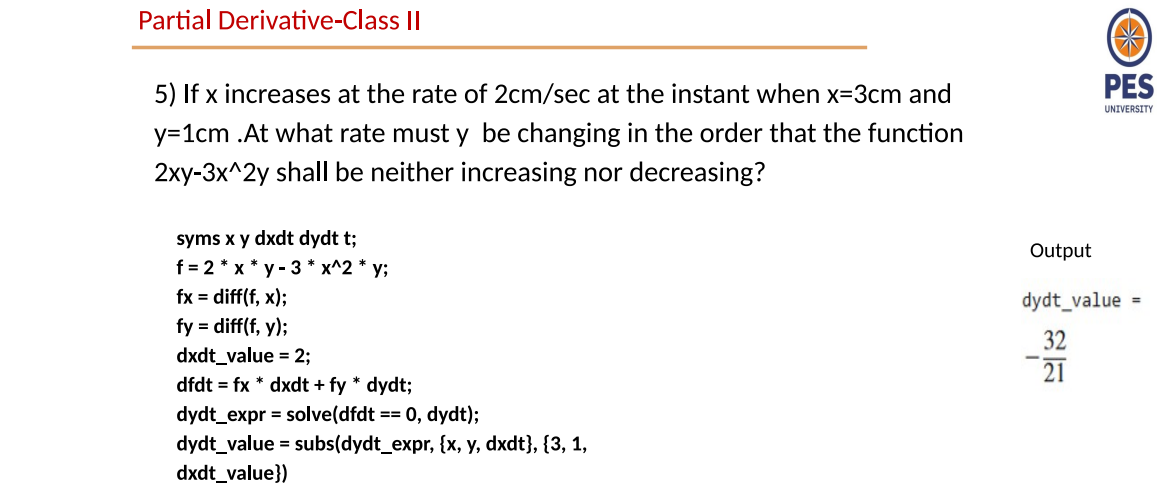

Figure G000200. UNIT 1 CONTINUED · source slide page 6.

EV0000692

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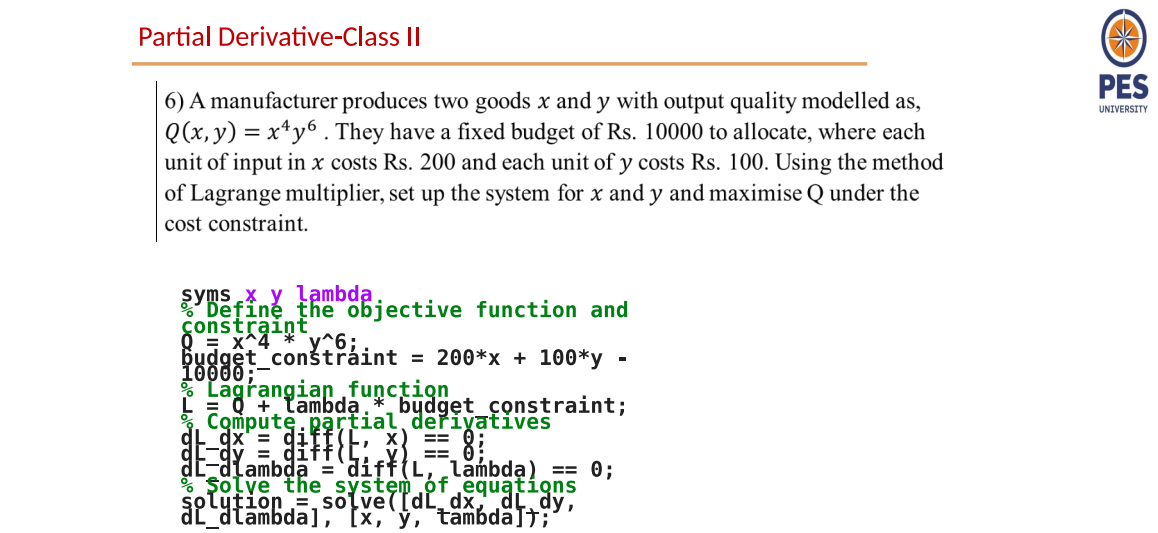

Figure G000201. UNIT 1 CONTINUED · source slide page 7.

EV0000695

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

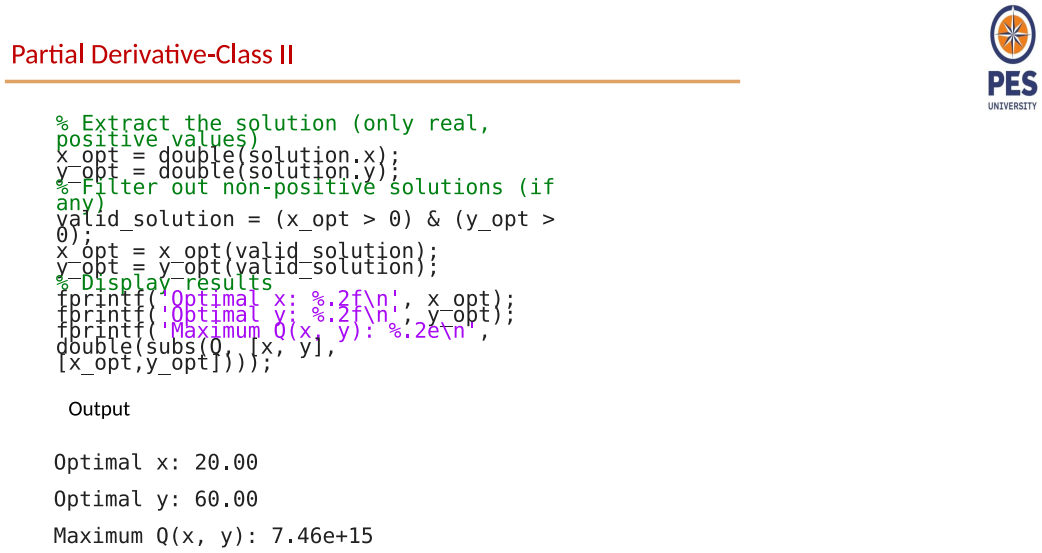

Figure G000202. UNIT 1 CONTINUED · source slide page 8.

EV0000698

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

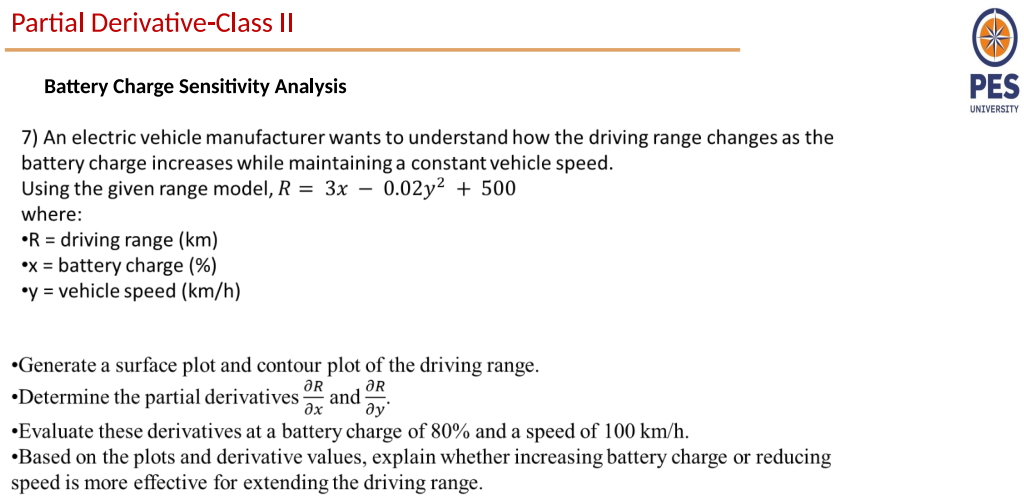

Figure G000203. UNIT 1 CONTINUED · source slide page 9.

EV0000701

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

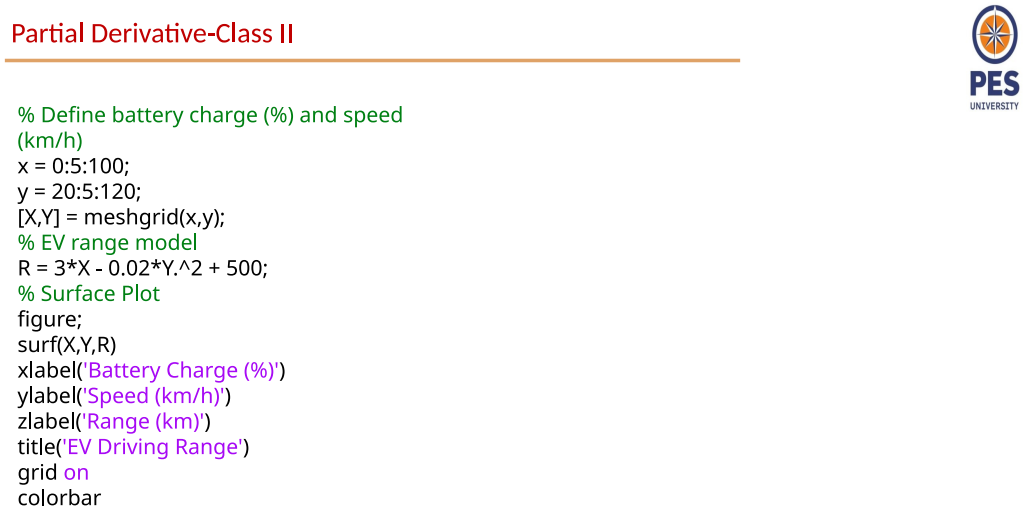

Figure G000204. UNIT 1 CONTINUED · source slide page 10.

EV0000704

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

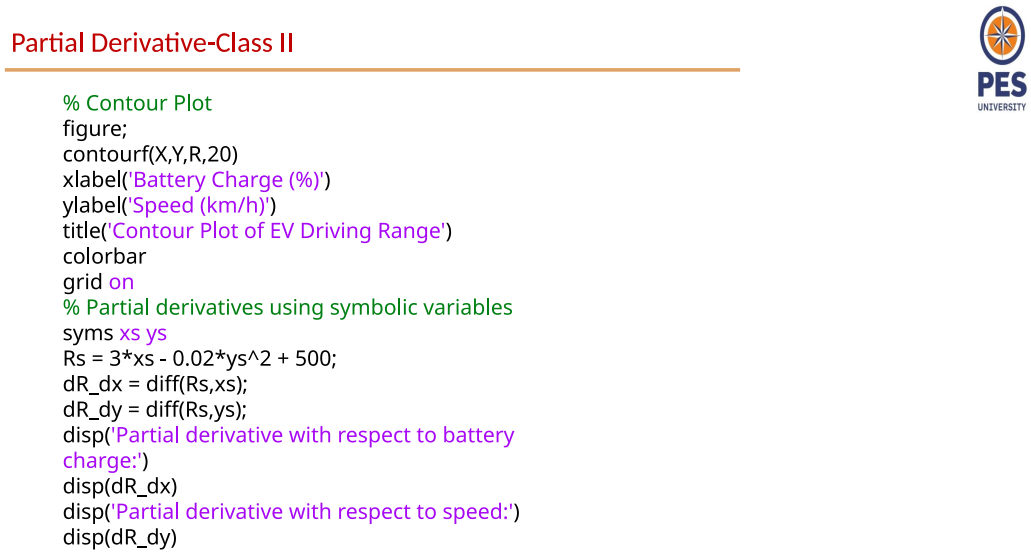

Figure G000205. UNIT 1 CONTINUED · source slide page 11.

EV0000707

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Partial Derivative-Class II

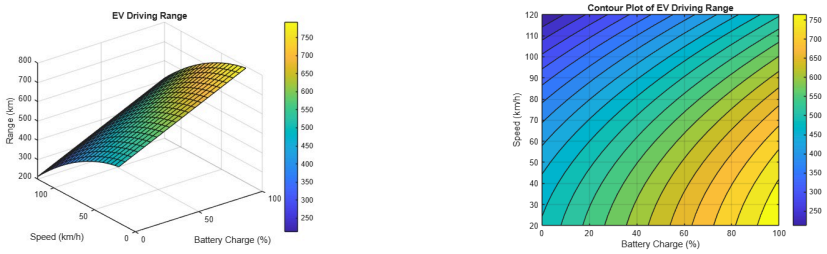
```
% Evaluate at x=80%, y=100 km/h
dR_dx_val = subs(dR_dx,[xs ys],[80 100]);
dR_dy_val = subs(dR_dy,[xs ys],[80 100]);
fprintf('\nAt Battery Charge = 80%% and Speed = 100 km/h:\n');
fprintf('dR/dx = %.2f km per %% battery
charge\n',double(dR_dx_val));
fprintf('dR/dy = %.2f km per km/h speed\n',double(dR_dy_val));
```

Figure G000206. UNIT 1 CONTINUED · source slide page 12.

EV0000710

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Partial Derivative-Class II



Partial derivative with respect to battery charge:

3

Partial derivative with respect to speed:

$$-\frac{ys}{25}$$

At Battery Charge = 80% and Speed = 100 km/h:
dR/dx = 3.00 km per % battery charge
dR/dy = -4.00 km per km/h speed

Figure G000207. UNIT 1 CONTINUED · source slide page 13.

EV0000713

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000208. UNIT 1 CONTINUED · source slide page 14.

EV0000716

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000676 · EV0000677 · EV0000678
2. Source slide page 2 (extracted text): Partial Derivative-Class II syms x y; T = 100 - 2*x^2 - 3*y^2; Tx = diff(T, x) Ty = diff(T, y) x_val = subs(Tx, [x, y], [2, 1]) y_val = subs(Ty, [x, y], [2, 1]) Output EV0000679 · EV0000680 · EV0000681
3. Source slide page 3 (extracted text): Partial Derivative-Class II syms m h; B = m / h^2; Bm = diff(B, m) Bh = diff(B, h) m_value = 64; h_value = 1.68; Bm_value = subs(Bm, {m, h}, {m_value, h_value}); Bh_value = subs(Bh, {m, h}, {m_value, h_value}); vpa(Bm_value, 4) vpa(Bh_value, 4) Output EV0000682 · EV0000683 · EV0000684
4. Source slide page 4 (extracted text): Partial Derivative-Class II 3) A balloon is in the form of right circular cylindrical of radius 1.5m and length 4m and is Surmounted by hemispherical end. If the radius is increased by 0.01 m and length by 0.05m, find the percentage change in volume of balloon. syms r h dr dh; V = pi * r^2 * h + (2/3) * pi * r^3; dV = diff(V, r) * dr + diff(V, h) * dh; r_val = 1.5; % meters h_val = 4; % meters dr_val = 0.01; % meters dh_val = 0.05; % meters V_value = double(subs(V, {r, h}, {r_val, h_val})); dV_value = double(subs(dV, {r, h, dr, dh}, {r_val, h_val, dr_val, dh_val})); percentage_change = (dV_value / V_value) * 100 Output EV0000685 · EV0000686 · EV0000687

5. Source slide page 5 (extracted text): Partial Derivative-Class II 4) Divide 24 into 3 parts such that the continued product of the first, square of the second and the cube of the third may be maximum. syms x y z lambda L = x * y^2 * z^3 + lambda * (24 - x - y - z); eq1 = diff(L, x) == 0; eq2 = diff(L, y) == 0; eq3 = diff(L, z) == 0; eq4 = diff(L, lambda) == 0; sol = solve([eq1, eq2, eq3, eq4], [x, y, z, lambda], 'Real', true); x_vals = double(sol.x); y_vals = double(sol.y); z_vals = double(sol.z); for i = 1:length(x_vals) if x_vals(i) > 0 && y_vals(i) > 0 && z_vals(i) > 0 x_sol = x_vals(i) y_sol = y_vals(i) z_sol = z_vals(i) break; end end max_product = x_sol * y_sol^2 * z_sol^3 Output: x_sol = 4 y_sol = 8 z_sol = 12 max_product = 442368 EV0000688 · EV0000689 · EV0000690

6. Source slide page 6 (extracted text): Partial Derivative-Class II 5) If x increases at the rate of 2cm/sec at the instant when x=3cm and y=1cm .At what rate must y be changing in the order that the function 2xy-3x^2y shall be neither increasing nor decreasing? syms x y dxdt dydt t; f = 2 * x * y - 3 * x^2 * y; fx = diff(f, x); fy = diff(f, y); dxdt_value = 2; dfdt = fx * dxdt + fy * dydt; dydt_expr = solve(dfdt == 0, dydt); dydt_value = subs(dydt_expr, {x, y, dxdt}, {3, 1, dxdt_value}) Output EV0000691 · EV0000692 · EV0000693

7. Source slide page 7 (extracted text): Partial Derivative-Class II syms x y lambda % Define the objective function and constraint Q = x^4 * y^6; budget_constraint = 200*x + 100*y - 10000; % Lagrangian function L = Q + lambda * budget_constraint; % Compute partial derivatives dL_dx = diff(L, x) == 0; dL_dy = diff(L, y) == 0; dL_dlambda = diff(L, lambda) == 0; % Solve the system of equations solution = solve([dL_dx, dL_dy, dL_dlambda], [x, y, lambda]); EV0000694 · EV0000695 · EV0000696

8. Source slide page 8 (extracted text): Partial Derivative-Class II % Extract the solution (only real, positive values) x_opt = double(solution.x); y_opt = double(solution.y); % Filter out non-positive solutions (if any) valid_solution = (x_opt > 0) & (y_opt > 0); x_opt = x_opt(valid_solution); y_opt = y_opt(valid_solution); % Display results fprintf('Optimal x: %.2f\n', x_opt); fprintf('Optimal y: %.2f\n', y_opt); fprintf('Maximum Q(x, y): %.2e\n', double(subs(Q, [x, y], [x_opt,y_opt]))); Output Optimal x: 20.00 Optimal y: 60.00 Maximum Q(x, y): 7.46e+15 EV0000697 · EV0000698 · EV0000699

9. Source slide page 9 (extracted text): Partial Derivative-Class II Battery Charge Sensitivity Analysis EV0000700 · EV0000701 · EV0000702

10. Source slide page 10 (extracted text): Partial Derivative-Class II % Define battery charge (%) and speed (km/h) x = 0:5:100; y = 20:5:120; [X,Y] = meshgrid(x,y); % EV range model R = 3*X - 0.02*Y.^2 + 500; % Surface Plot figure; surf(X,Y,R) xlabel('Battery Charge (%)') ylabel('Speed (km/h)') zlabel('Range (km)') title('EV Driving Range') grid on colorbar EV0000703 · EV0000704 · EV0000705

11. Source slide page 11 (extracted text): Partial Derivative-Class II % Contour Plot figure; contourf(X,Y,R,20) xlabel('Battery Charge (%)') ylabel('Speed (km/h)') title('Contour Plot of EV Driving Range') colorbar grid on % Partial derivatives using symbolic variables syms xs ys Rs = 3*xs - 0.02*ys^2 + 500; dR_dx = diff(Rs,xs); dR_dy = diff(Rs,ys); disp('Partial derivative with respect to battery charge:') disp(dR_dx) disp('Partial derivative with respect to speed:') disp(dR_dy) EV0000706 · EV0000707 · EV0000708

12. Source slide page 12 (extracted text): Partial Derivative-Class II % Evaluate at x=80%, y=100 km/h dR_dx_val = subs(dR_dx,[xs ys],[80 100]); dR_dy_val = subs(dR_dy,[xs ys],[80 100]); fprintf('\nAt Battery Charge = 80%% and Speed = 100 km/h:\n'); fprintf('dR/dx = %.2f km per %% battery charge\n',double(dR_dx_val)); fprintf('dR/dy = %.2f km per km/h speed\n',double(dR_dy_val)); EV0000709 · EV0000710 · EV0000711
13. Source slide page 13 (extracted text): Partial Derivative-Class II [Image text, OCR] ContourPlotofEVDrivingRange 120 750 110 700 100 650 90 600 80 550 70 500 pads 450 60 400 50 350 40 300 30 250 20 0 20 40 60 80 100 BatteryCharge(%) [Image text, OCR] EV DrivingRange 750 800 700 700 650 600 (y) 600 550 ①) 500 500 400 450 300 400 200 350 100 100 300 50 50 250 Speed (km/h) 0 0 Battery Charge (%) [Image text, OCR] Partial derivative with respect to battery charge: 3 Partial derivative with respect to speed： ys 25 At BatteryCharge=80%and Speed=100km/h: dR/dx=3.00 kmper%batterycharge dR/dy=-4.00kmper km/h speed EV0000712 · EV0000713 · EV0000714
14. Source slide page 14 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU EV0000715 · EV0000716 · EV0000717

SHORT-NOTE RECAP

Source deck preserved: UNIT 1 CONTINUED. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000676

11 Source slide deck: UNIT 2

This entry preserves the complete slide deck titled UNIT 2 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000718 · EV0000719

Graphs and diagrams

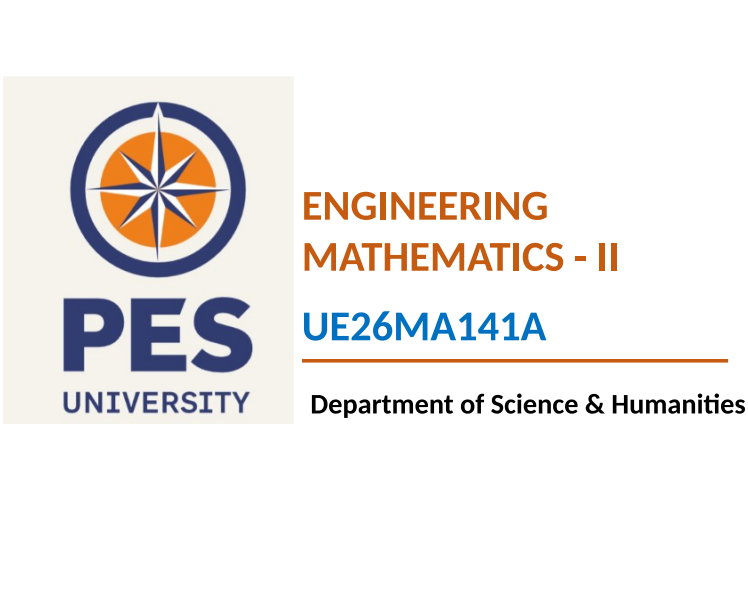


Figure G000209. UNIT 2 · source slide page 1.

EV0000719

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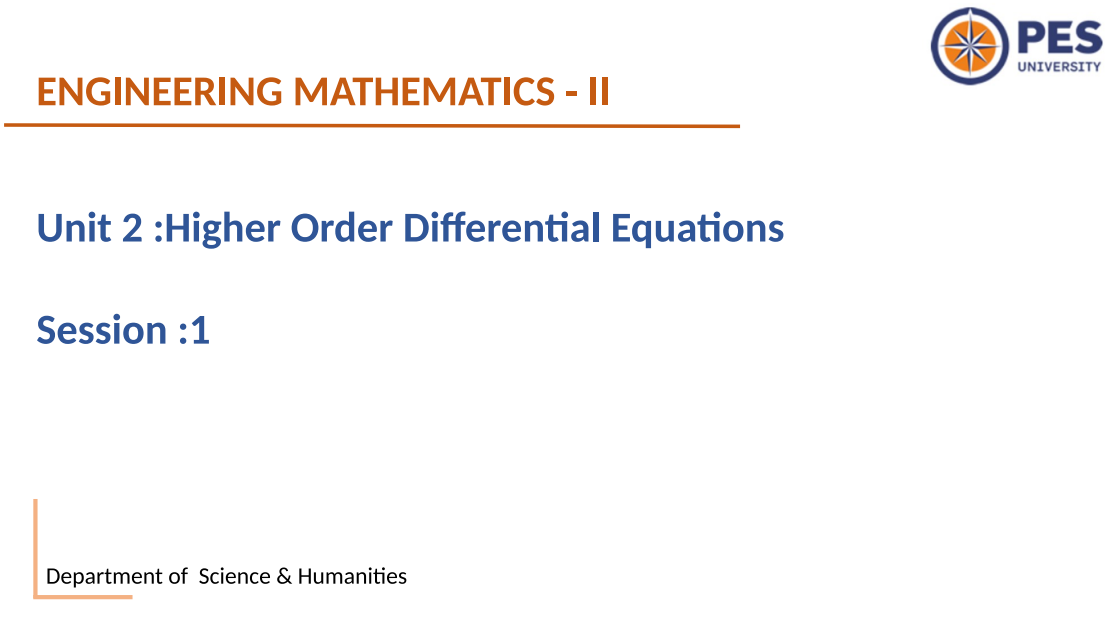

Figure G000210. UNIT 2 · source slide page 2.

EV0000722

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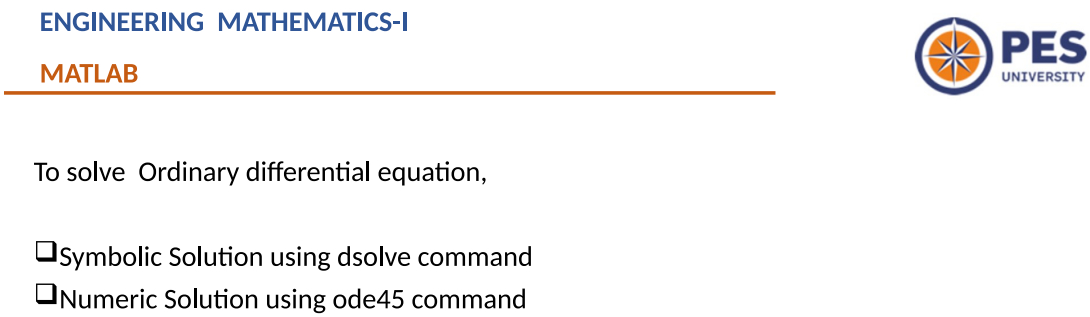

Figure G000211. UNIT 2 · source slide page 3.

EV0000725

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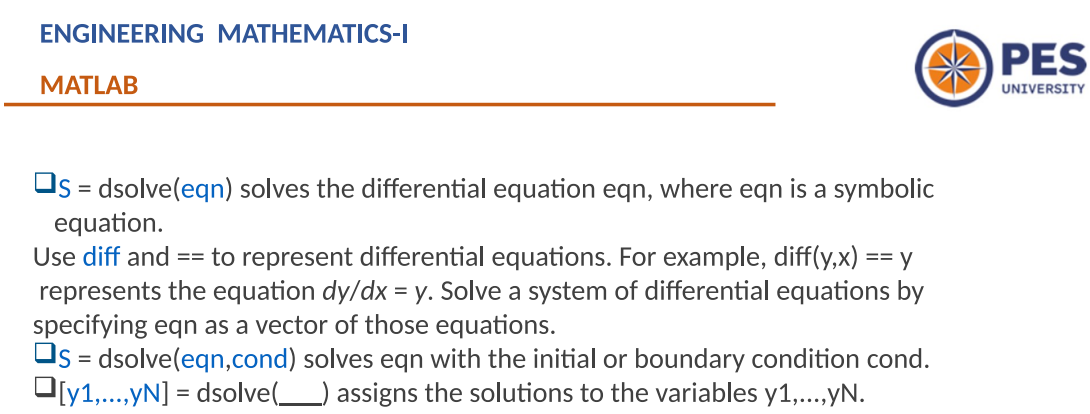

Figure G000212. UNIT 2 · source slide page 4.

EV0000728

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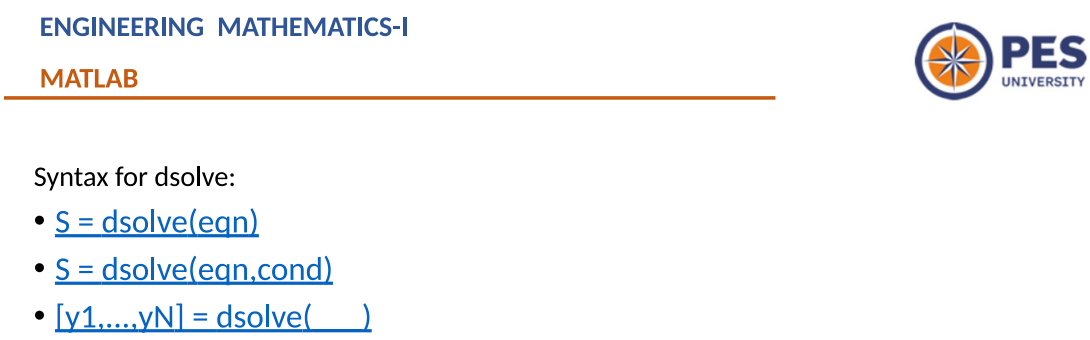

Figure G000213. UNIT 2 · source slide page 5.

EV0000731

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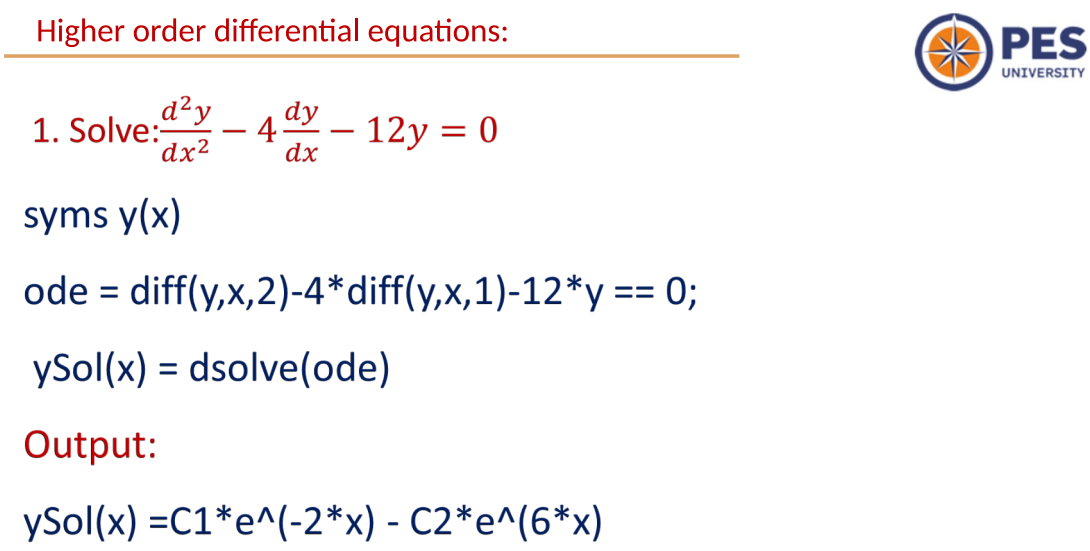

Figure G000214. UNIT 2 · source slide page 6.

EV0000734

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

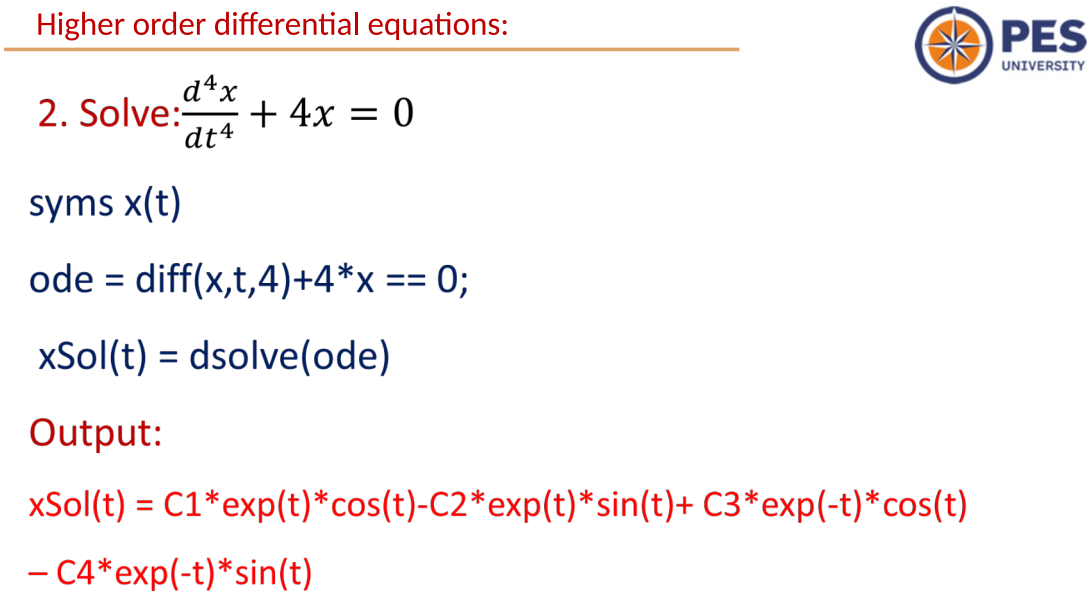

Figure G000215. UNIT 2 · source slide page 7.

EV0000737

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations, Continued...

3. Solve:$4y'' - 8y' + 3y = 0, y(0) = 1, y'(0) = 3$.
Plot its solution.

```
syms y(x)
Dy = diff(y);
D2y = diff(y,2);
ode = 4*D2y - 8*Dy + 3*y == 0;
ySol = dsolve(ode, y(0) == 1, Dy(0) == 3)
ezplot(ySol)
```

Figure G000216. UNIT 2 · source slide page 8.

EV0000740

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations, Continued...

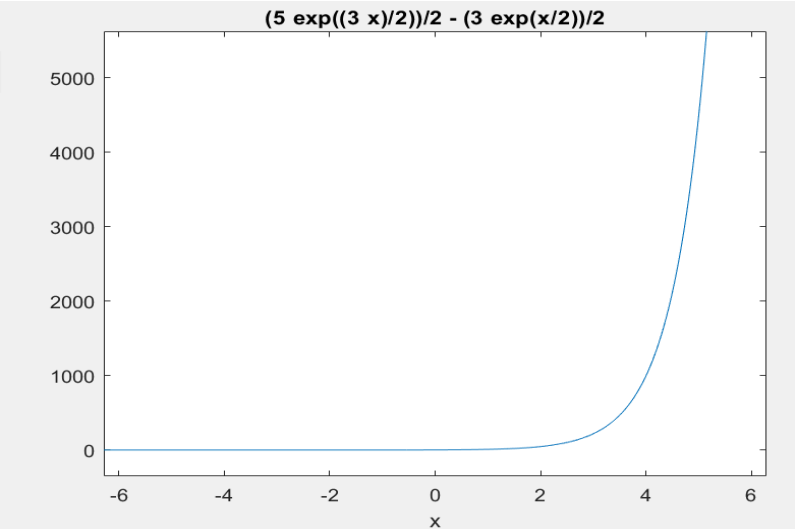

Figure G000217. UNIT 2 · source slide page 9.

EV0000743

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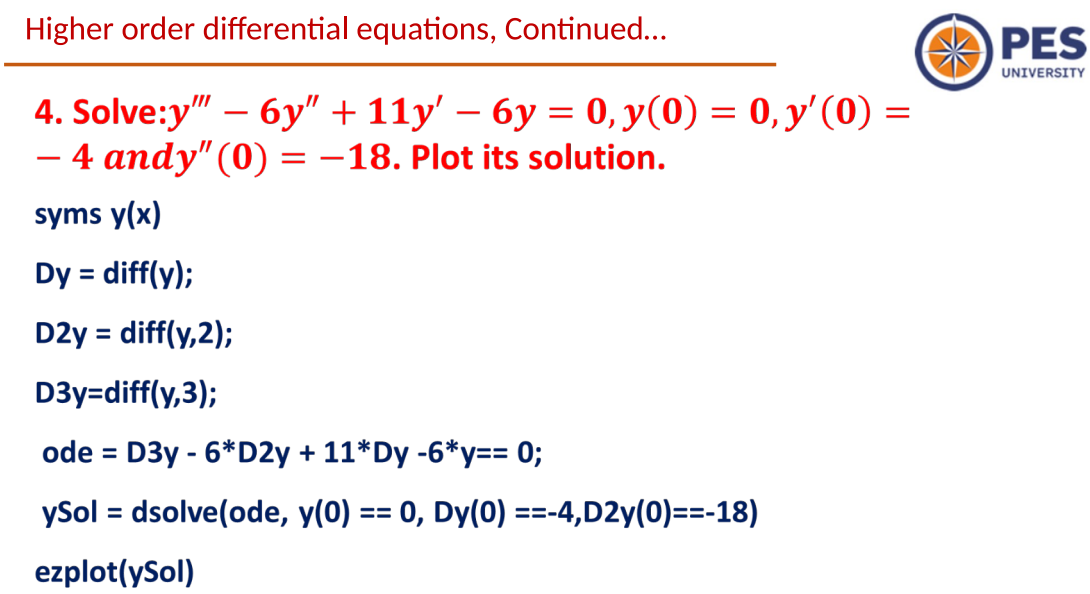

Figure G000218. UNIT 2 · source slide page 10.

EV0000746

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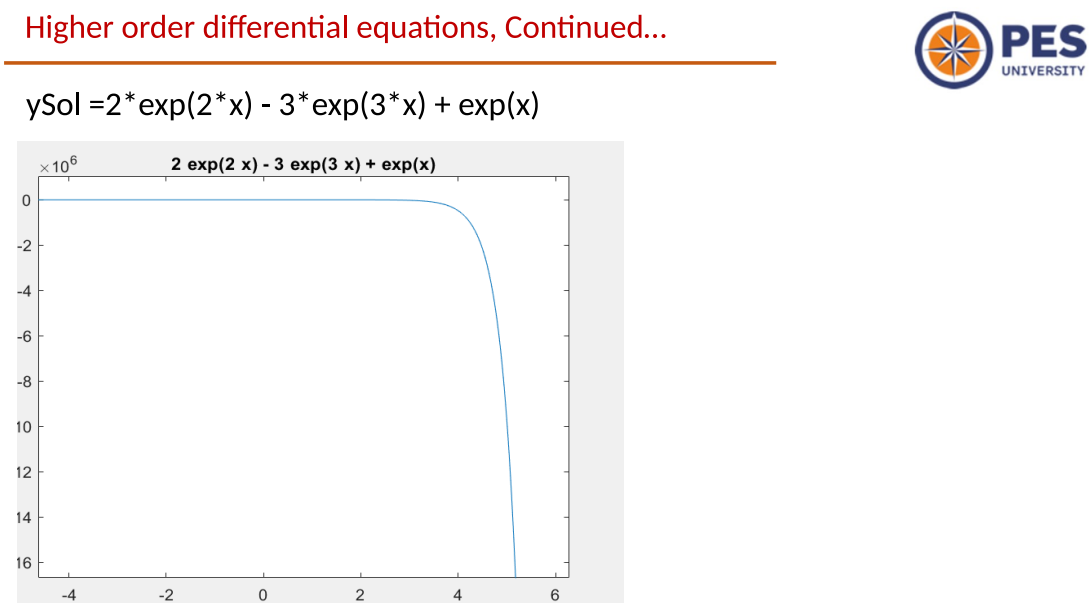

Figure G000219. UNIT 2 · source slide page 11.

EV0000749

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

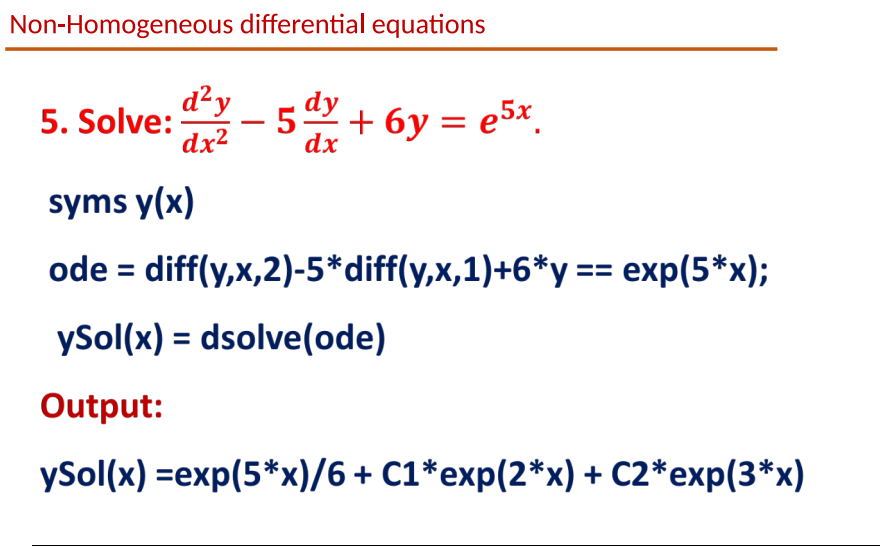

Figure G000220. UNIT 2 · source slide page 12.

EV0000752

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

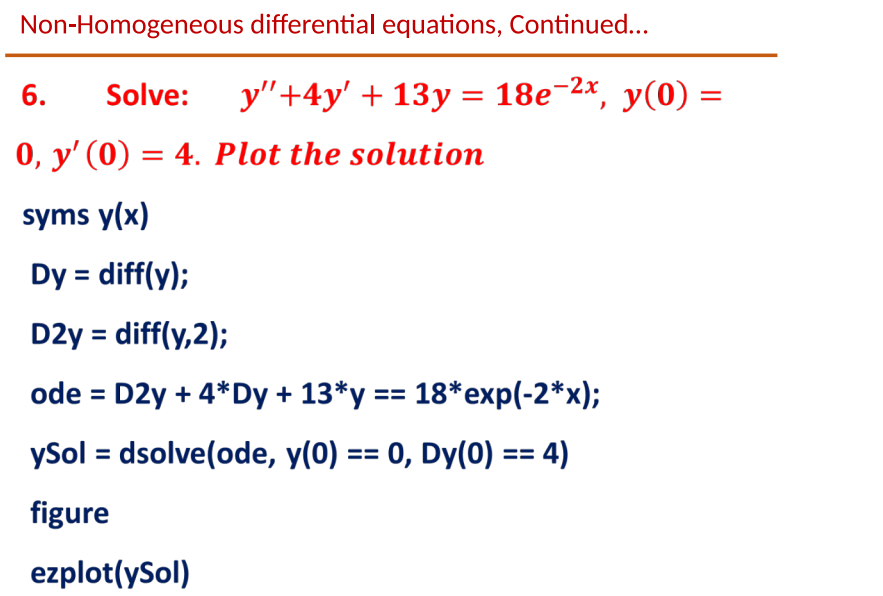

Figure G000221. UNIT 2 · source slide page 13.

EV0000755

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Non-Homogeneous differential equations, Continued...

Output:

```
Sol =exp(-2*x)*(2*sin(3*x) - 9*cos(3*x) + 6))/3
```

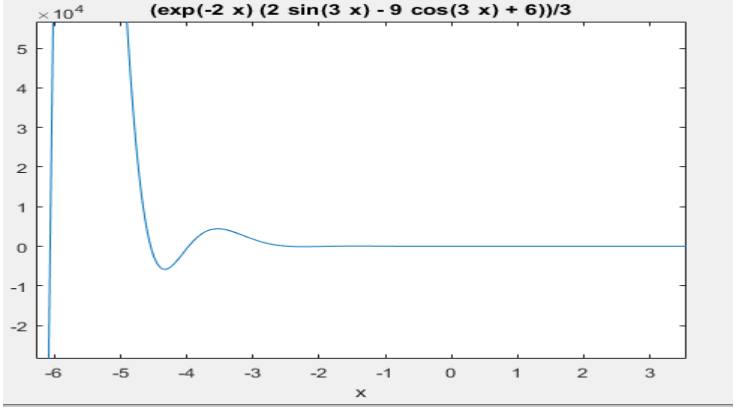

Figure G000222. UNIT 2 · source slide page 14.

EV0000758

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Non-Homogeneous differential equations, Continued...

7. Solve: $(D^3 + D^2 - D - 1)y = cos2x.$

```
>> syms y(x)
>> ode= diff(y,x,3)+diff(y,x,2)-diff(y,x,1)-1*y == cos(2*x);
>> ySol(x) = dsolve(ode)
```

Output:

```
ySol(x) =
C3*exp(x) - (5^(1/2)*cos(2*x - atan(2)))/25 + C1*exp(-x) +
C2*x*exp(-x)
```

Figure G000223. UNIT 2 · source slide page 15.

EV0000761

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

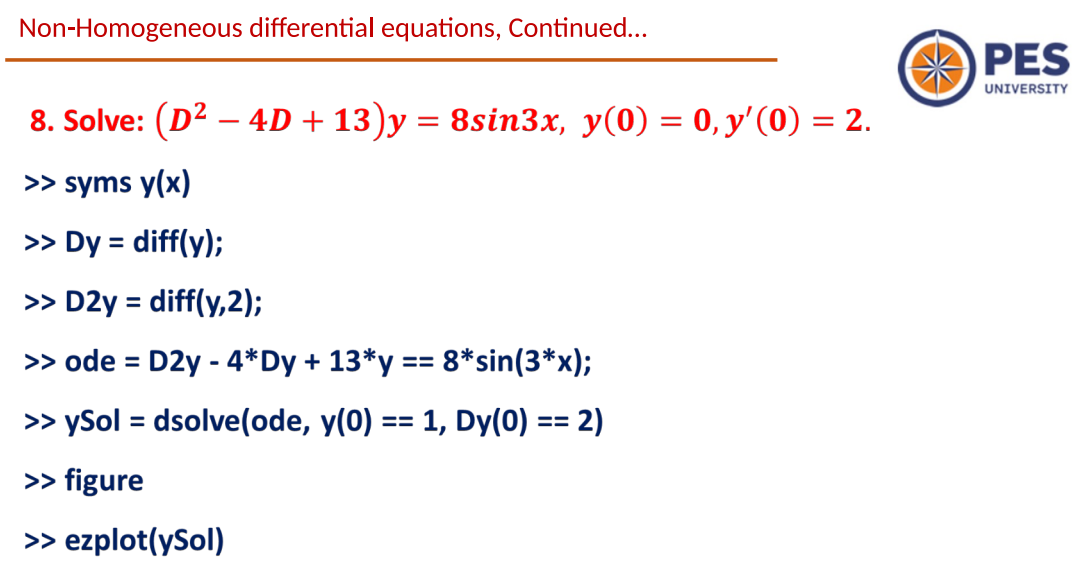

Figure G000224. UNIT 2 · source slide page 16.

EV0000764

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

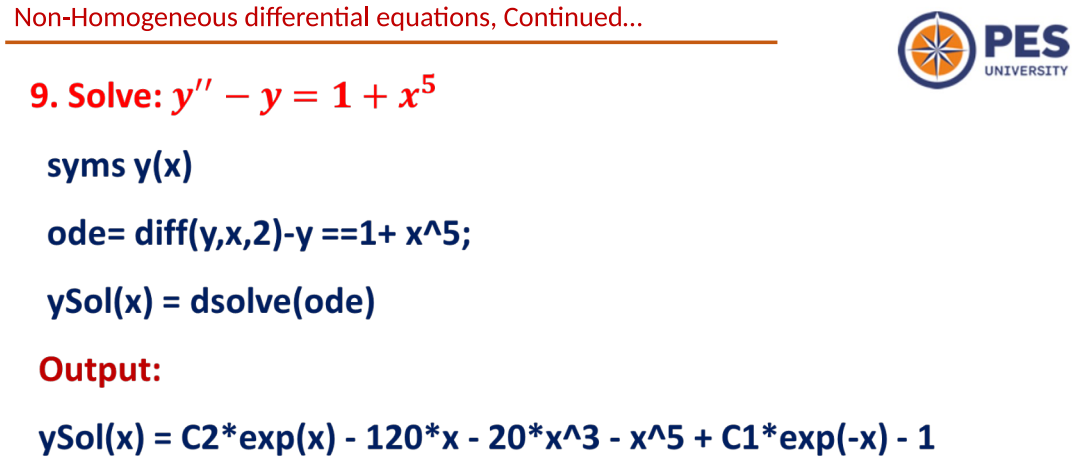

Figure G000225. UNIT 2 · source slide page 17.

EV0000767

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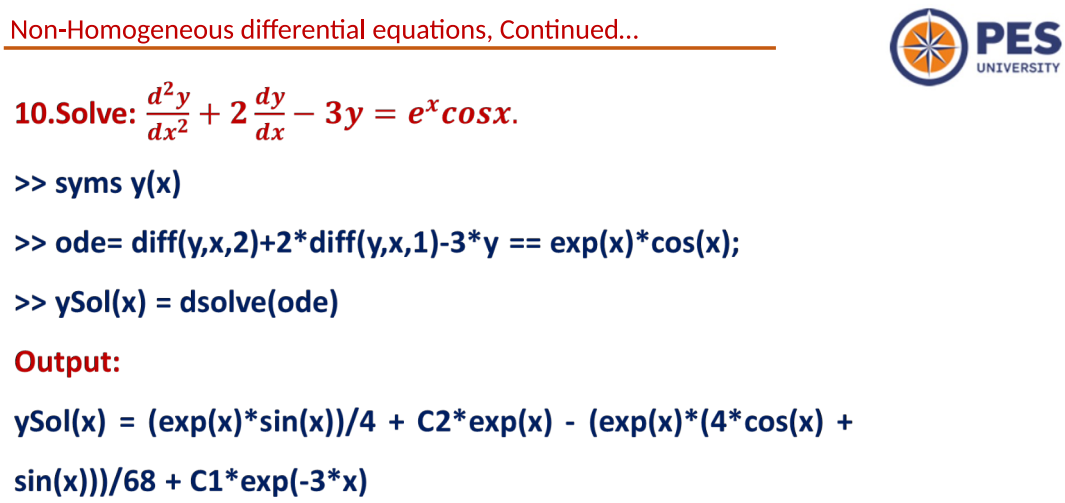

Figure G000226. UNIT 2 · source slide page 18.

EV0000770

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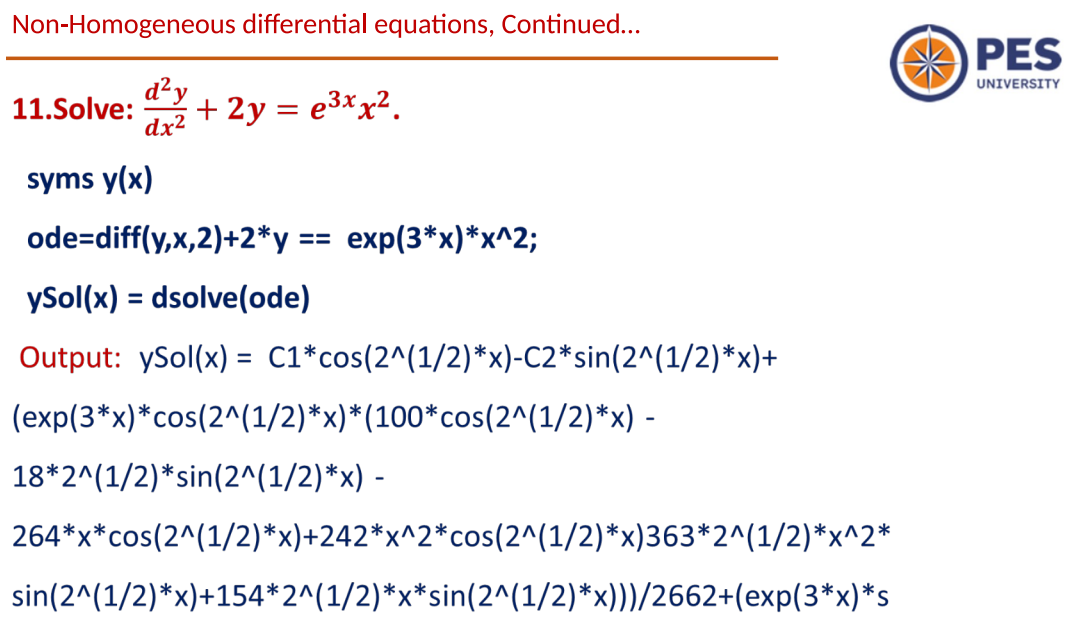

Figure G000227. UNIT 2 · source slide page 19.

EV0000773

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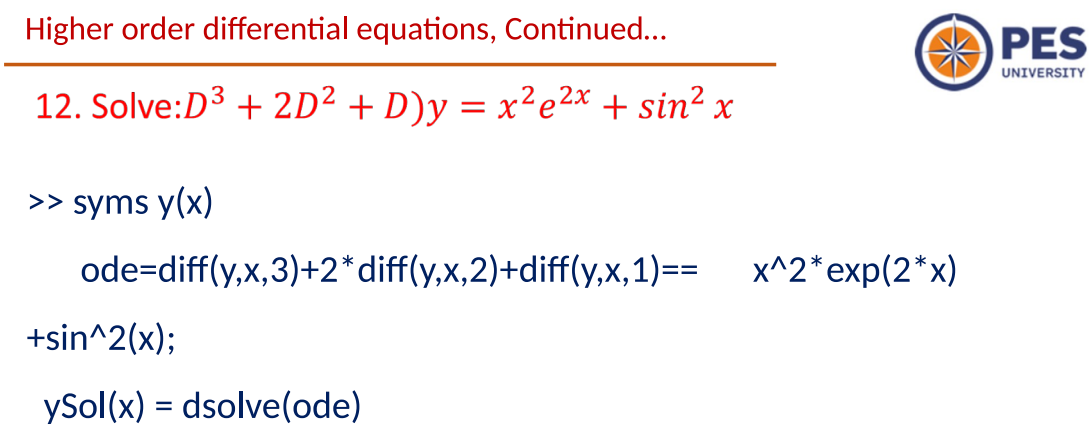

Figure G000228. UNIT 2 · source slide page 20.

EV0000776

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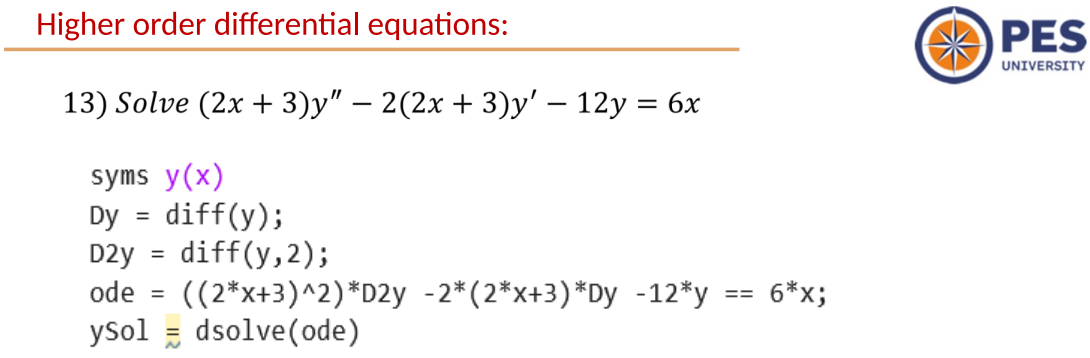

Figure G000229. UNIT 2 · source slide page 21.

EV0000779

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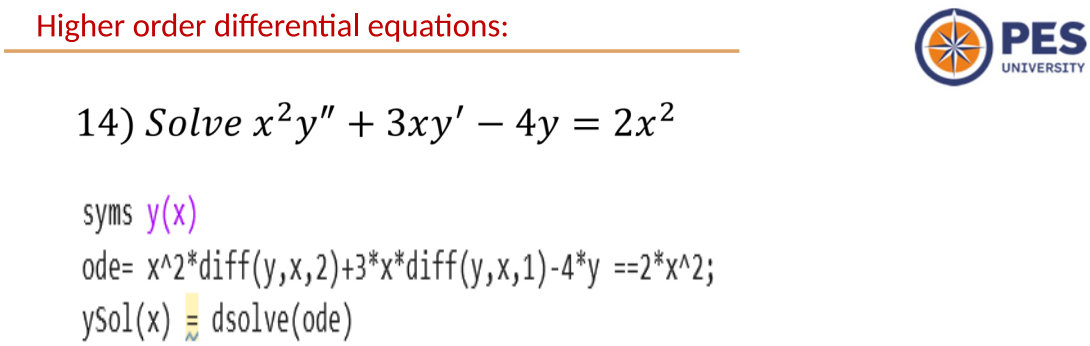

Figure G000230. UNIT 2 · source slide page 22.

EV0000782

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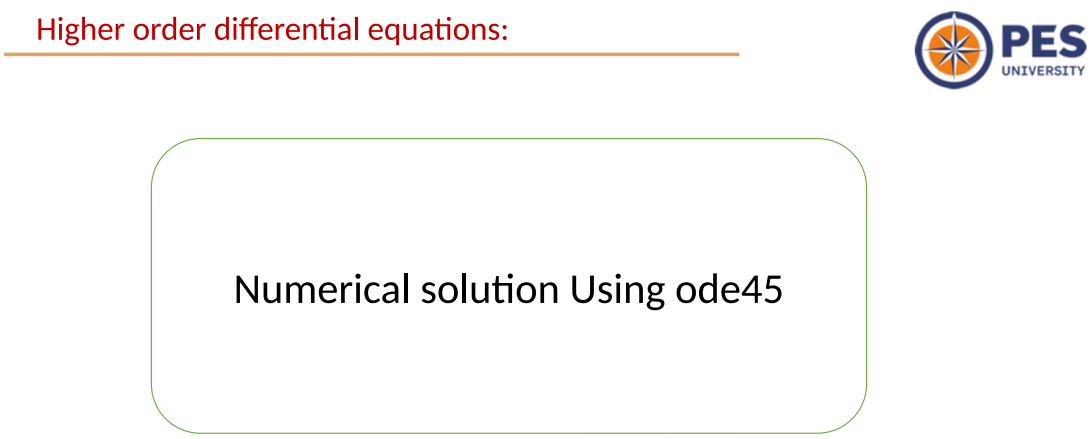

Figure G000231. UNIT 2 · source slide page 23.

EV0000785

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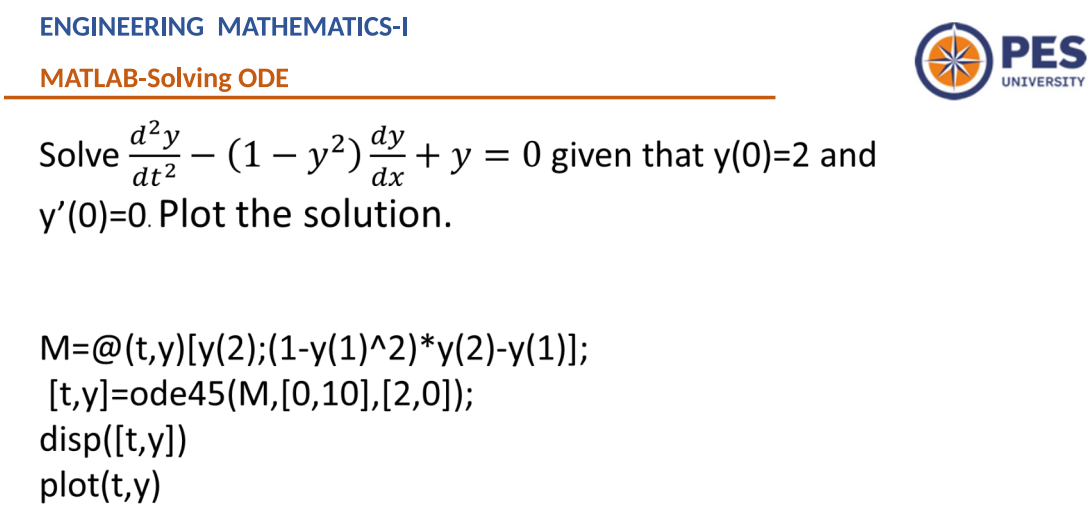

Figure G000232. UNIT 2 · source slide page 24.

EV0000788

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Figure G000233. UNIT 2 · source slide page 25.

EV0000791

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - II UE26MA141A Department of Science & Humanities EV0000718 · EV0000719 · EV0000720
2. Source slide page 2 (extracted text): Engineering MATHEMATICS - II Unit 2 :Higher Order Differential Equations Session :1 Department of Science & Humanities EV0000721 · EV0000722 · EV0000723
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS-I MATLAB To solve Ordinary differential equation, Symbolic Solution using dsolve command Numeric Solution using ode45 command EV0000724 · EV0000725 · EV0000726
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS-I MATLAB S = dsolve(eqn) solves the differential equation eqn, where eqn is a symbolic equation. Use diff and == to represent differential equations. For example, diff(y,x) == y represents the equation dy/dx = y. Solve a system of differential equations by specifying eqn as a vector of those equations. S = dsolve(eqn,cond) solves eqn with the initial or boundary condition cond. [y1,...,yN] = dsolve(___) assigns the solutions to the variables y1,...,yN. EV0000727 · EV0000728 · EV0000729
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS-I MATLAB Syntax for dsolve: S = dsolve(eqn) S = dsolve(eqn,cond) [y1,...,yN] = dsolve(___) EV0000730 · EV0000731 · EV0000732
6. Source slide page 6 (extracted text): Higher order differential equations: EV0000733 · EV0000734 · EV0000735
7. Source slide page 7 (extracted text): Higher order differential equations: EV0000736 · EV0000737 · EV0000738
8. Source slide page 8 (extracted text): Higher order differential equations, Continued... EV0000739 · EV0000740 · EV0000741
9. Source slide page 9 (extracted text): Higher order differential equations, Continued... [Image text, OCR] (5 exp((3 x)/2))/2 - (3 exp(x/2))/2 5000 4000 3000 2000 1000 0 -6 -4 -2 0 2 6 X EV0000742 · EV0000743 · EV0000744
10. Source slide page 10 (extracted text): Higher order differential equations, Continued... EV0000745 · EV0000746 · EV0000747
11. Source slide page 11 (extracted text): Higher order differential equations, Continued... ySol =2*exp(2*x) - 3*exp(3*x) + exp(x) [Image text, OCR] X106 2 exp(2 x) -3 exp(3 x) + exp(x) 0 -2 6 8 10 12 14 16 -2 0 2 6 EV0000748 · EV0000749 · EV0000750
12. Source slide page 12 (extracted text): Non-Homogeneous differential equations EV0000751 · EV0000752 · EV0000753
13. Source slide page 13 (extracted text): Non-Homogeneous differential equations, Continued... EV0000754 · EV0000755 · EV0000756
14. Source slide page 14 (extracted text): Non-Homogeneous differential equations, Continued... Output: Sol =exp(-2*x)*(2*sin(3*x) - 9*cos(3*x) + 6))/3 [Image text, OCR] 104 (exp (-2x) (2sin(3x)-9cos(3x)+6))/3 5 4 3 2 0 2 -6 -5 -4 -3 -2 -1 0 2 3 X EV0000757 · EV0000758 · EV0000759

15. Source slide page 15 (extracted text): Non-Homogeneous differential equations, Continued... EV0000760 · EV0000761 · EV0000762
16. Source slide page 16 (extracted text): Non-Homogeneous differential equations, Continued... EV0000763 · EV0000764 · EV0000765
17. Source slide page 17 (extracted text): Non-Homogeneous differential equations, Continued... EV0000766 · EV0000767 · EV0000768
18. Source slide page 18 (extracted text): Non-Homogeneous differential equations, Continued... EV0000769 · EV0000770 · EV0000771
19. Source slide page 19 (extracted text): Non-Homogeneous differential equations, Continued... EV0000772 · EV0000773 · EV0000774
20. Source slide page 20 (extracted text): Higher order differential equations, Continued... >> syms y(x) ode=diff(y,x,3)+2*diff(y,x,2)+diff(y,x,1)== x^2*exp(2*x)+sin^2(x); ySol(x) = dsolve(ode) EV0000775 · EV0000776 · EV0000777
21. Source slide page 21 (extracted text): Higher order differential equations: [Image text, OCR] syms y(x) Dy = d diff(y); D2y=C diff(y,2) ; ode = ((2*x+3)^2)*D2y y-2*(2*x+3)*Dy-12*y== 6*x; ySol dsolve(ode) EV0000778 · EV0000779 · EV0000780
22. Source slide page 22 (extracted text): Higher order differential equations: EV0000781 · EV0000782 · EV0000783
23. Source slide page 23 (extracted text): Higher order differential equations: Numerical solution Using ode45 EV0000784 · EV0000785 · EV0000786
24. Source slide page 24 (extracted text): ENGINEERING MATHEMATICS-I MATLAB-Solving ODE EV0000787 · EV0000788 · EV0000789
25. Source slide page 25 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS THANK YOU EV0000790 · EV0000791 · EV0000792

SHORT-NOTE RECAP

Source deck preserved: UNIT 2. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000718

12 Source slide deck: UNIT 2 CONTINUED

This entry preserves the complete slide deck titled UNIT 2 CONTINUED as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000793 · EV0000794

Graphs and diagrams

Figure G000234. UNIT 2 CONTINUED · source slide page 1.

EV0000794

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

1. **Assume an object weighing 2 lb stretches a spring 6 in. The spring is released from the equilibrium position with an upward velocity of 16 ft/sec. Form the differential equation governing this motion and solve it. Find the period of the motion?**

```
syms x(t)
Dx = diff(x)
D2x = diff(x,2)
ode = D2x + 64*x==0
dsolve(ode, x(0)==0,Dx(0)==-16)
```

Dx(t) =

$$\frac{\partial}{\partial t}x(t)$$

D2x(t) =

$$\frac{\partial^2}{\partial t^2}x(t)$$

ode(t) =

$$\frac{\partial^2}{\partial t^2}x(t) + 64\,x(t) = 0$$

ans = $-2\sin(8t)$

Figure G000235. UNIT 2 CONTINUED · source slide page 2.

EV0000797

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

2. At t=0, a current of 2 amperes flows in an RLC circuit with resistance R=40 ohms, inductance L=0.2 henrys, and capacitance C=10^{-5} farads. Find the current flowing in the circuit at t>0, if the initial charge on the capacitor is 1 coulomb. Assume that E(t)=0 for t>0.

```
syms Q(x)
DQ = diff(Q)
D2Q = diff(Q,2)
ode = D2Q + 200*DQ +50000*Q==0
Qsol(x)= dsolve(ode, Q(0)==1,DQ(0)==2)
I(x)= diff(Qsol,x)
```

DQ(x) =

$$\frac{\partial}{\partial x}Q(x)$$

D2Q(x) =

$$\frac{\partial^2}{\partial x^2}Q(x)$$

ode(x) =

$$\frac{\partial^2}{\partial x^2}Q(x) + 200\frac{\partial}{\partial x}Q(x) + 50000\,Q(x) = 0$$

Qsol(x) =

$$\frac{e^{-100x}\,(100\cos(200x) + 51\sin(200x))}{100}$$

I(x) =

$$\frac{e^{-100x}\,(10200\cos(200x) - 20000\sin(200x))}{100} - e^{-100x}\,(100\cos(200x) + 51\sin(200x))$$

Figure G000236. UNIT 2 CONTINUED · source slide page 3.

EV0000800

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

3.A spring with a mass of 2 kg has natural length 0.5m. A force of 25.6N is required to maintain it stretched to a length of 0.7m. The spring is immersed in a fluid with damping constant c = 40. Find the position of the mass at any time if it is pushed from the equilibrium position with an initial velocity of 0.6 m/s.

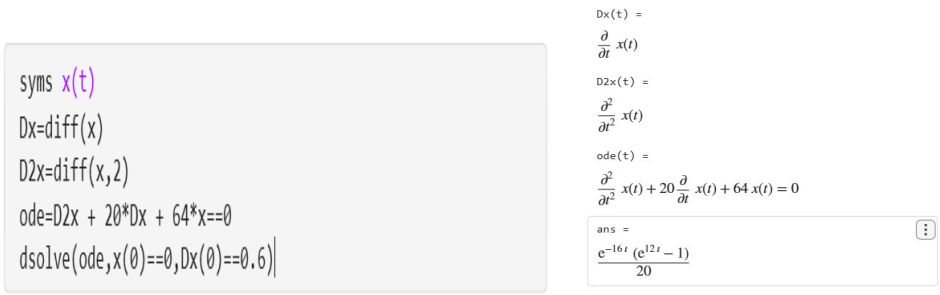


Figure G000237. UNIT 2 CONTINUED · source slide page 4.

EV0000803

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

4. A series circuit contains a resistor with R=24ohms, an inductor with L=2H, a capacitor with C=0.005F, and a 12V battery. The initial charge is Q=0.001C and the initial current is 0.
Find the charge and current at time t.

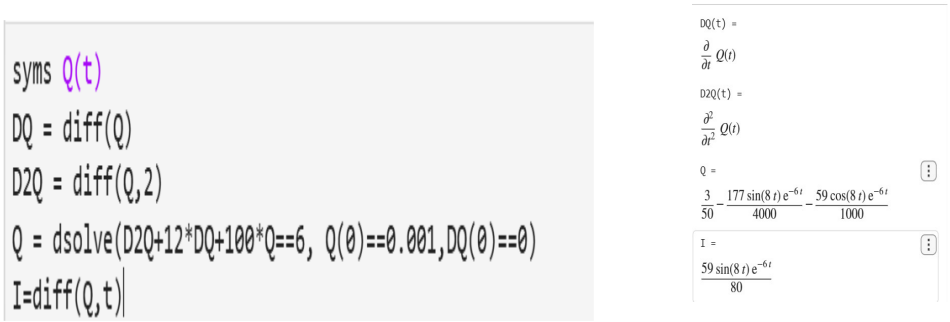


Figure G000238. UNIT 2 CONTINUED · source slide page 5.

EV0000806

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

5. Consider an RLC series circuit with a resistor of 0.06ohm and an inductor of 0.01H, and a capacitor of 50/89 F powered by a voltage source E(t) = 0.1sin(10t)V . Initially, the current and charge on the capacitor are zero. Determine the current in the circuit as a function of time.

```
syms I(t)
DI = diff(I)
D2I = diff(I,2)
dsolve (D2I+6*DI+178*I==100*cos(10*t), I(0)==0,DI(0)==0)
```

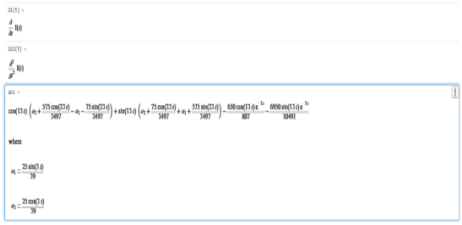


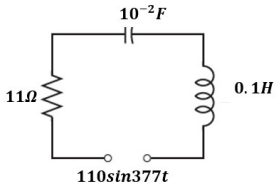
Figure G000239. UNIT 2 CONTINUED · source slide page 6.

EV0000809

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

6. Find the current $I(t)$ in a R-L-C circuit given below. Assume that current and capacitor charge are 0 when $t = 0$.



Solution:

The differential equation is:

$$0.1I'' + 11I' + 100I = 100.377cos377t$$

Initial conditions:

$$I(0) = 0, I'(0) = 0$$

Figure G000240. UNIT 2 CONTINUED · source slide page 7.

EV0000812

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

```
syms I(t)
D1(t)=diff(I,1);
D2(t)=diff(I,2);
f=0.1*D2(t)+11*D1(t)+100*I==100*377*cos(377*t);
dsolve(f,I(0)==0,D1(0)==0)
```

```
ans =
```

$$\frac{3770000\,e^{-100t}}{1369161}-\frac{377000\,e^{-10t}}{1280061}-\frac{377000\,\sqrt{21637155541}\,\cos\left(377\,t+\operatorname{atan}\left(\frac{41470}{141129}\right)\right)}{21637155541}$$

Figure G000241. UNIT 2 CONTINUED · source slide page 8.

EV0000815

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

7. Determine the charge on the capacitor in an R-L-C series circuit where $L=\frac{5}{3}H, R=10\Omega, C=\frac{1}{30}F$ and $E(t)=300V$. Assume the initial charge on the capacitor is 0C and the initial current is 9A. What happens to the charge on the capacitor over time?

Solution:

The differential equation is:

$$Q''+6Q'+18Q=180$$

Initial conditions:

$$Q(0)=0, Q'(0)=9$$

```
syms Q(t)
D1(t)=diff(Q,1);
D2(t)=diff(Q,2);
f=D2(t)+6*D1(t)+18*Q==180;
qsol(t)=dsolve(f,Q(0)==0,D1(0)==9)
```

```
qsol(t) =
```

$$10-7\sin(3t)\,e^{-3t}-10\cos(3t)\,e^{-3t}$$

Figure G000242. UNIT 2 CONTINUED · source slide page 9.

EV0000818

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

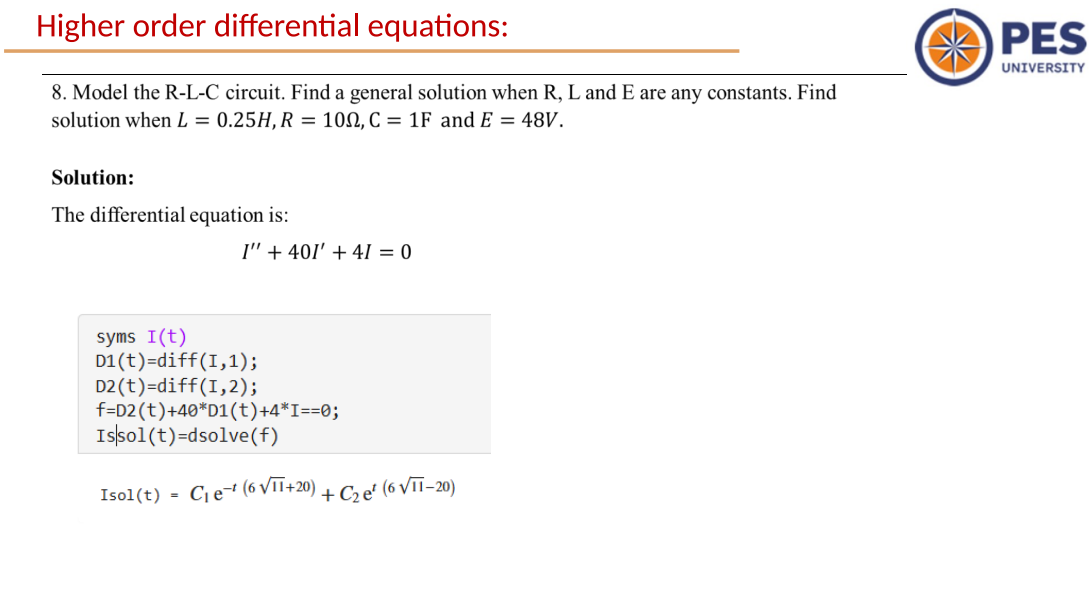


Figure G000243. UNIT 2 CONTINUED · source slide page 10.

EV0000821

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

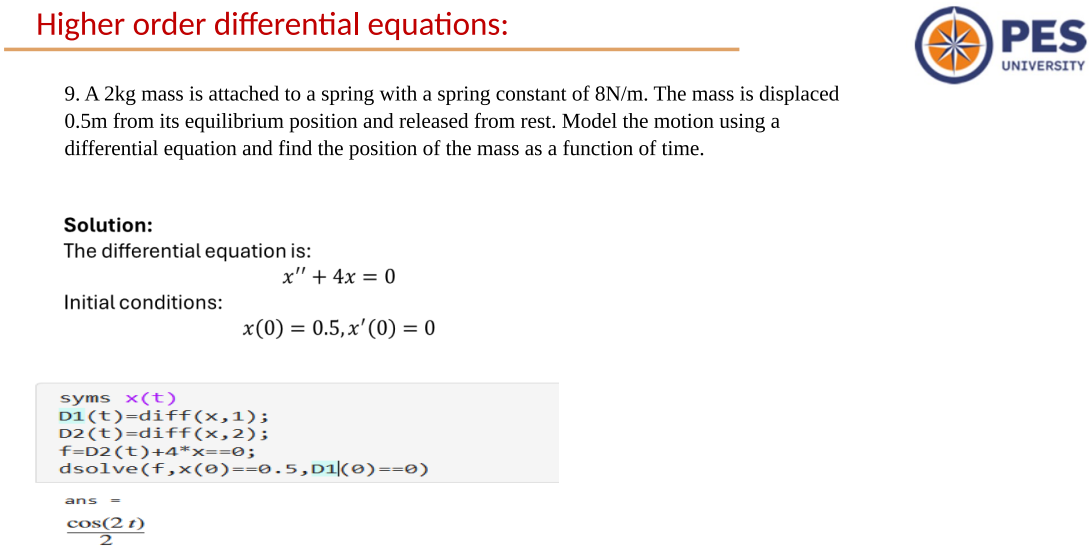


Figure G000244. UNIT 2 CONTINUED · source slide page 11.

EV0000824

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

10. In a spring-mass system, a mass attached to a spring oscillates when displaced from its equilibrium position. Without damping, the system would oscillate indefinitely (simple harmonic motion). However, in reality, a damping force (e.g., viscous friction proportional to velocity) opposes the motion, reducing the amplitude of oscillations over time. The system eventually comes to rest at the equilibrium position.

A car suspension system can be modelled as a spring-mass system with a 100 kg mass (quarter-car model), a spring constant of 40000 N/m, and a damping coefficient of 2000 Ns/m. The car hits a bump, displacing the suspension 0.1 m upward, with an initial velocity of 0 m/s. Model the motion and find the time it takes for the amplitude to reduce to 0.01 m due to damping.

Figure G000245. UNIT 2 CONTINUED · source slide page 12.

EV0000827

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

Solution:

The differential equation is:

$$x'' + 20x' + 400x = 0$$

Initial conditions:

$x(0) = 0.1, x'(0) = 0$

```
syms x(t)
D1(t)=diff(x,1);
D2(t)=diff(x,2);
f=D2(t)+20*D1(t)+400*x==0;
dsolve(f,x(0)==0.1,D1(0)==0)
```

```
ans =
```

$$\frac{e^{-10t}\left(3\cos\left(10\sqrt{3}\,t\right)+\sqrt{3}\sin\left(10\sqrt{3}\,t\right)\right)}{30}$$

Figure G000246. UNIT 2 CONTINUED · source slide page 13.

EV0000830

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

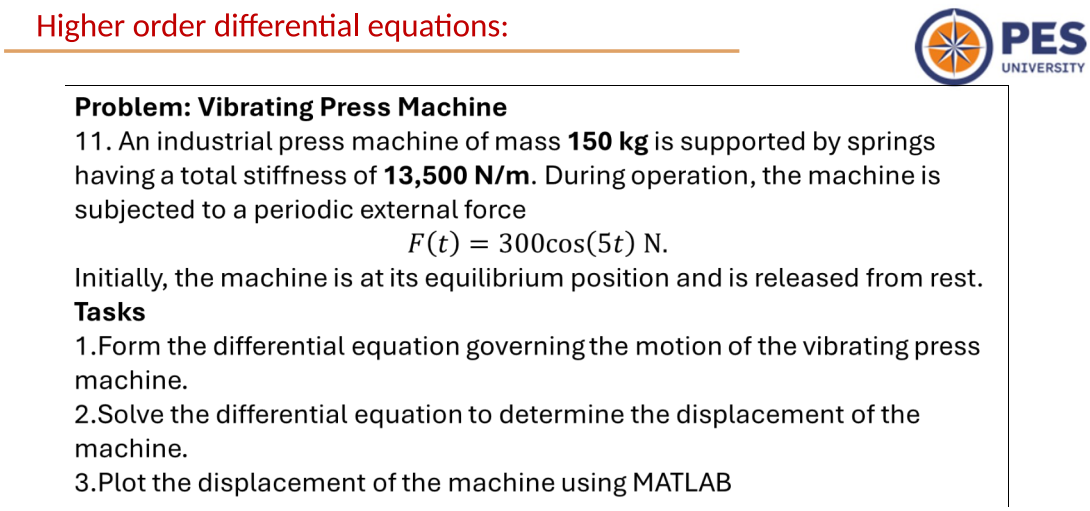


Figure G000247. UNIT 2 CONTINUED · source slide page 14.

EV0000833

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

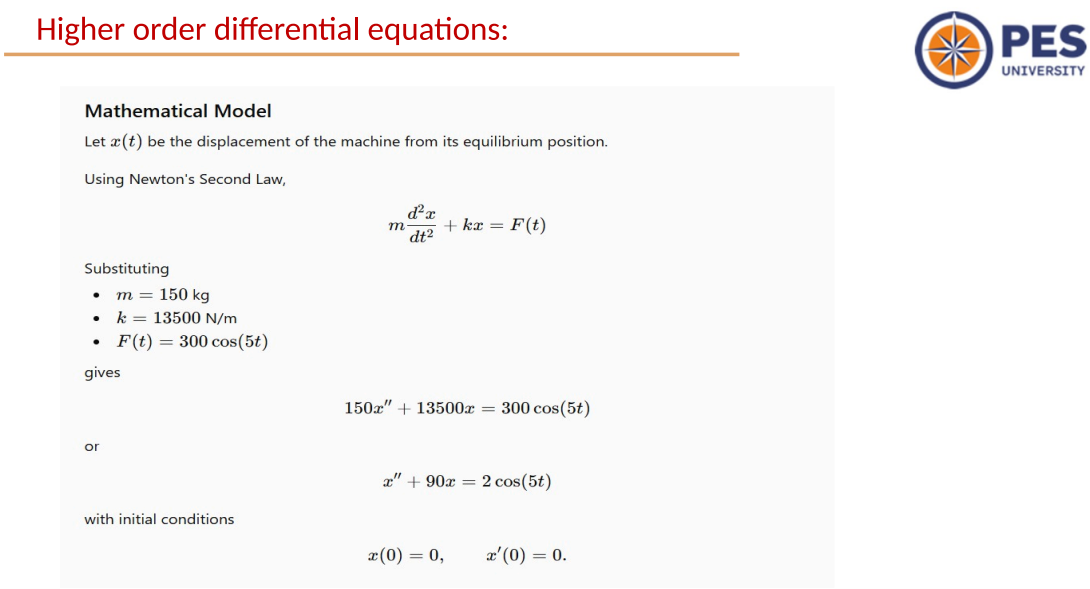


Figure G000248. UNIT 2 CONTINUED · source slide page 15.

EV0000836

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

```
syms x(t)

Dx = diff(x,t);

ode = diff(x,t,2) + 90*x == 2*cos(5*t);

cond1 = x(0)==0;
cond2 = Dx(0)==0;

xSol = dsolve(ode,cond1,cond2)

figure
fplot(xSol,[0 10],'LineWidth',2)
grid on
xlabel('Time (s)')
ylabel('Displacement (m)')
title('Displacement of Vibrating Press Machine')
```

Figure G000249. UNIT 2 CONTINUED · source slide page 16.

EV0000839

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

$$\sigma_5\left(\frac{\sqrt{10}\cos(\sigma_3)}{\sigma_1}+\frac{\sqrt{10}\cos(\sigma_4)}{\sigma_2}\right)+\sin(3\sqrt{10}\,t)\left(\frac{\sqrt{10}\sin(\sigma_3)}{\sigma_1}+\frac{\sqrt{10}\sin(\sigma_4)}{\sigma_2}\right)-\sigma_5\left(\frac{\sqrt{10}}{\sigma_1}+\frac{\sqrt{10}}{\sigma_2}\right)$$

where

$$\sigma_1 = 15\,(6\sqrt{10}-10)$$

$$\sigma_2 = 15\,(6\sqrt{10}+10)$$

$$\sigma_3 = t\,(3\sqrt{10}-5)$$

$$\sigma_4 = t\,(3\sqrt{10}+5)$$

$$\sigma_5 = \cos(3\sqrt{10}\,t)$$

Figure G000250. UNIT 2 CONTINUED · source slide page 17.

EV0000842

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

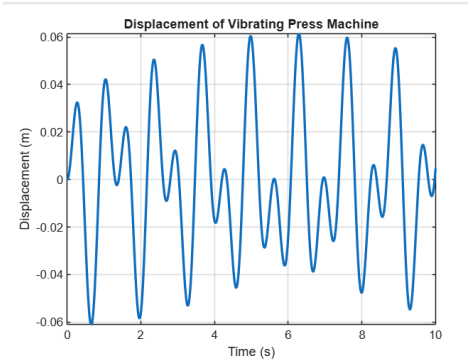


Figure G000251. UNIT 2 CONTINUED · source slide page 18.

EV0000845

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

Problem : High-Speed Railway Coach Suspension

12. A railway coach of effective mass **1200 kg** is supported by suspension springs having total stiffness **96,000 N/m**. As the train moves over periodically spaced rail joints, the suspension experiences an external force $F(t) = 2400\cos(6t)$ N.

Initially, the coach is displaced **0.05 m** downward from equilibrium and released with an upward velocity of **0.3 m/s**.

Tasks

1.Develop the governing differential equation.
2.Solve for the displacement of the coach.
3.Plot the displacement for the first 20 seconds using MATLAB.
4.Determine the natural frequency of the suspension.
5.Discuss the effect of the external periodic force on the motion.

Figure G000252. UNIT 2 CONTINUED · source slide page 19.

EV0000848

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

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Mathematical Model

Using Newton's Second Law,

$$1200x'' + 96000x = 2400\cos(6t)$$

or

$$x'' + 80x = 2\cos(6t)$$

with

$$x(0) = 0.05, \qquad x'(0) = -0.3.$$

Figure G000253. UNIT 2 CONTINUED · source slide page 20.

EV0000851

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

PES UNIVERSITY

```
syms x(t)
Dx = diff(x,t);

% Differential equation
ode = diff(x,t,2) + 80*x == 2*cos(6*t);
% Initial conditions
cond1 = x(0) == 0.05;
cond2 = Dx(0) == -0.3;
% Solve the differential equation
xSol = dsolve(ode,cond1,cond2)
% Plot the solution
figure
fplot(xSol,[0 20],'LineWidth',2)
grid on
xlabel('Time (s)')
ylabel('Displacement (m)')
title('Response of Railway Coach Suspension')
```

Figure G000254. UNIT 2 CONTINUED · source slide page 21.

EV0000854

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

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$$\sigma_6\left(\frac{\sqrt{5}\cos(\sigma_1)}{\sigma_3}+\frac{\sqrt{5}\cos(\sigma_2)}{\sigma_4}\right)-\frac{3\sqrt{5}\,\sigma_5}{200}+\sigma_5\left(\frac{\sqrt{5}\sin(\sigma_1)}{\sigma_3}+\frac{\sqrt{5}\sin(\sigma_2)}{\sigma_4}\right)+\frac{\sigma_6}{20\,(2\sqrt{5}-3)\,(2\sqrt{5}+3)}$$

where

$\sigma_1 = t\,(4\sqrt{5}-6)$

$\sigma_2 = t\,(4\sqrt{5}+6)$

$\sigma_3 = 10\,(8\sqrt{5}-12)$

$\sigma_4 = 10\,(8\sqrt{5}+12)$

$\sigma_5 = \sin(4\sqrt{5}\,t)$

$\sigma_6 = \cos(4\sqrt{5}\,t)$

Figure G000255. UNIT 2 CONTINUED · source slide page 22.

EV0000857

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Higher order differential equations:

PES UNIVERSITY

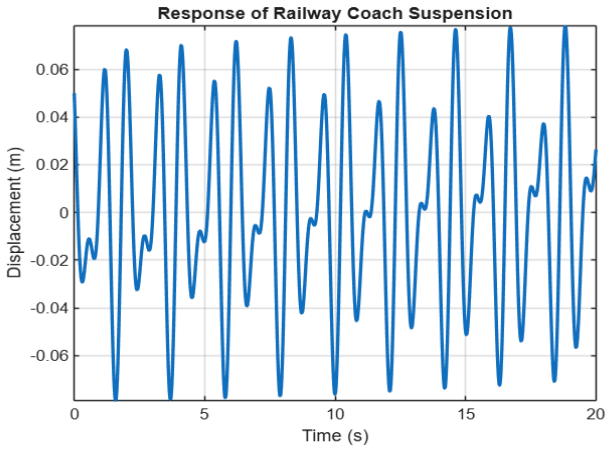


Figure G000256. UNIT 2 CONTINUED · source slide page 23.

EV0000860

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000257. UNIT 2 CONTINUED · source slide page 24.

EV0000863

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - II UE26MA141A Department of Science & Humanities EV0000793 · EV0000794 · EV0000795
2. Source slide page 2 (extracted text): Higher order differential equations: Assume an object weighing 2 lb stretches a spring 6 in. The spring is released from the equilibrium position with an upward velocity of 16 ft/sec. Form the differential equation governing this motion and solve it. Find the period of the motion? [Image text, OCR] Dx(t) 0 x(t) D2x(t) = x(t) dr2 ode(t)= 0r2 x(t)+ 64 x(t) = 0 ans = -2 sin(8 t) [Image text, OCR] syms x(t) Dx= diff(x) D2x = diff(x,2) ode = D2x+64*x==0 dsolve(ode, x(0)==0,Dx(0)==-16) EV0000796 · EV0000797 · EV0000798
3. Source slide page 3 (extracted text): Higher order differential equations: 2. At t=0, a current of 2 amperes flows in an RLC circuit with resistance R=40 ohms, inductance L=0.2 henrys, and capacitance C=10−5 farads. Find the current flowing in the circuit at t>0, if the initial charge on the capacitor is 1 coulomb. Assume that E(t)=0 for t>0. [Image text, OCR] Syms Q(x) DQ = diff(Q) D2Q = diff(Q,2) ode=D2Q+200*DQ+50000*Q==0 Qsol(x) = dsolve(ode,Q(0)==1,DQ(0)==2) I(x)= diff(Qsol,x) [Image text, OCR] DQ(x) = d d Q(x) D2Q(x) Q(x) dx² ode(x)= (x)+200 d dx Q(x)+50000 Q(x)=0 dx2 Qso1(x)= e100 x (100 cos(200 x) + 51 sin(200 x)) 100 I(x)= e100x(10200 cos(200 x) 2000sin(200 x) _e-100 (100 cos(200 x)+ 51 sin(200 x) 100 EV0000799 · EV0000800 · EV0000801

4. Source slide page 4 (extracted text): Higher order differential equations: 3.A spring with a mass of 2 kg has natural length 0.5m. A force of 25.6N is required to maintain it stretched to a length of 0.7m. The spring is immersed in a fluid with damping constant c = 40. Find the position of the mass at any time if it is pushed from the equilibrium position with an initial velocity of 0.6 m/s. [Image text, OCR] Dx(t) = 0 x(t) dt D2x(t) = x(t) dr2 ode(t) = x(t)+20 x(t)+ 64 x(t)=0 dr dt ans (e12t - 1) 20 [Image text, OCR] syms x(t) Dx=diff(x) D2x=diff(x,2) ode=D2x+20*Dx+64*x==0 dsolve(ode,x(0)==0,Dx(0)==0.6) EV0000802 · EV0000803 · EV0000804

5. Source slide page 5 (extracted text): Higher order differential equations: 4. A series circuit contains a resistor with R=24ohms, an inductor with L=2H, a capacitor with C=0.005F, and a 12V battery. The initial charge is Q=0.001C and the initial current is 0. Find the charge and current at time t. [Image text, OCR] DQ(t) = d Q(t) dt D2Q(t) = Q(t) dr2 Q= 3 177 sin(8 t)e-6t 59 cos(8 t)e-6t 50 4000 1000 I 59 sin(8 t) e-6t 80 [Image text, OCR] syms Q(t) DQ=C diff(Q) D2Q = diff(Q,2) Q=dso1ve(D2Q+12*DQ+100*Q==6, Q(0)==0.001,DQ(0)==0) I=diff(Q,t) EV0000805 · EV0000806 · EV0000807

6. Source slide page 6 (extracted text): Higher order differential equations: 5. Consider an RLC series circuit with a resistor of 0.06ohm and an inductor of 0.01H, and a capacitor of 50/89 F powered by a voltage source E(t) = 0.1sin(10t)V . Initially, the current and charge on the capacitor are zero. Determine the current in the circuit as a function of time. [Image text, OCR] 21(t) cpa(13r) (23.0) + xs(130 (3497 349T 75om(23.0 + 4 + 3497 575 ain(23 r)) 3-97 600 ee(13)e b MDT 6950xin(13)e 10491 25 sin(3) 39 25 em(3) 39 [Image text, OCR] syms I(t) DI= (diff(I) D2I=(diff (I,2) dsolve (D2I+6*DI+178*I==100*c0s(10*t), I(0)==0,DI(0)==0) EV0000808 · EV0000809 · EV0000810

7. Source slide page 7 (extracted text): Higher order differential equations: [Image text, OCR] 10-2F 0.1H 11Ω 110sin377t EV0000811 · EV0000812 · EV0000813

8. Source slide page 8 (extracted text): Higher order differential equations: [Image text, OCR] syms I(t) D1(t)=diff(I,1); D2(t)=diff(I,2); f=0.1*D2(t)+11*D1(t)+100*I==100*377*c0s(377*t); dsolve(f,I(0)==0,D1(0)==0) ans 37700021637155541 377t+ atan 41470 3770000e-100t 377000e-10t COS 141129 1369161 1280061 21637155541 EV0000814 · EV0000815 · EV0000816

9. Source slide page 9 (extracted text): Higher order differential equations: [Image text, OCR] syms Q(t) D1(t)=diff(Q,1); D2(t)=diff(Q,2); f=D2(t)+6*D1(t)+18*Q==180; qsol(t)=dsolve(f,Q(0)==0,D1(0)==9) qsol(t) = 10 - 7 sin(3 t)e-3t - 10 cos(3t)e-3t EV0000817 · EV0000818 · EV0000819

10. Source slide page 10 (extracted text): Higher order differential equations: [Image text, OCR] syms I(t) D1(t)=diff(I,1); D2(t)=diff(I,2); f=D2(t)+40*D1(t)+4*I==0; Issol(t)=dsolve(f) EV0000820 · EV0000821 · EV0000822

11. Source slide page 11 (extracted text): Higher order differential equations: 9. A 2kg mass is attached to a spring with a spring constant of 8N/m. The mass is displaced 0.5m from its equilibrium position and released from rest. Model the motion using a differential equation and find the position of the mass as a function of time. [Image text, OCR] syms x(t) D1(t)=diff(x,1); D2(t)=diff(x,2); f=D2(t)+4*x==0; dsolve(f,x(0)==0.5,D1(0)==0) ans cos(2 t) 2 EV0000823 · EV0000824 · EV0000825

12. Source slide page 12 (extracted text): Higher order differential equations: 10. In a spring-mass system, a mass attached to a spring oscillates when displaced from its equilibrium position. Without damping, the system would oscillate indefinitely (simple harmonic motion). However, in reality, a damping force (e.g., viscous friction proportional to velocity) opposes the motion, reducing the amplitude of oscillations over time. The system eventually comes to rest at the equilibrium position. A car suspension system can be modelled as a spring-mass system with a 100 kg mass (quarter-car model), a spring constant of 40000 N/m, and a damping coefficient of 2000 Ns/m. The car hits a bump, displacing the suspension 0.1 m upward, with an initial velocity of 0 m/s. Model the motion and find the time it takes for the amplitude to reduce to 0.01 m due to damping. EV0000826 · EV0000827 · EV0000828
13. Source slide page 13 (extracted text): Higher order differential equations: [Image text, OCR] syms x(t) D1(t)=diff(x,1); D2(t)=diff(x,2); f=D2(t)+20*D1(t)+400*x==0; dsolve(f,x(0)==0.1,D1(0)==0) ans e-10r (3 cos(10 √3 t) + V3 sin(10 √3 t) 30 EV0000829 · EV0000830 · EV0000831
14. Source slide page 14 (extracted text): Higher order differential equations: EV0000832 · EV0000833 · EV0000834
15. Source slide page 15 (extracted text): Higher order differential equations: [Image text, OCR] Mathematical Model Let c(t) be the displacement of the machine from its equilibrium position. Using Newton's Second Law, d²c m + kc = F(t) dt2 Substituting m = 150 kg k = 13500 N/m F(t) = 300 cos(5t) gives 150x" + 13500c = 300 cos(5t) or α" + 90 = 2 cos(5t) with initial conditions α(0) = 0, α'(0) = 0. EV0000835 · EV0000836 · EV0000837
16. Source slide page 16 (extracted text): Higher order differential equations: syms x(t) Dx = diff(x,t); ode = diff(x,t,2) + 90*x == 2*cos(5*t); cond1 = x(0)==0; cond2 = Dx(0)==0; xSol = dsolve(ode,cond1,cond2) figure fplot(xSol,[0 10],'LineWidth',2) grid on xlabel('Time (s)') ylabel('Displacement (m)') title('Displacement of Vibrating Press Machine') EV0000838 · EV0000839 · EV0000840
17. Source slide page 17 (extracted text): Higher order differential equations: [Image text, OCR] 10 cos(o3) 10 cos(o4) V10 sin(o3) 10 sin(o4) 10 10 + sin(3 √10 t) 05 05 01 02 01 02 01 02 where 01 = 15 (6 √10 - 10) 02 = 15 (6 √10 + 10) 03 = t (3 √10 - 5) 04 = t (3 √10+ 5) 05 = cos(3 10 t) EV0000841 · EV0000842 · EV0000843
18. Source slide page 18 (extracted text): Higher order differential equations: [Image text, OCR] DisplacementofVibratingPressMachine 0.06 0.04 w) 0.02 0 -0.02 -0.04 -0.06 0 2 4 6 8 10 Time (s) EV0000844 · EV0000845 · EV0000846
19. Source slide page 19 (extracted text): Higher order differential equations: EV0000847 · EV0000848 · EV0000849
20. Source slide page 20 (extracted text): Higher order differential equations: [Image text, OCR] Mathematical Model Using Newton's Second Law, 1200x" + 96000c = 2400 cos(6t) or α"+ 80c = 2 cos(6t) with c(0) = 0.05, α'(0) = -0.3. EV0000850 · EV0000851 · EV0000852

21. Source slide page 21 (extracted text): Higher order differential equations: syms x(t) Dx = diff(x,t); % Differential equation ode = diff(x,t,2) + 80*x == 2*cos(6*t); % Initial conditions cond1 = x(0) == 0.05; cond2 = Dx(0) == -0.3; % Solve the differential equation xSol = dsolve(ode,cond1,cond2) % Plot the solution figure fplot(xSol,[0 20],'LineWidth',2) grid on xlabel('Time (s)') ylabel('Displacement (m)') title('Response of Railway Coach Suspension') EV0000853 · EV0000854 · EV0000855
22. Source slide page 22 (extracted text): Higher order differential equations: [Image text, OCR] 5 cos(o1). V5 cos(o2) 3 √5 o5 √5 sin(o1) . V5 sin(o2) 06 06 +05 03 04 200 03 04 20 (2 5 -3) (2 √5 + 3) where 01 =t (4 v5 - 6) 02 = t (4 √5 + 6) 03 = 10 (8 <5 - 12) 04 = 10 (8 √5 + 12) os = sin(4 √5 t) 06 = cos(4 √5 t) EV0000856 · EV0000857 · EV0000858
23. Source slide page 23 (extracted text): Higher order differential equations: [Image text, OCR] ResponseofRailwayCoachSuspension 0.06 0.04 0.02 -0.02 -0.04 -0.06 5 10 15 20 Time (s) EV0000859 · EV0000860 · EV0000861
24. Source slide page 24 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000862 · EV0000863 · EV0000864

SHORT-NOTE RECAP

Source deck preserved: UNIT 2 CONTINUED. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000793

13 Source slide deck: UNIT 3

This entry preserves the complete slide deck titled UNIT 3 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000865 · EV0000866

Graphs and diagrams

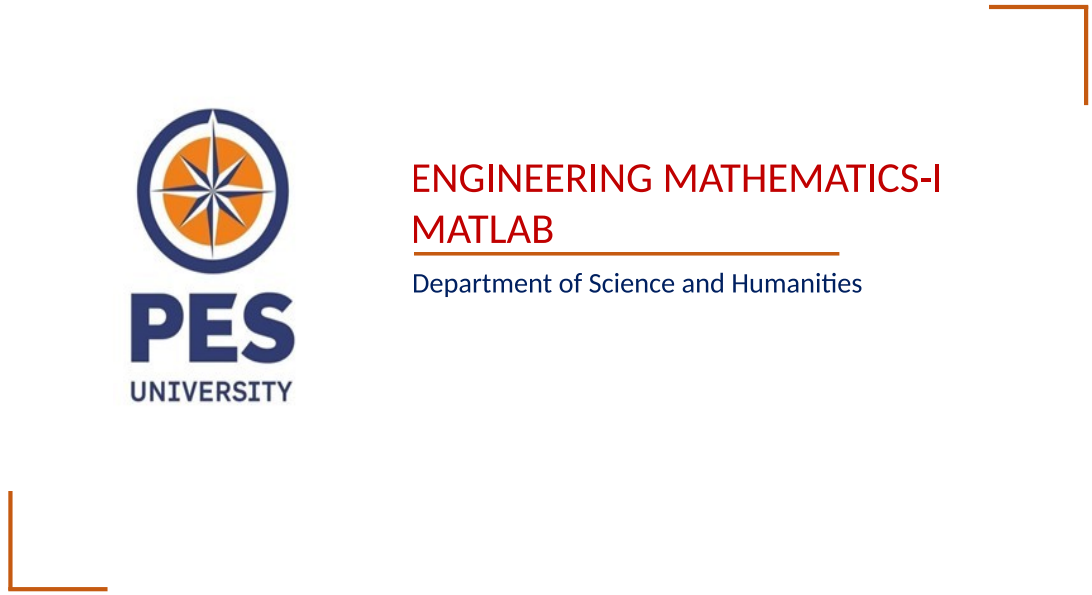
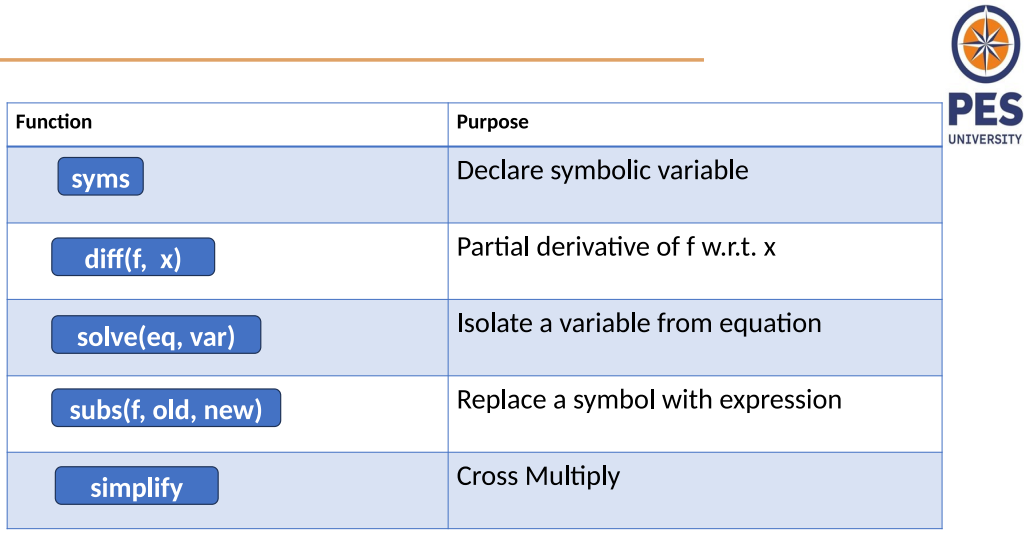


Figure G000258. UNIT 3 · source slide page 1.

EV0000866

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.



Function	Purpose
syms	Declare symbolic variable
diff(f, x)	Partial derivative of f w.r.t. x
solve(eq, var)	Isolate a variable from equation
subs(f, old, new)	Replace a symbol with expression
simplify	Cross Multiply

Figure G000259. UNIT 3 · source slide page 2.

EV0000869

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

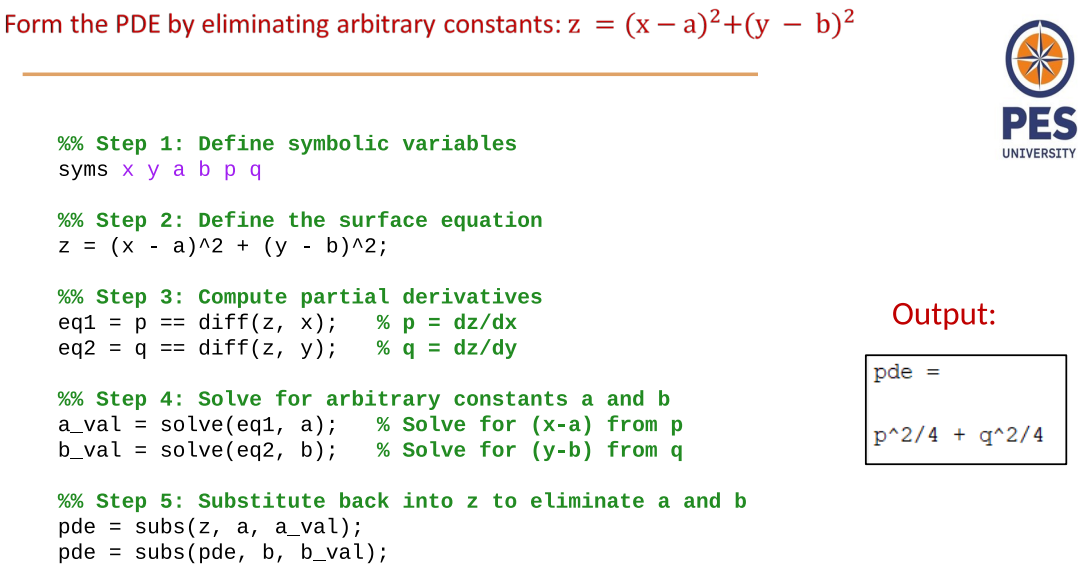


Figure G000260. UNIT 3 · source slide page 3.

EV0000872

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

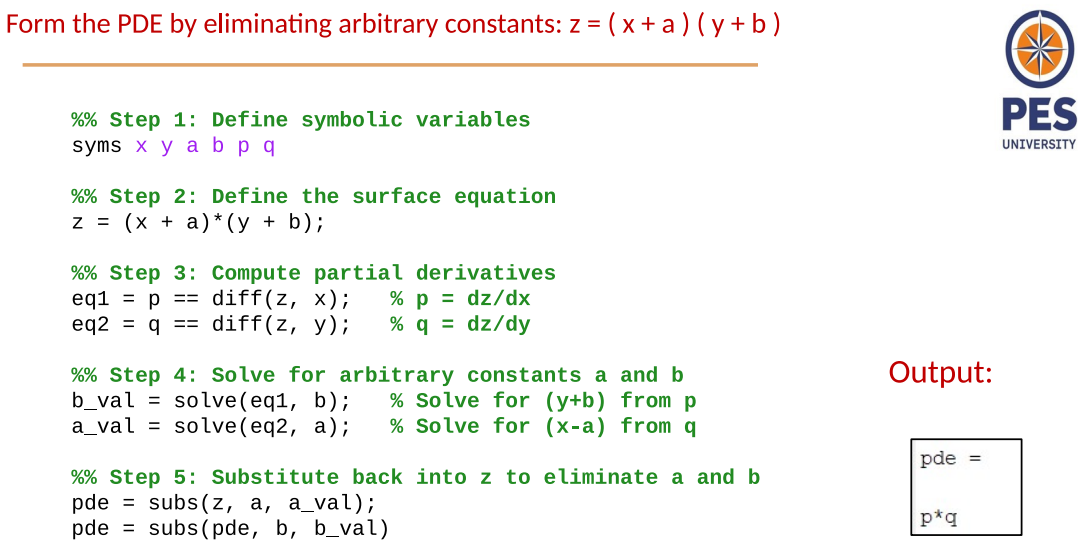


Figure G000261. UNIT 3 · source slide page 4.

EV0000875

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

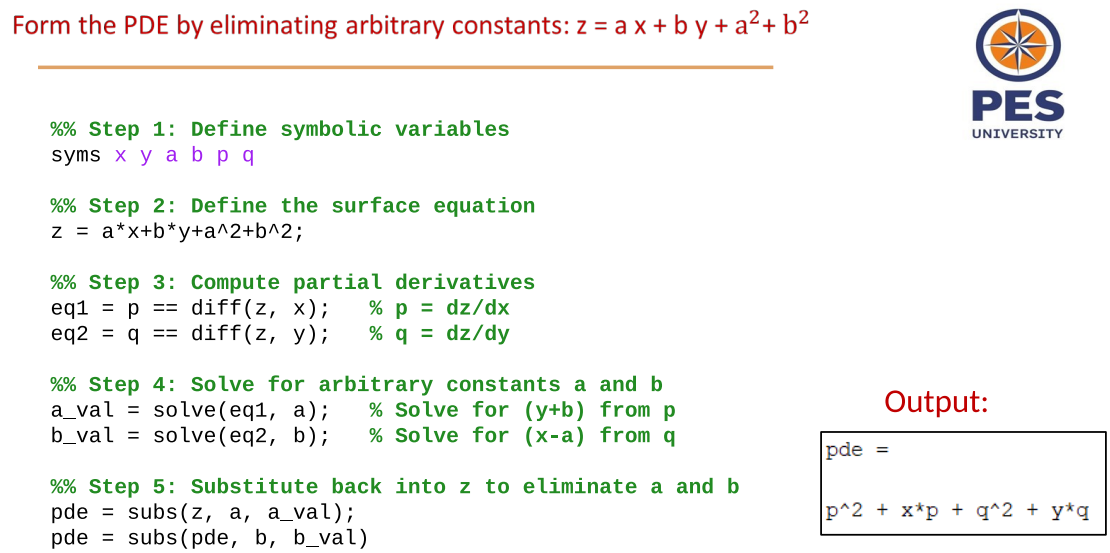


Figure G000262. UNIT 3 · source slide page 5.

EV0000878

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Form the PDE by eliminating arbitrary constants: $z = ax^2 + by^2$

```
%% Step 1: Define symbolic variables
syms x y a b p q

%% Step 2: Define the surface equation
z = a*x+b*y+a^2+b^2;

%% Step 3: Compute partial derivatives
eq1 = p == diff(z, x);   % p = dz/dx
eq2 = q == diff(z, y);   % q = dz/dy

%% Step 4: Solve for arbitrary constants a and b
a_val = solve(eq1, a);   % Solve for (y+b) from p
b_val = solve(eq2, b);   % Solve for (x-a) from q

%% Step 5: Substitute back into z to eliminate a and b
pde = subs(z, a, a_val);
pde = subs(pde, b, b_val)
```

Output:

```
pde =

(p*x)/2 + (q*y)/2
```

Figure G000263. UNIT 3 · source slide page 6.

EV0000881

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Form the PDE by eliminating the arbitrary function from the $z = xy + f(u)$ where $u(x,y) = x^2 + y^2$

```
%% Step 1: Define symbolic variables
syms x y f(u) u(x,y) p q

%% Step 2: Define the surface equation
u(x,y) = x^2 + y^2;
z = x*y + f(u);

%% Step 3: Compute partial derivatives
eq1 = p == diff(z, x);   % p = dz/dx
eq2 = q == diff(z, y);   % q = dz/dy

%% Step 4: Solve for function f'(u)
f1 = (p - y)/2*x;   % Calculating f' from eq1
f2 = (q - x)/2*y;   % Calculating f' from eq2

%% Step 5: Equating both expressions
pde = simplify(f1 == f2)
```

Output:

```
equate =

x*(p/2 - y/2) == y*(q/2 - x/2)
```

Figure G000264. UNIT 3 · source slide page 7.

EV0000884

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Form the PDE by eliminating the arbitrary function from the $z = f(u)$ where $u(x,y) = x^2 + y^2$

```
%% Step 1: Define symbolic variables
syms x y f(u) u(x,y) p q

%% Step 2: Define the surface equation
u(x,y) = x^2 + y^2;
z = f(u);

%% Step 3: Compute partial derivatives
eq1 = p == diff(z, x);   % p = dz/dx
eq2 = q == diff(z, y);   % q = dz/dy

%% Step 4: Solve for function f'(u)
f1 = p/2*x;   % Calculating f' from eq1
f2 = q/2*y;   % Calculating f' from eq2

%% Step 5: Equating both expressions
pde = simplify(f1 == f2)
```

Output:

```
equate =

p*x == q*y
```

Figure G000265. UNIT 3 · source slide page 8.

EV0000887

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Form the PDE by eliminating the arbitrary function from the $z = f(x + at) + g(x - at)$

```
%% Step 1: Define symbolic variables
syms x a t f(u) g(v)
%% Step 2: Define the surface equation
u = x + a*t;
v = x - a*t;
z = f(u) + g(v);
%% Step 3: Compute first order partial derivatives
z_x = diff(z, x);   % dz/dx
z_t = diff(z, t);   % dz/dt
%% Step 4: Compute second order partial derivatives
z_xx = diff(z, x, 2);    % d2z/dx2
z_tt = diff(z, t, 2);    % d2z/dt2
%% Step 5: Equating both expressions
pde = simplify(z_tt == a^2 * z_xx);
PDE = sym('z_tt') == a^2 * sym('z_xx')
```

Output:

```
PDE =

z_tt == a^2*z_xx
```

Figure G000266. UNIT 3 · source slide page 9.

EV0000890

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

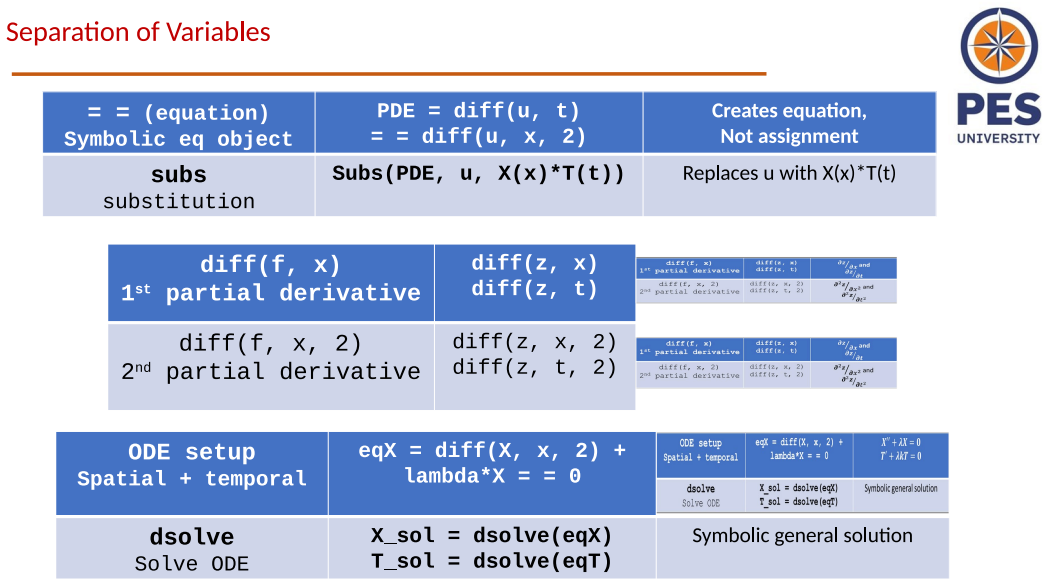


Figure G000267. UNIT 3 · source slide page 10.

EV0000893

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

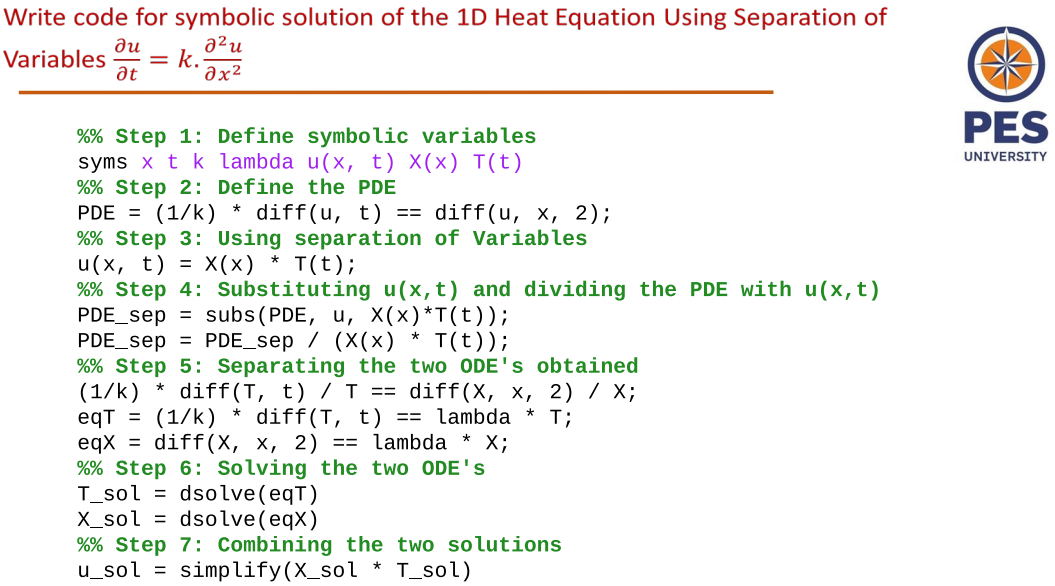


Figure G000268. UNIT 3 · source slide page 11.

EV0000896

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Write code for symbolic solution of the 1D Heat Equation Using Separation of Variables $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$

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Output:

```
T_sol =

C3*exp(k*lambda*t)
```

```
X_sol =

C4*exp(-lambda^(1/2)*x) + C5*exp(lambda^(1/2)*x)
```

```
u_sol =

C3*exp(k*lambda*t)*(C4*exp(-lambda^(1/2)*x) + C5*exp(lambda^(1/2)*x))
```

Figure G000269. UNIT 3 · source slide page 12.

EV0000899

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Write code for solution of PDE using Separation of Variables

$$3u_x + 2u_y = 0$$

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```
%% Step 1: Define symbolic variables
syms x y lambda u(x, y) X(x) Y(y)
%% Step 2: Define the PDE
PDE = 3 * diff(u, x) + 2 * diff(u, y) == 0;
%% Step 3: Using separation of Variables
u(x, y) = X(x) * Y(y);
%% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y)
PDE_sep = subs(PDE, u, X(x) * Y(y));
PDE_sep = PDE_sep / (X(x) * Y(y));
%% Step 5: Separating the two ODE's obtained
- 2 * diff(Y, y) / Y == 3 * diff(X, x) / X;
eqY = - 2 * diff(Y, y) == lambda * Y;
eqX = 3 * diff(X, x) == lambda * X;
%% Step 6: Solving the two ODE's
Y_sol = dsolve(eqY)
X_sol = dsolve(eqX)
%% Step 7: Combining the two solutions
u_sol = simplify(X_sol * Y_sol)
```

Figure G000270. UNIT 3 · source slide page 13.

EV0000902

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

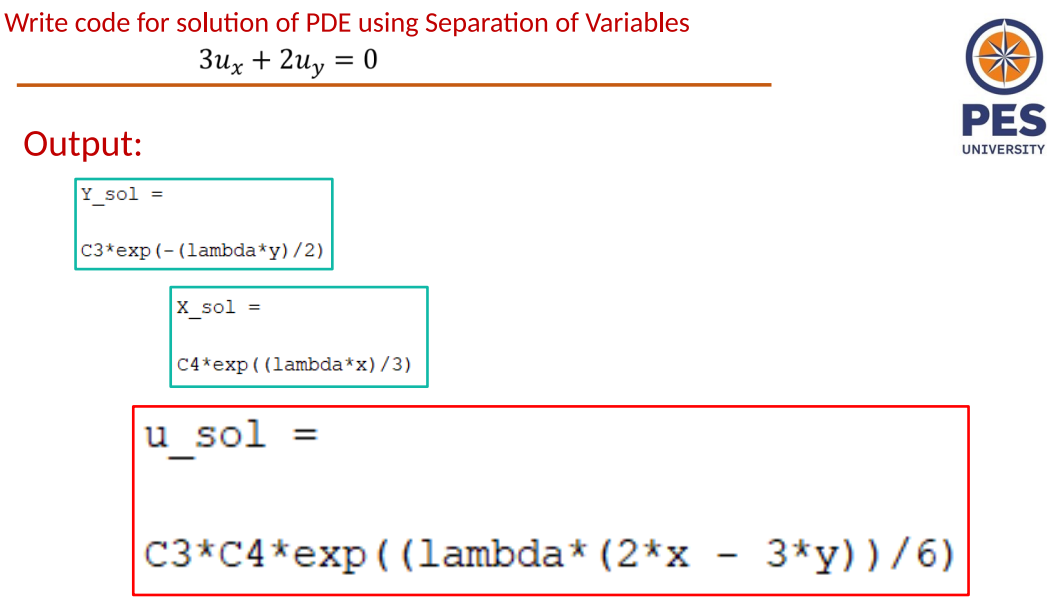


Figure G000271. UNIT 3 · source slide page 14.

EV0000905

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

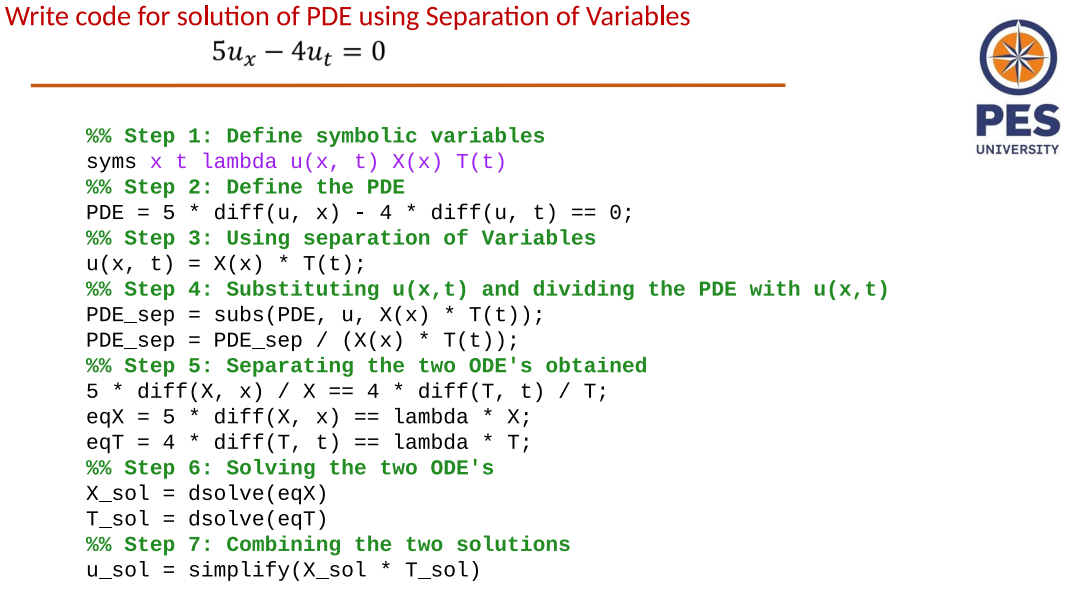


Figure G000272. UNIT 3 · source slide page 15.

EV0000908

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

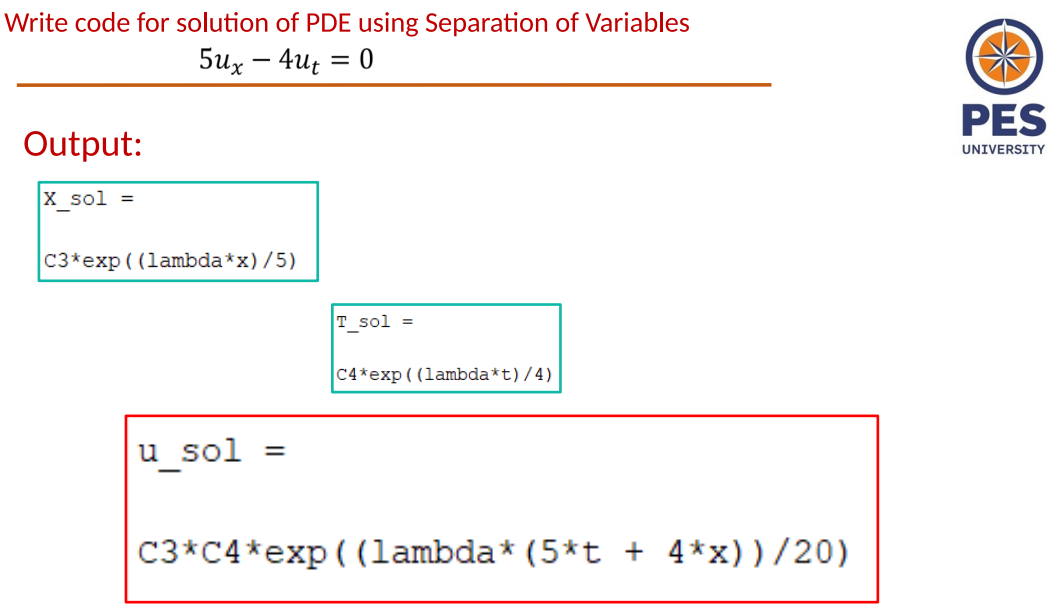
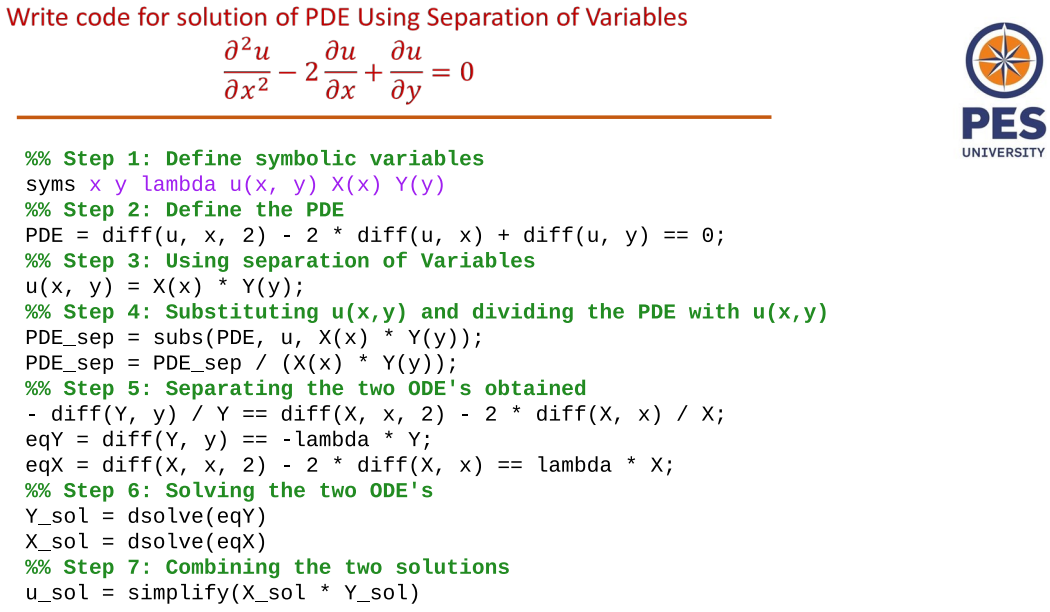


Figure G000273. UNIT 3 · source slide page 16.

EV0000911

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.




```
%% Step 1: Define symbolic variables
syms x y lambda u(x, y) X(x) Y(y)
%% Step 2: Define the PDE
PDE = diff(u, x, 2) - 2 * diff(u, x) + diff(u, y) == 0;
%% Step 3: Using separation of Variables
u(x, y) = X(x) * Y(y);
%% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y)
PDE_sep = subs(PDE, u, X(x) * Y(y));
PDE_sep = PDE_sep / (X(x) * Y(y));
%% Step 5: Separating the two ODE's obtained
- diff(Y, y) / Y == diff(X, x, 2) - 2 * diff(X, x) / X;
eqY = diff(Y, y) == -lambda * Y;
eqX = diff(X, x, 2) - 2 * diff(X, x) == lambda * X;
%% Step 6: Solving the two ODE's
Y_sol = dsolve(eqY)
X_sol = dsolve(eqX)
%% Step 7: Combining the two solutions
u_sol = simplify(X_sol * Y_sol)
```

Figure G000274. UNIT 3 · source slide page 17.

EV0000914

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

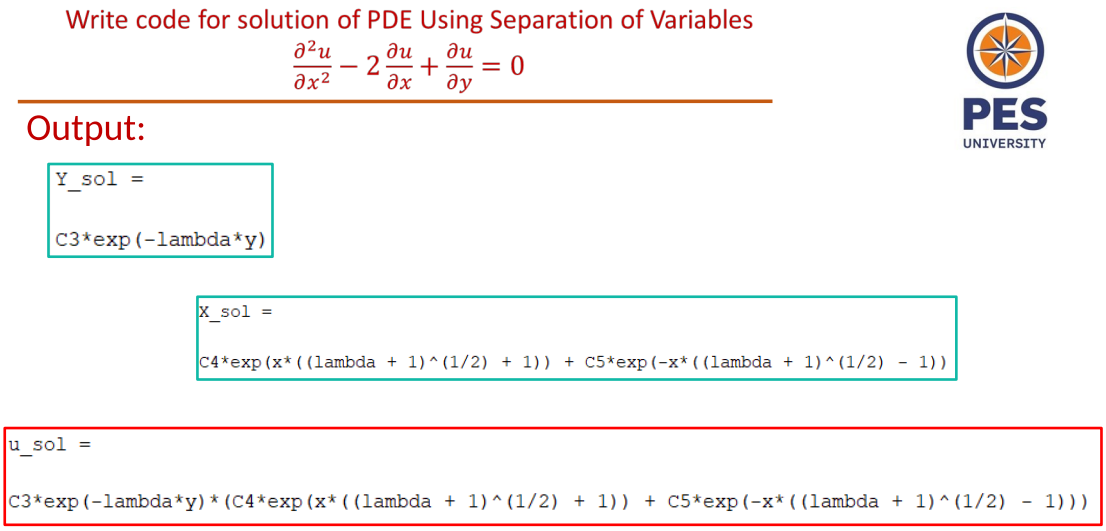


Figure G000275. UNIT 3 · source slide page 18.

EV0000917

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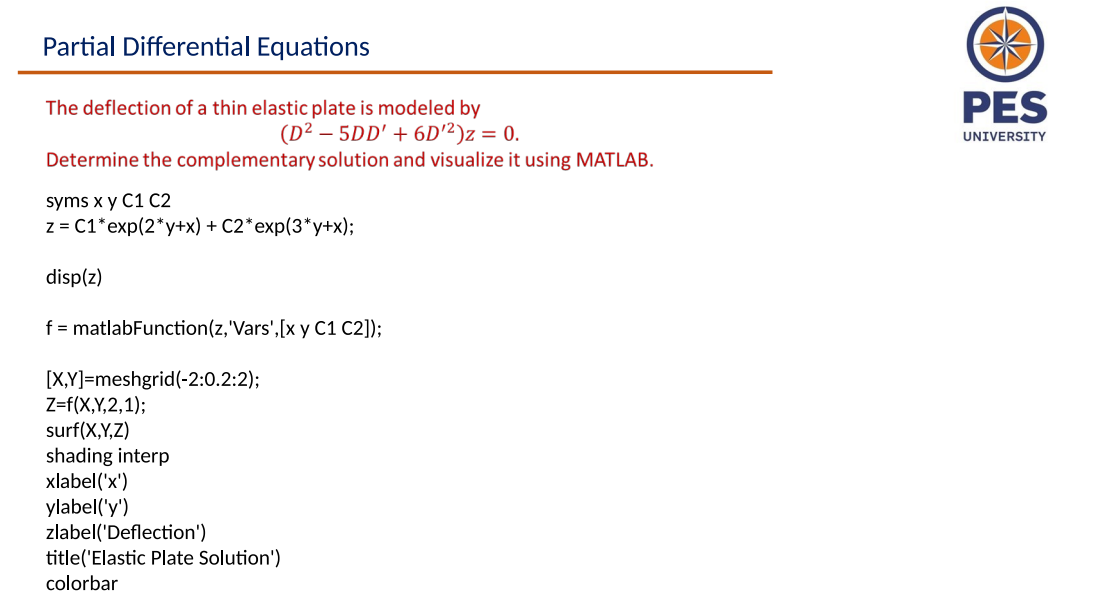

Figure G000276. UNIT 3 · source slide page 19.

EV0000920

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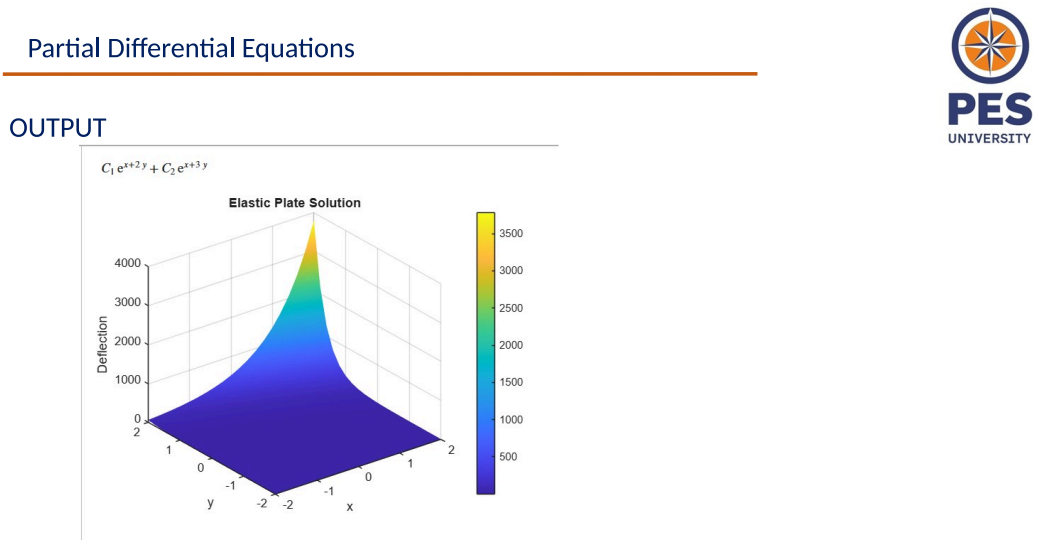

Figure G000277. UNIT 3 · source slide page 20.

EV0000923

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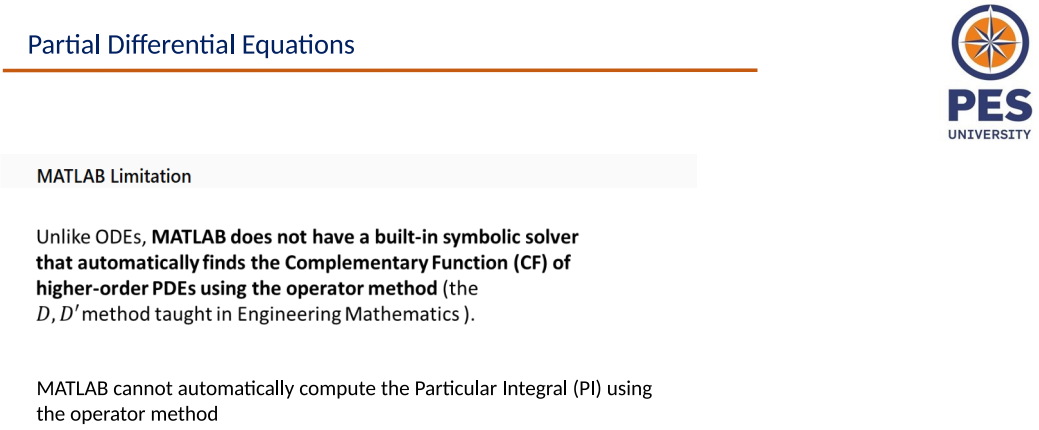

Figure G000278. UNIT 3 · source slide page 21.

EV0000926

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000279. UNIT 3 · source slide page 22.

EV0000929

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000865 · EV0000866 · EV0000867
2. Source slide page 2 (extracted text): Function | Purpose | Declare symbolic variable | Partial derivative of f w.r.t. x | Isolate a variable from equation | Replace a symbol with expression | Cross Multiply syms diff(f, x) solve(eq, var) subs(f, old, new) simplify EV0000868 · EV0000869 · EV0000870
3. Source slide page 3 (extracted text): %% Step 1: Define symbolic variables syms x y a b p q %% Step 2: Define the surface equation z = (x - a)^2 + (y - b)^2; %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for arbitrary constants a and b a_val = solve(eq1, a); % Solve for (x-a) from p b_val = solve(eq2, b); % Solve for (y-b) from q %% Step 5: Substitute back into z to eliminate a and b pde = subs(z, a, a_val); pde = subs(pde, b, b_val); Output: EV0000871 · EV0000872 · EV0000873
4. Source slide page 4 (extracted text): Form the PDE by eliminating arbitrary constants: z = (x + a) (y + b) %% Step 1: Define symbolic variables syms x y a b p q %% Step 2: Define the surface equation z = (x + a)*(y + b); %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for arbitrary constants a and b b_val = solve(eq1, b); % Solve for (y+b) from p a_val = solve(eq2, a); % Solve for (x-a) from q %% Step 5: Substitute back into z to eliminate a and b pde = subs(z, a, a_val); pde = subs(pde, b, b_val) Output: EV0000874 · EV0000875 · EV0000876

5. Source slide page 5 (extracted text): %% Step 1: Define symbolic variables syms x y a b p q %% Step 2: Define the surface equation z = a*x+b*y+a^2+b^2; %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for arbitrary constants a and b a_val = solve(eq1, a); % Solve for (y+b) from p b_val = solve(eq2, b); % Solve for (x-a) from q %% Step 5: Substitute back into z to eliminate a and b pde = subs(z, a, a_val); pde = subs(pde, b, b_val) Output: EV0000877 · EV0000878 · EV0000879
6. Source slide page 6 (extracted text): %% Step 1: Define symbolic variables syms x y a b p q %% Step 2: Define the surface equation z = a*x+b*y+a^2+b^2; %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for arbitrary constants a and b a_val = solve(eq1, a); % Solve for (y+b) from p b_val = solve(eq2, b); % Solve for (x-a) from q %% Step 5: Substitute back into z to eliminate a and b pde = subs(z, a, a_val); pde = subs(pde, b, b_val) Output: EV0000880 · EV0000881 · EV0000882
7. Source slide page 7 (extracted text): %% Step 1: Define symbolic variables syms x y f(u) u(x,y) p q %% Step 2: Define the surface equation u(x,y) = x^2 + y^2; z = x*y + f(u); %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for function f'(u) f1 = (p - y)/2*x; % Calculating f' from eq1 f2 = (q - x)/2*y; % Calculating f' from eq2 %% Step 5: Equating both expressions pde = simplify(f1 == f2) Output: EV0000883 · EV0000884 · EV0000885
8. Source slide page 8 (extracted text): %% Step 1: Define symbolic variables syms x y f(u) u(x,y) p q %% Step 2: Define the surface equation u(x,y) = x^2 + y^2; z = f(u); %% Step 3: Compute partial derivatives eq1 = p == diff(z, x); % p = dz/dx eq2 = q == diff(z, y); % q = dz/dy %% Step 4: Solve for function f'(u) f1 = p/2*x; % Calculating f' from eq1 f2 = q/2*y; % Calculating f' from eq2 %% Step 5: Equating both expressions pde = simplify(f1 == f2) Output: EV0000886 · EV0000887 · EV0000888
9. Source slide page 9 (extracted text): %% Step 1: Define symbolic variables syms x a t f(u) g(v) %% Step 2: Define the surface equation u = x + a*t; v = x - a*t; z = f(u) + g(v); %% Step 3: Compute first order partial derivatives z_x = diff(z, x); % dz/dx z_t = diff(z, t); % dz/dt %% Step 4: Compute second order partial derivatives z_xx = diff(z, x, 2); % d2z/dx2 z_tt = diff(z, t, 2); % d2z/dt2 %% Step 5: Equating both expressions pde = simplify(z_tt == a^2 * z_xx); PDE = sym('z_tt') == a^2 * sym('z_xx') Output: EV0000889 · EV0000890 · EV0000891
10. Source slide page 10 (extracted text): Separation of Variables = = (equation) Symbolic eq object | PDE = diff(u, t) = = diff(u, x, 2) | Creates equation, Not assignment subs substitution | Subs(PDE, u, X(x)*T(t)) | Replaces u with X(x)*T(t) EV0000892 · EV0000893 · EV0000894
11. Source slide page 11 (extracted text): %% Step 1: Define symbolic variables syms x t k lambda u(x, t) X(x) T(t) %% Step 2: Define the PDE PDE = (1/k) * diff(u, t) == diff(u, x, 2); %% Step 3: Using separation of Variables u(x, t) = X(x) * T(t); %% Step 4: Substituting u(x,t) and dividing the PDE with u(x,t) PDE_sep = subs(PDE, u, X(x)*T(t)); PDE_sep = PDE_sep / (X(x) * T(t)); %% Step 5: Separating the two ODE's obtained (1/k) * diff(T, t) / T == diff(X, x, 2) / X; eqT = (1/k) * diff(T, t) == lambda * T; eqX = diff(X, x, 2) == lambda * X; %% Step 6: Solving the two ODE's T_sol = dsolve(eqT) X_sol = dsolve(eqX) %% Step 7: Combining the two solutions u_sol = simplify(X_sol * T_sol) EV0000895 · EV0000896 · EV0000897

12. Source slide page 12 (extracted text): Write code for symbolic solution of the 1D Heat Equation Using Separation of Variables Output: EV0000898 · EV0000899 · EV0000900

13. Source slide page 13 (extracted text): Write code for solution of PDE using Separation of Variables %% Step 1: Define symbolic variables syms x y lambda u(x, y) X(x) Y(y) %% Step 2: Define the PDE PDE = 3 * diff(u, x) + 2 * diff(u, y) == 0; %% Step 3: Using separation of Variables u(x, y) = X(x) * Y(y); %% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y) PDE_sep = subs(PDE, u, X(x) * Y(y)); PDE_sep = PDE_sep / (X(x) * Y(y)); %% Step 5: Separating the two ODE's obtained - 2 * diff(Y, y) / Y == 3 * diff(X, x) / X; eqY = - 2 * diff(Y, y) == lambda * Y; eqX = 3 * diff(X, x) == lambda * X; %% Step 6: Solving the two ODE's Y_sol = dsolve(eqY) X_sol = dsolve(eqX) %% Step 7: Combining the two solutions u_sol = simplify(X_sol * Y_sol) EV0000901 · EV0000902 · EV0000903

14. Source slide page 14 (extracted text): Write code for solution of PDE using Separation of Variables Output: EV0000904 · EV0000905 · EV0000906

15. Source slide page 15 (extracted text): Write code for solution of PDE using Separation of Variables %% Step 1: Define symbolic variables syms x t lambda u(x, t) X(x) T(t) %% Step 2: Define the PDE PDE = 5 * diff(u, x) - 4 * diff(u, t) == 0; %% Step 3: Using separation of Variables u(x, t) = X(x) * T(t); %% Step 4: Substituting u(x,t) and dividing the PDE with u(x,t) PDE_sep = subs(PDE, u, X(x) * T(t)); PDE_sep = PDE_sep / (X(x) * T(t)); %% Step 5: Separating the two ODE's obtained 5 * diff(X, x) / X == 4 * diff(T, t) / T; eqX = 5 * diff(X, x) == lambda * X; eqT = 4 * diff(T, t) == lambda * T; %% Step 6: Solving the two ODE's X_sol = dsolve(eqX) T_sol = dsolve(eqT) %% Step 7: Combining the two solutions u_sol = simplify(X_sol * T_sol) EV0000907 · EV0000908 · EV0000909

16. Source slide page 16 (extracted text): Write code for solution of PDE using Separation of Variables Output: EV0000910 · EV0000911 · EV0000912

17. Source slide page 17 (extracted text): %% Step 1: Define symbolic variables syms x y lambda u(x, y) X(x) Y(y) %% Step 2: Define the PDE PDE = diff(u, x, 2) - 2 * diff(u, x) + diff(u, y) == 0; %% Step 3: Using separation of Variables u(x, y) = X(x) * Y(y); %% Step 4: Substituting u(x,y) and dividing the PDE with u(x,y) PDE_sep = subs(PDE, u, X(x) * Y(y)); PDE_sep = PDE_sep / (X(x) * Y(y)); %% Step 5: Separating the two ODE's obtained - diff(Y, y) / Y == diff(X, x, 2) - 2 * diff(X, x) / X; eqY = diff(Y, y) == -lambda * Y; eqX = diff(X, x, 2) - 2 * diff(X, x) == lambda * X; %% Step 6: Solving the two ODE's Y_sol = dsolve(eqY) X_sol = dsolve(eqX) %% Step 7: Combining the two solutions u_sol = simplify(X_sol * Y_sol) EV0000913 · EV0000914 · EV0000915

18. Source slide page 18 (extracted text): Output: EV0000916 · EV0000917 · EV0000918

19. Source slide page 19 (extracted text): Partial Differential Equations syms x y C1 C2 z = C1*exp(2*y+x) + C2*exp(3*y+x); disp(z) f = matlabFunction(z,'Vars',[x y C1 C2]); [X,Y]=meshgrid(-2:0.2:2); Z=f(X,Y,2,1); surf(X,Y,Z) shading interp xlabel('x') ylabel('y') zlabel('Deflection') title('Elastic Plate Solution') colorbar EV0000919 · EV0000920 · EV0000921

20. Source slide page 20 (extracted text): Partial Differential Equations OUTPUT [Image text, OCR] Cex+2y + C2 ex+3 y ElasticPlateSolution 3500 4000 3000 3000 2500 2000 2000 D 1000 1500 0 1000 2 2 500 0 y 2 -2 X EV0000922 · EV0000923 · EV0000924

21. Source slide page 21 (extracted text): Partial Differential Equations MATLAB cannot automatically compute the Particular Integral (PI) using the operator method EV0000925 · EV0000926 · EV0000927

22. Source slide page 22 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU
EV0000928 · EV0000929 · EV0000930

SHORT-NOTE RECAP

Source deck preserved: UNIT 3. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.
EV0000865

14 Source slide deck: UNIT 3 CONTINUED

This entry preserves the complete slide deck titled UNIT 3 CONTINUED as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000931 · EV0000932

Graphs and diagrams

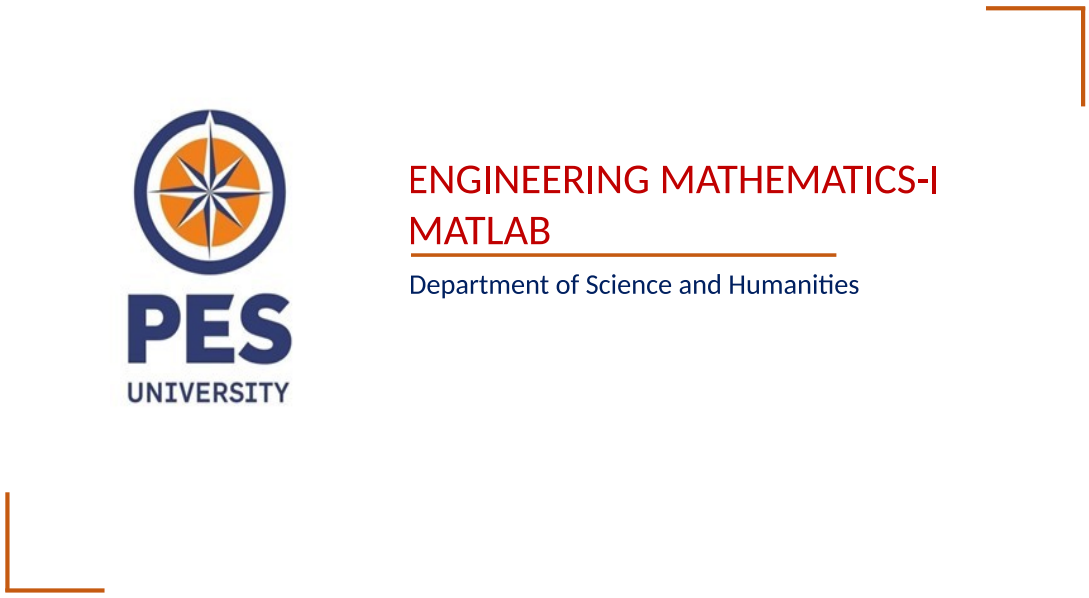

Figure G000280. UNIT 3 CONTINUED · source slide page 1.
EV0000932

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Applications of Partial Differential Equation:

Equations	Applications	PDE
Heat Equation (one dimensional)	Temperature flow	$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$
Wave equation	Vibration and signals	
Laplace equation	Equilibrium systems	
Diffusion equations	Spread of substances	
Traffic PDE	Movement of vehicles	
Weather PDE	Climate and forecasting	

Figure G000281. UNIT 3 CONTINUED · source slide page 2.

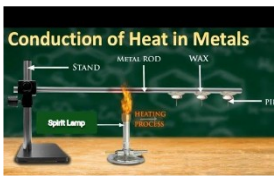
EV0000935

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

One-Dimensional Heat Equation

The heat equation is a partial differential equation that models the distribution and flow of heat (or temperature) in a given region over time.

lone-dimensional heat equation helps to find distribution of heat in metal rod, cooling of electronic devices, heat flow in buildings, etc,.



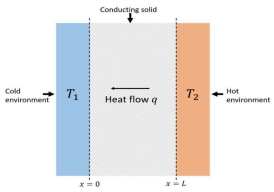


Figure G000282. UNIT 3 CONTINUED · source slide page 3.

EV0000938

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

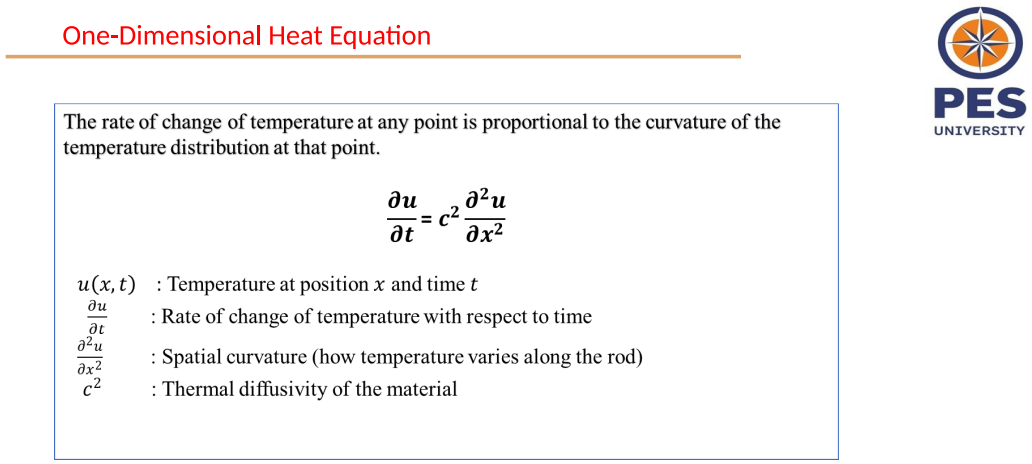


Figure G000283. UNIT 3 CONTINUED · source slide page 4.

EV0000941

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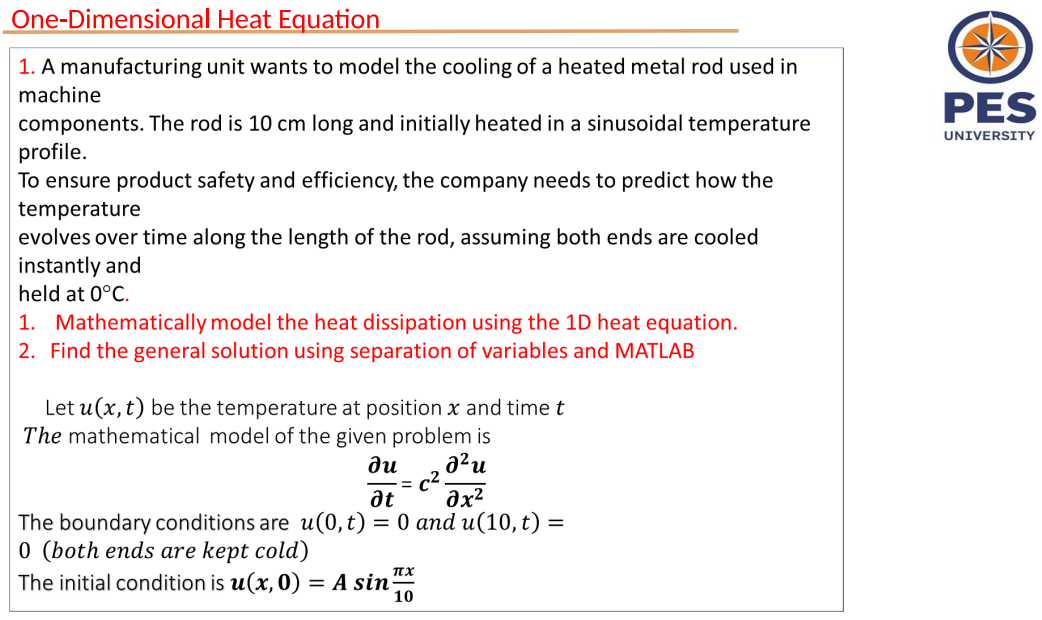

Figure G000284. UNIT 3 CONTINUED · source slide page 5.

EV0000944

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB code for the 1-D heat equation

```
clc;
clear;
syms x t u
syms lambda positive
syms X(x) T(t)
U= X(x)*T(t);
PDE = diff(u,t) == diff(u,x,2);
PDE_sep = simplify(PDE/(X(x)*T(t)));

% Solve for T(t)
eqT = diff(T(t), t) == - lambda*T(t);
T_sol = dsolve(eqT);
disp('T(t) = ')
disp(T_sol)
```

Figure G000285. UNIT 3 CONTINUED · source slide page 6.

EV0000947

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

MATLAB code for the 1-D heat equation

```
% Solve for X(x)
eqX = diff(X(x), x, 2) == - lambda*X(x);
X_sol = dsolve(eqX);

% Final solution
U_sol= simplify(X_sol*T_sol);
disp('U(x, t) = ')
disp(U_sol)

Final Output:
T(t) =
C1*exp(-lambda*t)
X(x) =
C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x)
U(x, t) =
C1*exp(-lambda*t)*(C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x))
```

Figure G000286. UNIT 3 CONTINUED · source slide page 7.

EV0000950

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

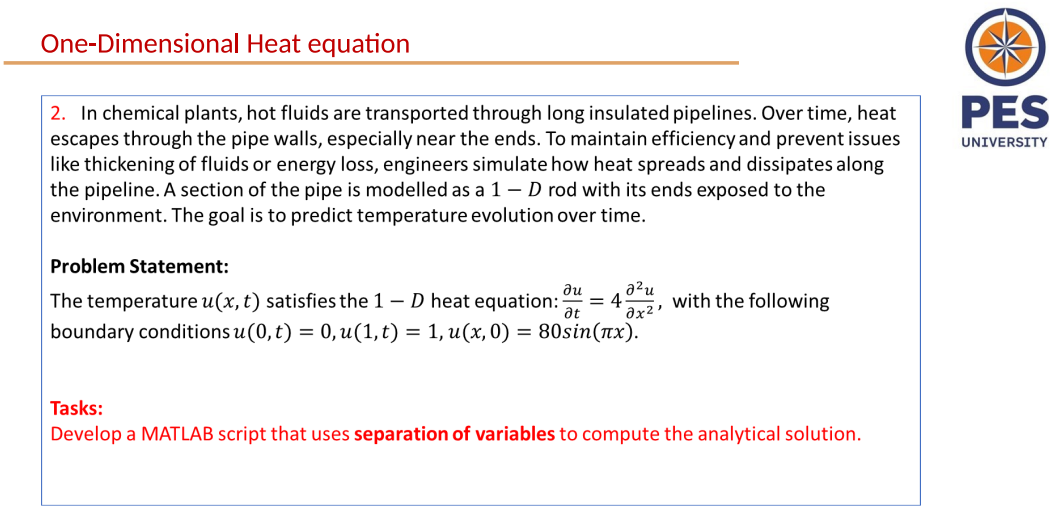

Figure G000287. UNIT 3 CONTINUED · source slide page 8.

EV0000953

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

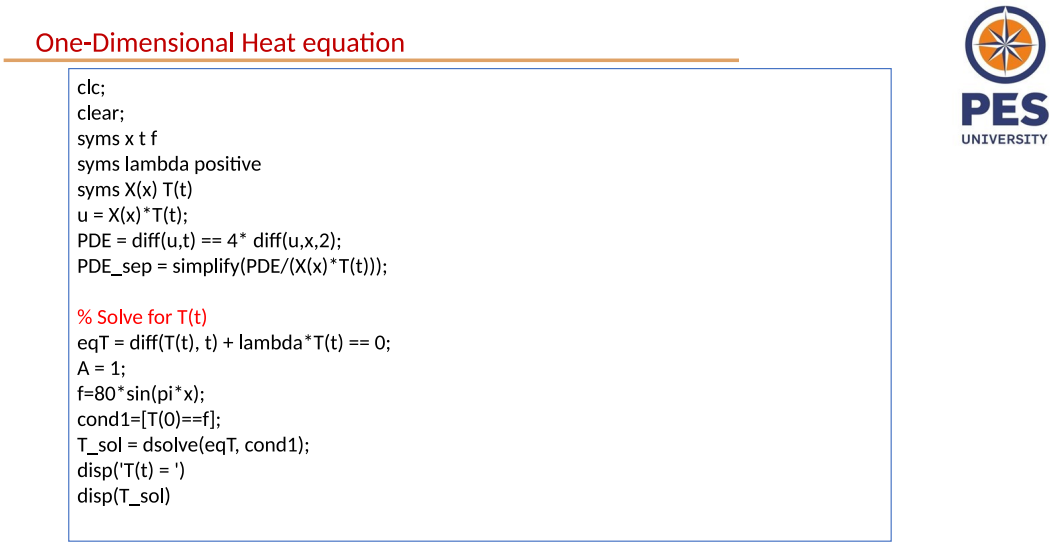

Figure G000288. UNIT 3 CONTINUED · source slide page 9.

EV0000956

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

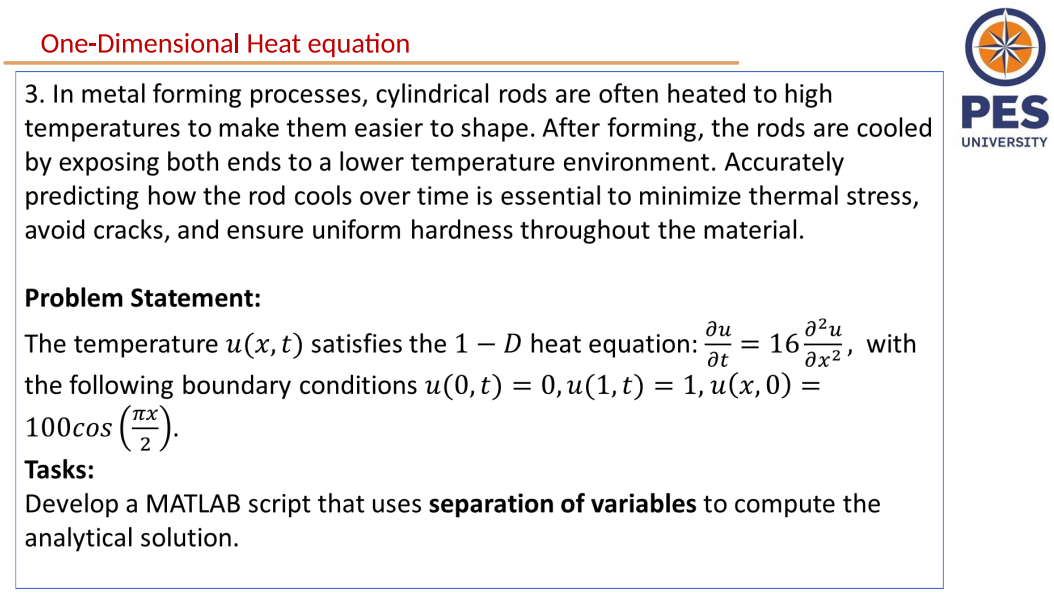

Figure G000289. UNIT 3 CONTINUED · source slide page 10.

EV0000959

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

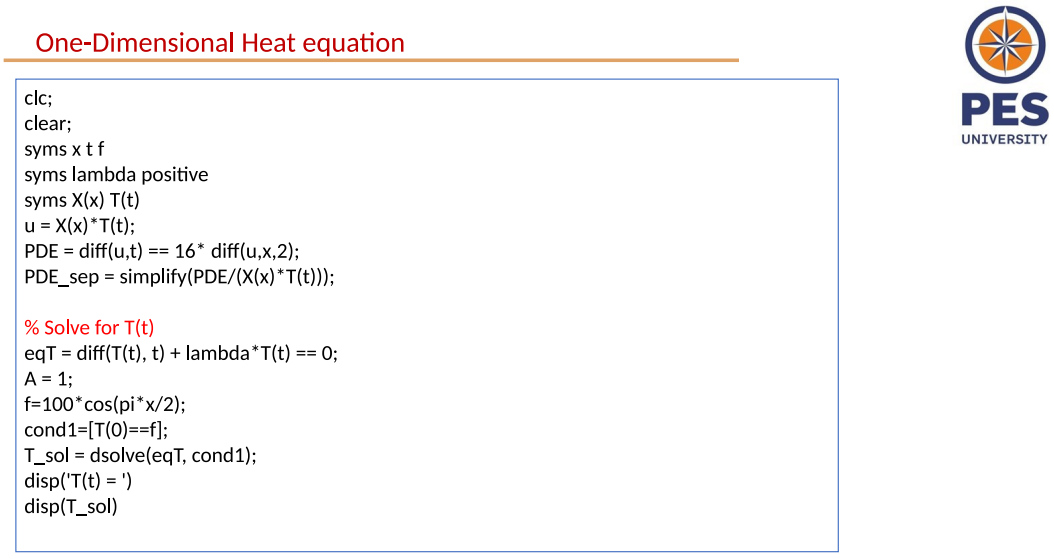

Figure G000290. UNIT 3 CONTINUED · source slide page 11.

EV0000962

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

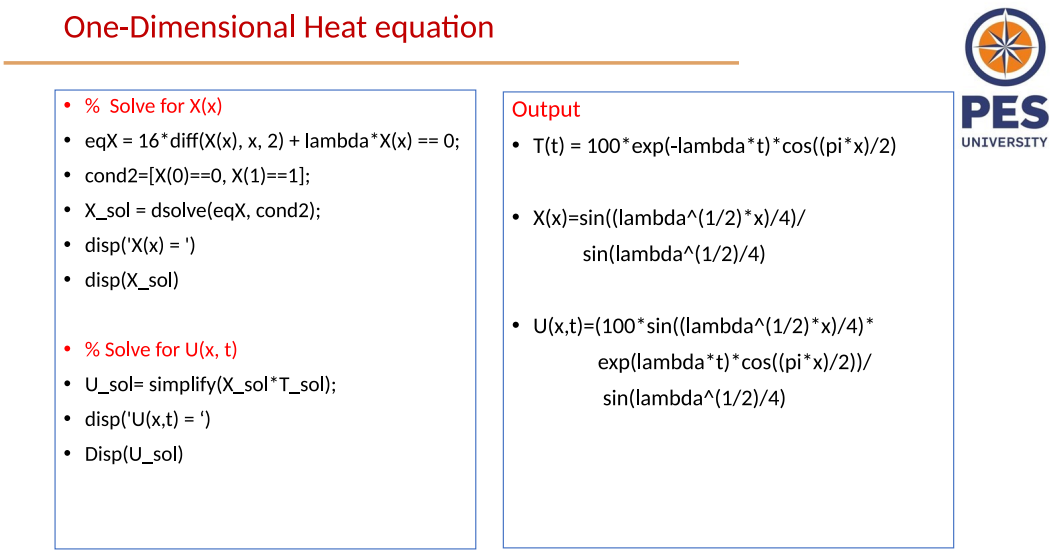

Figure G000291. UNIT 3 CONTINUED · source slide page 12.

EV0000965

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

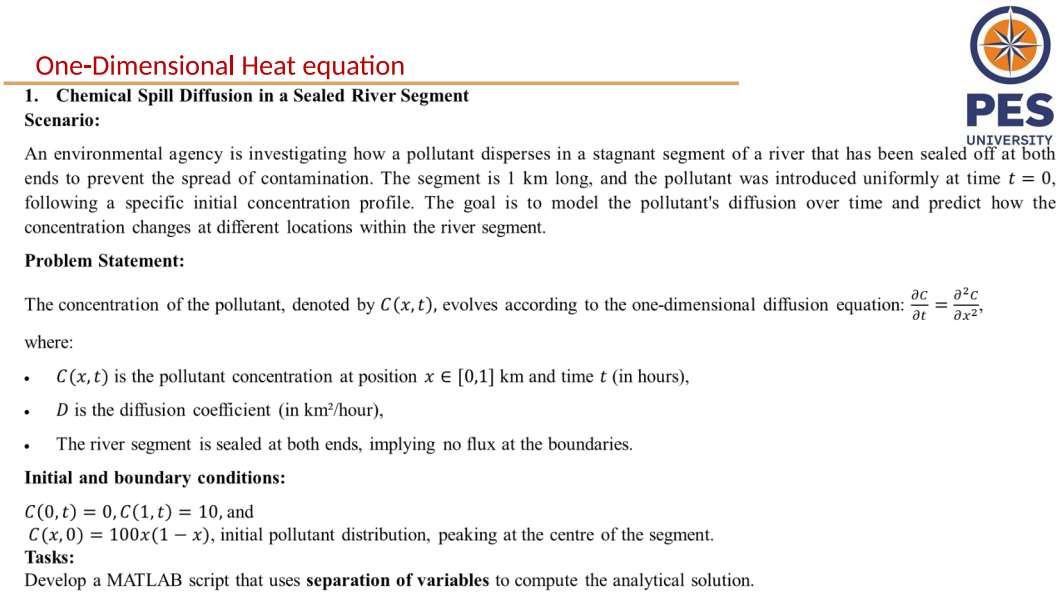

Figure G000292. UNIT 3 CONTINUED · source slide page 13.

EV0000968

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

One-Dimensional Heat equation

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```
syms u(x, t) x t    % Symbolic variables and functions
PDE = diff(u, t) == diff(u, x, 2);
syms X(x) T(t)
u(x, t) = X(x) * T(t);  % Assume u(x,t) = X(x)T(t)
PDE_sep = subs(PDE, u, X(x)*T(t)); % Substitute into
the PDE
PDE_sep = PDE_sep / (X(x) * T(t));
eq1 = diff(T, t) / T == diff(X, x, 2) / X; % Separate
variables
syms lambda
eqT = diff(T, t) == -lambda * T;   % Set equal to
separation constant -lambda
eqX = diff(X, x, 2) == -lambda * X;
% Solve the ODEs
T_sol = dsolve(eqT, T(0) == 100*x*(1-x))% Arbitrary
initial condition for T
X_sol = dsolve(eqX, X(0) == 0, X(1) == 10)
```

Figure G000293. UNIT 3 CONTINUED · source slide page 14.

EV0000971

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

One-Dimensional Heat equation

PES UNIVERSITY

OUTPUT:

$$T_sol = -100\,x\,e^{-\lambda t}\,(x-1)$$

$$X_sol = \frac{10\,e^{-\sqrt{-\lambda}\,x}\,e^{\sqrt{-\lambda}}\,(e^{2\sqrt{-\lambda}\,x}-1)}{e^{2\sqrt{-\lambda}}-1}$$

Figure G000294. UNIT 3 CONTINUED · source slide page 15.

EV0000974

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000295. UNIT 3 CONTINUED · source slide page 16.

EV0000977

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000931 · EV0000932 · EV0000933
2. Source slide page 2 (extracted text): Applications of Partial Differential Equation: EV0000934 · EV0000935 · EV0000936
3. Source slide page 3 (extracted text): One-Dimensional Heat Equation The heat equation is a partial differential equation that models the distribution and flow of heat (or temperature) in a given region over time. Ione-dimensional heat equation helps to find distribution of heat in metal rod, cooling of electronic devices, heat flow in buildings, etc,. [Image text, OCR] Conductingsolid Cold T1 T2 Hot Heat flow q environment environment x=0 x=L EV0000937 · EV0000938 · EV0000939
4. Source slide page 4 (extracted text): One-Dimensional Heat Equation EV0000940 · EV0000941 · EV0000942
5. Source slide page 5 (extracted text): One-Dimensional Heat Equation EV0000943 · EV0000944 · EV0000945
6. Source slide page 6 (extracted text): MATLAB code for the 1-D heat equation clc; clear; syms x t u syms lambda positive syms X(x) T(t) U= X(x)*T(t); PDE = diff(u,t) == diff(u,x,2); PDE_sep = simplify(PDE/(X(x)*T(t))); % Solve for T(t) eqT = diff(T(t), t) == - lambda*T(t); T_sol = dsolve(eqT); disp('T(t) = ') disp(T_sol) EV0000946 · EV0000947 · EV0000948

7. Source slide page 7 (extracted text): MATLAB code for the 1-D heat equation % Solve for X(x) eqX = diff(X(x), x, 2) == - lambda*X(x); X_sol = dsolve(eqX); % Final solution U_sol= simplify(X_sol*T_sol); disp('U(x, t) = ') disp(U_sol) Final Output: T(t) = C1*exp(-lambda*t) X(x) = C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x) U(x, t) = C1*exp(-lambda*t)* (C1*cos(lambda^(1/2)*x) - C2*sin(lambda^(1/2)*x)) EV0000949 · EV0000950 · EV0000951
8. Source slide page 8 (extracted text): One-Dimensional Heat equation EV0000952 · EV0000953 · EV0000954
9. Source slide page 9 (extracted text): One-Dimensional Heat equation clc; clear; syms x t f syms lambda positive syms X(x) T(t) u = X(x)*T(t); PDE = diff(u,t) == 4* diff(u,x,2); PDE_sep = simplify(PDE/(X(x)*T(t))); % Solve for T(t) eqT = diff(T(t), t) + lambda*T(t) == 0; A = 1; f=80*sin(pi*x); cond1=[T(0)==f]; T_sol = dsolve(eqT, cond1); disp('T(t) = ') disp(T_sol) EV0000955 · EV0000956 · EV0000957
10. Source slide page 10 (extracted text): One-Dimensional Heat equation EV0000958 · EV0000959 · EV0000960
11. Source slide page 11 (extracted text): One-Dimensional Heat equation clc; clear; syms x t f syms lambda positive syms X(x) T(t) u = X(x)*T(t); PDE = diff(u,t) == 16* diff(u,x,2); PDE_sep = simplify(PDE/(X(x)*T(t))); % Solve for T(t) eqT = diff(T(t), t) + lambda*T(t) == 0; A = 1; f=100*cos(pi*x/2); cond1=[T(0)==f]; T_sol = dsolve(eqT, cond1); disp('T(t) = ') disp(T_sol) EV0000961 · EV0000962 · EV0000963
12. Source slide page 12 (extracted text): One-Dimensional Heat equation % Solve for X(x) eqX = 16*diff(X(x), x, 2) + lambda*X(x) == 0; cond2=[X(0)==0, X(1)==1]; X_sol = dsolve(eqX, cond2); disp('X(x) = ') disp(X_sol) % Solve for U(x, t) U_sol= simplify(X_sol*T_sol); disp('U(x,t) = ‘) Disp(U_sol) Output T(t) = 100*exp(-lambda*t)*cos((pi*x)/2) X(x)=sin((lambda^(1/2)*x)/4)/ sin(lambda^(1/2)/4) U(x,t)=(100*sin((lambda^(1/2)*x)/4)* exp(lambda*t)*cos((pi*x)/2))/ sin(lambda^(1/2)/4) EV0000964 · EV0000965 · EV0000966
13. Source slide page 13 (extracted text): One-Dimensional Heat equation EV0000967 · EV0000968 · EV0000969
14. Source slide page 14 (extracted text): One-Dimensional Heat equation syms u(x, t) x t % Symbolic variables and functions PDE = diff(u, t) == diff(u, x, 2); syms X(x) T(t) u(x, t) = X(x) * T(t); % Assume u(x,t) = X(x)T(t) PDE_sep = subs(PDE, u, X(x)*T(t)); % Substitute into the PDE PDE_sep = PDE_sep / (X(x) * T(t)); eq1 = diff(T, t) / T == diff(X, x, 2) / X; % Separate variables syms lambda eqT = diff(T, t) == -lambda * T; % Set equal to separation constant - lambda eqX = diff(X, x, 2) == -lambda * X; % Solve the ODEs T_sol = dsolve(eqT, T(0) == 100*x*(1-x))% Arbitrary initial condition for T X_sol = dsolve(eqX, X(0) == 0, X(1) == 10) EV0000970 · EV0000971 · EV0000972
15. Source slide page 15 (extracted text): One-Dimensional Heat equation OUTPUT: T_sol = X_sol = EV0000973 · EV0000974 · EV0000975
16. Source slide page 16 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU EV0000976 · EV0000977 · EV0000978

SHORT-NOTE RECAP

Source deck preserved: UNIT 3 CONTINUED. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000931

15 Source slide deck: UNIT 4

This entry preserves the complete slide deck titled UNIT 4 as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0000979 · EV0000980

Graphs and diagrams

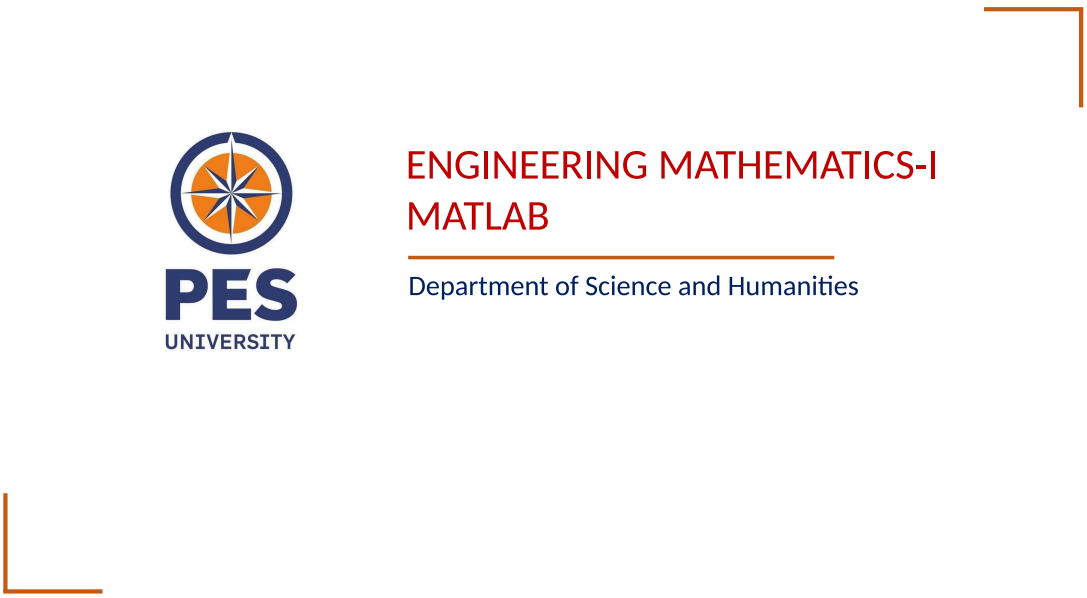

Figure G000296. UNIT 4 · source slide page 1.

EV0000980

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

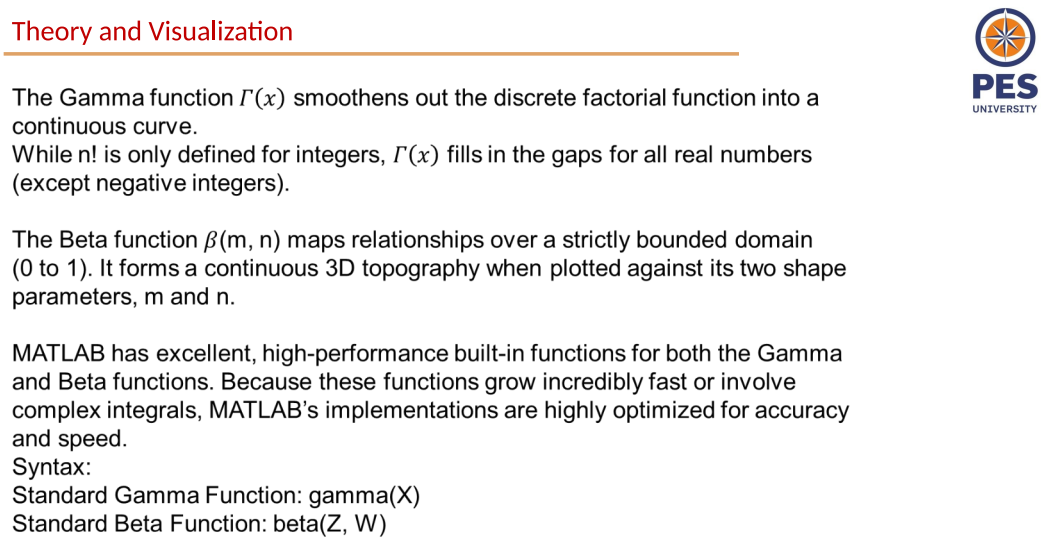

Figure G000297. UNIT 4 · source slide page 2.

EV0000983

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

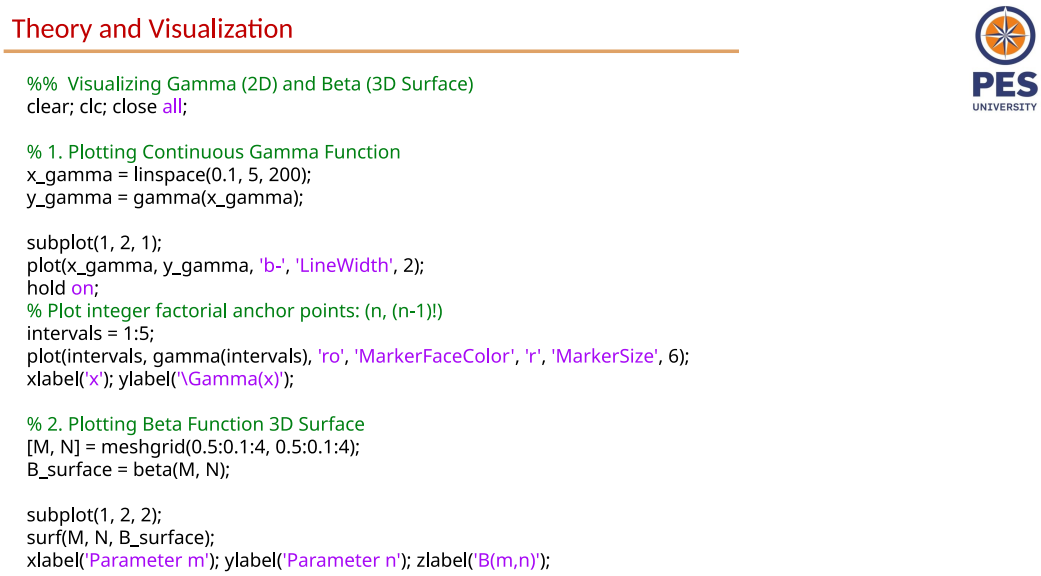

Figure G000298. UNIT 4 · source slide page 3.

EV0000986

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Theory and Visualization

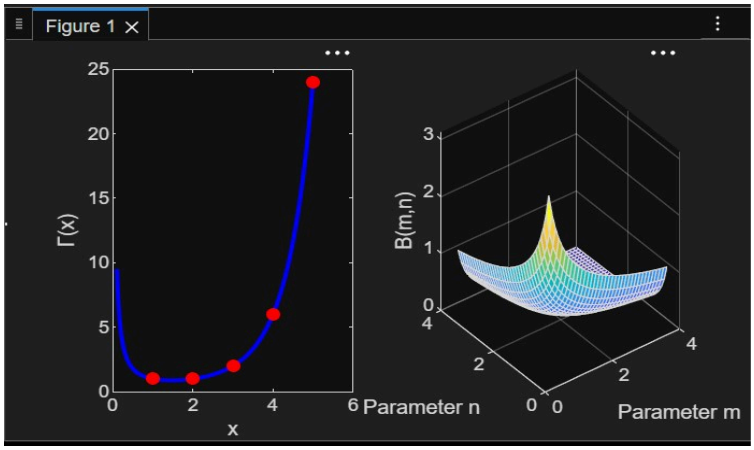

Figure G000299. UNIT 4 · source slide page 4.

EV0000989

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Evaluation of Gamma function

1. gamma(0.5)

ans = 1.7725

2. gamma(1)

ans = 1

3. gamma(3/2)

ans = 0.8862

4. gamma(7/2)

ans = 3.3234

5. gamma(1/4)*gamma(3/4)

ans = 4.4429

Figure G000300. UNIT 4 · source slide page 5.

EV0000992

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

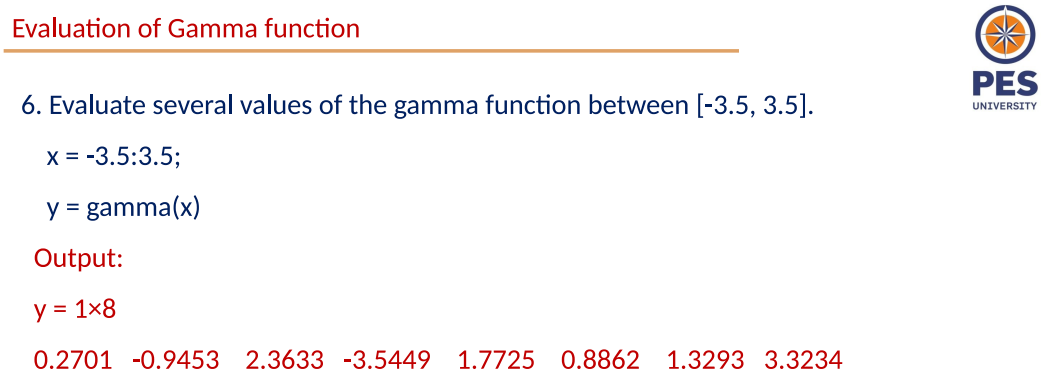

Figure G000301. UNIT 4 · source slide page 6.

EV0000995

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

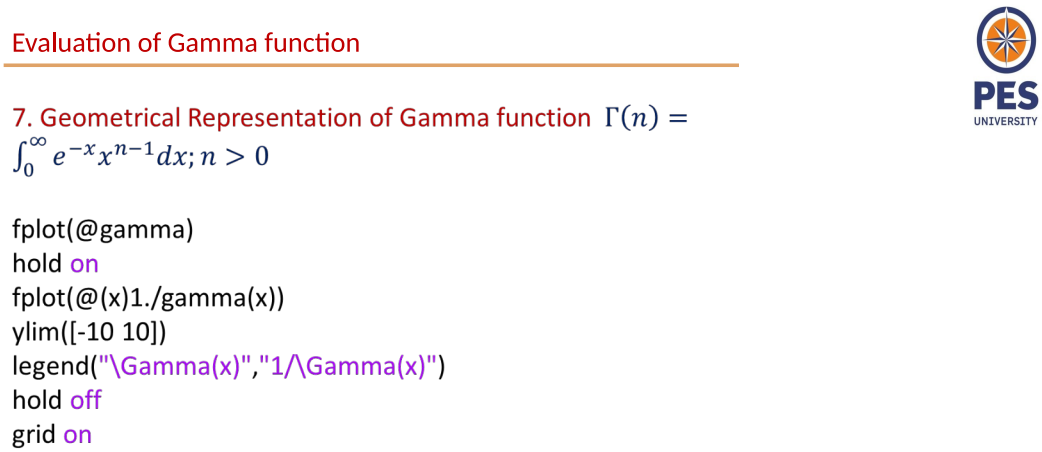

Figure G000302. UNIT 4 · source slide page 7.

EV0000998

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Gamma function

Geometrical representation of Gamma function:

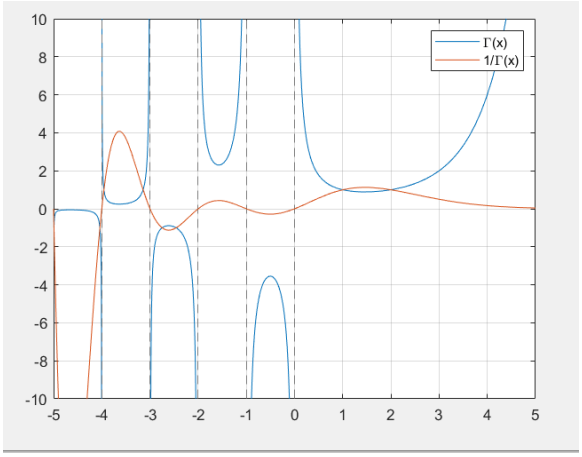

Figure G000303. UNIT 4 · source slide page 8.

EV0001001

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Evaluation of Beta function

1. beta(2,3)
ans = 0.0833

2. beta(2,5)
ans = 0.0333

3. beta(0,1)
ans = Inf

4. beta(3,4)
ans=0.0167

Figure G000304. UNIT 4 · source slide page 9.

EV0001004

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

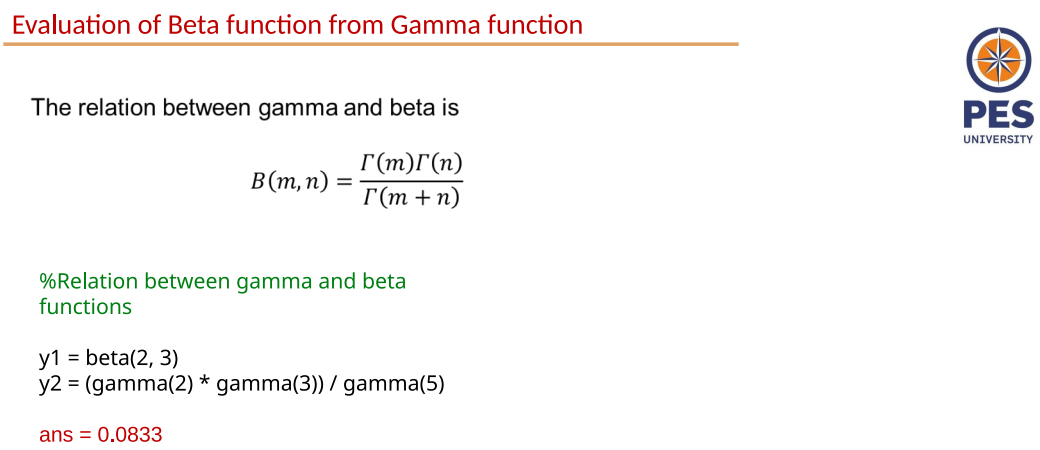

Figure G000305. UNIT 4 · source slide page 10.

EV0001007

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

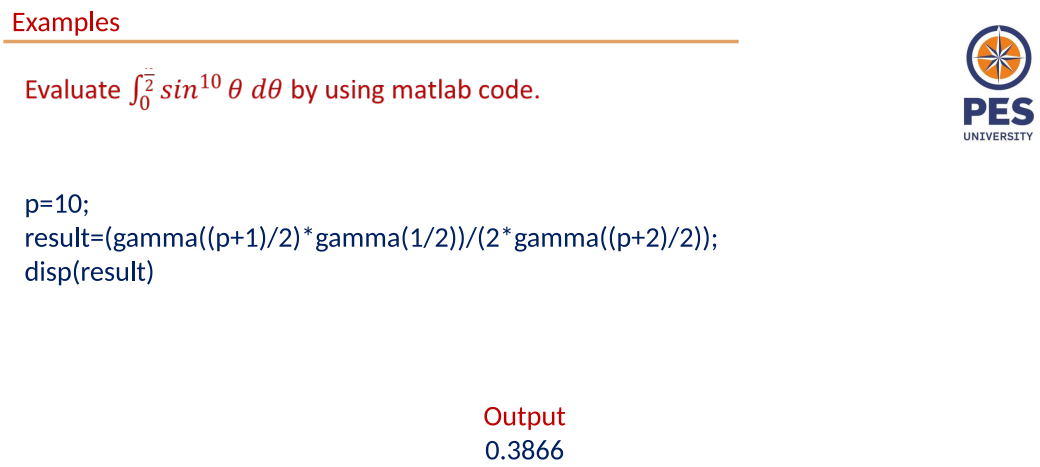

Figure G000306. UNIT 4 · source slide page 11.

EV0001010

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

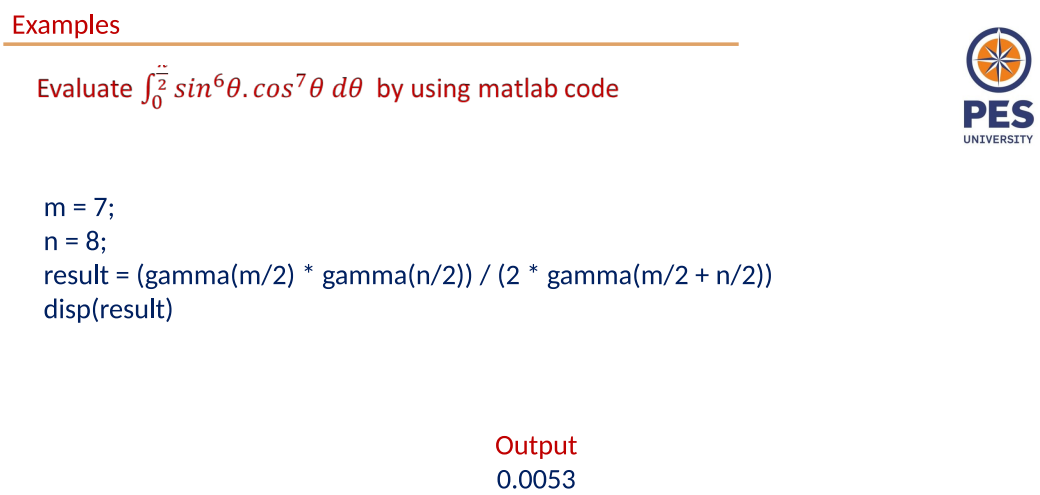

Figure G000307. UNIT 4 · source slide page 12.

EV0001013

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

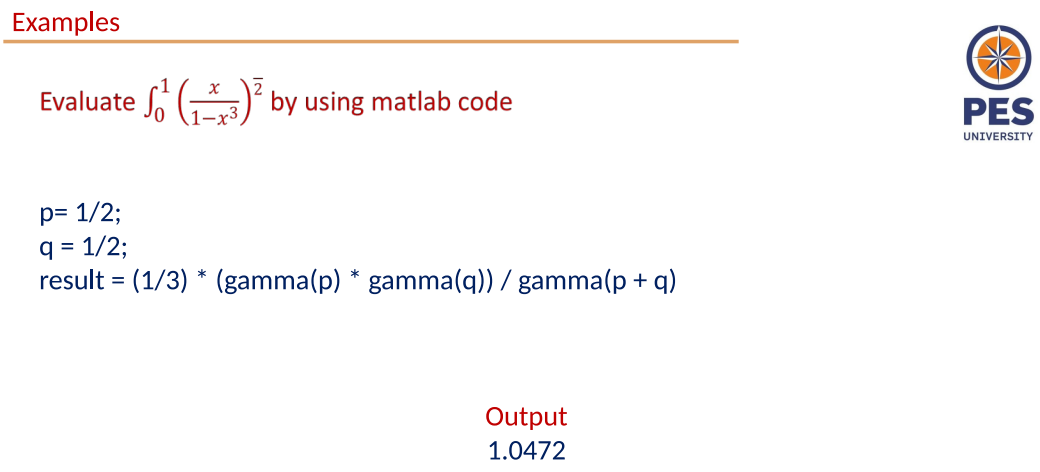

Figure G000308. UNIT 4 · source slide page 13.

EV0001016

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem 1

The density of a thin rod is given by $\rho(x) = x^{-\frac{5}{2}}e^{-x}$ for $x > 0$. Write a MATLAB to find the mass of the rod.
Answer :

density = mass* volume= $\int_0^\infty x^{-\frac{5}{2}}e^{-x}dx = \Gamma\left(\frac{7}{2}\right)$

Figure G000309. UNIT 4 · source slide page 14.

EV0001019

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Problem 1

```
% Define the integrand
f = @(x) x.^(5/2).*exp(-x);
% Compute mass using Gamma function
M_gamma = gamma(7/2);
```

OR

```
% Define the integrand
syms x
f = x^(5/2)*exp(-x)
% Area using MATLAB beta function
Mass = gamma(7/2)
```

OUTPUT:

$f = x^{5/2}e^{-x}$

```
Mass = 3.3234
```

Figure G000310. UNIT 4 · source slide page 15.

EV0001022

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

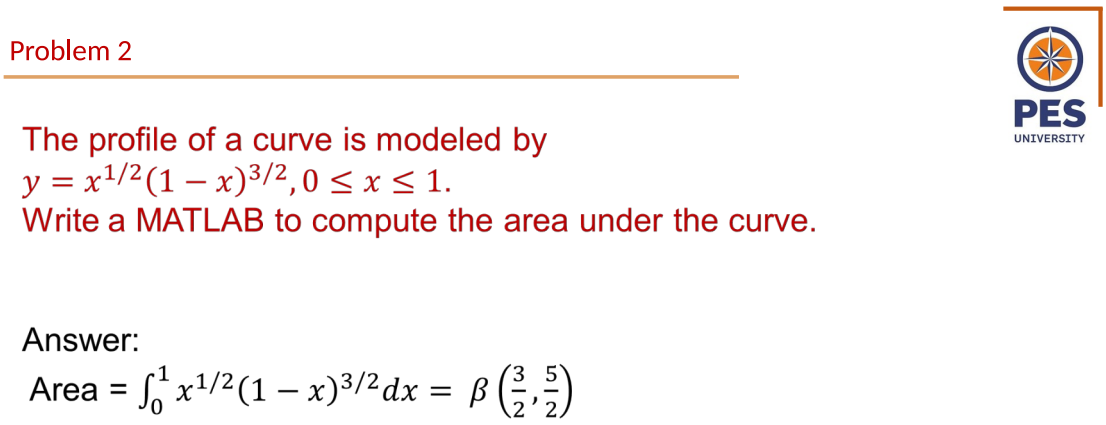

Figure G000311. UNIT 4 · source slide page 16.

EV0001025

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

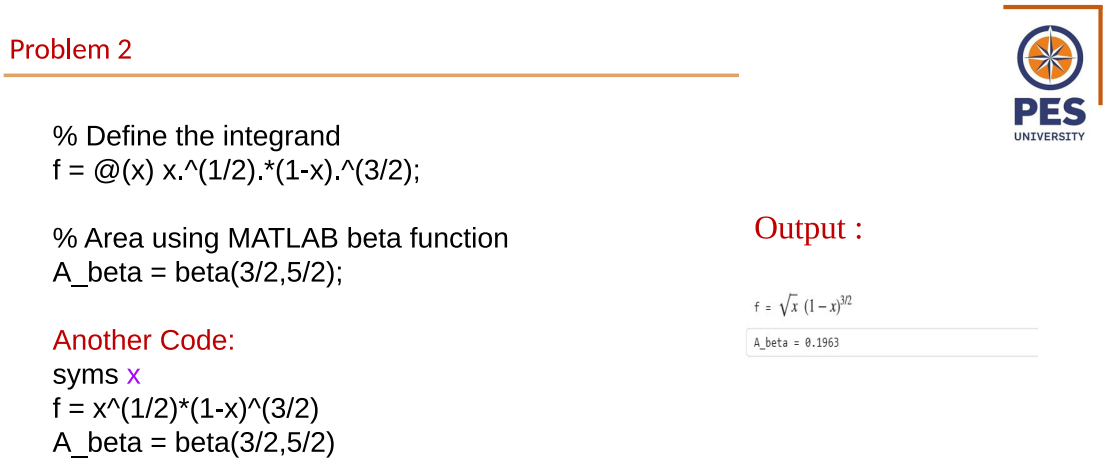

Figure G000312. UNIT 4 · source slide page 17.

EV0001028

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000313. UNIT 4 · source slide page 18.

EV0001031

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0000979 · EV0000980 · EV0000981
2. Source slide page 2 (extracted text): Theory and Visualization EV0000982 · EV0000983 · EV0000984
3. Source slide page 3 (extracted text): Theory and Visualization %% Visualizing Gamma (2D) and Beta (3D Surface) clear; clc; close all; % 1. Plotting Continuous Gamma Function x_gamma = linspace(0.1, 5, 200); y_gamma = gamma(x_gamma); subplot(1, 2, 1); plot(x_gamma, y_gamma, 'b-', 'LineWidth', 2); hold on; % Plot integer factorial anchor points: (n, (n-1)!) intervals = 1:5; plot(intervals, gamma(intervals), 'ro', 'MarkerFaceColor', 'r', 'MarkerSize', 6); xlabel('x'); ylabel('\Gamma(x)'); % 2. Plotting Beta Function 3D Surface [M, N] = meshgrid(0.5:0.1:4, 0.5:0.1:4); B_surface = beta(M, N); subplot(1, 2, 2); surf(M, N, B_surface); xlabel('Parameter m'); ylabel('Parameter n'); zlabel('B(m,n)'); EV0000985 · EV0000986 · EV0000987
4. Source slide page 4 (extracted text): Theory and Visualization [Image text, OCR] Figure 1 x 25 20 15 (u'w)g 2 10 D 5 4 2 2 2 4 6Parametern 00 Parameter m X EV0000988 · EV0000989 · EV0000990
5. Source slide page 5 (extracted text): Evaluation of Gamma function gamma(0.5) ans = 1.7725 2. gamma(1) ans = 1 3. gamma(3/2) ans = 0.8862 4. gamma(7/2) ans = 3.3234 5. gamma(1/4)*gamma(3/4) ans = 4.4429 EV0000991 · EV0000992 · EV0000993
6. Source slide page 6 (extracted text): Evaluation of Gamma function 6. Evaluate several values of the gamma function between [-3.5, 3.5]. x = -3.5:3.5; y = gamma(x) Output: y = 1×8 0.2701 -0.9453 2.3633 -3.5449 1.7725 0.8862 1.3293 3.3234 EV0000994 · EV0000995 · EV0000996

7. Source slide page 7 (extracted text): Evaluation of Gamma function EV0000997 · EV0000998 · EV0000999
8. Source slide page 8 (extracted text): Gamma function Geometrical representation of Gamma function: [Image text, OCR] 10 F(x) 8 1/T(x) 6 4 2 0 -2 -4 -6 -8 -10 -5 -4 -3 -2 -1 0 1 2 3 4 5 EV0001000 · EV0001001 · EV0001002
9. Source slide page 9 (extracted text): Evaluation of Beta function 1. beta(2,3) ans = 0.0833 2. beta(2,5) ans = 0.0333 3. beta(0,1) ans = Inf 4. beta(3,4) ans=0.0167 EV0001003 · EV0001004 · EV0001005
10. Source slide page 10 (extracted text): Evaluation of Beta function from Gamma function %Relation between gamma and beta functions y1 = beta(2, 3) y2 = (gamma(2) * gamma(3)) / gamma(5) ans = 0.0833 EV0001006 · EV0001007 · EV0001008
11. Source slide page 11 (extracted text): Examples p=10; result= (gamma((p+1)/2)*gamma(1/2))/(2*gamma((p+2)/2)); disp(result) Output 0.3866 EV0001009 · EV0001010 · EV0001011
12. Source slide page 12 (extracted text): Examples m = 7; n = 8; result = (gamma(m/2) * gamma(n/2)) / (2 * gamma(m/2 + n/2)) disp(result) Output 0.0053 EV0001012 · EV0001013 · EV0001014
13. Source slide page 13 (extracted text): Examples p= 1/2; q = 1/2; result = (1/3) * (gamma(p) * gamma(q)) / gamma(p + q) Output 1.0472 EV0001015 · EV0001016 · EV0001017
14. Source slide page 14 (extracted text): Problem 1 EV0001018 · EV0001019 · EV0001020
15. Source slide page 15 (extracted text): Problem 1 % Define the integrand f = @(x) x.^(5/2).*exp(-x); % Compute mass using Gamma function M_gamma = gamma(7/2); OR % Define the integrand syms x f = x^(5/2)*exp(-x) % Area using MATLAB beta function Mass = gamma(7/2) OUTPUT: EV0001021 · EV0001022 · EV0001023
16. Source slide page 16 (extracted text): Problem 2 EV0001024 · EV0001025 · EV0001026
17. Source slide page 17 (extracted text): Problem 2 % Define the integrand f = @(x) x.^(1/2).*(1-x).^(3/2); % Area using MATLAB beta function A_beta = beta(3/2,5/2); Another Code: syms x f = x^(1/2)*(1-x)^(3/2) A_beta = beta(3/2,5/2) Output : EV0001027 · EV0001028 · EV0001029
18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU EV0001030 · EV0001031 · EV0001032

SHORT-NOTE RECAP

Source deck preserved: UNIT 4. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000979

16 Source slide deck: UNIT 4 CONTINUED

This entry preserves the complete slide deck titled UNIT 4 CONTINUED as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_05, to avoid an unsupported match. EV0001033 · EV0001034

Graphs and diagrams

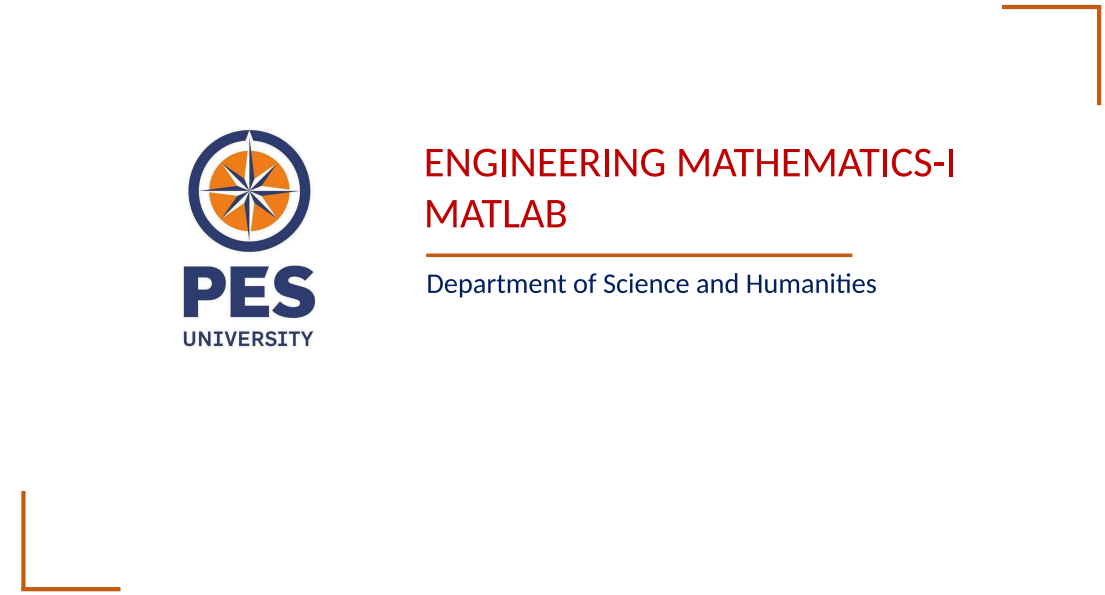

Figure G000314. UNIT 4 CONTINUED · source slide page 1.

EV0001034

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

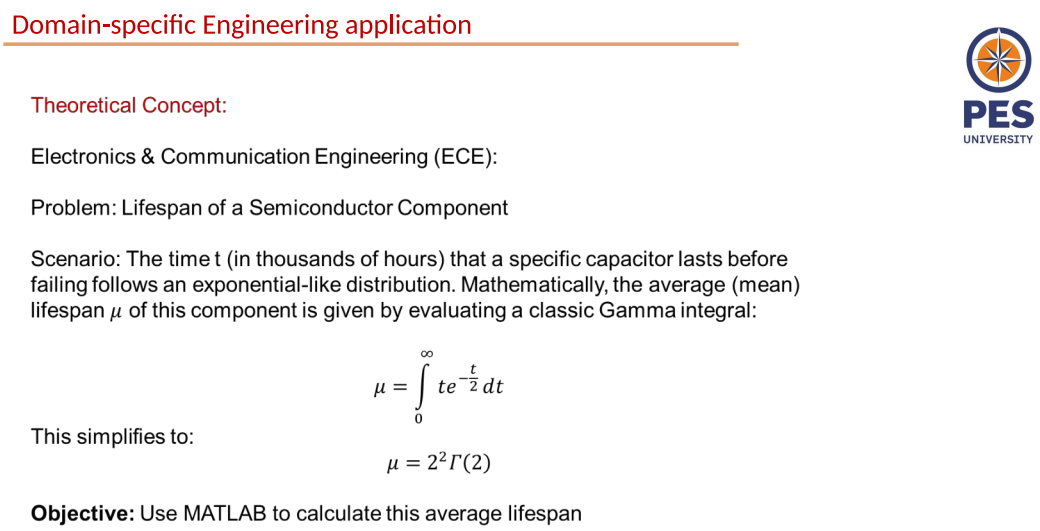

Figure G000315. UNIT 4 CONTINUED · source slide page 2.

EV0001037

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

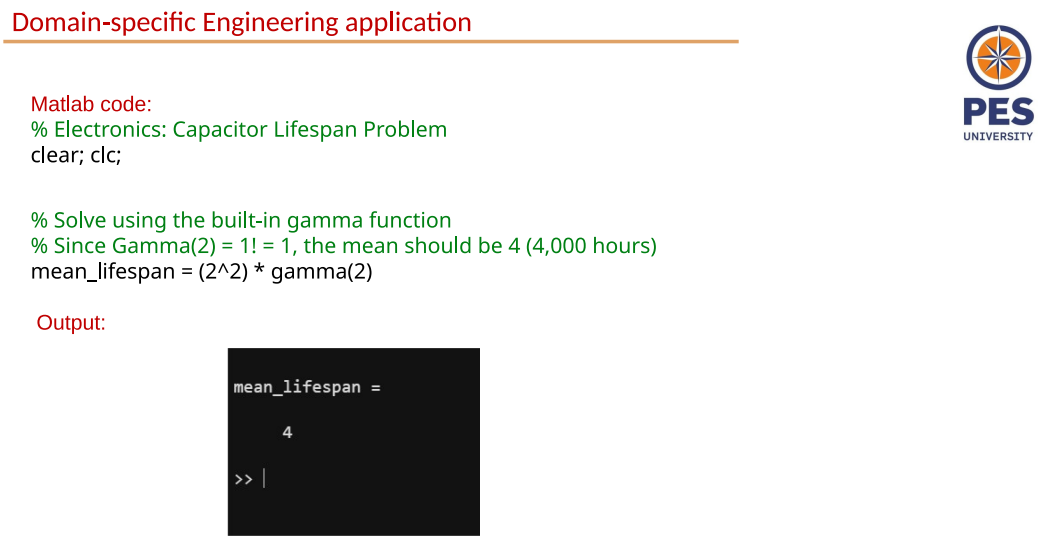

Figure G000316. UNIT 4 CONTINUED · source slide page 3.

EV0001040

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

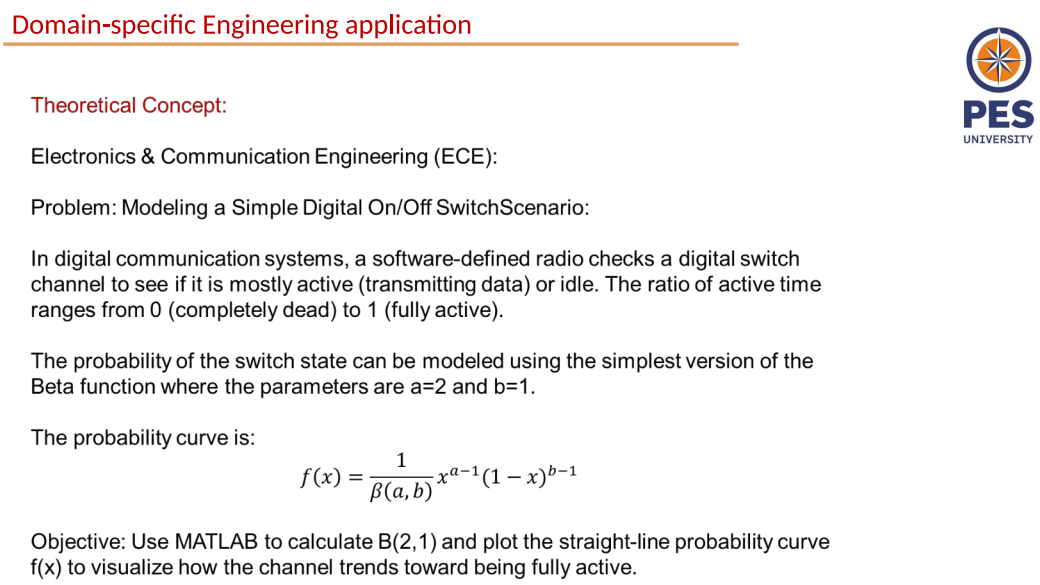

Figure G000317. UNIT 4 CONTINUED · source slide page 4.

EV0001043

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

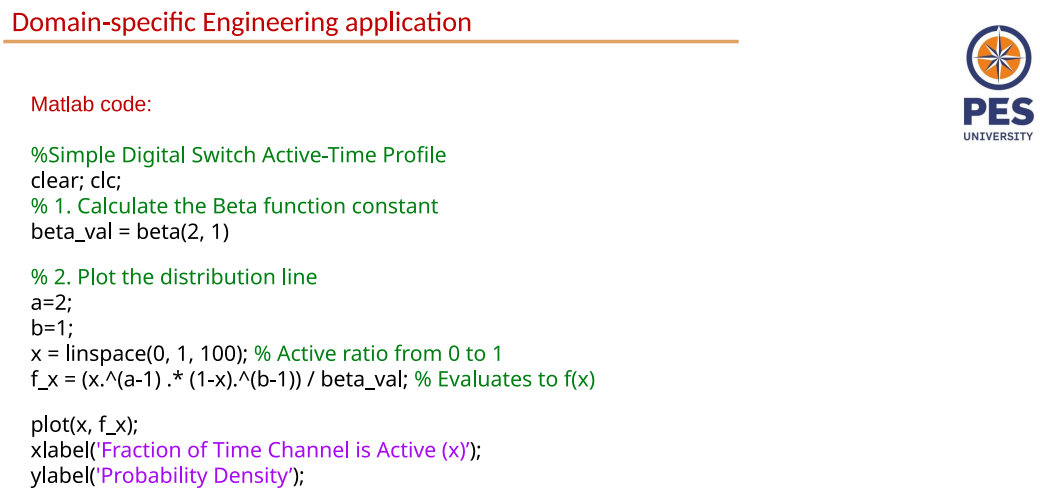

Figure G000318. UNIT 4 CONTINUED · source slide page 5.

EV0001046

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Output:

```
beta_val =

    0.5000

>>
```

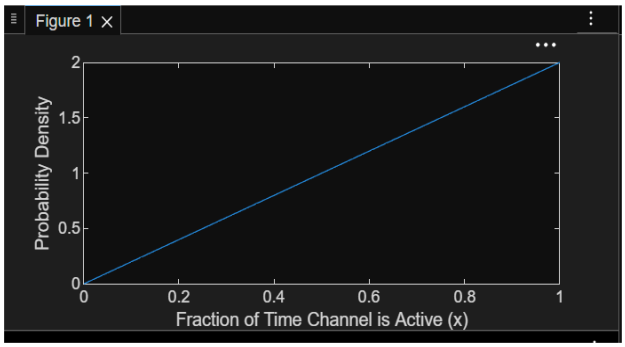

Figure G000319. UNIT 4 CONTINUED · source slide page 6.

EV0001049

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Theoretical Concept:

Computer Science

Problem: Time Taken to Compile 2 Software Packages

ScenarioA software compiler takes an average of 1 second to compile a single package of code. If a computer science student submits 2 identical packages to be compiled one after another, the total time t (in seconds) taken to finish the entire job follows a basic Gamma distribution where the shape parameter k = 2. The equation relies on the continuous Gamma function $\Gamma(k)$

$$f(t) = \frac{t \cdot e^{-t}}{\Gamma(k)}$$

Objective: Use MATLAB to verify $\Gamma(2)$ and plot the compilation time curve.

Figure G000320. UNIT 4 CONTINUED · source slide page 7.

EV0001052

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Matlab code:

```
% CS: 2-Package Software Compilation Delay
clear; clc;

% 1. Calculate the Gamma function constant
gamma_val = gamma(2)

% 2. Plot the compilation time curve
t = linspace(0, 8, 200); % Time scale from 0 to 8 seconds
f_t = (t .* exp(-t)) / gamma_val; % Evaluates to t*e^(-t)

plot(t, f_t, 'm-');
xlabel('Total Time (t) in Seconds');
ylabel('Probability Density');
```

Figure G000321. UNIT 4 CONTINUED · source slide page 8.

EV0001055

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Output:

```
gamma_val =

     1
```

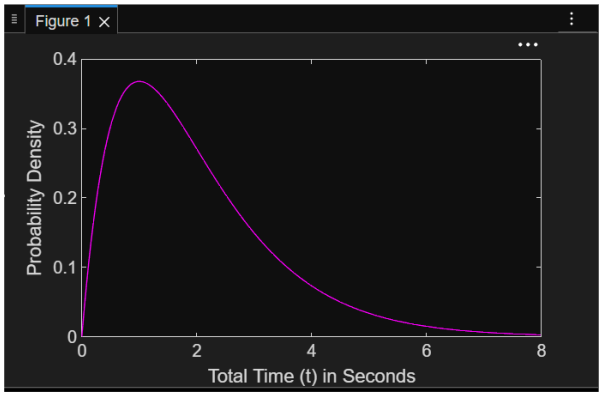

Figure G000322. UNIT 4 CONTINUED · source slide page 9.

EV0001058

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

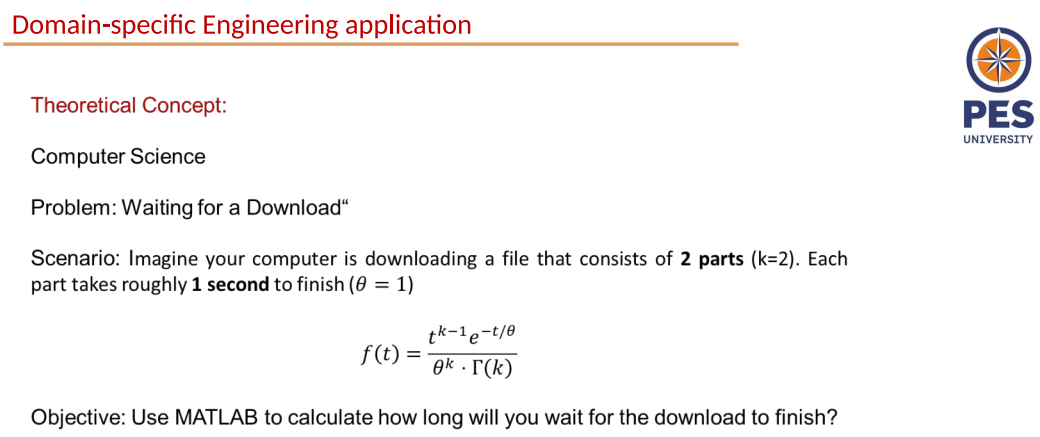

Figure G000323. UNIT 4 CONTINUED · source slide page 10.

EV0001061

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

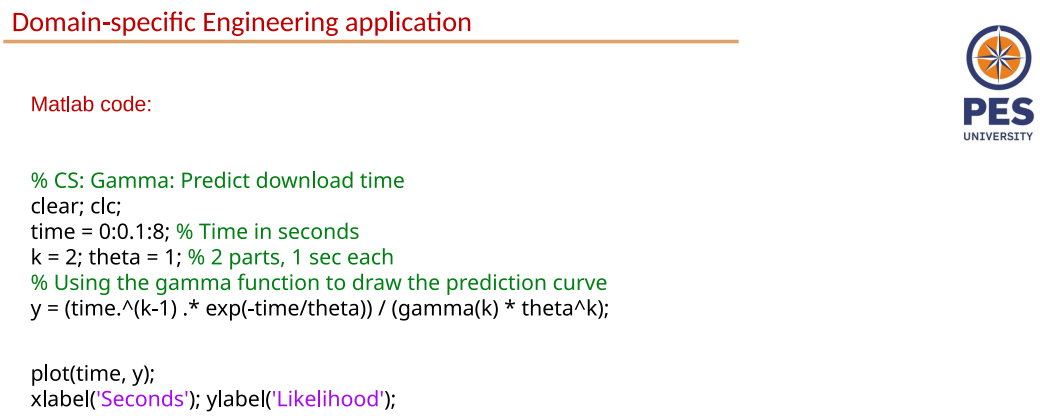

Figure G000324. UNIT 4 CONTINUED · source slide page 11.

EV0001064

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Output:

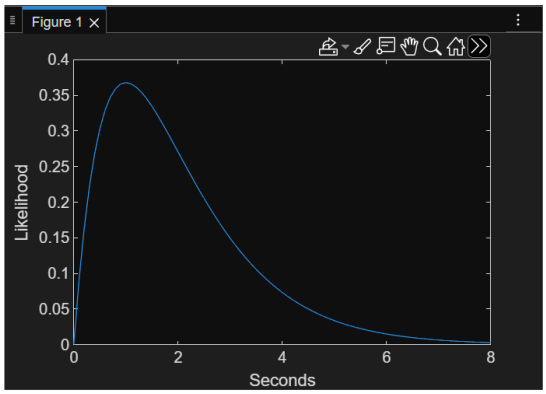

Figure G000325. UNIT 4 CONTINUED · source slide page 12.

EV0001067

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Theoretical Concept:

Computer Science

Problem: Will the User Click?

Scenario: You have a website with a "Buy Now" button. You want to predict the percentage of users who will click it (a number between 0% and 100%). You have seen 2 people click (a=2) and 2 people ignore it (b=2). The distribution is given by the Beta function ,

$$f(x) = \frac{1}{\beta(\mathrm{a},\mathrm{b})} x^{\mathrm{a}-1}(1-x)^{\mathrm{b}-1}$$

Objective: Use MATLAB to calculate the most likely click-rate for this button?

Figure G000326. UNIT 4 CONTINUED · source slide page 13.

EV0001070

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Matlab code:

```
% Beta: Predict click percentage
x= linspace(0, 1, 100);
a = 2; b = 2; % 2 clicks, 2 ignores

% Using the beta function to draw the likelihood curve
y = (x.^(a-1) .* (1-x).^(b-1)) / beta(a, b);

plot(x, y);
xlabel('Click Percentage (0 to 1)'); ylabel('Likelihood');
```

Figure G000327. UNIT 4 CONTINUED · source slide page 14.

EV0001073

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Domain-specific Engineering application

Output:

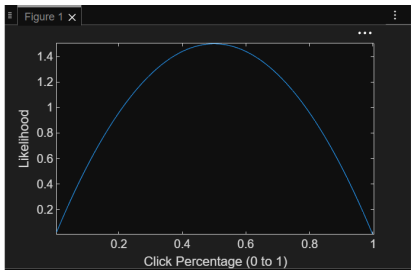

Figure G000328. UNIT 4 CONTINUED · source slide page 15.

EV0001076

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000329. UNIT 4 CONTINUED · source slide page 16.

EV0001079

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY ENGINEERING MATHEMATICS-I MATLAB Department of Science and Humanities EV0001033 · EV0001034 · EV0001035
2. Source slide page 2 (extracted text): Domain-specific Engineering application EV0001036 · EV0001037 · EV0001038
3. Source slide page 3 (extracted text): Domain-specific Engineering application Matlab code: % Electronics: Capacitor Lifespan Problem clear; clc; % Solve using the built-in gamma function % Since Gamma(2) = 1! = 1, the mean should be 4 (4,000 hours) mean_lifespan = (2^2) * gamma(2) Output: [Image text, OCR] mean_lifespan >X EV0001039 · EV0001040 · EV0001041
4. Source slide page 4 (extracted text): Domain-specific Engineering application EV0001042 · EV0001043 · EV0001044
5. Source slide page 5 (extracted text): Domain-specific Engineering application Matlab code: %Simple Digital Switch Active-Time Profile clear; clc; % 1. Calculate the Beta function constant beta_val = beta(2, 1) % 2. Plot the distribution line a=2; b=1; x = linspace(0, 1, 100); % Active ratio from 0 to 1 f_x = (x.^(a-1) .* (1-x).^(b-1)) / beta_val; % Evaluates to f(x) plot(x, f_x); xlabel('Fraction of Time Channel is Active (x)'); ylabel('Probability Density'); EV0001045 · EV0001046 · EV0001047
6. Source slide page 6 (extracted text): Domain-specific Engineering application Output: [Image text, OCR] Figure 1 x 1.5 0.5 0.2 0.4 0.6 0.8 FractionofTimeChannelisActive (× EV0001048 · EV0001049 · EV0001050

7. Source slide page 7 (extracted text): Domain-specific Engineering application EV0001051 · EV0001052 · EV0001053
8. Source slide page 8 (extracted text): Domain-specific Engineering application Matlab code: % CS: 2-Package Software Compilation Delay clear; clc; % 1. Calculate the Gamma function constant gamma_val = gamma(2) % 2. Plot the compilation time curve t = linspace(0, 8, 200); % Time scale from 0 to 8 seconds f_t = (t .* exp(-t)) / gamma_val; % Evaluates to t*e^(-t) plot(t, f_t, 'm-'); xlabel('Total Time (t) in Seconds'); ylabel('Probability Density'); EV0001054 · EV0001055 · EV0001056
9. Source slide page 9 (extracted text): Domain-specific Engineering application Output: [Image text, OCR] Figure 1 x 0.4 0.3 0.2 0.1 2 4 Total Time (t) in Seconds EV0001057 · EV0001058 · EV0001059
10. Source slide page 10 (extracted text): Domain-specific Engineering application EV0001060 · EV0001061 · EV0001062
11. Source slide page 11 (extracted text): Domain-specific Engineering application Matlab code: % CS: Gamma: Predict download time clear; clc; time = 0:0.1:8; % Time in seconds k = 2; theta = 1; % 2 parts, 1 sec each % Using the gamma function to draw the prediction curve y = (time.^(k-1) .* exp(-time/theta)) / (gamma(k) * theta^k); plot(time, y); xlabel('Seconds'); ylabel('Likelihood'); EV0001063 · EV0001064 · EV0001065
12. Source slide page 12 (extracted text): Domain-specific Engineering application Output: [Image text, OCR] 目 Figure 1 ✕ - 0.4 0.35 0.3 0.25 0.2 0.15 0.1 0.05 0 2 4 6 8 Seconds EV0001066 · EV0001067 · EV0001068
13. Source slide page 13 (extracted text): Domain-specific Engineering application EV0001069 · EV0001070 · EV0001071
14. Source slide page 14 (extracted text): Domain-specific Engineering application Matlab code: % Beta: Predict click percentage x= linspace(0, 1, 100); a = 2; b = 2; % 2 clicks, 2 ignores % Using the beta function to draw the likelihood curve y = (x.^(a-1) .* (1-x).^(b-1)) / beta(a, b); plot(x, y); xlabel('Click Percentage (0 to 1)'); ylabel('Likelihood'); EV0001072 · EV0001073 · EV0001074
15. Source slide page 15 (extracted text): Domain-specific Engineering application Output: [Image text, OCR] 三 Figure 1 x 1.4 1.2 0.8 0.6 0.4 0.2 0.2 0.4 0.6 0.8 ClickPercentage(0 to1) EV0001075 · EV0001076 · EV0001077
16. Source slide page 16 (extracted text): [Image text, OCR] PES UNIVERSITY THANK YOU EV0001078 · EV0001079 · EV0001080

SHORT-NOTE RECAP

Source deck preserved: UNIT 4 CONTINUED. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0001033

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Introduction to MATLAB Software transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Introduction to MATLAB Software transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/001_Introduction_to_MATLAB_Software__60472b361e5bb782.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Data types; relational operators; logical operators transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Data types; relational operators; logical operators transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/002_Data_types_relational_operators_logical_operators__915d8f716427b954.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
M-Files; Loop Control Statements; Conditional Statements transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9986; only the JSON segmentation supplies the displayed timestamps.
M-Files; Loop Control Statements; Conditional Statements transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/003_M-Files_Loop_Control_Statements_Conditional_Statements__d8fc5a0a5b8f6dbe.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Plotting of Curves (2D and 3D), Plot Labeling, Curve Labeling transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/004_Plotting_of_Curves_2D_and_3D_Plot_Labeling_Curve_Labeling__e96c8141e2597d5f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9998; only the JSON segmentation supplies the displayed timestamps.
Video transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/006_Video__bb969ddbec162496.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Video transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Video transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9997; only the JSON segmentation supplies the displayed timestamps.
Video transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/AV_Summaries/008_Video__c789c482b6a9141a.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Slides (converted PDF; original hash f50acf5797d9d8245e41ff64a366d774cd7e3e344e8cf08a17f91ff715d950f6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Slides (converted PDF; original hash 6ae797d02b605192a59363cd34046d94d555b15ec6da51461b7dc3575b6bf74f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
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Page-render source for Slides (converted PDF; original hash 9988a37be0a34576dd47bea022ec4b865fabb5177da0c2acbb6534e78056d068)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Slides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Slides (converted PDF; original hash 2e5770ab70ab393896a1c030e9188dd0b9bff0b94ee6896928faa6ebc98aaf4e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: MATLAB INTRODUCTION	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: MATLAB INTRODUCTION	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for MATLAB INTRODUCTION (converted PDF; original hash e6958cd9f18100fdd7d6aeaa16622f9e9dc13fd36b1538aeb854e3b5f44de850)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UNIT 1	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 1 (converted PDF; original hash b9b09a103cf726f6b5ce82506486e7950cc58cc08192d2b0282d209386130451)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 1 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 1 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 1 CONTINUED (converted PDF; original hash cbb14f9556b8abbb9d56b7ef343a57527b2bc4fdc9129f71b84d71b9a2af6bff)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UNIT 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 2 (converted PDF; original hash 125940f7b2e0b5146087b4d7dac7a42562e2a5ac72fcf50b71e36f58a1c3335f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 2 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 2 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 2 CONTINUED (converted PDF; original hash cc656367591e70ecee8c8c2bed5168c4be97bb5375bb0cd112c313c359a66be8)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UNIT 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 3 (converted PDF; original hash 878ecae3697008466dc626559d5d483e7430220a9c80ff9b52d3d11c0f58d20c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 3 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 3 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 3 CONTINUED (converted PDF; original hash 9700aa5026207dd1afba49fa12f067da4dd78a34edac2c8722499d9e00cf13c6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UNIT 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 4 (converted PDF; original hash 3b995ad93f2cb3ec614459bbc2a4ac4770165b8c527997a60e8257c0bb457442)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UNIT 4 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UNIT 4 CONTINUED	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UNIT 4 CONTINUED (converted PDF; original hash 1d2300b5e26a0738a13cd90afdb8aece95f5fb6bab9de84ca9db21e34a148039)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 4 rendered pages in 001_Slides__7ac00f797de85ee4.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 002_Slides__ef919841425356a4.slides.md have no readable `.slides.md` text; their original page images are included.

3. 1 rendered pages in 003_Slides__3d7af4a88d5b4490.slides.md have no readable `.slides.md` text; their original page images are included.
4. 1 rendered pages in 004_Slides__4150cd4c33be9a2c.slides.md have no readable `.slides.md` text; their original page images are included.
5. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
6. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
7. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
8. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
9. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
10. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV00000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV00000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0061 to s0119. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0120 to s0176. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0177 to s0235. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0177 to s0235. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0177 to s0235. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0177 to s0235. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0236 to s0291. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0236 to s0291. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0236 to s0291. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0236 to s0291. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00002/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0292 to s0303. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000023	SRC00003/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0292 to s0303. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000024	SRC00004/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0292 to s0303. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000025	SRC00005/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0292 to s0303. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000026	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000027	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000028	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000029	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000030	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000031	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000032	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000033	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000034	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000035	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000036	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000037	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0056. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000040	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0061. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0057 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000044	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000045	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000046	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000047	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000048	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000049	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000051	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000052	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000053	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0035 to s0051. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000054	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0052 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000055	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0052 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000056	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0052 to s0069. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000057	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0052 to s0069. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000058	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0070 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000059	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0070 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000060	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0070 to s0070. Sidecar variants compare at similarity 0.9986; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000061	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0070 to s0070. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000068	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000070	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000071	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000072	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000075	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0058 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00014/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000079	SRC00015/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000080	SRC00016/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0088. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000081	SRC00017/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0088. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000082	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000089	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000090	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000091	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000092	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000093	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000096	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0073. Sidecar variants compare at similarity 0.9998; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0073. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000102	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000103	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000104	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000105	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000106	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000107	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare at similarity 0.9997; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0055. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000110	SRC00026/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000111	SRC00028/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000112	SRC00027/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000113	SRC00026/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000114	SRC00028/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000115	SRC00027/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000116	SRC00026/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000117	SRC00028/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000118	SRC00027/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000119	SRC00026/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000120	SRC00028/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000121	SRC00027/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000122	SRC00026/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000123	SRC00028/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000124	SRC00027/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000125	SRC00026/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000126	SRC00028/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000127	SRC00027/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000128	SRC00026/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000129	SRC00028/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000130	SRC00027/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000131	SRC00026/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000132	SRC00028/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000133	SRC00027/p8	covered	Alternate extracted slide text page 8; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000134	SRC00026/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000135	SRC00028/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000136	SRC00027/p9	covered	Alternate extracted slide text page 9; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000137	SRC00026/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000138	SRC00028/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000139	SRC00027/p10	covered	Alternate extracted slide text page 10; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000140	SRC00026/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000141	SRC00028/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000142	SRC00027/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000143	SRC00026/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000144	SRC00028/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00001145	SRC00027/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00001146	SRC00026/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001147	SRC00028/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001148	SRC00027/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00001149	SRC00026/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001150	SRC00028/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001151	SRC00027/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000152	SRC00026/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000153	SRC00028/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000154	SRC00027/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000155	SRC00026/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000156	SRC00028/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000157	SRC00027/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000158	SRC00026/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00001 59	SRC00028/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001 60	SRC00027/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 61	SRC00026/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 62	SRC00028/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00001 63	SRC00027/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV00001 64	SRC00026/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 65	SRC00028/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000166	SRC00027/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000167	SRC00026/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000168	SRC00028/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000169	SRC00027/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000170	SRC00026/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000171	SRC00028/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000172	SRC00027/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000173	SRC00026/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000174	SRC00028/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000175	SRC00027/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000176	SRC00026/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000177	SRC00028/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000178	SRC00027/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV0000179	SRC00026/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000180	SRC00028/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000181	SRC00027/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000182	SRC00026/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000183	SRC00028/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000184	SRC00027/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000185	SRC00026/p26	covered	Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000186	SRC00028/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000187	SRC00027/p26	covered	Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending.
EV0000188	SRC00026/p27	covered	Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000189	SRC00028/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000190	SRC00027/p27	covered	Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending.
EV0000191	SRC00026/p28	covered	Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000192	SRC00028/p28	covered	Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000193	SRC00027/p28	covered	Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000194	SRC00026/p29	covered	Extracted slide page 29 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000195	SRC00028/p29	covered	Original slide page 29 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000196	SRC00027/p29	covered	Alternate extracted slide text page 29; normalized text matches the .slides.md page. Human comparison is pending.
EV0000197	SRC00026/p30	covered	Extracted slide page 30 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000198	SRC00028/p30	covered	Original slide page 30 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000199	SRC00027/p30	covered	Alternate extracted slide text page 30; normalized text matches the .slides.md page. Human comparison is pending.
EV0000200	SRC00026/p31	covered	Extracted slide page 31 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00002 01	SRC00028/p31	covered	Original slide page 31 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 02	SRC00027/p31	covered	Alternate extracted slide text page 31; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 03	SRC00026/p32	covered	Extracted slide page 32 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 04	SRC00028/p32	covered	Original slide page 32 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 05	SRC00027/p32	covered	Alternate extracted slide text page 32; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 06	SRC00026/p33	covered	Extracted slide page 33 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 07	SRC00028/p33	covered	Original slide page 33 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000208	SRC00027/p33	covered	Alternate extracted slide text page 33; normalized text matches the .slides.md page. Human comparison is pending.
EV0000209	SRC00026/p34	covered	Extracted slide page 34 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000210	SRC00028/p34	covered	Original slide page 34 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000211	SRC00027/p34	covered	Alternate extracted slide text page 34; normalized text matches the .slides.md page. Human comparison is pending.
EV0000212	SRC00026/p35	covered	Extracted slide page 35 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000213	SRC00028/p35	covered	Original slide page 35 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000214	SRC00027/p35	covered	Alternate extracted slide text page 35; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000215	SRC00026/p36	covered	Extracted slide page 36 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000216	SRC00028/p36	covered	Original slide page 36 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000217	SRC00027/p36	covered	Alternate extracted slide text page 36; normalized text matches the .slides.md page. Human comparison is pending.
EV0000218	SRC00026/p37	covered	Extracted slide page 37 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000219	SRC00028/p37	covered	Original slide page 37 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000220	SRC00027/p37	covered	Alternate extracted slide text page 37; normalized text matches the .slides.md page. Human comparison is pending.
EV0000221	SRC00026/p38	covered	Extracted slide page 38 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000222	SRC00028/p38	covered	Original slide page 38 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000223	SRC00027/p38	covered	Alternate extracted slide text page 38; normalized text matches the .slides.md page. Human comparison is pending.
EV0000224	SRC00026/p39	covered	Extracted slide page 39 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000225	SRC00028/p39	covered	Original slide page 39 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000226	SRC00027/p39	covered	Alternate extracted slide text page 39; normalized text matches the .slides.md page. Human comparison is pending.
EV0000227	SRC00026/p40	covered	Extracted slide page 40 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000228	SRC00028/p40	covered	Original slide page 40 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002 29	SRC00027/p40	covered	Alternate extracted slide text page 40; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 30	SRC00026/p41	covered	Extracted slide page 41 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 31	SRC00028/p41	covered	Original slide page 41 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 32	SRC00027/p41	covered	Alternate extracted slide text page 41; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 33	SRC00028/p42	covered	Original slide page 42 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 34	SRC00028/p43	covered	Original slide page 43 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002 35	SRC00028/p44	covered	Original slide page 44 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000236	SRC00028/p45	covered	Original slide page 45 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/001_Slides__7ac00f797de85ee4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000237	SRC00029/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000238	SRC00031/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000239	SRC00030/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000240	SRC00029/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000241	SRC00031/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000242	SRC00030/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000243	SRC00029/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000244	SRC00031/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000245	SRC00030/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000246	SRC00029/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000247	SRC00031/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000248	SRC00030/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000249	SRC00029/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000250	SRC00031/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000251	SRC00030/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000252	SRC00029/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000253	SRC00031/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000254	SRC00030/p6	covered	Alternate extracted slide text page 6; text differs from or is absent in the .slides.md page; alternate text is preserved separately. Human comparison is pending.
EV0000255	SRC00029/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000256	SRC00031/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000257	SRC00030/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000258	SRC00029/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000259	SRC00031/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000260	SRC00030/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000261	SRC00029/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000262	SRC00031/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000263	SRC00030/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000264	SRC00029/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000265	SRC00031/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000266	SRC00030/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000267	SRC00029/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000268	SRC00031/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000269	SRC00030/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000270	SRC00029/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000271	SRC00031/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000272	SRC00030/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000273	SRC00029/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000274	SRC00031/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000275	SRC00030/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000276	SRC00029/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000277	SRC00031/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00002278	SRC00030/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00002279	SRC00029/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002280	SRC00031/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002281	SRC00030/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00002282	SRC00029/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002283	SRC00031/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00002284	SRC00030/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000285	SRC00029/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000286	SRC00031/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000287	SRC00030/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000288	SRC00029/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000289	SRC00031/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000290	SRC00030/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000291	SRC00029/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000292	SRC00031/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000293	SRC00030/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000294	SRC00029/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000295	SRC00031/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000296	SRC00030/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000297	SRC00029/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000298	SRC00031/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000299	SRC00030/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000300	SRC00029/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000301	SRC00031/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000302	SRC00030/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000303	SRC00029/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000304	SRC00031/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000305	SRC00030/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000306	SRC00029/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000307	SRC00031/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000308	SRC00030/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000309	SRC00029/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000310	SRC00031/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00030/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000312	SRC00031/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000313	SRC00031/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/002_Slides__ef919841425356a4.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00032/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000315	SRC00034/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000316	SRC00033/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000317	SRC00032/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000318	SRC00034/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000319	SRC00033/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003320	SRC00032/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003321	SRC00034/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003322	SRC00033/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003323	SRC00032/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003324	SRC00034/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003325	SRC00033/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003326	SRC00032/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003327	SRC00034/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003328	SRC00033/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003329	SRC00032/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003330	SRC00034/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003331	SRC00033/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003332	SRC00032/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003333	SRC00034/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000334	SRC00033/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000335	SRC00032/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000336	SRC00034/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000337	SRC00033/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000338	SRC00032/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000339	SRC00034/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000340	SRC00033/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000341	SRC00032/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000342	SRC00034/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000343	SRC00033/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000344	SRC00032/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000345	SRC00034/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000346	SRC00033/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000347	SRC00032/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000348	SRC00034/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000349	SRC00033/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000350	SRC00032/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000351	SRC00034/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000352	SRC00033/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000353	SRC00032/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000354	SRC00034/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000355	SRC00033/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000356	SRC00032/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000357	SRC00034/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000358	SRC00033/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000359	SRC00032/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000360	SRC00034/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000361	SRC00033/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000362	SRC00032/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000363	SRC00034/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000364	SRC00033/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000365	SRC00032/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000366	SRC00034/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000367	SRC00033/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000368	SRC00032/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000369	SRC00034/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000370	SRC00033/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000371	SRC00032/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000372	SRC00034/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000373	SRC00033/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000374	SRC00032/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000375	SRC00034/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000376	SRC00033/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000377	SRC00032/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000378	SRC00034/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000379	SRC00033/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000380	SRC00032/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000381	SRC00034/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000382	SRC00033/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003383	SRC00034/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/003_Slides__3d7af4a88d5b4490.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003384	SRC00035/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003385	SRC00037/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003386	SRC00036/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003387	SRC00035/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003388	SRC00037/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003389	SRC00036/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003390	SRC00035/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003391	SRC00037/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003392	SRC00036/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003393	SRC00035/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003394	SRC00037/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003395	SRC00036/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003396	SRC00035/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003 97	SRC00037/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00003 98	SRC00036/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 99	SRC00035/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00037/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00036/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00035/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 03	SRC00037/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000404	SRC00036/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000405	SRC00035/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000406	SRC00037/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00036/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000408	SRC00035/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000409	SRC00037/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00036/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000411	SRC00035/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000412	SRC00037/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00036/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000414	SRC00035/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00037/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00036/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000417	SRC00035/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000418	SRC00037/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00036/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00035/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00037/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00036/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000423	SRC00035/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000424	SRC00037/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000425	SRC00036/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00035/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00037/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00036/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000429	SRC00035/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000430	SRC00037/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00036/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000432	SRC00035/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00037/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00036/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000435	SRC00035/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000436	SRC00037/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00036/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00035/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000439	SRC00037/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00036/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00035/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00037/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00036/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000444	SRC00035/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00037/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000446	SRC00036/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000447	SRC00035/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00037/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00036/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00035/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000451	SRC00037/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00036/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000453	SRC00035/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000454	SRC00037/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00036/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00035/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00037/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00036/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000459	SRC00035/p26	covered	Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000460	SRC00037/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000461	SRC00036/p26	covered	Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending.
EV0000462	SRC00035/p27	covered	Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00037/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00036/p27	covered	Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending.
EV0000465	SRC00035/p28	covered	Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000466	SRC00037/p28	covered	Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00004067	SRC00036/p28	covered	Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending.
EV00004068	SRC00035/p29	covered	Extracted slide page 29 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004069	SRC00037/p29	covered	Original slide page 29 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004070	SRC00036/p29	covered	Alternate extracted slide text page 29; normalized text matches the .slides.md page. Human comparison is pending.
EV00004071	SRC00037/p30	covered	Original slide page 30 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/004_Slides__4150cd4c33be9a2c.pdf; visual page included as vector figure. Human layout/content review is pending.
EV00004072	SRC00038/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004073	SRC00040/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000474	SRC00039/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000475	SRC00038/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000476	SRC00040/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000477	SRC00039/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000478	SRC00038/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000479	SRC00040/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000480	SRC00039/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000481	SRC00038/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000482	SRC00040/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000483	SRC00039/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000484	SRC00038/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000485	SRC00040/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000486	SRC00039/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000487	SRC00038/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000488	SRC00040/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000489	SRC00039/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000490	SRC00038/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000491	SRC00040/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000492	SRC00039/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000493	SRC00038/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000494	SRC00040/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000495	SRC00039/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000496	SRC00038/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000497	SRC00040/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000498	SRC00039/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000499	SRC00038/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000500	SRC00040/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000501	SRC00039/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000502	SRC00038/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000503	SRC00040/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000504	SRC00039/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000505	SRC00038/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000506	SRC00040/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000507	SRC00039/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000508	SRC00038/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000509	SRC00040/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000510	SRC00039/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 11	SRC00038/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 12	SRC00040/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 13	SRC00039/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 14	SRC00038/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 15	SRC00040/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 16	SRC00039/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000517	SRC00038/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000518	SRC00040/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000519	SRC00039/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000520	SRC00038/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000521	SRC00040/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000522	SRC00039/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000523	SRC00038/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000524	SRC00040/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000525	SRC00039/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000526	SRC00038/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000527	SRC00040/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000528	SRC00039/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000529	SRC00038/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000530	SRC00040/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000531	SRC00039/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000532	SRC00038/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000533	SRC00040/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000534	SRC00039/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000535	SRC00038/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000536	SRC00040/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000537	SRC00039/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000538	SRC00038/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000539	SRC00040/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000540	SRC00039/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000541	SRC00038/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000542	SRC00040/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000543	SRC00039/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000544	SRC00038/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000545	SRC00040/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000546	SRC00039/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000547	SRC00038/p26	covered	Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000548	SRC00040/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000549	SRC00039/p26	covered	Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending.
EV0000550	SRC00038/p27	covered	Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000551	SRC00040/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000552	SRC00039/p27	covered	Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000553	SRC00038/p28	covered	Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000554	SRC00040/p28	covered	Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000555	SRC00039/p28	covered	Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending.
EV0000556	SRC00038/p29	covered	Extracted slide page 29 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000557	SRC00040/p29	covered	Original slide page 29 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000558	SRC00039/p29	covered	Alternate extracted slide text page 29; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000559	SRC00038/p30	covered	Extracted slide page 30 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000560	SRC00040/p30	covered	Original slide page 30 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/005_MATLAB_INTRODUCTION__fd21aaadeb7e04a5.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000561	SRC00039/p30	covered	Alternate extracted slide text page 30; normalized text matches the .slides.md page. Human comparison is pending.
EV0000562	SRC00041/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000563	SRC00043/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000564	SRC00042/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000565	SRC00041/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000566	SRC00043/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000567	SRC00042/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000568	SRC00041/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000569	SRC00043/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000570	SRC00042/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000571	SRC00041/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000572	SRC00043/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000573	SRC00042/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000574	SRC00041/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000575	SRC00043/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000576	SRC00042/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000577	SRC00041/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000578	SRC00043/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000579	SRC00042/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000580	SRC00041/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000581	SRC00043/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000582	SRC00042/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000583	SRC00041/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000584	SRC00043/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000585	SRC00042/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000586	SRC00041/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000587	SRC00043/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000588	SRC00042/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000589	SRC00041/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000590	SRC00043/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000591	SRC00042/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000592	SRC00041/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000593	SRC00043/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000594	SRC00042/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000595	SRC00041/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000596	SRC00043/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000597	SRC00042/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000598	SRC00041/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000599	SRC00043/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000600	SRC00042/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000601	SRC00041/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000602	SRC00043/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000603	SRC00042/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000604	SRC00041/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000605	SRC00043/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000606	SRC00042/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000607	SRC00041/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000608	SRC00043/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000609	SRC00042/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000610	SRC00041/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000611	SRC00043/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000612	SRC00042/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000613	SRC00041/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000614	SRC00043/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000615	SRC00042/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000616	SRC00041/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000617	SRC00043/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000618	SRC00042/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000619	SRC00041/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000620	SRC00043/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000621	SRC00042/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000622	SRC00041/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000623	SRC00043/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000624	SRC00042/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000625	SRC00041/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000626	SRC00043/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000627	SRC00042/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000628	SRC00041/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000629	SRC00043/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000630	SRC00042/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV0000631	SRC00041/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000632	SRC00043/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000633	SRC00042/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000634	SRC00041/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000635	SRC00043/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000636	SRC00042/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000637	SRC00041/p26	covered	Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000638	SRC00043/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000639	SRC00042/p26	covered	Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending.
EV0000640	SRC00041/p27	covered	Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000641	SRC00043/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000642	SRC00042/p27	covered	Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000643	SRC00041/p28	covered	Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000644	SRC00043/p28	covered	Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000645	SRC00042/p28	covered	Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending.
EV0000646	SRC00041/p29	covered	Extracted slide page 29 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000647	SRC00043/p29	covered	Original slide page 29 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000648	SRC00042/p29	covered	Alternate extracted slide text page 29; normalized text matches the .slides.md page. Human comparison is pending.
EV0000649	SRC00041/p30	covered	Extracted slide page 30 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00006 50	SRC00043/p30	covered	Original slide page 30 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 51	SRC00042/p30	covered	Alternate extracted slide text page 30; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 52	SRC00041/p31	covered	Extracted slide page 31 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 53	SRC00043/p31	covered	Original slide page 31 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 54	SRC00042/p31	covered	Alternate extracted slide text page 31; normalized text matches the .slides.md page. Human comparison is pending.
EV00006 55	SRC00041/p32	covered	Extracted slide page 32 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 56	SRC00043/p32	covered	Original slide page 32 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000657	SRC00042/p32	covered	Alternate extracted slide text page 32; normalized text matches the .slides.md page. Human comparison is pending.
EV0000658	SRC00041/p33	covered	Extracted slide page 33 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000659	SRC00043/p33	covered	Original slide page 33 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000660	SRC00042/p33	covered	Alternate extracted slide text page 33; normalized text matches the .slides.md page. Human comparison is pending.
EV0000661	SRC00041/p34	covered	Extracted slide page 34 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000662	SRC00043/p34	covered	Original slide page 34 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000663	SRC00042/p34	covered	Alternate extracted slide text page 34; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000664	SRC00041/p35	covered	Extracted slide page 35 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000665	SRC00043/p35	covered	Original slide page 35 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000666	SRC00042/p35	covered	Alternate extracted slide text page 35; normalized text matches the .slides.md page. Human comparison is pending.
EV0000667	SRC00041/p36	covered	Extracted slide page 36 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000668	SRC00043/p36	covered	Original slide page 36 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000669	SRC00042/p36	covered	Alternate extracted slide text page 36; normalized text matches the .slides.md page. Human comparison is pending.
EV0000670	SRC00041/p37	covered	Extracted slide page 37 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000671	SRC00043/p37	covered	Original slide page 37 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000672	SRC00042/p37	covered	Alternate extracted slide text page 37; normalized text matches the .slides.md page. Human comparison is pending.
EV0000673	SRC00041/p38	covered	Extracted slide page 38 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000674	SRC00043/p38	covered	Original slide page 38 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/006_UNIT_1__f222ddf91e985905.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000675	SRC00042/p38	covered	Alternate extracted slide text page 38; normalized text matches the .slides.md page. Human comparison is pending.
EV0000676	SRC00044/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000677	SRC00046/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000678	SRC00045/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000679	SRC00044/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000680	SRC00046/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000681	SRC00045/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000682	SRC00044/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000683	SRC00046/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000684	SRC00045/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000685	SRC00044/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000686	SRC00046/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000687	SRC00045/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000688	SRC00044/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000689	SRC00046/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000690	SRC00045/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000691	SRC00044/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000692	SRC00046/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000693	SRC00045/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000694	SRC00044/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000695	SRC00046/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000696	SRC00045/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00006 97	SRC00044/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006 98	SRC00046/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006 99	SRC00045/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 00	SRC00044/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 01	SRC00046/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 02	SRC00045/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000703	SRC00044/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000704	SRC00046/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000705	SRC00045/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000706	SRC00044/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000707	SRC00046/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000708	SRC00045/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 09	SRC00044/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 10	SRC00046/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 11	SRC00045/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00007 12	SRC00044/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007 13	SRC00046/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007 14	SRC00045/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000715	SRC00044/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000716	SRC00046/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/007_UNIT_1__CONTINUED__0383b93ca2115cf4.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000717	SRC00045/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000718	SRC00047/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000719	SRC00049/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000720	SRC00048/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000721	SRC00047/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000722	SRC00049/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000723	SRC00048/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000724	SRC00047/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000725	SRC00049/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000726	SRC00048/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000727	SRC00047/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000728	SRC00049/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00007729	SRC00048/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00007730	SRC00047/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007731	SRC00049/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007732	SRC00048/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00007733	SRC00047/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007734	SRC00049/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007735	SRC00048/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007736	SRC00047/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007737	SRC00049/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007738	SRC00048/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00007739	SRC00047/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007740	SRC00049/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007741	SRC00048/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00007742	SRC00047/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000743	SRC00049/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000744	SRC00048/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000745	SRC00047/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000746	SRC00049/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000747	SRC00048/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000748	SRC00047/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000749	SRC00049/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000750	SRC00048/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000751	SRC00047/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000752	SRC00049/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000753	SRC00048/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000754	SRC00047/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000755	SRC00049/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000756	SRC00048/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000757	SRC00047/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000758	SRC00049/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000759	SRC00048/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000760	SRC00047/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000761	SRC00049/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000762	SRC00048/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000763	SRC00047/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000764	SRC00049/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000765	SRC00048/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000766	SRC00047/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000767	SRC00049/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000768	SRC00048/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000769	SRC00047/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000770	SRC00049/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000771	SRC00048/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000772	SRC00047/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000773	SRC00049/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000774	SRC00048/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000775	SRC00047/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000776	SRC00049/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000777	SRC00048/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000778	SRC00047/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000779	SRC00049/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000780	SRC00048/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV0000781	SRC00047/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000782	SRC00049/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000783	SRC00048/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV0000784	SRC00047/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000785	SRC00049/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000786	SRC00048/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV0000787	SRC00047/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000788	SRC00049/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000789	SRC00048/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000790	SRC00047/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000791	SRC00049/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/008_UNIT_2__f6f6cdca7604931f.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000792	SRC00048/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000793	SRC00050/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000794	SRC00052/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000795	SRC00051/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000796	SRC00050/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000797	SRC00052/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000798	SRC00051/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007 99	SRC00050/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 00	SRC00052/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 01	SRC00051/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 02	SRC00050/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 03	SRC00052/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008 04	SRC00051/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 05	SRC00050/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00008006	SRC00052/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008007	SRC00051/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00008008	SRC00050/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008009	SRC00052/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008010	SRC00051/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00008011	SRC00050/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008012	SRC00052/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000813	SRC00051/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000814	SRC00050/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000815	SRC00052/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000816	SRC00051/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000817	SRC00050/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000818	SRC00052/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000819	SRC00051/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00008020	SRC00050/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008021	SRC00052/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008022	SRC00051/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00008023	SRC00050/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008024	SRC00052/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008025	SRC00051/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00008026	SRC00050/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000827	SRC00052/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000828	SRC00051/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000829	SRC00050/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000830	SRC00052/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000831	SRC00051/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000832	SRC00050/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000833	SRC00052/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000834	SRC00051/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000835	SRC00050/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000836	SRC00052/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000837	SRC00051/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000838	SRC00050/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000839	SRC00052/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000840	SRC00051/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000841	SRC00050/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000842	SRC00052/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000843	SRC00051/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000844	SRC00050/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000845	SRC00052/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000846	SRC00051/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000847	SRC00050/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000848	SRC00052/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000849	SRC00051/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000850	SRC00050/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000851	SRC00052/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000852	SRC00051/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000853	SRC00050/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000854	SRC00052/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00008555	SRC00051/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV00008556	SRC00050/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008557	SRC00052/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008558	SRC00051/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV00008559	SRC00050/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008560	SRC00052/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008561	SRC00051/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000862	SRC00050/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000863	SRC00052/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/009_UNIT_2_CONTINUED__8cc940fbfbd000f3.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000864	SRC00051/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.
EV0000865	SRC00053/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000866	SRC00055/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000867	SRC00054/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000868	SRC00053/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00008069	SRC00055/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008070	SRC00054/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00008071	SRC00053/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008072	SRC00055/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008073	SRC00054/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00008074	SRC00053/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008075	SRC00055/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00008076	SRC00054/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00008077	SRC00053/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008078	SRC00055/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008079	SRC00054/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00008080	SRC00053/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008081	SRC00055/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008082	SRC00054/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000883	SRC00053/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000884	SRC00055/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000885	SRC00054/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000886	SRC00053/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000887	SRC00055/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000888	SRC00054/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000889	SRC00053/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00008090	SRC00055/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008091	SRC00054/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00008092	SRC00053/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008093	SRC00055/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00008094	SRC00054/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00008095	SRC00053/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008096	SRC00055/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00008 97	SRC00054/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00008 98	SRC00053/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00008 99	SRC00055/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 00	SRC00054/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00009 01	SRC00053/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009 02	SRC00055/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 03	SRC00054/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000904	SRC00053/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000905	SRC00055/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000906	SRC00054/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000907	SRC00053/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000908	SRC00055/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000909	SRC00054/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000910	SRC00053/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000911	SRC00055/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000912	SRC00054/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000913	SRC00053/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000914	SRC00055/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000915	SRC00054/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000916	SRC00053/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000917	SRC00055/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000918	SRC00054/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000919	SRC00053/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000920	SRC00055/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000921	SRC00054/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000922	SRC00053/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000923	SRC00055/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000924	SRC00054/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00009325	SRC00053/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009326	SRC00055/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009327	SRC00054/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV00009328	SRC00053/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009329	SRC00055/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/010_UNIT_3__f7e5447dacd289b1.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009330	SRC00054/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.
EV00009331	SRC00056/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000932	SRC00058/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000933	SRC00057/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000934	SRC00056/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000935	SRC00058/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000936	SRC00057/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000937	SRC00056/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000938	SRC00058/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000939	SRC00057/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000940	SRC00056/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000941	SRC00058/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000942	SRC00057/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000943	SRC00056/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000944	SRC00058/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000945	SRC00057/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00009946	SRC00056/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009947	SRC00058/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009948	SRC00057/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00009949	SRC00056/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009950	SRC00058/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009951	SRC00057/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00009952	SRC00056/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00009 53	SRC00058/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 54	SRC00057/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00009 55	SRC00056/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009 56	SRC00058/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 57	SRC00057/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00009 58	SRC00056/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009 59	SRC00058/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000960	SRC00057/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000961	SRC00056/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000962	SRC00058/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000963	SRC00057/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000964	SRC00056/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000965	SRC00058/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000966	SRC00057/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00009967	SRC00056/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009968	SRC00058/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009969	SRC00057/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00009970	SRC00056/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009971	SRC00058/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009972	SRC00057/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00009973	SRC00056/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00009974	SRC00058/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009975	SRC00057/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV00009976	SRC00056/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009977	SRC00058/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/011_UNIT_3_CONTINUED__536b5cc4f2ea6730.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009978	SRC00057/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00009979	SRC00059/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009980	SRC00061/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00009081	SRC00060/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00009082	SRC00059/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009083	SRC00061/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009084	SRC00060/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00009085	SRC00059/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009086	SRC00061/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009087	SRC00060/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00009988	SRC00059/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009989	SRC00061/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009990	SRC00060/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00009991	SRC00059/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009992	SRC00061/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009993	SRC00060/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00009994	SRC00059/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00009 95	SRC00061/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 96	SRC00060/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00009 97	SRC00059/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00009 98	SRC00061/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00009 99	SRC00060/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00010 00	SRC00059/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00010 01	SRC00061/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0001002	SRC00060/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0001003	SRC00059/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001004	SRC00061/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001005	SRC00060/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0001006	SRC00059/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001007	SRC00061/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001008	SRC00060/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0001009	SRC00059/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001010	SRC00061/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001011	SRC00060/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0001012	SRC00059/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001013	SRC00061/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001014	SRC00060/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0001015	SRC00059/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001016	SRC00061/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001017	SRC00060/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0001018	SRC00059/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001019	SRC00061/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001020	SRC00060/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0001021	SRC00059/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001022	SRC00061/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0001023	SRC00060/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0001024	SRC00059/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001025	SRC00061/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001026	SRC00060/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0001027	SRC00059/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001028	SRC00061/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001029	SRC00060/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00010030	SRC00059/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00010031	SRC00061/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/012_UNIT_4__ebad8f42d7e98f9e.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010032	SRC00060/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV00010033	SRC00062/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00010034	SRC00064/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010035	SRC00063/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00010036	SRC00062/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001037	SRC00064/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001038	SRC00063/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0001039	SRC00062/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001040	SRC00064/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001041	SRC00063/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0001042	SRC00062/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001043	SRC00064/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001044	SRC00063/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0001045	SRC00062/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001046	SRC00064/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001047	SRC00063/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0001048	SRC00062/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001049	SRC00064/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001050	SRC00063/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0001051	SRC00062/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001052	SRC00064/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001053	SRC00063/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0001054	SRC00062/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001055	SRC00064/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001056	SRC00063/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0001057	SRC00062/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001058	SRC00064/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001059	SRC00063/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0001060	SRC00062/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00010061	SRC00064/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010062	SRC00063/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00010063	SRC00062/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00010064	SRC00064/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010065	SRC00063/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00010066	SRC00062/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00010067	SRC00064/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010068	SRC00063/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV00010069	SRC00062/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00010070	SRC00064/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010071	SRC00063/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV00010072	SRC00062/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0001073	SRC00064/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001074	SRC00063/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0001075	SRC00062/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0001076	SRC00064/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0001077	SRC00063/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0001078	SRC00062/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00010079	SRC00064/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_05/Slides/013_UNIT_4__CONTINUED__c425089f8dc0fd73.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00010080	SRC00063/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

C Revision and retrieval practice